Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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Abstract

[To be continued. . .]

- Appendix on s-wave SC ghost simulations on HM
- Appendix on triplet pairing in MFT self-consistent scheme
- Set ComputerModern font in all CairoMakie figures

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List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
$\Delta^{(\gamma)}$	γ-wave component of the gap function
DoF	Degree of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
iHF	Iterative Hartree-Fock
LRT	Linear Response Theory
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
PH	Particle-hole
PHS	Particle-hole symmetry
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Introduction

This thesis project is about my favorite ice cream flavor. [To be continued...]

Chapter 1

Introduction to the Extended Hubbard Model

A major breakthrough in Condensed Matter Physics was the discovery of superconductivity in two-dimensional copper-oxide based compounds (often referred to as "cuprates"). Until 1986, when Bednorz and Muller found evidence of a superconducting transition in Ba-La-Cu-O systems [5], superconductivity had been observed up to critical temperatures (T_c) around 30 K. The new class of copper oxides rapidly took over as highest-T_c class of superconductors, dramatically increasing critical temperatures to around 130 K at ambient pressure. Today, the highest observed value is $T_c \approx 151$ K at ambient pressure on the cuprate $HgBa_2Ca_2Cu_3O_{8+\delta}$ [6]. It is worth mentioning that this result was obtained in recent months after almost 30 years from the previous record.

Cuprates Physics is particularly interesting and surprising also considering their very poor conducting properties at normal phase. Copper oxides are basically ceramics, highly prone to establishing insulating antiferromagnetic phases. Antiferromagnetism and superconductivity are apparently antithetical orderings: the first arises from electronic repulsion, while the second from attractive interactions [7]. It is widely observed, however, how in cuprates the two phases can compete **and even coexist** [8].

Moreover, cuprates superconduct through a new, not-yet understood mechanism. Classical Bardeen-Cooper-Shrieffer (BCS) theory of superconductivity describes the arising of a phonon-induced effective electron-electron attractive interaction as a source of so-called "conventional superconductivity". It is the interplay of electrons with lattice vibrations that couples them in Cooper pairs. The same theory fails completely in explaining this new "unconventional superconductivity". All of these observation have gathered, in years, a lot of interest in these materials.

Such complex materials are surprisingly well described in many features by the simple (but rich) single-band Hubbard model (HM) [8],

$$\hat{H}_{\text{HM}} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \qquad t, U > 0 \tag{1.1}$$

being t the hopping energy and U the on-site repulsion. Sum is intended on the spin index $\sigma \in \{\uparrow, \downarrow\}$ and the sites i of a two-dimensional square lattice. The notation $\langle ij \rangle$ indicates nearest neighbours (NNs). Recent developments [9] have shown, however, that a more accurate description of these materials could be obtained by adding a non-local electron-electron attraction. This leads us to a widely studied modified version of Eq. (1.1),

$$\hat{H}_{\text{EHM}} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{n}_{i\sigma} \hat{n}_{j\sigma'} \qquad t, U, V > 0 \tag{1.2}$$

commonly referred to as "Extended Hubbard model" (EHM). The difference with the conventional HM is the presence of a NNs attractive interaction, independent of site and spin indices. In this thesis, we study the insurgence of antiferromagnetism and superconductivity in the EHM due to the presence of this new interaction. To do so, we use numerical Hartree-Fock (HF) methods.

This introductory chapter is organised as follows: [To be continued...]

Chapter 2

Introduction to Hartree-Fock methods

In this chapter we give a technical introduction to the Hartree-Fock (HF) method and its numerical implementation. We will keep this discussion as general as possible, without referring directly to the EHM.

The HF method is a variational approach to approximate the ground-state wavefunction of a many-body hamiltonian. The wavefunction is determined by minimising energy, limiting our search to a specific subspace of states. The general idea is to approximate the real ground-state with the best possible many-body state that is an eigenstate of a non-interacting fermionic hamiltonian.

To start, we need to define what Hilbert subspace we work with. Then we need to identify a way of parametrising the possible non-interacting hamiltonians and their respective ground-states, and finally a method to choose the best approximation to our original model. This is done in Sec. 2.2, largely based on [29, 31].

Variational algorithms generally work by numeric minimisation of some given functional. This is not the case. We implement HF theory starting from a theoretical minimisation procedure, with some derivative being set to zero, and then prove that this procedure leads to solving a self-consistency problem. We end up with a set of equations where the right-hand side depends non-trivially on the left-hand side, and through an iterative approach we determine the solution. This procedure is sketched in detail in Sec. 2.3.

2.1 Prerequisites and generalities

In this short section we give some motivation to the choice of the HF method, before moving to a technical explanation about how it works. As we said before, we are performing a constrained search for an approximation to the ground-state wavefunction.

Let $\hat{H}$ be a generic hamiltonian, and $\mathcal{H}$ the Hilbert space. For any many-body state $|\Psi\rangle \in \mathcal{H}$, the energy is given by the expectation value (EV)

$$E[\Psi] = \langle\Psi|\hat{H}|\Psi\rangle$$

Let $\mathcal{S} \subset \mathcal{H}$ be a subspace of states, ad let $|\Psi\rangle$ be a state inside that locally minimises energy,

$$|\Psi\rangle \in \mathcal{S} \qquad \frac{\delta E[\Psi]}{\delta\,|\Psi\rangle} = 0$$

If the true ground state does not belong to $\mathcal{S}$, the found local minimum $|\Psi\rangle$ approximates it. The energy $E[\Psi]$ is, in this case, bigger than the actual ground-state energy. In Fig. 2.1 we give a pictorial representation of this idea: the state $|\Psi\rangle \in \mathcal{S}$ is an approximation, found by constrained minimisation, of the true ground-state.

In HF methods, we identify $\mathcal{S}$ with the set of so-called “gaussian states”. These are all the ground-states of non-interacting hamiltonians. We give a complete description of them in App. B. We restrict minimisation to gaussian states because we can use Wick’s theorem to simplify significantly the computation of expectation values. Indeed, by averaging a quartic operator over a

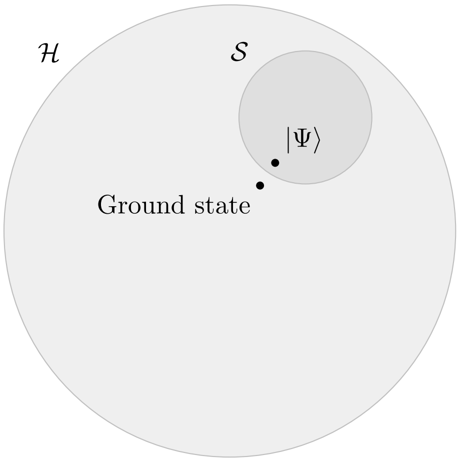


Figure 2.1 Pictorial representation of the search for the best approximation $|\Psi\rangle$ to the true ground-state inside the subspace $\mathcal{S} \subset \mathcal{H}$.

gaussian state, we get the exact decomposition

$$\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta \rangle \overset{\mathrm{W}}{=} \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}_\delta \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\gamma \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \langle \hat{c}^\dagger_\gamma \hat{c}_\delta \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \langle \hat{c}_\gamma \hat{c}_\delta \rangle}_{\text{Cooper}} \tag{2.1}$$

The subscripts indicate the names we will give to these terms throughout the text. The notation $\overset{\mathrm{W}}{=}$ indicates that we are using Wick's theorem.

Let us rephrase: the HF method is a minimisation constrained to all states that satisfy exactly Eq. (2.1). While this method simplifies a lot the computation of expectation values (reducing them to products of quadratic operators), the drawback is that the ground-state approximation is rather rough. Despite the roughness, HF methods are able to tackle various physical phenomena.

Operatively, we are approximating the hamiltonian with a non-interacting hamiltonian

$$\hat{H} \overset{\mathrm{HF}}{\simeq} \hat{H}_{\mathrm{HF}} = \sum_\alpha E_\alpha \hat{\gamma}^\dagger_\alpha \hat{\gamma}_\alpha$$

The notation $\overset{\mathrm{HF}}{\simeq}$ indicates that this operatorial approximation only holds inside the subspace of gaussian states, and is explained rigorously in next paragraph. To determine the energies E_α as well as the fermionic operators $\hat{\gamma}_\alpha$ is the point of the HF method, and the result of minimisation procedure. These operators satisfy Fermi anti-commutation rules,

$$\left\{ \hat{\gamma}_\alpha, \hat{\gamma}^\dagger_\beta \right\} = \delta_{\alpha\beta}$$

and are connected to the conventional electronic operators $\hat{c}_\alpha$ through an unitary map. This is discussed in detail in Sec. B.3.

Let us explain the approximation $\overset{\mathrm{HF}}{\simeq}$. When applying the HF method, we are not approximating an operator, but a wavefunction. The operatorial approximation is a shorthand notation that must be intended as follows: $\hat{H}_{\mathrm{HF}}$ is that non-interacting operator whose ground-state is the best approximation to the ground-state of $\hat{H}$. This does not mean that in general the two operators are related.

2.2 Sketch of HF variational method

We have identified the features of the states we use to approximate the model ground-state. The central property of these states is expressed by Eq. (2.1). Now, let us get to the core of HF method and give a technical introduction on how it works. Consider the generic quartic hamiltonian:

$$\hat{H} = \sum_{\alpha,\beta} A_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}_\beta + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta \tag{2.2}$$

where

$$A_{\alpha\beta} = A^*_{\beta\alpha} \qquad B_{\alpha\beta\gamma\delta} = B^*_{\delta\gamma\beta\alpha} \tag{2.3}$$

to ensure hermiticity. Moreover, the model must show the following symmetry:

$$B_{\alpha\beta\gamma\delta} = B_{\beta\alpha\delta\gamma} \tag{2.4}$$

because exchanging the position of the two annihilations and the two creations, the operator must remain the same.

Our aim is to approximate the above hamiltonian with a non-interacting hamiltonian. The most general quadratic hamiltonian is given by:

$$\hat{H} \simeq \hat{H}_{\text{HF}} \equiv E_0 + \sum_{\alpha,\beta} k_{\alpha\beta}\hat{\gamma}^\dagger_\alpha \hat{\gamma}_\beta \tag{2.5}$$

As is specified in Sec. B.3, γ and c operators must be linked by an unitary map. To find the best approximation to the original hamiltonian means to identify:

- the offset E_0;
- the fermionic operators $\hat{\gamma}_\alpha$, $\hat{\gamma}^\dagger_\alpha$;
- the $k_{\alpha\beta}$ coefficients.

Note that, in terms of algebraic solution, E_0 is irrelevant. To determine it correctly is only relevant in order to get the correct approximation of the actual ground-state energy. Let us discuss briefly the features of $\hat{H}_{\text{HF}}$ and then propose a solution.

2.2.1 Properties of the target quadratic hamiltonian

In this short section we enumerate some useful properties of $\hat{H}_{\text{HF}}$. We will use also the following notation:

$$\hat{a}^-_\alpha \equiv \hat{c}_\alpha \qquad \hat{a}^+_\alpha \equiv \hat{c}^\dagger_\alpha$$

Due to linearity of the unitary mapping, the operator $\hat{H}_{\text{HF}}$ of Eq. (2.5) can also be expressed as the most general quadratic hamiltonians for electronic operators:

$$\hat{H}_{\text{HF}} = E_0 + \sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta}\hat{a}^p_\alpha \hat{a}^q_\beta \tag{2.6}$$

By changing basis from $\hat{\gamma}$ to $\hat{c}$ operators, we moved all our ignorance into the new $h^{pq}_{\alpha\beta}$ fields.

We can infer some mathematical properties of the fields $h^{pq}_{\alpha\beta}$. First, note that the hamiltonian can be represented through Nambu spinors as

$$\begin{aligned} \hat{H}_{\text{HF}} &= E_0 + \hat{\Phi}^\dagger \mathbf{h} \hat{\Phi} \\ &= E_0 + \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^\dagger \end{bmatrix}^\dagger \begin{bmatrix} h^{+-} & h^{++} \\ h^{--} & h^{-+} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^\dagger \end{bmatrix} \end{aligned}$$

Our hamiltonian must be hermitian. We conclude that:

$$\hat{H}^\dagger_{\text{HF}} = \hat{H}_{\text{HF}} \qquad \Longrightarrow \qquad h^{++} = \left[h^{--}\right]^\dagger$$

which reads, element-wise:

$$h^{++}_{\alpha\beta} = \left(h^{--}_{\beta\alpha}\right)^* \tag{2.7}$$

Same-sign off-diagonal fields are linked.

We can set, without loss of generality,

$$\operatorname{Tr} h^{+-} + \operatorname{Tr} h^{-+} = 0$$

This passage is guaranteed by the fact that if $\mathbf{h}$ is not traceless, we can redefine it by subtracting its trace and including the latter in the energy shift E_0. Then

$$\begin{aligned}
\hat{H}_{\mathrm{HF}} &= E_0 + \sum_{\alpha,\beta} \left[h^{+-}_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}_\beta + h^{-+}_{\alpha\beta} \hat{c}_\alpha \hat{c}^\dagger_\beta \right] + (\text{same sign terms}) \\
&= E_0 + \sum_{\alpha,\beta} \left[h^{+-}_{\alpha\beta} \left(\delta_{\alpha\beta} - \hat{c}_\beta \hat{c}^\dagger_\alpha \right) + h^{-+}_{\alpha\beta} \left(\delta_{\alpha\beta} - \hat{c}^\dagger_\beta \hat{c}_\alpha \right) \right] + (\text{same sign terms}) \\
&= E_0 - \sum_{\alpha,\beta} \left[h^{+-}_{\beta\alpha} \hat{c}_\alpha \hat{c}^\dagger_\beta + h^{-+}_{\beta\alpha} \hat{c}^\dagger_\alpha \hat{c}_\beta \right] + (\text{same sign terms})
\end{aligned}$$

In second row we got rid of the traces sum and exchanged indices $\alpha \leftrightarrow \beta$. Confronting last and first row we conclude:

$$h^{+-}_{\beta\alpha} = -h^{-+}_{\alpha\beta} \tag{2.8}$$

thus, also on-diagonal fields are linked. We are allowed to determine, say, just h^{+-}: its counterpart h^{-+} is completely determined. These properties were all we needed to get to the main argument of the method.

2.2.2 Self-consistent variational solution

We follow and extend the variational scheme proposed by Fabrizio [31]. Let:

$$\Delta^{pq}_{\alpha\beta} \equiv \left\langle \hat{a}^p_\alpha \hat{a}^q_\beta \right\rangle$$

thus, explicitly

$$\Delta^{+-}_{\alpha\beta} = \left\langle \hat{c}^\dagger_\alpha \hat{c}_\beta \right\rangle \qquad \Delta^{-+}_{\alpha\beta} = \left\langle \hat{c}_\alpha \hat{c}^\dagger_\beta \right\rangle \qquad \Delta^{++}_{\alpha\beta} = \left\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \right\rangle \qquad \Delta^{--}_{\alpha\beta} = \left\langle \hat{c}_\alpha \hat{c}_\beta \right\rangle \tag{2.9}$$

The first two are “normal densities”; the second two are “anomalous densities”. Conjugation of the last two gives a symmetry relation similar to Eq. (2.7):

$$\left(\Delta^{++}_{\alpha\beta} \right)^\dagger = \Delta^{--}_{\beta\alpha} \tag{2.10}$$

The HF ground-state energy is given by:

$$\begin{aligned}
E_{\mathrm{HF}} &= \left\langle \hat{H}_{\mathrm{HF}} \right\rangle \\
&= E_0 + \sum_{\alpha,\beta} \sum_{p,q} h^{pq}_{\alpha\beta} \Delta^{pq}_{\alpha\beta}
\end{aligned}$$

Here $\Delta^{pq}_{\alpha\beta}$ are functions of the fields $h^{pq}_{\alpha\beta}$. Derive energy:

$$\begin{aligned}
\frac{\partial E_{\mathrm{HF}}}{\partial h^{rs}_{\mu\nu}} &= \frac{\partial}{\partial h^{rs}_{\mu\nu}} \sum_{\alpha,\beta} \sum_{p,q} h^{pq}_{\alpha\beta} \Delta^{pq}_{\alpha\beta} \\
&= \Delta^{rs}_{\mu\nu} + \sum_{\alpha,\beta} \sum_{p,q} h^{pq}_{\alpha\beta} \frac{\partial \Delta^{pq}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}
\end{aligned}$$

Note that by construction E_{HF} is greater than the actual ground-state energy, since our approximated wavefunction is not the actual ground-state wavefunction. Now, from Hellman-Feynman theorem [29], we know that for a parametric hamiltonian it holds:

$$\begin{aligned}
\frac{\partial E_{\mathrm{HF}}}{\partial h^{rs}_{\mu\nu}} &= \langle \Psi | \frac{\partial \hat{H}_{\mathrm{HF}}}{\partial h^{rs}_{\mu\nu}} | \Psi \rangle \\
&= \langle \Psi | \frac{\partial}{\partial h^{rs}_{\mu\nu}} \sum_{\alpha,\beta} \sum_{p,q} h^{pq}_{\alpha\beta} \hat{a}^p_\alpha \hat{a}^q_\beta | \Psi \rangle \\
&= \left\langle \hat{a}^r_\mu \hat{a}^s_\nu \right\rangle \\
&= \Delta^{rs}_{\mu\nu}
\end{aligned}$$

The last two results imply:

$$\sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta}\frac{\partial \Delta^{pq}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} = 0 \tag{2.11}$$

This must hold for any possible choice of fields $h^{pq}_{\alpha\beta}$. Note that we did not impose the energy derivative to be zero: it is not the HF energy we are minimising.

Let us get back to the actual hamiltonian $\hat{H}$. To reduce it to a quadratic form is equivalent to assume that its ground-state has the properties described in previous sections. Then the following decomposition must hold:

$$\begin{aligned}
E &= \langle \hat{H} \rangle \\
&= \sum_{\alpha,\beta} A_{\alpha\beta}\langle \hat{c}^\dagger_\alpha \hat{c}_\beta\rangle + \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta}\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta\rangle \\
&= \sum_{\alpha,\beta} A_{\alpha\beta}\langle \hat{a}^+_\alpha \hat{a}^-_\beta\rangle + \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta}\left(\langle \hat{a}^+_\alpha \hat{a}^+_\beta\rangle\langle \hat{a}^-_\gamma \hat{a}^-_\delta\rangle - \langle \hat{a}^+_\alpha \hat{a}^-_\gamma\rangle\langle \hat{a}^+_\beta \hat{a}^-_\delta\rangle + \langle \hat{a}^+_\alpha \hat{a}^-_\delta\rangle\langle \hat{a}^+_\beta \hat{a}^-_\gamma\rangle\right) \\
&= \sum_{\alpha,\beta} A_{\alpha\beta}\Delta^{+-}_{\alpha\beta} + \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta}\left(\Delta^{++}_{\alpha\beta}\Delta^{--}_{\gamma\delta} - \Delta^{+-}_{\alpha\gamma}\Delta^{+-}_{\beta\delta} + \Delta^{+-}_{\alpha\delta}\Delta^{+-}_{\beta\gamma}\right) \\
&= \sum_{\alpha,\beta} A_{\alpha\beta}\Delta^{+-}_{\alpha\beta} + \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta}\left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}\right)\Delta^{+-}_{\alpha\beta}\Delta^{+-}_{\gamma\delta} + \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta}\Delta^{++}_{\alpha\beta}\Delta^{--}_{\gamma\delta}
\end{aligned}$$

In second passage we inserted the the relations (2.9). Take the derivative:

$$\begin{aligned}
\frac{\partial E}{\partial h^{rs}_{\mu\nu}} = \sum_{\alpha,\beta} A_{\alpha\beta}\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} &+ \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta}\left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}\right)\left(\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\Delta^{+-}_{\gamma\delta} + \Delta^{+-}_{\alpha\beta}\frac{\partial \Delta^{+-}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}}\right) \\
&+ \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta}\left(\frac{\partial \Delta^{++}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\Delta^{--}_{\gamma\delta} + \Delta^{++}_{\alpha\beta}\frac{\partial \Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}}\right)
\end{aligned}$$

Using Eq. (2.4) we end up with:

$$\begin{aligned}
\frac{\partial E}{\partial h^{rs}_{\mu\nu}} = \sum_{\alpha,\beta}&\left[A_{\alpha\beta} + \sum_{\gamma,\delta}\left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}\right)\Delta^{+-}_{\gamma\delta}\right]\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} \\
&+ \sum_{\alpha,\beta,\gamma,\delta}\frac{B_{\alpha\beta\gamma\delta}\Delta^{--}_{\gamma\delta}}{2}\frac{\partial \Delta^{++}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + \sum_{\alpha,\beta,\gamma,\delta}\frac{B_{\alpha\beta\gamma\delta}\Delta^{++}_{\alpha\beta}}{2}\frac{\partial \Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}}
\end{aligned}$$

Let now:

$$C_{\alpha\beta} \equiv \frac{1}{2}\left[A_{\alpha\beta} + \sum_{\gamma,\delta}\left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}\right)\Delta^{+-}_{\gamma\delta}\right] \qquad D_{\alpha\beta} \equiv \sum_{\gamma,\delta}\frac{B_{\alpha\beta\gamma\delta}\Delta^{--}_{\gamma\delta}}{2} \tag{2.12}$$

From Fermi anticommutation rules, we have:

$$\Delta^{+-}_{\alpha\beta} = \delta_{\alpha\beta} - \Delta^{-+}_{\beta\alpha} \quad\Longrightarrow\quad \frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} = -\frac{\partial \Delta^{-+}_{\beta\alpha}}{\partial h^{rs}_{\mu\nu}}$$

which implies:

$$\begin{aligned}
\sum_{\alpha,\beta} 2C_{\alpha\beta}\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} &= \sum_{\alpha,\beta}\left(C_{\alpha\beta}\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + C_{\alpha\beta}\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\right) \\
&= \sum_{\alpha,\beta}\left(C_{\alpha\beta}\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} - C_{\beta\alpha}\frac{\partial \Delta^{-+}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\right)
\end{aligned}$$

Moreover, using Eqns. (2.3), (2.10),

$$\sum_{\alpha,\beta} \frac{B_{\alpha\beta\gamma\delta}\Delta^{++}_{\alpha\beta}}{2} = \sum_{\alpha,\beta} \left(\frac{B_{\delta\gamma\beta\alpha}\Delta^{--}_{\beta\alpha}}{2} \right)^*$$

Using these two relations, we conclude:

$$\sum_{\alpha,\beta,\gamma,\delta} \frac{B_{\alpha\beta\gamma\delta}\Delta^{++}_{\alpha\beta}}{2} \frac{\partial \Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}} = \sum_{\gamma,\delta} D^*_{\delta\gamma} \frac{\partial \Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}}$$

Recollect everything and impose the solution to be a stationary point of energy,

$$\frac{\partial E}{\partial h^{rs}_{\mu\nu}} \stackrel{!}{=} 0$$

The notation "$\stackrel{!}{=}$" indicates, throughout the thesis, that we are imposing some condition. In the end we have:

$$\frac{\partial E}{\partial h^{rs}_{\mu\nu}} = \sum_{\alpha,\beta} \left(C_{\alpha\beta} \frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} - C_{\beta\alpha} \frac{\partial \Delta^{-+}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + D_{\alpha\beta} \frac{\partial \Delta^{++}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + D^*_{\beta\alpha} \frac{\partial \Delta^{--}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} \right) \stackrel{!}{=} 0 \tag{2.13}$$

Now compare Eqns. (2.11) and (2.13). We recognise $h^{+-}_{\alpha\beta} = C_{\alpha\beta}$ and $h^{++}_{\alpha\beta} = D_{\alpha\beta}$; the remaining fields are determined by the symmetry relations (2.8) and (2.7). Explicitly, the fields are self-consistently determined by the following two equations:

$$h^{+-}_{\alpha\beta} = \frac{1}{2} \left[A_{\alpha\beta} + \sum_{\gamma,\delta} \left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta} \right) \Delta^{+-}_{\gamma\delta} \right] \tag{2.14}$$

$$h^{++}_{\alpha\beta} = \sum_{\gamma,\delta} \frac{B_{\alpha\beta\gamma\delta}\Delta^{--}_{\gamma\delta}}{2} \tag{2.15}$$

The right-hand side of both equations depends on the fields themselves: $\Delta^{pq}_{\alpha\beta}$ are defined as expectation values of two-points fermionic operators over the ground-state $|\Psi\rangle$. For a given set of fields $h^{pq}_{\alpha\beta}$, the latter is determined (up to some unitary transformation) as the ground-state of a fermionic non-interacting model. In other words, being $\mathbf{h}$ the tensor containing all fields,

$$\Delta^{pq}_{\alpha\beta}[\mathbf{h}] = \langle \Psi[\mathbf{h}] | \hat{a}^p_\alpha \hat{a}^q_\beta | \Psi[\mathbf{h}] \rangle \tag{2.16}$$

This set of equations defines our computational problem. We are searching a set of parameters $\{\Delta^{pq}_{\alpha\beta}\}$ such that, determining the fields via Eqns. (2.14), (2.15), then Eq. (2.16) is respected. The great advantage of this scheme is that, with an appropriate choice of the Hilbert space basis (for example, moving to reciprocal space for a translationally-invariant model), we can implement easily model symmetries.

The list of parameters $\{\Delta^{pq}_{\alpha\beta}\}$ to be determined self-consistently may be not-so-short: by looking at Eq. (2.16), the indices α, β, p, q are spanning a large Hilbert space. However, this equation can be manipulated and transformed, reducing the set of parameters to be determined self-consistently to a very manageable number. For each phase we define these last as "Hartree-Fock parameters" (HFPs): a set of objects self-consistently determined by a set equivalent, but transformed, version of Eq. (2.16).

2.2.3 Operative hamiltonian approximation

The derived Eqns. (2.14), (2.15), although general and correct, are a little unpractical to use. To derive them was useful in order to prove that the self-consistent method is a variational search in disguise. Operatively, they can be rewritten in a simpler way. For a generic two-body quartic operator, the HF approximation is given by the decomposition

$$\begin{aligned}
\hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\beta \hat{c}_\alpha \overset{\text{HF}}{\simeq} & \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \langle \hat{c}_\beta \hat{c}_\alpha \rangle + \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \hat{c}_\beta \hat{c}_\alpha - \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \langle \hat{c}_\beta \hat{c}_\alpha \rangle && \text{(Hartree)} \\
& - \hat{c}^\dagger_\alpha \hat{c}_\beta \langle \hat{c}^\dagger_\beta \hat{c}_\alpha \rangle - \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \hat{c}^\dagger_\beta \hat{c}_\alpha + \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\alpha \rangle && \text{(Fock)} \\
& + \hat{c}^\dagger_\alpha \hat{c}_\alpha \langle \hat{c}^\dagger_\beta \hat{c}_\beta \rangle + \langle \hat{c}^\dagger_\alpha \hat{c}_\alpha \rangle \hat{c}^\dagger_\beta \hat{c}_\beta - \langle \hat{c}^\dagger_\alpha \hat{c}_\alpha \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\beta \rangle && \text{(Cooper)}
\end{aligned}$$

The proof of this statement is given in App. C, where we also argue the interpretation of HF method as a mean-field theory (MFT). Let us summarise this section through a practical rule to decompose the quartic part of $\hat{H}_U$ and $\hat{H}_V$ in the EHM, by applying Eq. (C.9).

Local repulsion. Applying the HF method, the local repulsion $\hat{H}_U$ is approximated by the quadratic operator:

$$\begin{aligned}
\hat{H}_U &= U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} && (2.17) \\
&\simeq U \sum_i \Big(\hat{c}^\dagger_{i\uparrow} \hat{c}_{i\uparrow} \langle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\downarrow} \rangle + \langle \hat{c}^\dagger_{i\uparrow} \hat{c}_{i\uparrow} \rangle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\downarrow} - \langle \hat{c}^\dagger_{i\uparrow} \hat{c}_{i\uparrow} \rangle \langle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\downarrow} \rangle && \text{(Hartree)} \\
&\qquad - \hat{c}^\dagger_{i\uparrow} \hat{c}_{i\downarrow} \langle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\uparrow} \rangle - \langle \hat{c}^\dagger_{i\uparrow} \hat{c}_{i\downarrow} \rangle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\uparrow} + \langle \hat{c}^\dagger_{i\uparrow} \hat{c}_{i\downarrow} \rangle \langle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\uparrow} \rangle && \text{(Fock)} \\
&\qquad + \hat{c}^\dagger_{i\uparrow} \hat{c}^\dagger_{i\downarrow} \langle \hat{c}_{i\downarrow} \hat{c}_{i\uparrow} \rangle + \langle \hat{c}^\dagger_{i\uparrow} \hat{c}^\dagger_{i\downarrow} \rangle \hat{c}_{i\downarrow} \hat{c}_{i\uparrow} - \langle \hat{c}^\dagger_{i\uparrow} \hat{c}^\dagger_{i\downarrow} \rangle \langle \hat{c}_{i\downarrow} \hat{c}_{i\uparrow} \rangle \Big) && \text{(Cooper)}
\end{aligned}$$

Non-local attraction. Applying the HF method, the local repulsion $\hat{H}_V$ is approximated by the quadratic operator:

$$\begin{aligned}
\hat{H}_V &= -V \sum_{\langle ij \rangle} \sum_{\sigma,\sigma'} \hat{n}_{i\sigma} \hat{n}_{j\sigma'} && (2.18) \\
&\simeq -V \sum_{\langle ij \rangle} \sum_{\sigma,\sigma'} \Big(\hat{c}^\dagger_{i\sigma} \hat{c}_{i\sigma} \langle \hat{c}^\dagger_{j\sigma'} \hat{c}_{j\sigma'} \rangle + \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{i\sigma} \rangle \hat{c}^\dagger_{j\sigma'} \hat{c}_{j\sigma'} - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{i\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma'} \hat{c}_{j\sigma'} \rangle && \text{(Hartree)} \\
&\qquad - \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma'} \langle \hat{c}^\dagger_{j\sigma'} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma'} \rangle \hat{c}^\dagger_{j\sigma'} \hat{c}_{i\sigma} + \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma'} \rangle \langle \hat{c}^\dagger_{j\sigma'} \hat{c}_{i\sigma} \rangle && \text{(Fock)} \\
&\qquad + \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{j\sigma'} \langle \hat{c}_{j\sigma'} \hat{c}_{i\sigma} \rangle + \langle \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{j\sigma'} \rangle \hat{c}_{j\sigma'} \hat{c}_{i\sigma} - \langle \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{j\sigma'} \rangle \langle \hat{c}_{j\sigma'} \hat{c}_{i\sigma} \rangle \Big) && \text{(Cooper)}
\end{aligned}$$

We will use these decomposition referring to each sector in dedicated chapters.

2.3 Sketch of HF iterative algorithm

In this section we give a practical sketch of the algorithm we use to compute Eq. (2.16) for all given phases. We refer to this technique as "iterative Hartree Fock" (iHF). Let $\mathbf{v}$ be the vector containing all HFPs. Suppose we have N HFPs v_i: then

$$\mathbf{v} \in \mathbb{C}^N \qquad \mathbf{v} \equiv \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix}$$

In order to get a physical solution, each HFP must satisfy a self-consistency equation similar to Eq. (2.16). Each equation has the structure:

$$v_i = (\text{Some non-linear function of } \mathbf{v})$$

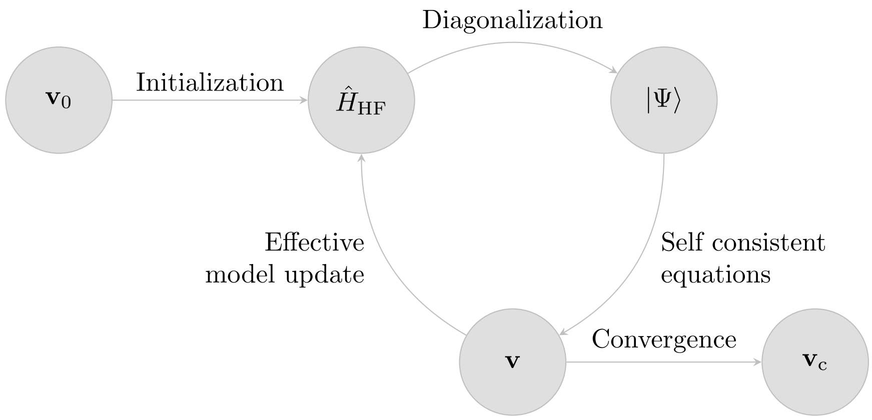


Figure 2.2 | Schematisation of the self-consistent Hartree Fock method described in Sec. 2.3. The effective model is solved, the solution is used to estimate self-consistently the Hartree Fock vector $\mathbf{v}$, which is then used to update iteratively the model itself. The notation $\mathbf{v}_0$ indicates the initialiser HFPs vector, $\mathbf{v}_\mathrm{c}$ the converged HFPs vector.

Let A be the non-linear operator collecting all these functions. The problem of finding a self-consistent solution can be reformulated as follows: $\mathbf{v}$ represents a physical solution if it is a fixed point of the map A,

$$\mathrm{A}\colon \mathbb{C}^N \to \mathbb{C}^N \qquad \mathrm{A}(\mathbf{v}) = \mathbf{v} \qquad \Longrightarrow \qquad \mathbf{v} \text{ is a solution.} \tag{2.19}$$

We implement a rather easy scheme: first, we obtain an analytic expression for A. Then we choose an initialiser $\mathbf{v}_0$ and apply A to it, obtaining a new HFPs vector. Iterating this procedure, we "follow" the non-linear map A until we claim to have converged. Let us explain in detail:

1. Start with a generic quartic hamiltonian:

$$\hat{H} = \sum_{\alpha\beta} A_{\alpha\beta}\hat{c}^\dagger_\alpha \hat{c}_\beta + \sum_{\alpha\beta\gamma\delta} B_{\alpha\beta\gamma\delta}\hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta$$

   Choose the model parameters and set an electronic density;

2. Following the discussion of Sec. 2.2, approximate analytically $\hat{H}$ with a quadratic hamiltonian

$$\hat{H} \simeq \hat{H}_{\mathrm{HF}} \equiv \sum_{\alpha\beta} \left[h^{+-}_{\alpha\beta}\hat{c}^\dagger_\alpha \hat{c}_\beta - h^{+-}_{\beta\alpha}\hat{c}_\alpha \hat{c}^\dagger_\beta + h^{++}_{\alpha\beta}\hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta + \left(h^{++}_{\beta\alpha}\right)^* \hat{c}_\alpha \hat{c}_\beta \right]$$

   with the fields given by Eqns. (2.14), (2.15);

3. Impose selection rules on the fields $h^{+-}_{\alpha\beta}$, $h^{++}_{\alpha\beta}$, eventually eliminating those not respecting the symmetry sector we are working in. As an example: if we choose not to break total charge conservation, set $\forall\alpha,\beta\colon h^{++}_{\alpha\beta} = 0$;

4. Choose a starting value for the HFPs vector $\mathbf{v}_0$;

5. Set $\mathbf{v} = \mathbf{v}_0$ and enter the iterative scheme of Fig. 2.2:

   a) Get a numeric approximation of the hamiltonian for the current value of $\mathbf{v}$, $\hat{H}_{\mathrm{HF}}[\mathbf{v}]$;
   b) Determine the value $\mu[\mathbf{v}]$ giving the desired density;
   c) Diagonalise the effective hamiltonian $\hat{H}_{\mathrm{HF}}[\mathbf{v}] - \mu[\mathbf{v}]\hat{N}$, and obtain the approximated ground-state $|\Psi[\mathbf{v}]\rangle$;
   d) Compute $\mathrm{A}(\mathbf{v})$ using self-consistency equations;
   e) If the algorithm converged, exit; otherwise, update the value of $\mathbf{v}$ repeat from point a).

We voluntarily left unspecified three details:

- How is the chemical potential μ determined?
- What criterion do we use to claim that the algorithm "converged"?
- How is the HFPs vector $\mathbf{v}$ updated?

The sections that follows deal with these questions.

2.3.1 Fixed density simulations and the role of μ

In this thesis all derivations and simulations are carried out at fixed density. This requires a way to fix μ during iterations of the algorithm. This must be done iteratively: changing at each step the HFPs vector, we are actively modifying the approximate hamiltonian. Chemical potential must be determined accordingly to maintain the required density.

For a given choice of HFPs vector and chemical potential, the ground state $|\Psi[\mathbf{v},\mu]\rangle$ is uniquely determined; then we compute density as

$$n(\mathbf{v},\mu) \equiv \frac{1}{\mathcal{V}} \sum_{\alpha} \langle \Psi[\mathbf{v},\mu] | \hat{c}_{\alpha}^{\dagger} \hat{c}_{\alpha} | \Psi[\mathbf{v},\mu] \rangle$$

being $\mathcal{V}$ the volume. Suppose we fix density to a value $n_0 \in [0,1]$. At step i, naming the current HFPs vector $\mathbf{v}_i$, μ_i (the chemical potential at current step) is determined as

$$\mu_i \equiv \min_{\mu} |n(\mathbf{v}_i,\mu) - n_0| \tag{2.20}$$

The density n must be a monotonically increasing function of μ: then we can translate the above equation, which would inspire a minimisation procedure, to a much simpler node-search rule:

$$\mu_i = \mu \,|_{\delta n(\mathbf{v}_i,\mu)=0} \qquad \text{being} \qquad \delta n(\mathbf{v}_i,\mu) \equiv n(\mathbf{v}_i,\mu) - n_0$$

In order to where the function changes sign (which is, the position of μ_i) we use a basic bisection algorithm:

1. Set an arbitrary tolerance $\Delta n > 0$ and a shifting parameter $\Delta\mu > 0$;
2. Choose starting lower and upper bounds $\mu_{\mathrm{L}} < \mu_{\mathrm{U}}$;
3. Compute $\delta n(\mathbf{v}_i, \mu_{\mathrm{L}})$. Evaluate:
   - if $|\delta n(\mathbf{v}_i,\mu_{\mathrm{L}})| \leq \Delta n$, set $\mu_i = \mu_{\mathrm{L}}$ and halt;
   - if $\delta n(\mathbf{v}_i,\mu_{\mathrm{L}}) < -\Delta n$ do nothing;
   - if $\delta n(\mathbf{v}_i,\mu_{\mathrm{L}}) > \Delta n$, shift the lower bound down, $\mu_{\mathrm{L}} \to \mu_{\mathrm{L}} - \Delta\mu$ and repeat from step 3;
4. Compute $\delta n(\mathbf{v}_i, \mu_{\mathrm{U}})$. Evaluate:
   - if $|\delta n(\mathbf{v}_i,\mu_{\mathrm{U}})| \leq \Delta n$, set $\mu_i = \mu_{\mathrm{U}}$ and halt;
   - if $\delta n(\mathbf{v}_i,\mu_{\mathrm{U}}) > \Delta n$ do nothing;
   - if $\delta n(\mathbf{v}_i,\mu_{\mathrm{U}}) < -\Delta n$, shift the upper bound up, $\mu_{\mathrm{U}} \to \mu_{\mathrm{U}} + \Delta\mu$ and repeat from step 4;
5. At this point, the the lower bound underestimates density while the upper bound overestimates. Start iterations:

   a) Compute the midpoint,
   $$\mu_{\mathrm{M}} = \frac{\mu_{\mathrm{L}} + \mu_{\mathrm{U}}}{2}$$

   b) Evaluate:
   - If $|\delta n(\mathbf{v}_i,\mu_{\mathrm{M}})| \leq \Delta n$, set $\mu_i = \mu_{\mathrm{M}}$ and halt;
   - If $\delta n(\mathbf{v}_i,\mu_{\mathrm{M}}) > \Delta n$, set $\mu_{\mathrm{U}} = \mu_{\mathrm{M}}$ and repeat from point a);
   - If $\delta n(\mathbf{v}_i,\mu_{\mathrm{M}}) < -\Delta n$, set $\mu_{\mathrm{L}} = \mu_{\mathrm{M}}$ and repeat from point a);

The bottleneck of the procedure is our ability to compute expectation values. This depends on how much we are able to simplify analytically the computations for each phase.

As will turn out in the derivation, chemical potential will be renormalised by interaction. This means that applying self-consistent HF methods to the EHM will produce an effective hamiltonian containing a shift to the chemical potential,

$$\mu \to \mu + F[\mathbf{v}] \equiv \tilde{\mu}[\mathbf{v}]$$

being F some function of the HFPs. The chemical potential μ is an example of Renormalised Model Parameter (RMP). As we will see, also the hopping t will turn out to be renormalised by interactions.

Note that at step i of the iHF algorithm, our bisection procedure determines $\tilde{\mu}_i$, not μ_i. This consideration is relevant when dealing with ground state energy estimation. Let us abstract from the EHM and iHF, and consider a generic hamiltonian $\hat{H}$. To find the ground state $|\Psi\rangle$ means precisely to solve

$$\frac{\delta}{\delta\langle\Psi|}\langle\Psi|\hat{H}|\Psi\rangle = 0$$

Now, working with a fixed particles number N means we are performing a constrained minimisation (which is, taking the functional derivative) on the composite operator

$$\hat{H} - \mu(\hat{N} - N) \equiv \hat{K} + \mu N$$

being μ the Lagrange multiplier, $\hat{N}$ the number operator and $\hat{K} \equiv \hat{H} - \mu\hat{N}$. Finding the constrained ground state parametrised by a vector $\mathbf{v}$ now means to solve

$$\mathbf{v} \implies |\Psi[\mathbf{v}]\rangle \quad : \quad \begin{cases} \dfrac{\delta}{\delta\langle\Psi|}\langle\Psi|\hat{K} + \mu N|\Psi\rangle\Big|_{|\Psi\rangle = |\Psi[\mathbf{v}]\rangle} = 0 \\ \langle\Psi[\mathbf{v}]|\hat{N}|\Psi[\mathbf{v}]\rangle = N \end{cases}$$

In the first line μN is derived to zero; the constraint is imposed in the second line. Energy is given by:

$$\begin{aligned}\langle\hat{H}\rangle &= \langle\Psi[\mathbf{v}]|\hat{K}|\Psi[\mathbf{v}]\rangle + \mu N \\ &= \Omega[\mathbf{v}] + (\tilde{\mu}[\mathbf{v}] - F[\mathbf{v}])N\end{aligned}$$

being Ω the expectation value (EV) of $\hat{K}$. To correctly compute energy, we need to add $+\mu N$; since we determine $\tilde{\mu}[\mathbf{v}] \neq \mu$, everything needs to be expressed in terms of it. Using this relation, when the algorithm has converged, we can estimate ground-state energy.

2.3.2 The reject-mix-iterate convergence scheme

Let us discuss briefly the convergence criterion of our algorithm and the iteration mechanism. The iHF scheme we design is a sequential loop that runs up to a logical halt condition. In order to specify such condition, we need to define our notion of "convergence". Recall the definition of $\mathrm{A}(\mathbf{v})$, the non-linear map in HFPs space whose component are given by a set of self-consistency equations: we claim to have reached convergence when $\mathbf{v}$ comes "close enough" to a fixed point of the map A, as in Eq. (2.19). Explicitly, we stop whenever the following logic condition is satisfied for each component of $\mathbf{v} = (v_1, v_2, \cdots)$:

$$\forall v_j \quad : \quad |v_j - \mathrm{A}_j(\mathbf{v})| \leq \Delta v_j$$

being $\mathrm{A}(\mathbf{v})$ the map image of the HFP vector $\mathbf{v}$, and A_j its j-th component. The tolerance Δv_j is arbitrary. If just one component of $\mathbf{v}$ is mapped "too far" (with respect to Δv_j) the step is rejected and iterations continue. In this text we deal in with quantities of order 10^{-1}, so convergence values of order $10^{-3} \div 10^{-4}$ will be in general sufficient.

At step i, we check the distance of the two vectors $\mathrm{A}(\mathbf{v}_i)$ and $\mathbf{v}_i$ and if possible, halt. If instead convergence is not reached, we need repeat the cycle with a new initialiser $\mathbf{v}_{i+1}$. A very simple choice would be to set $\mathbf{v}_{i+1} = \mathbf{v}_i$. We do something slightly different:

$$\mathbf{v}_{i+1} \equiv g\mathrm{A}(\mathbf{v}_i) + (1-g)\mathbf{v}_i \tag{2.21}$$

The "mixing parameter" g is a simple but useful knob to tune algorithmic evolution. Its usefulness comes from this observation: even if a fixed point exists, we don't know how easy it is to actually get there following A. Setting $g = 1$, the new initialiser is just the image of the previous initialiser. By moving "slower" ($g < 1$) with an accurate optimisation of the mixing, it is possible to reach convergence in a controlled way.

Let us give some concrete justification about the relevance of controlling g. When it comes to HF iterative schemes of this kind, it can sometimes happen for the algorithm to remain stuck in infinite loops. An example of this behaviour is given in Sec. 4.6.3. A typical situation is the following: during algorithm evolution we reach a couple of points $\mathbf{v}^{(a)}$, $\mathbf{v}^{(b)}$ such that

$$\dots \xrightarrow{\mathrm{A},g} \mathbf{v}^{(a)} \xrightarrow{\mathrm{A},g} \mathbf{v}^{(b)} \xrightarrow{\mathrm{A},g} \mathbf{v}^{(a)} \xrightarrow{\mathrm{A},g} \mathbf{v}^{(b)} \xrightarrow{\mathrm{A},g} \dots$$

With the notation "$\xrightarrow{\mathrm{A},g}$" we intend one algorithmic step at mixing parameter g. We are stuck forever in this loop. Now, to predict if couples like $(\mathbf{v}^{(a)}, \mathbf{v}^{(b)})$ exist, how they depend on g, where they are located in HFPs space... is impossible at this level. Thus there is no immediate way to avoid these loops *a priori*. One workaround we found, however, was to just make g smaller. We gained empirical evidence that to "slow down" in HFPs space makes numeric instabilities less frequent. The trade-off, however, is that if we move too slow, runtime could get very large. Note that inaccurate setup of the algorithm could lead to resources-intensive runs also in "ordinary" situations, when we are not stuck in a loop.

To keep the number of iterations under control, we also defined an upper bound p to the total number of steps. If after p steps we algorithm failed to converge, it halts. In that case, $\mathbf{v}$ is saved as a "`NaN`" (Not a number) vector and we move to another simulation. Introducing this simple rule, runtime is controlled and we gain an easy method to identify the regions of parameter space requiring more care – just read from the raw data where `NaN`s have been saved and repeat simulations fine-tuning the parameters. This basically means to lower g and increase p. These considerations end this technical introductory chapter on iHF methods.

Chapter 3

Phase transitions and SSB in EHM

The general idea of this project is to apply HF methods to the EHM, derive analytically the approximated quadratic hamiltonian of Eq. (2.5) and then apply the iHF procedure sketched in Sec. 2.3 to extract the self-consistent values of the various HFPs. One of the advantages of using iHF methods is that we can impose rather simply the symmetries of the HF approximated wavefunction.

This is most useful when dealing with competing phases. A phase transition is basically described by some spontaneous symmetry breaking (SSB) and the rising to a non-zero value of an order parameter. We will deal with three phases:

1. the normal phase, where no symmetry is broken;

2. the antiferromagnetic phase, where we allow for spin density to be non-uniform across sites;

3. the superconducting phase, where we do not conserve particle number.

Each phase is discussed in a separate chapter. Many other "phases" are possible, based on which symmetries are respected and which are not.

For each phase, we will extract the best approximation to the true ground state wavefunction. In computational terms, we will find three solutions to the general self-consistency problem of Eq. (2.16). In the end we want to compare the free energy of these three solutions, assessing which one is the most stable. Any HF approximation, being by construction not the true ground state, overestimates energy. Thus, comparing energies of our solutions we get a hint of the features of the true ground state.

3.1 The EHM symmetries

In this section we present the symmetries of the extended Hubbard model introduced in Eq. (1.2)

$$\hat{H}_{\mathrm{EHM}} = \underbrace{-t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}}_{\hat{H}_t} \underbrace{+U\sum_{i}\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}}_{\hat{H}_U} \underbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{n}_{i\sigma}\hat{n}_{j\sigma'}}_{\hat{H}_V}$$

The symmetries of the model can be divided in two groups: the ones related to the geometry of the square lattice, and those related to the algebraic properties of $\hat{H}_{\mathrm{EHM}}$. We start by the lattice symmetries.

3.1.1 Lattice symmetries

We study the EHM on the planar square lattice, with one orbital per site. We indicate with L_x the number of sites in direction x, and L_y in direction y. The hamiltonian of Eq. (1.2) respects three fundamental lattice symmetries:

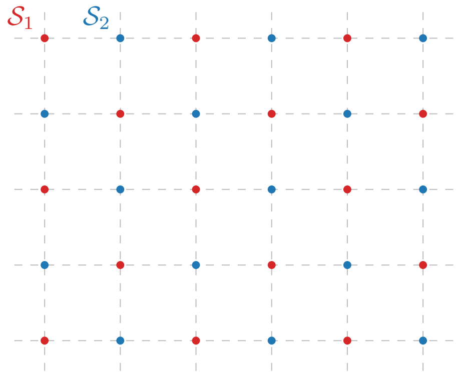


Figure 3.1 | Square lattice $\mathcal{L}$ bipartition in two sublattices $\mathcal{S}_1$ (red), $\mathcal{S}_2$ (blue).

1. Discrete translations symmetry in both the x and y directions. We refer to the groups of one-site translations as as $\mathrm{T}_1^{(x)}$, $\mathrm{T}_1^{(x)}$. It holds:

$$\mathrm{T}_n^{(\ell)} \supset \mathrm{T}_{kn}^{(\ell)} \qquad k \in \mathbb{N} \tag{3.1}$$

   A shift in direction ℓ by kn sites can be decomposed in k shift by n sites. Invariance by n-sites shifts, thus, implies invariance also for kn-sites shifts.

2. Discrete $\pi/2$ rotations symmetry. This symmetry is described by the four-fold cyclic point group C_4, containing rotations by angle $\pi/2, \pi, 3\pi/2$ and identity. A similar property as for the translations holds for cyclic point groups,

$$\mathrm{C}_n \supset \mathrm{C}_{n/k} \qquad k, n/k \in \mathbb{N} \tag{3.2}$$

   Invariance for rotations of angle $2\pi/n$ implies invariance for rotations of angle $k \times 2\pi/n$.

3. Inversion symmetry. We define I_2 as the inversion group in two-dimensional space, containing the reflection operator and identity. This point may appear redundant, because

$$\mathrm{I}_2 \equiv \mathrm{C}_2 \tag{3.3}$$

   In 2D space, to rotate by π around the origin and to mirror the position of a point are the same operation.

The planar square lattice is bipartite. Let us give a definition. Consider a graph $\mathcal{L}$ and the set of its vertices $V(\mathcal{L}) = \{v_i\}$. Let $E(\mathcal{L})$ be the set of graph edges,

$$E(\mathcal{L}) = \{(v, w) \mid v, w \in V(\mathcal{L}) \wedge v,\ w \text{ are connected}\}$$

The graph is said to be bipartite if two sub-graphs $\mathcal{S}_1$, $\mathcal{S}_2$ exist such that:

- The vertices of $\mathcal{L}$ split in the two sub-graphs,

$$V(\mathcal{L}) = V(\mathcal{S}_1) \cup V(\mathcal{S}_2)$$

- Vertices of one sub-graph are linked only to vertices of the other sub-graph. In other words

$$v, w \in \mathcal{S}_i \quad \Longrightarrow \quad (v, w) \notin E(\mathcal{L})$$

The planar square lattice is bipartite: it can be divided in two square lattices, as represented in Fig. 3.1. Two sites are "connected" if linked by a non-zero hopping operator. Which is, if they are NNs. Note that one-site translations leave the hamiltonian invariant but exchange the two sublattices, while two-sites translations don't.

3.1.2 Spin and charge symmetries

The conventional HM shows a global U(2) symmetry, extended to $\mathrm{SU}(2) \times \mathrm{SU}(2)/\mathbb{Z}_2 = \mathrm{SO}(4)$ symmetry at half-filling on a bipartite lattice [14], such as the square lattice. For the EHM, we limit to prove that we can identify a $\mathrm{U}(2) \subset \mathrm{SO}(4)$ global symmetry.

Gauge symmetry. The EHM is symmetric under the global transformation

$$\hat{c}_{i\sigma} \to e^{-i\eta}\hat{c}_{i\sigma} \qquad \hat{c}^\dagger_{i\sigma} \to e^{i\eta}\hat{c}^\dagger_{i\sigma} \qquad \eta \in \mathbb{R} \tag{3.4}$$

To prove this, it suffices to note that

$$\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'} \quad \to \quad e^{i\eta}\hat{c}^\dagger_{i\sigma}e^{-i\eta}\hat{c}_{j\sigma'} = \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'} \tag{3.5}$$

As a consequence, any product of fermionic operators with an equal number of annihilations and creations is symmetric. Hence the kinetic part $\hat{H}_t$ and the interactions $\hat{H}_U$, $\hat{H}_V$ are invariant. As a consequence, the total number of particles (charge)

$$\hat{N} = \sum_{i\sigma} \hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}$$

is a good quantum number. The symmetry is described by the transformation of Eq. (3.4), which is a pure phase shift by η. The relative group is $\mathrm{U}^{\mathrm{c}}(1)$, with a superscript "c" indicating "charge".

Global spin symmetry The EHM is described by three operators: NN hopping, independent of σ; local repulsion between opposite-spin densities; non-local attraction, independent of σ. If we rotate all spins by an equal amount, the hamiltonian must remain identical. An identical observation holds for the number operator, $\hat{N}$. Thus $\hat{H}_{\mathrm{EHM}} - \mu\hat{N}$ must be exhibit a global SU(2) symmetry, related to the rotation in 3D space of all spins altogether.

Let us prove this rigorously. Let the on-site spinor:

$$\hat{\Psi}_i \equiv \begin{bmatrix} \hat{c}_{i\uparrow} \\ \hat{c}_{i\downarrow} \end{bmatrix} \tag{3.6}$$

and the Pauli matrices

$$\tau^x \equiv \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} \qquad \tau^y \equiv \begin{bmatrix} & -i \\ i & \end{bmatrix} \qquad \tau^z \equiv \begin{bmatrix} 1 & \\ & -1 \end{bmatrix}$$

We define the local spin operator as:

$$\hat{S}^\alpha_i \equiv \frac{1}{2}\hat{\Psi}^\dagger_i \tau^\alpha \hat{\Psi}_i \qquad \left[\hat{S}^\alpha_j, \hat{S}^\beta_j\right] = i\sum_\gamma \epsilon_{\alpha\beta\gamma}\hat{S}^\gamma_j \tag{3.7}$$

with $\epsilon_{\alpha\beta\gamma}$ the Levi-Civita tensor. It can be checked by explicit calculation that the $\mathfrak{su}(2)$ spin algebra is actually respected. For example, consider $\alpha = x$, $\beta = y$:

$$\begin{aligned}
\left[\hat{S}^x_j, \hat{S}^y_j\right] &= \frac{1}{4}\left[\hat{\Psi}^\dagger_j \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} \hat{\Psi}_j, \hat{\Psi}^\dagger_j \begin{bmatrix} & -i \\ i & \end{bmatrix} \hat{\Psi}_j\right] \\
&= \frac{1}{4}\left[\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow} + \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}, -i\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow} + i\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}\right] \\
&= \frac{i}{4}\left[\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}, \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}\right] - \frac{i}{4}\left[\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}, \hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}\right] \\
&= \frac{i}{2}\left[\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}, \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}\right] \\
&= \frac{i}{2}\left(\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow} - \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}\right) \\
&= \frac{i}{2}\left(-\hat{c}^\dagger_{j\uparrow}\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\downarrow}\hat{c}_{j\uparrow} + \hat{c}^\dagger_{j\uparrow}\hat{c}_{j\uparrow} + \hat{c}^\dagger_{j\downarrow}\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\uparrow}\hat{c}_{j\downarrow} - \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\downarrow}\right) \\
&= \frac{i}{2}\left(\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\uparrow} - \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\downarrow}\right) \\
&= i\hat{S}^z_j
\end{aligned}$$

We used Fermi anticommutation rules, $\{\hat{c}_{i\sigma}, \hat{c}^\dagger_{j\sigma'}\} = \delta_{ij}\delta_{\sigma\sigma'}$. Identical derivations can be carried out for all other components.

Let the raising and lowering operators:

$$\hat{S}^+_j \equiv \frac{\hat{S}^x_j + i\hat{S}^y_j}{2} \qquad \hat{S}^-_j \equiv \frac{\hat{S}^x_j - i\hat{S}^y_j}{2}$$

and the global spin operators

$$\hat{S}^\alpha \equiv \sum_{i\in\mathcal{L}} \hat{S}^\alpha_i \qquad \left[\hat{S}^\alpha, \hat{S}^\beta\right] = i\sum_\gamma \epsilon_{\alpha\beta\gamma}\hat{S}^\gamma$$

being $\mathcal{L}$ the lattice. Explicitly,

$$\hat{S}^z \equiv \frac{1}{2}\sum_{i\in\mathcal{L}} (\hat{n}_{i\uparrow} - \hat{n}_{i\downarrow}) \qquad \hat{S}^+ \equiv \frac{1}{2}\sum_{i\in\mathcal{L}} \hat{c}^\dagger_{i\uparrow}\hat{c}_{i\downarrow} \qquad \hat{S}^- \equiv \left(\hat{S}^+\right)^\dagger$$

The EHM hamiltonian and the number operator commutate with the global spin operator. For the kinetic part $\hat{H}_t$ and the local repulsion $\hat{H}_U$ this is a well known result [14, 27],

$$\left[\hat{H}_t + \hat{H}_U, \hat{\mathbf{S}}\right] = 0 \qquad \left[\hat{N}, \hat{\mathbf{S}}\right] = 0$$

For the non-local attraction, since it depends solely on total densities on neighbouring sites, spin-rotational invariance must hold. Anyway, let us give a short formal proof following this chain of results:

1. The non-local attraction can be written as:

$$\begin{aligned} \hat{H}_V &= V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'} \hat{n}_{i\sigma}\hat{n}_{j\sigma'} \\ &= V\sum_{\langle ij\rangle} \hat{\Psi}^\dagger_i\hat{\Psi}_i\hat{\Psi}^\dagger_j\hat{\Psi}_j \end{aligned} \tag{3.8}$$

   having we used $\hat{\Psi}^\dagger_i\hat{\Psi}_i = \hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow} + \hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}$.

2. Similarly, the number operator is given by:

$$\begin{aligned} \hat{N} &= \sum_{i\in\mathcal{L}}\sum_\sigma \hat{n}_{i\sigma} \\ &= \sum_{i\in\mathcal{L}} \hat{\Psi}^\dagger_i\hat{\Psi}_i \end{aligned} \tag{3.9}$$

3. Let now $\hat{R}(\boldsymbol{\theta})$ be any global spin rotation,

$$\hat{R}(\boldsymbol{\theta}) \equiv \exp\left(-i\boldsymbol{\theta}\cdot\hat{\mathbf{S}}\right) \quad \text{where} \quad \boldsymbol{\theta} = (\theta_x, \theta_y, \theta_z)$$

   Spinors transform as:

$$\hat{R}(\boldsymbol{\theta})\hat{\Psi}_i\hat{R}(\boldsymbol{\theta})^\dagger = \mathcal{R}(\boldsymbol{\theta})\hat{\Psi}_i$$

   with $\mathcal{R}(\boldsymbol{\theta}) \in \mathrm{SU}(2)$ the unitary matrix associated to said rotation,

$$\mathcal{R}(\boldsymbol{\theta}) = \exp\left(-\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\right) \quad \text{where} \quad \boldsymbol{\tau} = (\tau_x, \tau_y, \tau_z) \tag{3.10}$$

   Let us prove this. Let $\hat{A} = i\boldsymbol{\theta}\cdot\hat{\mathbf{S}}$ e $\hat{B} = \hat{\Psi}_i$. The following identity, a consequence of Baker-Campbell-Housdorff formula, holds:

$$e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + \left[\hat{A}, \hat{B}\right] + \frac{1}{2!}\left[\hat{A}, \left[\hat{A}, \hat{B}\right]\right] + \frac{1}{3!}\left[\hat{A}, \left[\hat{A}, \left[\hat{A}, \hat{B}\right]\right]\right] + \cdots$$

The first order of the expansion gives:

$$
\begin{aligned}
[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},\hat{\Psi}_i] &= i\sum_\alpha \theta_\alpha[\hat{S}^\alpha,\hat{\Psi}_i] \\
&= -\frac{i}{2}\sum_\alpha \theta_\alpha \tau^\alpha \hat{\Psi}_i \\
&= -\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\hat{\Psi}_i
\end{aligned}
$$

We omitted some purely algebraic passages. The second order:

$$
\begin{aligned}
\Big[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},\Big[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},\hat{\Psi}_i\Big]\Big] &= \Big[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},-\frac{i}{2}(\boldsymbol{\theta}\cdot\boldsymbol{\tau})\hat{\Psi}_i\Big] \\
&= -\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\Big[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},\Psi_i\Big] \\
&= \frac{1}{4}(\boldsymbol{\theta}\cdot\boldsymbol{\tau})^2\hat{\Psi}_i
\end{aligned}
$$

and so on. Collecting all terms,

$$
\begin{aligned}
\hat{R}(\boldsymbol{\theta})\hat{\Psi}_i\hat{R}(\boldsymbol{\theta})^\dagger &= \hat{\Psi}_i - \frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\hat{\Psi}_i + \frac{1}{2!\times 4}(\boldsymbol{\theta}\cdot\boldsymbol{\tau})^2\hat{\Psi}_i + \cdots \\
&= \sum_{k=1}^{\infty}\frac{1}{k!}\left(-\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\right)^k\hat{\Psi}_i \\
&= \mathcal{R}(\boldsymbol{\theta})\hat{\Psi}_i
\end{aligned}
$$

In last passage, we identified the series as the Taylor expansion of the exponential, recognising the expression for $\mathcal{R}(\boldsymbol{\theta})$ given in Eq. (3.10).

4. Then, the group acts on the product $\hat{\Psi}_i^\dagger\hat{\Psi}_i$ as

$$
\begin{aligned}
\hat{R}\hat{\Psi}_i^\dagger\hat{\Psi}_i\hat{R}^\dagger &= \hat{R}\hat{\Psi}_i^\dagger\hat{R}^\dagger\hat{R}\hat{\Psi}_i\hat{R}^\dagger \\
&= \hat{\Psi}_i^\dagger\mathcal{R}^\dagger\mathcal{R}\hat{\Psi}_i \\
&= \hat{\Psi}_i^\dagger\hat{\Psi}_i
\end{aligned}
$$

being $\mathcal{R}\in \mathrm{SU}(2)$. Recalling Eqns. (3.8), (3.9), both $\hat{H}_V$ and $\hat{N}$ are SU(2) symmetric.

The EHM is symmetric under global spin rotations. Note that $\hat{H}_V$ is actually symmetric under local spin rotations, meaning that we could rotate the spin at each site by a different amount and get the same interaction. The same holds for $\hat{H}_U$. It is the kinetic operator that makes the EHM globally (and not locally) SU(2) symmetric.

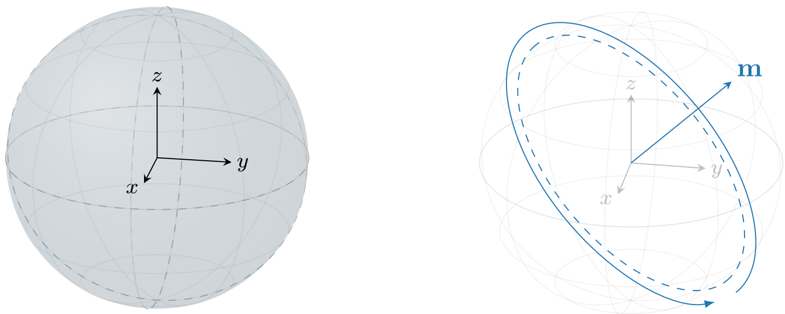
Note that the $U^{\mathrm{c}}(1)$ gauge symmetry and the SU(2) spin rotations symmetry can be combined,

$$
\mathrm{U}^{\mathrm{c}}(1)\otimes\mathrm{SU}(2)=\mathrm{U}(2)
$$

We will refer to U(2) symmetry to indicate full charge-spin conservation. Finally, note the group of 3D rotations contains the subgroup of 2D rotations around a given axis z,

$$
\mathrm{U}^z(1)\subset\mathrm{SU}(2)
$$

where the z superscript serves to distinguish from the gauge group. The z axis does not need to be identified with the "real" z axis, perpendicular with the lattice plane. Any given direction works. Inverting our logic, we can build the SU(2) generators, starting from $\hat{S}^z$, after we have chosen the direction z. In terms of commutation relations, a given operator can respect $\mathrm{U}^z(1)$ but not SU(2): in that case, it will commutate to zero with $\hat{S}^z$, and not the other spin operators. A graphic rendition of this SSB is given in Fig. 3.2, representing for a given vector $\mathbf{m}$ the passage from full SU(2) rotational symmetry around any direction to reduced U(1) rotational symmetry around just one direction.



a | SU(2) rotations domain before SSB.

b | U(1) rotations domain after SSB.

Figure 3.2 | Graphic representation of the concept of SU(2) → $U^z(1)$ SSB. "Before" SSB, we have full symmetry under any possible rotation (solid color sphere, left panel). "After" SSB we have symmetry only for rotations around some vector **m** (dashed tilted circumference, right panel).

3.1.3 Particle-hole symmetry at half-filling

The half-filled EHM on the square lattice exhibits particle-hole symmetry (PHS). The presence of this additional symmetry is a consequence of the fact that the square lattice is bipartite. Let us prove this. Consider the generic unitary particle-hole (PH) transformation

$$\mathrm{P} \quad : \quad \hat{c}_{i\sigma} \quad \rightarrow \quad e^{i\eta_{i\sigma}} \hat{h}^{\dagger}_{i\sigma} \tag{3.11}$$

being in general η_i a phase dependent on the fermionic quantum state $(i\sigma)$. Inserting the above transformation in the EHM, we get

$$\mathrm{P}\{\hat{H}\} = -t \sum_{\langle ij \rangle} \sum_{\sigma} e^{i(\eta_{i\sigma}+\eta_{j\sigma})} \hat{h}_{i\sigma} \hat{h}^{\dagger}_{j\sigma} + U \sum_{i} \hat{h}_{i\uparrow} \hat{h}_{i\downarrow} \hat{h}^{\dagger}_{i\downarrow} \hat{h}^{\dagger}_{i\uparrow} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma} \hat{h}_{j\sigma'} \hat{h}^{\dagger}_{j\sigma'} \hat{h}^{\dagger}_{i\sigma}$$

For the two interactions $\hat{H}_U$, $\hat{H}_V$ the phase factors cancelled out. We aim to find the conditions under which the above hamiltonian is mapped onto the EHM of (1.2),

$$\mathrm{P}\{\hat{H}\} \stackrel{?}{=} -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{h}^{\dagger}_{i\sigma} \hat{h}_{j\sigma} + U \sum_{i \in \mathcal{L}} \hat{h}^{\dagger}_{i\uparrow} \hat{h}^{\dagger}_{i\downarrow} \hat{h}_{i\downarrow} \hat{h}_{i\uparrow} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}^{\dagger}_{i\sigma} \hat{h}_{j\sigma'} \hat{h}_{j\sigma'} \hat{h}_{i\sigma}$$

Before starting the derivation, which is an extension of the argument carried out for the conventional HM [14], let us discuss some properties of the hole operators. Due to the presence of the phase factors in the PHS operative definition, the Fermi anticommutation rules hold regardlessly of the phase factor:

$$\begin{aligned} \{\hat{h}_{i\sigma}, \hat{h}^{\dagger}_{j\sigma'}\} &= e^{i(\eta_{i\sigma}-\eta_{j\sigma'})} \{\hat{c}^{\dagger}_{i\sigma}, \hat{c}_{j\sigma'}\} \\ &= e^{i(\eta_{i\sigma}-\eta_{j\sigma'})} \delta_{ij}\delta_{\sigma\sigma'} \\ &= \delta_{ij}\delta_{\sigma\sigma'} \end{aligned}$$

The on-site density operator is given by:

$$\hat{h}_{i\sigma} \hat{h}^{\dagger}_{i\sigma} = \hat{n}_{i\sigma}$$

being $\hat{n}_{i\sigma}$ the fermionic number operator on site i with spin σ. It follows, for quartic interactions

$$
\begin{aligned}
\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^{\dagger}_{j\sigma'}\hat{h}^{\dagger}_{i\sigma} &= -\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^{\dagger}_{i\sigma}\hat{h}^{\dagger}_{j\sigma'} \\
&= \hat{h}_{i\sigma}\hat{h}^{\dagger}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^{\dagger}_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^{\dagger}_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= -\hat{h}^{\dagger}_{i\sigma}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^{\dagger}_{j\sigma'} + \hat{h}_{j\sigma'}\hat{h}^{\dagger}_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^{\dagger}_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= \hat{h}^{\dagger}_{i\sigma}\hat{h}_{i\sigma}\hat{h}^{\dagger}_{j\sigma'}\hat{h}_{j\sigma'} - \hat{h}^{\dagger}_{i\sigma}\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}^{\dagger}_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^{\dagger}_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= -\hat{h}^{\dagger}_{i\sigma}\hat{h}^{\dagger}_{j\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'} - \hat{h}^{\dagger}_{i\sigma}\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}^{\dagger}_{j\sigma'} - \left(\hat{h}_{i\sigma}\hat{h}^{\dagger}_{j\sigma'} - \hat{h}^{\dagger}_{i\sigma}\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'} \\
&= \hat{h}^{\dagger}_{i\sigma}\hat{h}^{\dagger}_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} + (\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) - 1 - \left(\hat{h}_{i\sigma}\hat{h}^{\dagger}_{j\sigma'} - \hat{h}^{\dagger}_{i\sigma}\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'} \qquad (3.12)
\end{aligned}
$$

The first term of the last passage is mirrored with respect to the starting term. Now, let us deal with the kinetic, local and non-local interactions separately.

Kinetic term. Consider the kinetic term,

$$\hat{H}_t = -t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}$$

Apply the PH transform of Eq. (3.11)

$$\mathrm{P}\{\hat{H}_t\} = -t\sum_{\langle ij\rangle}\sum_{\sigma}e^{i(\eta_{i\sigma}+\eta_{j\sigma})}\hat{h}_{i\sigma}\hat{h}^{\dagger}_{j\sigma}$$

Anti-commutate the two fermionic operators, and exchange site labels $i \leftrightarrow j$. We get:

$$\mathrm{P}\{\hat{H}_t\} = -t\sum_{\langle ij\rangle}\sum_{\sigma}e^{i(\eta_{i\sigma}+\eta_{j\sigma}+\pi)}\hat{h}^{\dagger}_{i\sigma}\hat{h}_{j\sigma}$$

A necessary condition on the unitary operation P is the following,

$$\eta_{i\sigma} + \eta_{j\sigma} = (2k-1)\pi \quad \text{with} \quad k \in \mathbb{Z} \qquad (3.13)$$

valid for NN sites.

Local repulsion. Substitute Eq. (3.12) in $\hat{H}_U$,

$$
\begin{aligned}
\mathrm{P}\{\hat{H}_U\} &= U\sum_{i\in\mathcal{L}}\hat{h}_{i\uparrow}\hat{h}_{i\downarrow}\hat{h}^{\dagger}_{i\downarrow}\hat{h}^{\dagger}_{i\uparrow} \\
&= U\sum_{i\in\mathcal{L}}\hat{h}^{\dagger}_{i\uparrow}\hat{h}^{\dagger}_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} + U\sum_{i\in\mathcal{L}}(\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow}) - U\sum_{i\in\mathcal{L}}(1) \\
&= U\sum_{i}\hat{h}^{\dagger}_{i\uparrow}\hat{h}^{\dagger}_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} + U(\hat{N} - L_xL_y)
\end{aligned}
$$

For the half-filled state $\langle\hat{N}\rangle = L_xL_y$. The unitary map of Eq. (3.11), independently of the condition of Eq. (3.13), leaves the hamiltonian identical at half-filling.

Non-local attraction. Proceed identically for $\hat{H}_V$,

$$
\begin{aligned}
\mathrm{P}\{\hat{H}_V\} &= -V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^{\dagger}_{j\sigma'}\hat{h}^{\dagger}_{i\sigma} \\
&= -V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}^{\dagger}_{i\sigma}\hat{h}^{\dagger}_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} - V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}(\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) + V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}(1) \\
&= -V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}^{\dagger}_{i\sigma}\hat{h}^{\dagger}_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} - 2zV(\hat{N} - L_xL_y)
\end{aligned}
$$

Symmetry	Operations
$\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$	One-site discrete translations separately in both directions
C_4	Four-fold point group (real $\pi/2$ angle rotations)
$\mathrm{U}^{\mathrm{c}}(1)$	Charge-conserving global phase shifts
$\mathrm{SU}(2)$	Three-dimensional rotations in global spin space

Table 3.1 | Model symmetries and relative group operations. Note that $\mathrm{U}^{\mathrm{c}}(1)$ represents the gauge symmetry linked to charge conservation.

with $z = 4$ the lattice coordination number. In the last passage, we used that on the square lattice

$$\begin{aligned}\sum_{\langle ij\rangle}(1) &= \sum_{i\in\mathcal{L}/2} z \\ &= \frac{L_x L_y}{2} \times z \end{aligned} \tag{3.14}$$

As before, the non-local attraction is particle-hole symmetric at half-filling. As for the local repulsion, PHS is given just the operation of Eq. (3.11). The condition of Eq. (3.13) is irrelevant.

Recollecting everything, if P is defined by the unitary operation

$$\mathrm{P} \quad : \quad \begin{cases} \hat{c}_{i\sigma} \to \hat{h}^\dagger_{i\sigma} & \text{for} \quad i \in \mathcal{S}_1 \\ \hat{c}_{i\sigma} \to -\hat{h}^\dagger_{i\sigma} & \text{for} \quad i \in \mathcal{S}_2 \end{cases}$$

and the system is half-filled, PHS is respected. Any other phase, as long as it alternates by π on the two sublattices, is good enough. As for the conventional HM [14], this result can be generalised to any bipartite lattice. In turn, if a lattice is not bipartite, PHS is not present. To summarise, we report in Tab. 3.1 all identified symmetries.

3.2 Symmetry breaking channels in real space

In many-body Physics, a phase transition is described by some order parameter rising from zero to a non-zero value. This is associated with some spontaneous symmetry breaking (SSB). In order to simulate different phases, we need a way to distinguish them based on which symmetries they break (or not). We also need a way to work computationally in a given symmetry sector. HF methods, by reducing the hamiltonian to a non-interacting approximation, offer a simple framework to these ends. Indeed, once we identified all symmetries of a given phase, we just impose them on our approximation of the full hamiltonian. The found ground-state approximation will entail these symmetries by construction.

Recall now the discussion of the HF method of Sec. 2.2. The ending point were Eqns. (2.14), (2.15), which identify the self-consistent single-particle fields which approximate the original quartic hamiltonian. In order to get there, we applied Wick's theorem to the quartic part of the hamiltonian and derived the most general expressions of $\mathbf{h}$. In next sections here apply Wick's theorem to the quartic operators of $\hat{H}_{\mathrm{EHM}}$ and discuss the symmetry properties of the resulting decomposition.

Before starting, let us define precisely the three phases we will investigate in terms of symmetries:

- Normal phase: no broken symmetry.
- Antiferromagnetic phase: simultaneous $\mathrm{SU}(2) \overset{\mathrm{SSB}}{\to} \mathrm{U}^z(1)$ and $\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)} \overset{\mathrm{SSB}}{\to} \mathrm{T}_2^{(x)} \otimes \mathrm{T}_2^{(y)}$. Among all possible realisations of antiferromagnetism, we realise a SDW with alternating distribution of spin densities;
- Superconducting phase: simultaneous $\mathrm{U}^c(1) \overset{\mathrm{SSB}}{\to} 0$ and $\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$. We investigate anisotropic superconductivity, allowing for the particle number to be not conserved and for nematic effects.

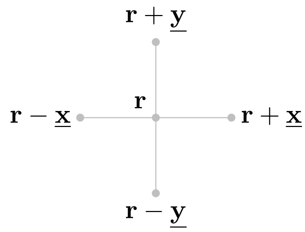


Figure 3.3 | Schematic representation of the four NNs of a given site i for a planar square lattice.

Nematic effects are observed in unconventional superconductors [23], and this is the reason of our interest in the possibility of rotational SSBs. We will theoretically discuss the nematic SSB $C_4 \to C_2$ also for normal and AF phases, as if discrete rotational symmetry was broken. As we will see, while it not trivial to exclude it *a priori*, it will actually turn out that the HF ground state preserves C_4 in both cases.

We now move to a more detailed discussion of each hamiltonian contribution. In next sections, we switch from the site index i to the vector notation $\mathbf{r}$, indicating the site 2D position on the lattice

$$i \quad \to \quad \mathbf{r} = (x, y)$$

with $x, y \in \mathbb{N}$. The lattice versors are:

$$\underline{\mathbf{x}} \equiv (1,0) \qquad \underline{\mathbf{y}} \equiv (0,1)$$

as in Fig. 3.3.

3.2.1 Local repulsion

The local repulsion $\hat{H}_U$ is the quartic operator of the conventional HM,

$$\begin{aligned}\hat{H}_U &= U \sum_{\mathbf{r}\in\mathcal{L}} \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\downarrow} \\ &= U \sum_{\mathbf{r}\in\mathcal{L}} \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\end{aligned}$$

In order to obtain the self-consistent values of $h^{+-}_{\alpha\beta}$, $h^{++}_{\alpha\beta}$ we need to compute an expectation value of the full hamiltonian. Wick's theorem guarantees that, for the wavefunctions we are using, the expectation value of a quartic fermionic operator is the sum of all fully contracted terms. Recalling the nomenclature of Eq. (2.1),

$$\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle = \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\rangle\langle \hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Cooper}} \tag{3.15}$$

where the equal sign holds as long as we perform the averaging operation on the subspace of gaussian states. Now we discuss the above EVs separately, identifying if any can be neglected based on selection rules in a given phase.

Hartree term. The Hartree part of the above decomposition is a product of density EVs,

$$\langle \hat{n}_{\mathbf{r}\uparrow}\rangle\langle \hat{n}_{\mathbf{r}\downarrow}\rangle$$

Recalling the groups of Tab. 3.1, we see that the density operator preserves all symmetries. Single-site density operators commutate with spin operators and are gauge-invariant. Moreover, shifting or rotating the lattice leaves density operators unchanged. Thus no selection rule applies, and the above EVs are in principle non-zero.

Fock terms. For the Fock part, notice that

$$\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow} = \hat{S}^+_{\mathbf{r}}$$

being $S^+_{\mathbf{r}}$ the local raising operator. From $\mathfrak{su}(2)$ algebra we have:

$$\left[\hat{S}^+_{\mathbf{r}}, \hat{S}^\alpha\right] \neq 0 \qquad \alpha \in \{x, y, z\}$$

If the ground-state is invariant under global spin rotations, these EVs must be zero. Gauge symmetry gives no selection rules, due to Eq. (3.5).

Cooper terms. The Cooper part can be non-zero only if charge is not conserved. In terms of the gauge symmetry, it suffices to apply the map of Eq. (3.4) to $\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}$

$$\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow} \quad \to \quad e^{2i\eta}\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}$$

to conclude that this operator is in general not gauge invariant. It is instead SU(2)-symmetric. To prove this, it suffices to compute commutators of $\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}$ with $\hat{S}^z$, $\hat{S}^+$:

$$\begin{aligned}
\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{S}^z\right] &= \left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \frac{1}{2}\sum_{\mathbf{r}'\in\mathcal{L}}(\hat{n}_{\mathbf{r}'\uparrow} - \hat{n}_{\mathbf{r}'\downarrow})\right] \\
&= \frac{1}{2}\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right] \\
&= \frac{1}{2}\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}, \hat{n}_{\mathbf{r}\uparrow}\right]\hat{c}^\dagger_{\mathbf{r}\downarrow} - \frac{1}{2}\hat{c}^\dagger_{\mathbf{r}\uparrow}\left[\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{n}_{\mathbf{r}\downarrow}\right]
\end{aligned}$$

Since:

$$\begin{aligned}
\left[\hat{c}^\dagger_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}\right] &= \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma} - \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\sigma} \\
&= -\hat{c}^\dagger_{\mathbf{r}\sigma}\left(1 - \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma}\right) \\
&= -\hat{c}^\dagger_{\mathbf{r}\sigma}
\end{aligned} \tag{3.16}$$

we have:

$$\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{S}^z\right] = -\frac{1}{2}\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow} + \frac{1}{2}\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow} = 0$$

Moreover, considering the raising operator:

$$\begin{aligned}
\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{S}^+\right] &= \left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \frac{1}{2}\sum_{\mathbf{r}'\in\mathcal{L}}\hat{c}^\dagger_{\mathbf{r}'\uparrow}\hat{c}_{\mathbf{r}'\downarrow}\right] \\
&= \left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow}\right] \\
&= \left[\hat{c}^\dagger_{\mathbf{r}\uparrow}, \hat{c}^\dagger_{\mathbf{r}\uparrow}\right]\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow} - \left(\hat{c}^\dagger_{\mathbf{r}\uparrow}\right)^2\left[\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{c}_{\mathbf{r}\downarrow}\right] \\
&= 0
\end{aligned}$$

where we used the facts that every operator commutates with itself, and a squared creation operator is null. These commutation relations, together, prove that Cooper EVs are legitimately non-zero for a SU(2)-preserving phase.

In Tab. 3.2 we collect the selection rules for the Hartree, Fock and Cooper terms of Eq. (3.15), based on the symmetry specifications given for each phase in above paragraphs. Note that Fock-like EVs are prohibited for all phases.

Term	Normal phase	Antiferromagnet	Superconductor
Hartree			
Fock	Prohibited	Prohibited	Prohibited
Cooper	Prohibited	Prohibited	

Table 3.2 Summary of selection rules for the Wick's decomposition of Eq. (3.15) of the local repulsion $\hat{H}_U$. An empty entry indicates no selection rule.

3.2.2 Non-local attraction

In order to analyse similarly the non-local attraction $\hat{H}_V$, it is useful to decompose the operator in spin channels:

$$\begin{aligned}\hat{H}_V &\equiv \underbrace{\hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow}}_{\text{Same spin}} + \underbrace{\hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow}}_{\text{Opposite spin}} \\ &= \hat{H}^{(\text{s.s.})} + \hat{H}^{(\text{o.s.})}\end{aligned} \tag{3.17}$$

being

$$\hat{H}_V^{\sigma\sigma'} \equiv -V \sum_{\langle \mathbf{r},\mathbf{r}'\rangle} \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}'\sigma'}$$

Let us start from the opposite spin sector. On the square lattice (as for any bipartite lattice) NNs belong to different sublattices. This implies that any NNs sum can be rewritten as:

$$\sum_{\langle \mathbf{r}\mathbf{r}'\rangle} f(\mathbf{r},\mathbf{r}') = \sum_{\mathbf{r}\in\mathcal{L}/2} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} f(\mathbf{r},\mathbf{r}+\boldsymbol{\delta}) \tag{3.18}$$

with f some function. We are using the notation of Fig. 3.3. The symbol "$\mathcal{L}/2$" indicates one of the two sublattices $\mathcal{S}_1$, $\mathcal{S}_2$ represented in Fig. 3.1. We can rewrite:

$$\hat{H}_V^{\sigma\sigma'} = -V \sum_{\mathbf{r}\in\mathcal{L}/2} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma'}\right)$$

Now, if we choose $\mathcal{L}/2 = \mathcal{S}_1$, this gives:

$$\begin{aligned}\hat{H}_V^{\uparrow\downarrow} &= -V \sum_{\mathbf{r}\in\mathcal{S}_1} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} + \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\downarrow}\right) \\ \hat{H}_V^{\downarrow\uparrow} &= -V \sum_{\mathbf{r}\in\mathcal{S}_1} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\downarrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\uparrow}\right)\end{aligned}$$

These two operators are "complementary": $\sigma =\uparrow$ densities act on $\mathcal{S}_1$ for $\hat{H}_V^{\uparrow\downarrow}$, on $\mathcal{S}_2$ for $\hat{H}_V^{\downarrow\uparrow}$; $\sigma =\downarrow$ densities, the opposite. Then their sum can be expressed as:

$$\begin{aligned}\hat{H}_V^{(\text{o.s.})} &\equiv \hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow} \\ &= -V \sum_{\mathbf{r}\in\mathcal{L}} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} + \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\downarrow}\right)\end{aligned} \tag{3.19}$$

Wick's theorem applied to this quartic operator gives:

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle &= \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle \\ &\simeq \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle\langle \hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Cooper}}\end{aligned} \tag{3.20}$$

For the same-spin sector we define

$$\begin{aligned}\hat{H}_V^{(\text{s.s.})} &\equiv \hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow} \\ &= -V \sum_{\mathbf{r}\in\mathcal{L}/2} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \sum_{\sigma\in\{\uparrow,\downarrow\}} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma}\right)\end{aligned} \tag{3.21}$$

Note that, with respect to the o.s. sector, we are summing on half the sites two times: one for spin $\uparrow$, one for spin $\downarrow$. Applying Wick's theorem:

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle &= \langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle \\ &\simeq \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle\langle \hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle}_{\text{Cooper}}\end{aligned} \tag{3.22}$$

As we did for $\hat{H}_U$, we analyse each separately.

Hartree terms. As it was for $\hat{H}_U$, Hartree EVs of Eqns. (3.19), (3.21) are not subject to selection rules for any symmetry of Tab. 3.1. Thus they are allowed in both sectors.

Fock terms. The Fock EVs from the two sectors behave differently. Let us compute the following generic commutation relations:

$$\begin{aligned}\left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{S}^z\right] &= \left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \frac{1}{2}\sum_{\mathbf{r}'\in\mathcal{L}}(\hat{n}_{\mathbf{r}'\uparrow}-\hat{n}_{\mathbf{r}'\downarrow})\right] \\ &= \frac{(-1)^{\delta_{\sigma=\downarrow}}}{2}\left[\hat{c}^\dagger_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}\right]\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2}\hat{c}^\dagger_{\mathbf{r}\sigma}\left[\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}\right] \\ &= \left(-\frac{(-1)^{\delta_{\sigma=\downarrow}}}{2} + \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2}\right)\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}\end{aligned} \tag{3.23}$$

where in last passage we used Eq. (3.16) and its conjugate version,

$$[\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma}] = \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}$$

Now, if $\sigma = \sigma'$ the last passage of Eq. (3.23) is zero; otherwise is not. This proves that the o.s. Fock EVs are non-zero only for a phase that completely breaks SU(2); instead, in the s.s. sector we can have non-zero EVs for a phase that reduces SU(2) to U^z(1). [Commutation with $\hat{S}^+$?]

Cooper terms. Cooper terms are not gauge invariant. The proof is analogous to the one given for $\hat{H}_U$. Instead, the o.s. and s.s. spin sectors behave differently under global spin rotations. With an identical procedure as for Eq. (3.23), we get:

$$\begin{aligned}\left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{S}^z\right] &= \left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \frac{1}{2}\sum_{\mathbf{r}'\in\mathcal{L}}(\hat{n}_{\mathbf{r}'\uparrow}-\hat{n}_{\mathbf{r}'\downarrow})\right] \\ &= \frac{(-1)^{\delta_{\sigma=\downarrow}}}{2}\left[\hat{c}^\dagger_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}\right]\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2}\hat{c}^\dagger_{\mathbf{r}\sigma}\left[\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}\right] \\ &= \left(-\frac{(-1)^{\delta_{\sigma=\downarrow}}}{2} - \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2}\right)\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\sigma'}\end{aligned} \tag{3.24}$$

We have a mirrored situation with respect to the Fock terms. For the Cooper channel, the s.s. EVs are zero if U^z(1) is broken, while the o.s. sector are not. [Commutation with $\hat{S}^+$?]

In Tab. 3.3 we summarise these considerations, giving the selection rules on the EVs of Eqns. (3.20), (3.22), dividing in the two sectors "o.s." and "s.s.". By identifying these selection rules, we set grounds for the detailed derivation of each chapter and heavily simplify application of HF model to the EHM. We now move to the last part of this chapter: reciprocal space description of the EHM.

3.3 The EHM in reciprocal space

The EHM is symmetric under discrete lattice translations. We will investigate phases that do not (completely) break translational invariance. It is thus convenient to move to reciprocal space, by

Sector	Term	Normal phase	Antiferromagnet	Superconductor
o.s.	Hartree			
	Fock	Prohibited	Prohibited	Prohibited
	Cooper	Prohibited	Prohibited	
s.s.	Hartree			
	Fock			
	Cooper	Prohibited	Prohibited	Prohibited

Table 3.3 Summary of selection rules for the Wick's decomposition of Eqns. (3.20), (3.22) of the non-local attraction $\hat{H}_V$. An empty entry indicates no selection rule.

Fourier-transforming the hamiltonian. We will use the following convention:

$$\hat{c}_{\mathbf{r}\sigma} \stackrel{\mathcal{F}}{=} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in \mathrm{BZ}} e^{-i\mathbf{k}\cdot\mathbf{r}} \hat{c}_{\mathbf{k}\sigma} \tag{3.25}$$

Throughout the text, $\stackrel{\mathcal{F}}{=}$ will indicate a passage where a Fourier transform is used. "BZ" is shorthand notation for the Brillouin Zone.

It is useful to recall that the following equation for Kronecker's delta holds:

$$\frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} e^{i\mathbf{k}\cdot\mathbf{r}} = \delta_{\mathbf{k}=\mathbf{0}} \tag{3.26}$$

On one sublattice:

$$\frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}/2} e^{i\mathbf{k}\cdot\mathbf{r}} = \frac{1}{2}\delta_{\mathbf{k}=\mathbf{0}} \tag{3.27}$$

The notation $\mathcal{L}/2$ is shorthand to indicate $\mathcal{S}_1$ or $\mathcal{S}_2$.

Kinetic part. Let us transform $\hat{H}_t$:

$$\begin{aligned}
\hat{H}_t &= -t \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}'\sigma} + \text{h.c.}\right] \\
&\stackrel{\mathcal{F}}{=} -t \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in\mathrm{BZ}} e^{i\mathbf{k}\cdot\mathbf{r}} \hat{c}^\dagger_{\mathbf{k}\sigma} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}'\in\mathrm{BZ}} e^{-i\mathbf{k}'\cdot\mathbf{r}'} \hat{c}_{\mathbf{k}'\sigma} + \text{h.c.} \\
&= -2t \sum_{\mathbf{k},\mathbf{k}'} \sum_{\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma} \times \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}/2} e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \times \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} e^{i\mathbf{k}'\cdot\boldsymbol{\delta}} \\
&= -2t \sum_{\mathbf{k},\mathbf{k}'} \sum_{\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \left[\frac{e^{ik_x} + e^{-ik_x}}{2} + \frac{e^{ik_y} + e^{-ik_y}}{2}\right] \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}
\end{aligned} \tag{3.28}$$

where we used Eqns. (3.18), (3.27). We defined the bare bands:

$$\epsilon_{\mathbf{k}} \equiv -2t\left(\cos k_x + \cos k_y\right) \tag{3.29}$$

Local repulsion. The local repulsion transforms as:

$$
\begin{aligned}
\hat{H}_U &= U \sum_{\mathbf{r}\in\mathcal{L}} \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow} \\
&\overset{\mathcal{F}}{=} U \sum_{\mathbf{r}\in\mathcal{L}} \frac{1}{(L_xL_y)^2} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i\mathbf{k}_1\cdot\mathbf{r}}\hat{c}^\dagger_{\mathbf{k}_1\uparrow} e^{i\mathbf{k}_2\cdot\mathbf{r}}\hat{c}^\dagger_{\mathbf{k}_2\downarrow} e^{-i\mathbf{k}_3\cdot\mathbf{r}}\hat{c}_{\mathbf{k}_3\downarrow} e^{-i\mathbf{k}_4\cdot\mathbf{r}}\hat{c}_{\mathbf{k}_4\uparrow} \\
&= \frac{U}{L_xL_y} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} \hat{c}^\dagger_{\mathbf{k}_1\uparrow}\hat{c}^\dagger_{\mathbf{k}_2\downarrow}\hat{c}_{\mathbf{k}_3\downarrow}\hat{c}_{\mathbf{k}_4\uparrow} \times \frac{1}{L_xL_y}\sum_{\mathbf{r}\in\mathcal{L}} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} \\
&= \frac{U}{L_xL_y} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} \hat{c}^\dagger_{\mathbf{k}_1\uparrow}\hat{c}^\dagger_{\mathbf{k}_2\downarrow}\hat{c}_{\mathbf{k}_3\downarrow}\hat{c}_{\mathbf{k}_4\uparrow}\delta_{\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4}
\end{aligned}
$$

where we used Eq. (3.26). Let $\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4=\mathbf{K}$, and define $\mathbf{k}$, $\mathbf{k}'$ such that

$$
\mathbf{k}_1 \equiv \mathbf{K}+\mathbf{k} \qquad \mathbf{k}_2 \equiv \mathbf{K}-\mathbf{k} \qquad \mathbf{k}_3 \equiv \mathbf{K}-\mathbf{k}' \qquad \mathbf{k}_4 \equiv \mathbf{K}+\mathbf{k}'
$$

Sums over these variables must be intended as over the Brillouin Zone (BZ). Hence:

$$
\hat{H}_U \overset{\mathcal{F}}{=} \frac{U}{L_xL_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{3.30}
$$

Non-local attraction. To transform the non-local attraction $\hat{H}_V$, it is useful to start by defining:

$$
h_{\mathbf{r}\sigma\sigma'} \equiv -V \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma'}\right)
$$

with:

$$
\hat{H}_V = \sum_{\mathbf{r}\in\mathcal{L}/2}\sum_{\sigma,\sigma'} h_{\mathbf{r}\sigma\sigma'}
$$

The product of densities on neighbouring sites transforms as:

$$
\begin{aligned}
\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'} &= \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\sigma} \\
&\overset{\mathcal{F}}{=} \frac{1}{(L_xL_y)^2} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} e^{\pm i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}} \hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}
\end{aligned}
$$

Then:

$$
\begin{aligned}
h_{\mathbf{r}\sigma\sigma'} &\overset{\mathcal{F}}{=} -\frac{V}{(L_xL_y)^2} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} \\
&\qquad\qquad \times \left(e^{i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}} + e^{-i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}}\right) \hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma} \\
&= -\frac{2V}{(L_xL_y)^2} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} \cos\left[(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}\right] \hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}
\end{aligned}
$$

The full non-local interaction is given by summing over all sites of $\mathcal{L}/2$ (which is, one sublattice). Applying Eq. (3.27) gives back momentum conservation

$$
\frac{1}{L_xL_y}\sum_{\mathbf{r}\in\mathcal{L}/2} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} = \frac{1}{2}\delta_{\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4}
$$

where we used Eq. (3.27). Define $\mathbf{k}$, $\mathbf{k}'$ and $\mathbf{K}$ as we did fo the local repulsion, and

$$
\delta\mathbf{k} \equiv \mathbf{k}-\mathbf{k}'
$$

Hence:

$$\begin{aligned}
\hat{H}_V &= \sum_{\mathbf{r}\in\mathcal{L}/2}\sum_{\sigma,\sigma'} h_{\mathbf{r}\sigma\sigma'} \\
&\overset{\mathcal{F}}{=} -\frac{V}{L_x L_y}\sum_{\sigma\sigma'}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \cos\left(\delta\mathbf{k}\cdot\boldsymbol{\delta}\right)\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \\
&= -\frac{V}{L_x L_y}\sum_{\sigma\sigma'}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}
\end{aligned} \tag{3.31}$$

Finally, we can separate in same spin and opposite spin channels. This gives:

$$\begin{aligned}
\hat{H}_V \overset{\mathcal{F}}{=} & \overbrace{-\frac{V}{L_x L_y}\sum_{\sigma}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}^{\text{s.s.}} \\
& \underbrace{-\frac{V}{L_x L_y}\sum_{\sigma}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}_{\text{o.s.}}
\end{aligned}$$

The o.s. part can be simplified: since

$$\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} = \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}$$

due to fermionic anti-commutation rules, and since the composite transform

$$\mathbf{k}\to-\mathbf{k} \qquad \mathbf{k}'\to-\mathbf{k}'$$

leaves the form factor $\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)$ invariant, we reduce the spin sum to two times the $\sigma=\uparrow$ part,

$$\hat{H}_V^{(\text{o.s.})} = -\frac{2V}{L_x L_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{3.32}$$

The $\mathcal{F}$-transform of the s.s. part is given explicitly by:

$$\begin{aligned}
\hat{H}_V^{(\text{s.s.})} = & -\frac{V}{L_x L_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right] \\
& \times\left[\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\uparrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\uparrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow}+\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\downarrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\downarrow}\right]
\end{aligned} \tag{3.33}$$

All these expressions are going to be used in the analysis of each phase. In reciprocal space, following Eq. (**??**) for a quartic operator $\hat{c}^\dagger_{\mathbf{k}_1}\hat{c}^\dagger_{\mathbf{k}_2}\hat{c}_{\mathbf{k}_3}\hat{c}_{\mathbf{k}_4}$ we distinguish contractions as follows:

- Hartree contractions:

$$\contraction{}{\hat{c}^\dagger_{\mathbf{k}_1}}{\hat{c}^\dagger_{\mathbf{k}_2}\hat{c}_{\mathbf{k}_3}}{\hat{c}_{\mathbf{k}_4}}\hat{c}^\dagger_{\mathbf{k}_1}\hat{c}^\dagger_{\mathbf{k}_2}\hat{c}_{\mathbf{k}_3}\hat{c}_{\mathbf{k}_4} \qquad \contraction{\hat{c}^\dagger_{\mathbf{k}_1}}{\hat{c}^\dagger_{\mathbf{k}_2}}{}{\hat{c}_{\mathbf{k}_3}}\hat{c}^\dagger_{\mathbf{k}_1}\hat{c}^\dagger_{\mathbf{k}_2}\hat{c}_{\mathbf{k}_3}\hat{c}_{\mathbf{k}_4}$$

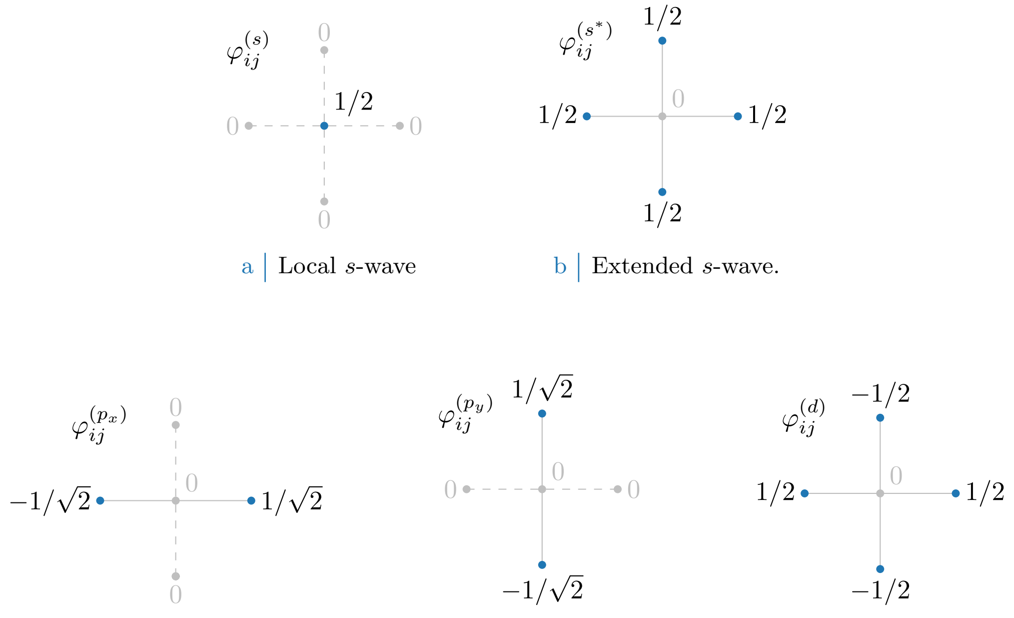
- Fock contractions:

$$\contraction{}{\hat{c}^\dagger_{\mathbf{k}_1}}{\hat{c}^\dagger_{\mathbf{k}_2}}{\hat{c}_{\mathbf{k}_3}}\hat{c}^\dagger_{\mathbf{k}_1}\hat{c}^\dagger_{\mathbf{k}_2}\hat{c}_{\mathbf{k}_3}\hat{c}_{\mathbf{k}_4} \qquad \contraction{\hat{c}^\dagger_{\mathbf{k}_1}}{\hat{c}^\dagger_{\mathbf{k}_2}}{\hat{c}_{\mathbf{k}_3}}{\hat{c}_{\mathbf{k}_4}}\hat{c}^\dagger_{\mathbf{k}_1}\hat{c}^\dagger_{\mathbf{k}_2}\hat{c}_{\mathbf{k}_3}\hat{c}_{\mathbf{k}_4}$$

- Cooper contractions:

$$\contraction{}{\hat{c}^\dagger_{\mathbf{k}_1}}{}{\hat{c}^\dagger_{\mathbf{k}_2}}\hat{c}^\dagger_{\mathbf{k}_1}\hat{c}^\dagger_{\mathbf{k}_2}\hat{c}_{\mathbf{k}_3}\hat{c}_{\mathbf{k}_4} \qquad \contraction{\hat{c}^\dagger_{\mathbf{k}_1}\hat{c}^\dagger_{\mathbf{k}_2}}{\hat{c}_{\mathbf{k}_3}}{}{\hat{c}_{\mathbf{k}_4}}\hat{c}^\dagger_{\mathbf{k}_1}\hat{c}^\dagger_{\mathbf{k}_2}\hat{c}_{\mathbf{k}_3}\hat{c}_{\mathbf{k}_4}$$

with appropriate sign factors. With the EHM expressed in reciprocal space, there is no general feature left to address. We can move to the detailed discussion of each phase.



a | Local s-wave b | Extended s-wave.

c | p_x-wave. d | p_y-wave. e | $d_{x^2-y^2}$-wave.

Figure 3.4 | Form factors at different topologies, as listed in Tab. 3.4. In figures five sites are represented: the hub site i and its four NN. Solid lines represent non-zero values for $\varphi_{\boldsymbol{\delta}}$, while dashed lines represent vanishing factors.

3.4 Square lattice spatial symmetry channels

This short conclusive section is apparently rather uncorrelated with all we said in the previous parts. We here introduce a set of orthonormal functions that results particularly useful to represent quantities defined on NNs sites on the square lattice. Let us start from a generic function of two variables $f(\mathbf{r},\mathbf{r}')$. Suppose that, for a given centre $\mathbf{r}$, the function is non-zero only on the site itself ($\mathbf{r}'=\mathbf{r}$) and its NNs $\mathbf{r}'=\mathbf{r}\pm\boldsymbol{\delta}$, with $\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}$. In other words,

$$f(\mathbf{r},\mathbf{r}')\neq 0 \qquad \Longleftrightarrow \qquad |\mathbf{r}-\mathbf{r}'|=0,1$$

in units of the lattice step. The following identity holds:

$$\begin{aligned} f(\mathbf{r},\mathbf{r}+\underline{\mathbf{x}}) &= f(\mathbf{r},\mathbf{r}')\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} \\ &= f(\mathbf{r},\mathbf{r}')\left(\varphi^{(s^*)}_{\mathbf{rr}'}+\varphi^{(d)}_{\mathbf{rr}'}+\sqrt{2}\varphi^{(p_x)}_{\mathbf{rr}'}\right) \end{aligned}$$

where we defined the set of orthonormal functions $\varphi^{(\gamma)}_{\mathbf{rr}'}$ listed explicitly in Tab. 3.4 and sketched in Fig. 3.4. The orthonormality condition reads:

$$\frac{1}{L_xL_y}\sum_{\mathbf{rr}'}\varphi^{(\gamma)}_{\mathbf{rr}'}\varphi^{(\gamma')*}_{\mathbf{rr}'}=\delta_{\gamma\gamma'}$$

Similarly, also the following relations hold:

$$\begin{aligned} f(\mathbf{r},\mathbf{r}) &= f(\mathbf{r},\mathbf{r}')\varphi^{(s)}_{\mathbf{rr}'} \\ f(\mathbf{r},\mathbf{r}-\underline{\mathbf{x}}) &= f(\mathbf{r},\mathbf{r}')\left(\varphi^{(s^*)}_{\mathbf{rr}'}+\varphi^{(d)}_{\mathbf{rr}'}-\sqrt{2}\varphi^{(p_x)}_{\mathbf{rr}'}\right) \\ f(\mathbf{r},\mathbf{r}+\underline{\mathbf{y}}) &= f(\mathbf{r},\mathbf{r}')\left(\varphi^{(s^*)}_{\mathbf{rr}'}-\varphi^{(d)}_{\mathbf{rr}'}+\sqrt{2}\varphi^{(p_y)}_{\mathbf{rr}'}\right) \\ f(\mathbf{r},\mathbf{r}-\underline{\mathbf{y}}) &= f(\mathbf{r},\mathbf{r}')\left(\varphi^{(s^*)}_{\mathbf{rr}'}-\varphi^{(d)}_{\mathbf{rr}'}-\sqrt{2}\varphi^{(p_y)}_{\mathbf{rr}'}\right) \end{aligned}$$

Structure	Form factor	Graph
s-wave	$\varphi^{(s)}_{\mathbf{r}\mathbf{r}'} = \delta_{\mathbf{r}\mathbf{r}'}$	Fig. 3.4a
Extended s-wave	$\varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} = \frac{1}{2}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 3.4b
p_x-wave	$\varphi^{(p_x)}_{\mathbf{r}\mathbf{r}'} = \frac{1}{\sqrt{2}}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}}\right]$	Fig. 3.4c
p_y-wave	$\varphi^{(p_y)}_{\mathbf{r}\mathbf{r}'} = \frac{1}{\sqrt{2}}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 3.4d
$d_{x^2-y^2}$-wave	$\varphi^{(d)}_{\mathbf{r}\mathbf{r}'} = \frac{1}{2}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} - \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 3.4e

Table 3.4 Form factors for a function defined on a square lattice, with a prefactor included for normalization. We use lattice vector notation here. The last column indicates the graph representation of each contribution given in Fig. 3.4. Subscript $x^2 - y^2$ is omitted for notational clarity.

Structure	Structure factor	Graph
s-wave	$\varphi^{(s)}_{\mathbf{k}} = 1$	Fig. 3.4a
Extended s-wave	$\varphi^{(s^*)}_{\mathbf{k}} = \cos k_x + \cos k_y$	Fig. 3.4b
p_x-wave	$\varphi^{(p_x)}_{\mathbf{k}} = i\sqrt{2}\sin k_x$	Fig. 3.4c
p_y-wave	$\varphi^{(p_y)}_{\mathbf{k}} = i\sqrt{2}\sin k_y$	Fig. 3.4d
$d_{x^2-y^2}$-wave	$\varphi^{(d)}_{\mathbf{k}} = \cos k_x - \cos k_y$	Fig. 3.4e

Table 3.5 Structure factors derived from the correlation structures of Tab. 3.4. The functions hereby defined are orthonormal, and are plotted in Fig. 3.5.

Thus, defining:

$$\begin{aligned}
f^{(s)} &\equiv f(\mathbf{r},\mathbf{r}) \\
f^{(s^*)} &\equiv f(\mathbf{r},\mathbf{r}+\underline{\mathbf{x}}) + f(\mathbf{r},\mathbf{r}-\underline{\mathbf{x}}) + f(\mathbf{r},\mathbf{r}+\underline{\mathbf{y}}) + f(\mathbf{r},\mathbf{r}-\underline{\mathbf{y}}) \\
f^{(p_x)} &\equiv f(\mathbf{r},\mathbf{r}+\underline{\mathbf{x}}) - f(\mathbf{r},\mathbf{r}-\underline{\mathbf{x}}) \\
f^{(p_y)} &\equiv f(\mathbf{r},\mathbf{r}+\underline{\mathbf{y}}) - f(\mathbf{r},\mathbf{r}-\underline{\mathbf{y}}) \\
f^{(d)} &\equiv f(\mathbf{r},\mathbf{r}+\underline{\mathbf{x}}) + f(\mathbf{r},\mathbf{r}-\underline{\mathbf{x}}) - f(\mathbf{r},\mathbf{r}+\underline{\mathbf{y}}) - f(\mathbf{r},\mathbf{r}-\underline{\mathbf{y}})
\end{aligned}$$

we get the decomposition:

$$f(\mathbf{r},\mathbf{r}') = \sum_{\gamma} f^{(\gamma)}\varphi^{(\gamma)}_{\mathbf{r}\mathbf{r}'}$$

We decomposed the original function in its harmonics. Moving to reciprocal space, we get the $\mathcal{F}$-transformed functions of Tab. 3.5, also satisfying the orthonormality condition:

$$\frac{1}{L_x L_y}\sum_{\mathbf{k}} \varphi^{(\gamma)}_{\mathbf{k}}\varphi^{(\gamma')*}_{\mathbf{k}} = \delta_{\gamma\gamma'}$$

The functions of Tab. 3.5 are plotted in the BZ in Fig. 3.5.

We will refer to these harmonics as "symmetry channels", indicating the various spatial components of the order parameter associated to a given phase transition. As we will see, indeed, we consider order parameters that have specific transformation properties under planar rotations. To distinguish the various spatial harmonics of the order parameter is useful in order to apply symmetry-based selection rules.

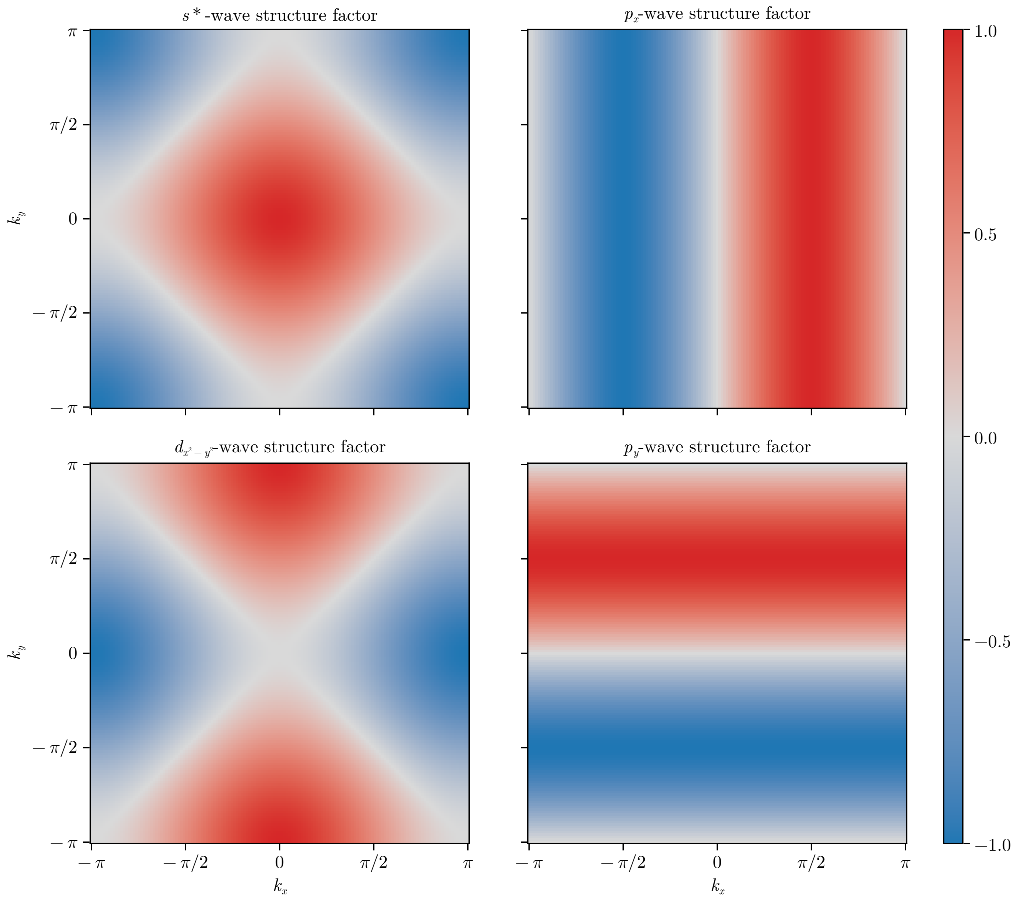


Figure 3.5 Representation of the various structure factors listed in Tab. 3.5 on the BZ. For the s^* and $d_{x^2-y^2}$-wave factors, the real part is plotted; for the p_ℓ factors, the imaginary part is plotted.

Finally, a very useful mathematical identity is given by the following argument. Consider Eq. (3.31), where is present the factor $\cos(\delta k_x) + \cos(\delta k_y)$ (with $\delta k_x = k_x - k'_x$ and $\delta k_y = k_y - k'_y$). Expanding,

$$\cos(\delta k_x) + \cos(\delta k_y) = c_x c'_x + c_y c'_y + s_x s'_x + s_y s'_y$$

In last passage, for the sake of readability, we used the notations

$$c_\ell \equiv \cos k_\ell \qquad s_\ell \equiv \sin k_\ell \qquad c'_\ell \equiv \cos k'_\ell \qquad s'_\ell \equiv \sin k'_\ell$$

Group the four terms,

$$\underbrace{\left(c_x c'_x + c_y c'_y\right)}_{\text{Symmetric}} + \underbrace{\left(s_x s'_x + s_y s'_y\right)}_{\text{Anti-symmetric}} \tag{3.34}$$

All four terms are invariant under simultaneous inversion of both momenta, $(\mathbf{k}, \mathbf{k}') \to (-\mathbf{k}, -\mathbf{k}')$. Instead, if we invert just one momentum of the pair, the first two are symmetric for both arguments $\mathbf{k}$, $\mathbf{k}'$; the second two are anti-symmetric. Decoupling the symmetric part, we obtain

$$c_x c'_x + c_y c'_y = \frac{1}{2}(c_x + c_y)(c'_x + c'_y) + \frac{1}{2}(c_x - c_y)(c'_x - c'_y)$$

which finally gives:

$$
\begin{aligned}
\cos(\delta k_x) + \cos(\delta k_y) = &\frac{1}{2}(c_x + c_y)(c'_x + c'_y) && (s^*\text{-wave}) \\
&+ s_x s'_x && (p_x\text{-wave}) \\
&+ s_y s'_y && (p_y\text{-wave}) \\
&+ \frac{1}{2}(c_x - c_y)(c'_x - c'_y) && (d_{x^2-y^2}\text{-wave})
\end{aligned}
$$

or, in a more compact form

$$
\cos(\delta k_x) + \cos(\delta k_y) = \frac{1}{2}\sum_\gamma \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}'}^{(\gamma)*} \quad \text{for } \gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\} \tag{3.35}
$$

$$
= \frac{1}{2}\sum_\gamma \varphi_{\mathbf{k}}^{(\gamma)*} \varphi_{\mathbf{k}'}^{(\gamma)} \tag{3.36}
$$

This decomposition will result particularly handy in later calculations.

Chapter 4

The normal phase

In this chapter we apply the HF methods sketched in Sec. 2.2 to the "normal phase". We are studying the model in a state were no symmetry of those enumerated in Tab. 3.1 is broken. Even though this phase may appear not so interesting, it actually shows a significant physical difference with respect to the conventional perfectly degenerate Fermi gas.

HF methods applied in this context will lead us to a self-consistent hopping renormalisation effect. For certain values of the non-local attraction V and the doping level, a complete Mott localisation effect takes place. The resulting bands shrinking and flattening is the main net effect of the presence of $\hat{H}_V$ in the EHM.

This chapter is divided in three main parts. First, we discuss the symmetries of the normal phase, following the general discussion of Chap. 3, and apply HF methods to derive the self-consistent approximated hamiltonian. Second, we impose some theoretical constraints on the self-consistent solution. Lastly, we present numeric results.

4.1 The "normal" phase and its symmetries

In the normal phase no symmetry is broken. Thus, the HF wavefunction approximating the ground state must exhibit the full $\mathrm{U}(2) \otimes \mathrm{C}_4 \otimes \mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$ symmetry discussed in Sec. 3.1. We take a slight detour from this, and instead allow for

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$$

This is the only potential SSB we consider. It will turn out in Sec. 4.5.2 that even allowing for breaking of rotational symmetry, the ground-state actually does not break C_4. This can be seen as a convolute way to confirm that there is not a normal-nematic phase. While this may seem as an useless complication, the derivation will set grounds for actual nematic effects found in the superconducting phase.

Our HF method relies on the application of Wick's theorem on the quartic part of the interacting hamiltonian, expressed in Eq. (2.1). We will decompose the quartic operators according to Eqns. (6.2), (2.18). Recalling Tabs. 3.2 and 3.3, for the normal phase we must consider the following contributes to the EVs of Eqns. (3.15), (3.20), (3.22):

$\hat{H}_U$	Hartree contraction
$\hat{H}_V$ (o.s. sector)	Hartree contraction
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions

As we will see in detail and is summarised in Tab. 4.1, Hartree contractions give a chemical potential shift, while the unique Fock contraction effectively renormalises the bare bands.

In the normal phase, spin is a pure physical degeneracy, being the $\uparrow$ and $\downarrow$ sectors separated; moreover, $\hat{H}$ is $\mathbb{Z}_2$ invariant with respect to spin inversion, $\sigma \leftrightarrow \overline{\sigma}$. This implies that all physical quantities, as is for the normal phase, should not depend on the spin index. We will perform all computations leaving σ explicit and using this consideration at the end.

Operator	Sector	RWCs	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
	s.s.	Hartree	μ shift by nzV
		Fock	Bands renormalization and resymmetrization

Table 4.1 | Relevant Wick's contractions and net effect in the normal phase.

4.2 Hartree effects: chemical potential renormalization

Let us start by the analysis of the Hartree channel. The Hartree channel is defined as operator resulting from summing Hartree-like terms originating from Wick's decomposition of the quartic parts of $\hat{H}$. Each quartic term decomposes as a sum of fully contracted pairs, as in Eq. (2.1).

In the final hamiltonian, the net effect originating from Hartree terms will be a renormalisation of μ. To derive analytically this effect is only important to the ends of computing correctly the free energy density of the approximated ground state. As is discussed in Sec. 2.3.1, in the context of numerical simulations at fixed density, the chemical potential is determined self-consistently at each step; thus any net effect that shifts μ is effectively ignored within (and limitedly to) the iterative algorithm.

4.2.1 Hartree channel in the normal phase: analytic derivation

The quartic part of $\hat{H}$ is given by the the local repulsion $\hat{H}_U$ and the non-local attraction $\hat{H}_V$. Applying HF methods to them, we get the decomposition of Eqns. (6.2), (2.18). Let us deal with the Hartree sector of both separately.

Local repulsion. In the normal phase, the local repulsion $\hat{H}_U$ decomposes in Hartree channel as specified in Eq. (6.2)

$$\hat{H}_U \simeq U \sum_i \left[\langle \hat{n}_{i\uparrow} \rangle \hat{n}_{i\downarrow} + \hat{n}_{i\uparrow} \langle \hat{n}_{i\downarrow} \rangle - \langle \hat{n}_{i\downarrow} \rangle \langle \hat{n}_{i\uparrow} \rangle \right]$$

For a translationally invariant phase such as the normal phase, it holds

$$\langle \hat{n}_{i\sigma} \rangle = n \tag{4.1}$$

which gives

$$\begin{aligned} \hat{H}_U &\simeq nU \sum_i \left[\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow} \right] - U \sum_i n^2 \\ &= nU \times \hat{N} - E^{(\mathrm{N})}_{\mathrm{H}/U} \end{aligned} \tag{4.2}$$

being $\hat{N}$ the total number operator and $E^{(\mathrm{N})}_{\mathrm{H}/U}$ the "contraction energy" (also known as double counting term). In Sec. 4.2.2 we deal with it. The chemical potential is corrected by

$$\mu \to \mu - nU \tag{4.3}$$

The non-local attraction further corrects μ.

Non-local attraction. The non-local attraction is divided in two sectors, as in Eq. (3.17). From Eq. (2.18), its Hartree-like terms are:

$$\hat{H}_V \simeq \overbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{n}_{i\sigma}\rangle\hat{n}_{j\sigma}+\hat{n}_{i\sigma}\langle\hat{n}_{j\sigma}\rangle-\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\sigma}\rangle\right]}^{\text{s.s.}}$$
$$\underbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{n}_{i\sigma}\rangle\hat{n}_{j\bar{\sigma}}+\hat{n}_{i\sigma}\langle\hat{n}_{j\bar{\sigma}}\rangle-\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\bar{\sigma}}\rangle\right]}_{\text{o.s.}}+(\text{all the rest}) \quad (4.4)$$

where "all the rest" collects all non-Hartree terms. Recalling the Ansatz of Eq. (4.1), we get

$$\hat{H}_V \simeq \underbrace{-nV\sum_{\langle ij\rangle}\sum_{\sigma}\left[\hat{n}_{j\sigma}+\hat{n}_{i\sigma}\right]}_{\text{s.s.}}\underbrace{-nV\sum_{\langle ij\rangle}\sum_{\sigma}\left[\hat{n}_{j\bar{\sigma}}+\hat{n}_{i\sigma}\right]}_{\text{o.s.}}-E^{(\mathrm{N})}_{\mathrm{H}/V}+(\text{all the rest})$$

with $E^{(\mathrm{N})}_{\mathrm{H}/V}$ the normal state shift to energy brought by $\hat{H}_V$. Both sums above give a number operator. Note that we sum over NNs: each site is connected to $z = 4$ sites. Including this combinatorial factor, we have

$$\hat{H}_V \simeq -2nzV\times\hat{N}-E^{(\mathrm{N})}_{\mathrm{H}/V}+(\text{all the rest}) \quad (4.5)$$

This accounts for the final Hartree shift of μ,

$$\mu \to \mu + 2znV \quad (4.6)$$

Considering the net shift to μ due to both interactions by combining Eqns. (4.3), (4.6), we get the final RMP anticipated in Tab. 4.1,

$$\tilde{\mu} \equiv \mu + n(2zV - U) \quad (4.7)$$

This result will remain valid throughout all phases discussed in this text. For the AF phase, Sec. 5.4.1, the SDW character leaves this shift untouched. The SC phase, Sec. 6.4.1, we will discuss is inherently translational invariant, thus the computation will be the same as here.

4.2.2 Hartree contraction energy

The double counting terms accounting for energy shifts in the Hartree channel are, as anticipated:

$$\begin{aligned} E^{(\mathrm{N})}_{\mathrm{H}/U} &= \sum_i \langle\hat{n}_{i\downarrow}\rangle\langle\hat{n}_{i\uparrow}\rangle \\ &= L_x L_y \times n^2 U \end{aligned} \quad (4.8)$$

for the local repulsion and

$$\begin{aligned} E^{(\mathrm{N})}_{\mathrm{H}/V} &= -V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\sigma}\rangle+\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\bar{\sigma}}\rangle\right] \\ &= -2V\sum_{\langle ij\rangle}\sum_{\sigma}n^2 \\ &= -L_x L_y \times 2zn^2 V \end{aligned} \quad (4.9)$$

for the non-local attraction. In the last passage, we used Eq. (3.14) and spin degeneracy. The contraction energy is an extensive quantity. When computing total free energy density, we will use the contraction energy per site.

4.3 Fock effects: bare bands renormalization

Analysis of Fock terms will lead us to the effect that makes the normal phase interesting, when compared to the free Fermi gas on the square lattice. From Wick's decomposition of $\hat{H}_V$, the unique allowed Fock term comes from the same-spin sector. This selection rules is implied by the SU(2) symmetry of the normal phase. Then, from Eq. (2.18):

$$\hat{H}_V \simeq (\text{Hartree channel}) + V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} \rangle \right] \tag{4.10}$$

Note the + sign in front of the displayed term: it originates from the − sign in Fock term of Eq. (2.1). A bond-wise hopping amplitude can be defined,

$$\tilde{t}_{ij\sigma} \equiv t - V \langle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} \rangle$$

The effective kinetic hamiltonian is given by

$$\begin{aligned} \hat{H}_{\tilde{t}} &= \hat{H}_t + V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} + \text{h.c.} \right] \\ &= - \sum_{\langle ij \rangle} \sum_{\sigma} \left[\tilde{t}_{ij\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] \end{aligned}$$

To proceed, it is useful to move to reciprocal space.

4.3.1 Fock channel in the normal phase: analytic derivation

In reciprocal space, the effective hopping must be transformed as well. Consider the Fourier Transform $\mathcal{F}$ given in Eq. (3.33), applied to the displayed part of Eq. (4.10),

$$\begin{aligned} & V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} \rangle \right] \\ & \qquad \stackrel{\mathcal{F}}{=} \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left[\cos(\delta k_x) + \cos(\delta k_y) \right] \langle \hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \end{aligned} \tag{4.11}$$

Here, the 2 prefactor comes from recognizing that the first two operators in square brackets in the first line generate identical contributions to the full sum (as is easy to see by performing trivial variables changes); the double counting energy shift is given by

$$E^{(\mathrm{N})}_{\mathrm{F}/V} \equiv \frac{V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left[\cos(\delta k_x) + \cos(\delta k_y) \right] \langle \hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \langle \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \rangle \tag{4.12}$$

Up to this point, this derivation is general and will be re-used in next chapters. Here the specificity of the physical phase settles in: if we assume the normal state to be a free degenerate Fermi gas with renormalized bands $\tilde{\epsilon}_{\mathbf{k}}$, we get

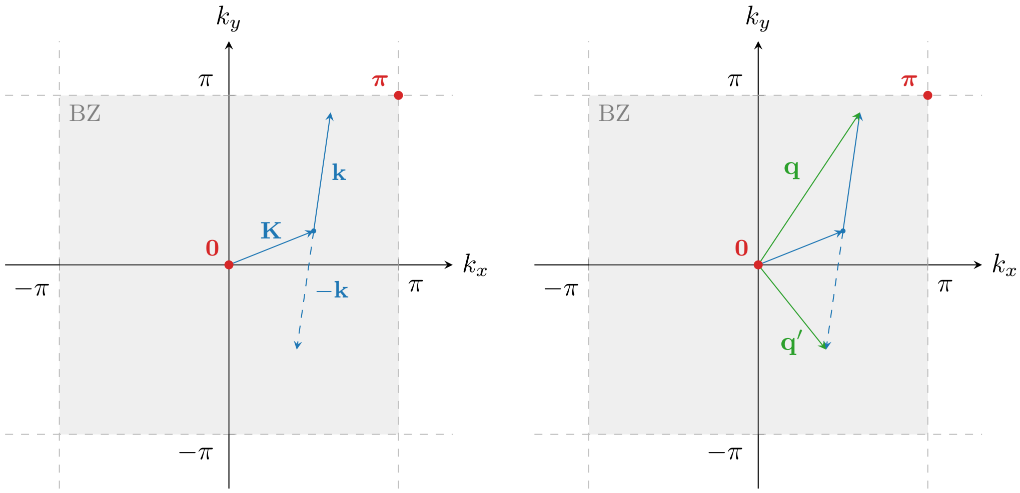
$$\langle \hat{c}^{\dagger}_{\mathbf{k}_1\sigma_1} \hat{c}_{\mathbf{k}_2\sigma_2} \rangle = \delta_{\mathbf{k}_1=\mathbf{k}_2} \delta_{\sigma_1=\sigma_2} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \tag{4.13}$$

being f the Fermi-Dirac distribution,

$$f(\epsilon; \beta, \mu) \equiv \frac{1}{e^{\beta(\epsilon-\mu)} + 1}$$

It follows that Eq. (4.11) becomes

$$\begin{aligned} & V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} \rangle \right] \\ & \qquad \stackrel{\mathcal{F}}{=} \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} \left[\cos(2k_x) + \cos(2k_y) \right] \langle \hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \end{aligned} \tag{4.14}$$



a | Generic vectors. b | Change of variables.

Figure 4.1 | Representation of the variable change employed to pass from Eq. (4.14) to (4.15). In Fig. 4.1a generic vectors are considered, cycling over all values of $\mathbf{K}, \mathbf{k} \in$ BZ. In Fig. 4.1b is depicted the variables change to the new vectors $\mathbf{q}, \mathbf{q}'$.

since the only contribution comes from $\mathbf{k}' = -\mathbf{k}$.

Let now $\mathbf{q} \equiv \mathbf{K} + \mathbf{k}$, $\mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$. A representation of this variables change is given in Fig. 4.1. Being for $\ell = x, y$

$$\begin{aligned} \delta q_\ell &\equiv q_\ell - q'_\ell \\ &= (K_\ell + k_\ell) - (K_\ell - k_\ell) \\ &= 2k_\ell \end{aligned}$$

we reduce Eq. (4.11) to

$$\begin{aligned} &V \sum_{\langle ij \rangle} \sum_\sigma \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \right] \\ &\qquad \overset{\mathcal{F}}{=} \frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_\sigma \left[\cos(\delta q_x) + \cos(\delta q_y) \right] \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \end{aligned} \tag{4.15}$$

Recall now the result of the symmetry channels decomposition of Eq. (3.35),

$$\cos(\delta q_x) + \cos(\delta q_y) = \frac{1}{2} \sum_\gamma \varphi^{(\gamma)}_{\mathbf{q}} \varphi^{(\gamma)*}_{\mathbf{q}'} \quad \text{for } \gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\}$$

Thanks to this handy property, we reduce Eq. (4.15) to

$$\begin{aligned} &V \sum_{\langle ij \rangle} \sum_\sigma \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \right] \\ &\qquad \overset{\mathcal{F}}{=} \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} u_{\mathbf{q}\sigma} \right] \times V \sum_{\mathbf{q}' \in \mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \end{aligned} \tag{4.16}$$

where we defined the EV:

$$u_{\mathbf{q}\sigma} \equiv \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \qquad u_{\mathbf{q}\sigma} = u^*_{\mathbf{q}\sigma} \tag{4.17}$$

The numeric value of $u_{\mathbf{q}\sigma}$ is given by the Fermi-Dirac distribution, but leaving it implicit we can re-use this computation later. Let $u^{(\gamma)}_\sigma$ be its γ-wave component,

$$u^{(\gamma)}_\sigma = \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle$$

Symmetry	Self-consistency equation	Graph
s^*-wave	$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$	Fig. 3.4b
$d_{x^2-y^2}$-wave	$u^{(d)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$	Fig. 3.4e

Table 4.2 | Symmetry resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$.

The bare bands $\epsilon_{\mathbf{k}}$ are s^*-wave symmetric,

$$\epsilon_{\mathbf{k}} = -2t(\cos k_x + \cos k_y) = -4t\varphi_{\mathbf{k}}^{(s^*)}$$

We used one function of those listed in Tab. 3.5. The renormalised bands $\tilde{\epsilon}_{\mathbf{k}}$ can exhibit a more complex symmetry. In the normal phase inversion symmetry is unbroken, thus

$$u_{\mathbf{q}\sigma} = u_{-\mathbf{q}\sigma}$$

It follows that antisymmetric harmonics are zero,

$$u_\sigma^{(p_\ell)} = 0 \quad \text{for} \quad \ell \in \{x, y\}$$

The remaining s^* and d symmetries give legitimate HFPs with the respective self-consistency equations, reported in Tab. 4.2.Spin index is omitted due to spin degeneracy.

The two HFPs $u_\sigma^{(s^*)}$ and $u_\sigma^{(d)}$ are in general non-zero, since we are allowing for $\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$. Note that when imposing C_4 invariance the $u_\sigma^{(d)}$ HFP vanishes, while in general $u_\sigma^{(s^*)}$ is non-zero. From these considerations Eq. (4.16) becomes

$$\begin{aligned} V \sum_{\langle ij \rangle} \sum_\sigma \left[\langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \rangle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} + \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \langle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} \rangle - \langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \rangle \langle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} \rangle \right] \\ \overset{\mathcal{F}}{=} V \sum_{\mathbf{q}\in\mathrm{BZ}} \sum_\sigma \left[u_\sigma^{(s^*)} \varphi_{\mathbf{q}}^{(s^*)*} + u_\sigma^{(d)} \varphi_{\mathbf{q}}^{(d)*} \right] \hat{c}_{\mathbf{q}\sigma}^\dagger \hat{c}_{\mathbf{q}\sigma} - E_{\mathrm{F}/V}^{(\mathrm{N})} \end{aligned} \tag{4.18}$$

Let

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u_\sigma^{(s^*)} V(\cos k_x + \cos k_y) + u_\sigma^{(d)} V(\cos k_x - \cos k_y) \tag{4.19}$$

This equation defines the bands renormalisation anticipated in Tab. 4.1, and is actually spin-independent. The fact that we have both s^* and a d-wave components introduces an anisotropy that is discussed in next section. However, we anticipate here that, as will result from Sec. 4.5.2, the d-wave harmonic will be suppressed. For now we keep this component ignoring these anticipations.

4.3.2 Fock contraction energy

Finally, let us derive the Fock contraction energy by expanding Eq. (4.12):

$$\begin{aligned} E_{\mathrm{F}/V}^{(\mathrm{N})} &= \frac{V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_\sigma [\cos(\delta q_x) + \cos(\delta q_y)] \langle \hat{c}_{\mathbf{q}\sigma}^\dagger \hat{c}_{\mathbf{q}'\sigma} \rangle \langle \hat{c}_{\mathbf{q}\sigma}^\dagger \hat{c}_{\mathbf{q}'\sigma} \rangle \\ &= L_x L_y \times \frac{V}{2} \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} u_{\mathbf{q}\sigma} \right] \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} u_{\mathbf{q}'\sigma}^* \right] \\ &= L_x L_y \times \frac{V}{2} \sum_\sigma \left[\left| u_\sigma^{(s^*)} \right|^2 + \left| u_\sigma^{(d)} \right|^2 \right] \end{aligned} \tag{4.20}$$

This expression gives the Fock contraction energy coming from the $\hat{H}_V$ operator. The squared spectral amplitude of $u_{\mathbf{k}\sigma}$ here is summed, and contributes to contraction energy linearly in V. This suggests that a mixed-symmetry band could be energetically preferred.

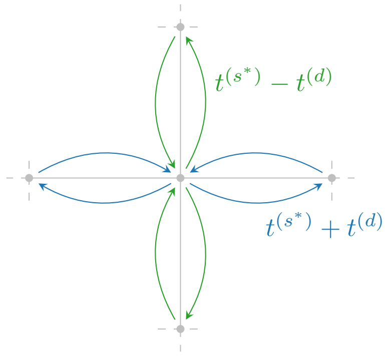


Figure 4.2 | Graphic rendition of anisotropic hopping, with a s^*-wave component (symmetric by $\pi/$ rotations) and a d-wave component (antisymmetric).

4.3.3 Nematicity: extending hopping to d-wave bands

For the normal phase, there is no reason to expect nematic effects, nor to exclude them. They are observed in superconducting phase [23]. Thus, we allow the possibility.

From Tabs. 3.4 and 3.5 we can inverse-Fourier transform the form factors to get the full effective hopping,

$$\tilde{t}_{ij\sigma} = \sum_{\gamma \in \{s^*, d\}} t_\sigma^{(\gamma)} \varphi_{ij}^{(\gamma)}$$

with

$$t_\sigma^{(s^*)} = t - \frac{u_\sigma^{(s^*)} V}{2} \qquad t_\sigma^{(d)} = -\frac{u_\sigma^{(d)} V}{2}$$

This is what we intend for "nematic effects". The possibility that planar rotational symmetry is broken in such a way that the kinetic operators in x and y directions differ. A schematics of anisotropic hopping is given in Fig. 4.2: for each bond linking two sites, the hopping parameter can be a positive or negative combination of the two components, whether the link direction is vertical or horizontal. The solution needs to be invariant under exchange $x \leftrightarrow y$, thus the actual SSB is $\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2/\mathbb{Z}_2$.

The fact that hopping is here anisotropic does not break full translational invariance $\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$. Imposing a free degenerate Fermi gas ground state in Eq. (4.13) leads to a uniform density. This is proven by Fourier-transforming the site number operator according to the convention of Eq. (3.25),

$$\hat{c}^\dagger_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}\sigma} \overset{\mathcal{F}}{=} \frac{1}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}'\sigma}$$

Then the on-site density EV is

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

A perfectly degenerate Fermi gas is always translational invariant and thus cannot exhibit non-uniform symmetry. The fact that hopping is allowed to be anisotropic does not interfere with that.

Note that for null HFPs,

$$u_\sigma^{(s^*)} = 0 \qquad u_\sigma^{(d)} = 0$$

we have, recalling Tab. 3.4:

$$\tilde{t}_{\mathbf{r}\mathbf{r}'\sigma} = \frac{t}{2} \left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}} \right]$$

The fact that $t/2$ appears here instead of t is a consequence of the equivalence:

$$-t \sum_{\langle ij \rangle} \sum_\sigma \left[\hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} + \mathrm{h.c.} \right] = -\frac{t}{2} \sum_{i,j \in \mathcal{S}} \sum_\sigma \left[\hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} + \mathrm{h.c.} \right]$$

where we including a prefactor that corrects double-counting. The RMP $\tilde{t}_{ij\sigma}$ must be interpreted like the right-hand side of the above equation, with the kinetic operator becoming:

$$\hat{H}_{\tilde{t}} \equiv -\frac{1}{2} \sum_{i,j \in \mathcal{S}} \sum_{\sigma} \left[\tilde{t}_{ij\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right]$$

To limit the sum to NNs is a matter of the symmetry structure of the function $\tilde{t}_{ij\sigma}$.

We can define the direction-dependent hopping as

$$\tilde{t}^{(x)} \equiv t - \frac{u^{(s^*)} - u^{(d)}}{2} V \tag{4.21}$$

$$\tilde{t}^{(y)} \equiv t - \frac{u^{(s^*)} + u^{(d)}}{2} V \tag{4.22}$$

Both components depend linearly on V. The HFP $u^{(s^*)}$ is a measure of the direction-averaged hopping shift with respect to V, while $u^{(d)}$ measures the anisotropy,

$$u^{(s^*)} = \frac{\left(t - \tilde{t}^{(x)}\right) + \left(t - \tilde{t}^{(y)}\right)}{V} \qquad u^{(d)} = \frac{\tilde{t}^{(y)} - \tilde{t}^{(x)}}{V}$$

This discussion is general, and not specific to the normal phase. In next chapters we will interpret phase renormalisation as in this section.

4.4 Normal free energy density

In the end, we want to compare phases. Thus we need a way to extract the free energy density of the approximated ground-state for each phase. The procedure sketched here will be re-used in next chapters. Let

$$f^{(\text{N})}_{\text{MFT}} \equiv \frac{F^{(\text{N})}_{\text{MFT}}}{L_x L_y}$$

being F_{MFT} the total free energy. Consider the complete hamiltonian

$$\hat{H} - \mu \hat{N} \simeq \sum_{\mathbf{k},\sigma} \tilde{\epsilon}_{\mathbf{k}\sigma} \hat{c}^{\dagger}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} + E^{(\text{N})}_{\text{c}} - \tilde{\mu} \hat{N}$$

where $E^{(\text{N})}_{\text{c}}$ is the (negative) total contraction energy for the normal phase,

$$E^{(\text{N})}_{\text{c}} \equiv -\left[E^{(\text{N})}_{\text{H}/U} + E^{(\text{N})}_{\text{H}/V} + E^{(\text{N})}_{\text{F}/V} \right]$$

having included the various contraction energies from Eqns. (4.8), (4.9) and (4.20).

Let us now take together all terms. From now on, we will omit spin index σ, making use of the always present $\mathbb{Z}_2$ spin inversion symmetry discussed in Sec. 4.1. For a free Fermi gas, the free energy density is made of four contributions:

1. The free particles ground state energy density,

$$e^{(\text{N})}_{\text{g}} = \frac{2}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

2. The contraction energy density,

$$\begin{aligned} e^{(\text{N})}_{\text{c}} &\equiv \frac{E^{(\text{N})}_{\text{c}}}{L_x L_y} \\ &= -n^2 U + V \left[2zn^2 - \left| u^{(s^*)} \right|^2 - \left| u^{(d)} \right|^2 \right] \end{aligned}$$

3. The entropic contribution to free energy

$$e_s^{(\mathrm{N})} \equiv -Ts$$

with $T = 1/k_B\beta$ and s the free Fermi gas entropy density for a single spin sector [29],

$$s = -\frac{k_B}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\left[(1-f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}))\ln(1-f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\right] \tag{4.23}$$

We use the identity:

$$\ln\left(1-\frac{1}{e^x+1}\right) = x + \ln\left(\frac{1}{e^x+1}\right)$$

and, substituting $x = \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu})$, reduce the above expression to:

$$\begin{aligned} s = -\frac{k_B}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\Big[&\ln(1-f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) \\ &- \beta(\tilde{\epsilon}_{\mathbf{k}}-\tilde{\mu})f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\Big] \end{aligned}$$

Second and fourth terms cancel out, and multiplying by $-2T$ (taking care of spin degeneracy) we have

$$e_s^{(\mathrm{N})} = \frac{2}{\beta L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\ln(1-f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) - \frac{2}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\tilde{\epsilon}_{\mathbf{k}}-\tilde{\mu})f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})$$

4. The chemical potential correction

$$e_n^{(\mathrm{N})} = \mu n$$

Note here that we are using the "physical" value of μ, not its RMP counterpart $\tilde{\mu}$. The reason for this is discussed in Sec. 2.3.1.

Thus, composing everything

$$\begin{aligned} f_{\mathrm{MFT}}^{(\mathrm{N})} &\equiv e_{\mathrm{g}}^{(\mathrm{N})} + e_{\mathrm{c}}^{(\mathrm{N})} + e_s^{(\mathrm{N})} + e_n^{(\mathrm{N})} \\ &= \frac{2}{\beta L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\ln(1-f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) - V\left[\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right] + [\mu + n(2zV-U)]\,n \end{aligned}$$

In Sec. 2.3.1 we argued that our iterative scheme computes $\tilde{\mu}$, not μ. Using Eq. (4.7), we see

$$\mu + n(2zV-U) = \tilde{\mu}$$

Then, the final expression is:

$$f_{\mathrm{MFT}}^{(\mathrm{N})} = \frac{2}{\beta L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\ln(1-f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) - V\left[\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right] + \tilde{\mu}n \tag{4.24}$$

including all effects.

4.5 Theoretical constraints on the normal phase HFPs

Within this section we discuss some theoretical constraints on the two HFPs of the normal phase. The discussion is divided in two parts: first we consider the simpler case of unviolated C_4 symmetry, then we move to anisotropic hopping and $\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$.

Before starting, it is useful to spend a few words on the spatial structure of a mixed $s^*\otimes d$-wave band. Let in general for notational simplicity

$$\tilde{\epsilon}_{\mathbf{k}} = \epsilon^{(s^*)}(\cos k_x + \cos k_y) + \epsilon^{(d)}(\cos k_x - \cos k_y)$$

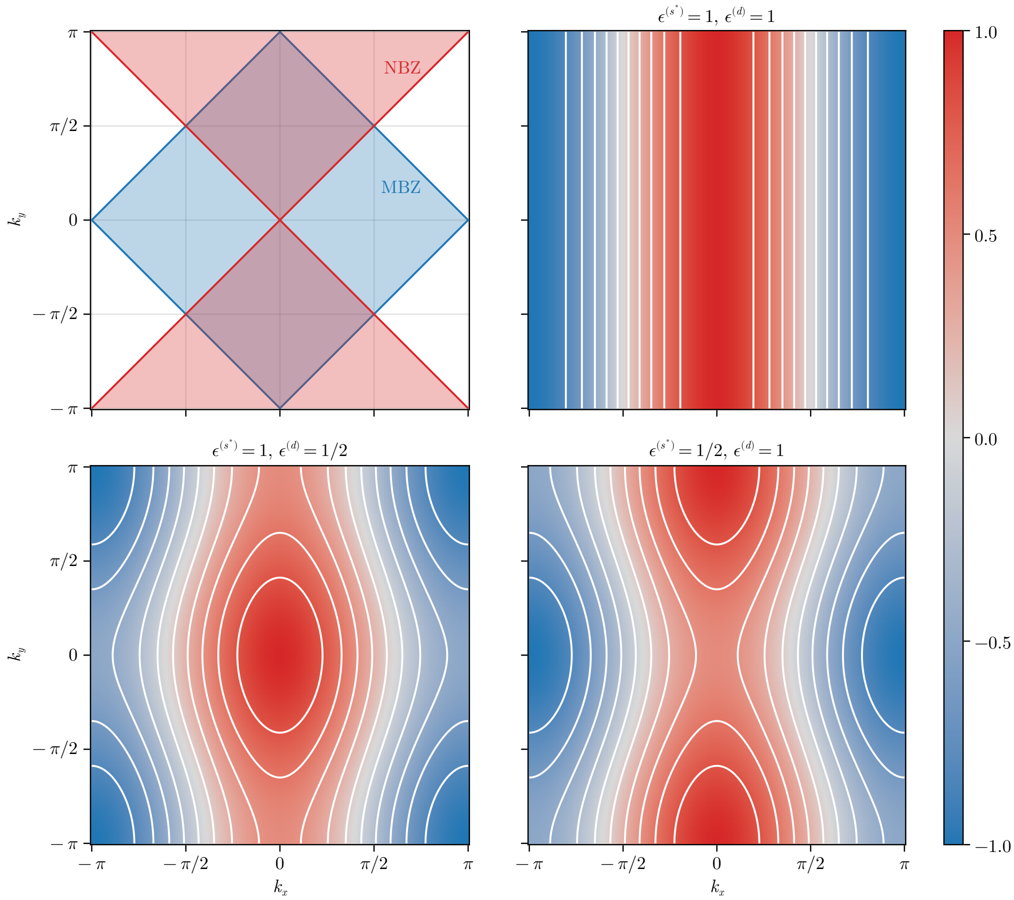


Figure 4.3 | Theoretical representation of mixed-symmetry nematic bands for the normal phase. The white contour represent level lines. In top-left panel, the MBZ and NBZ are depicted. While these plots only account for positive-defined components, in text is described how to map each one on its counterpart with different sign.

It is the balance between the two components $\epsilon^{(s^*)}$ and $\epsilon^{(d)}$ that gives the main features of the renormalised band. Consider the s^*-wave and d-wave structure factors already presented in Fig. 3.5, and define the Magnetic Brillouin Zone (MBZ) and Nematic Brillouin Zone (NBZ) as follows:

$$\mathrm{MBZ} \equiv \left\{ \mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \right\} \tag{4.25}$$

$$\mathrm{NBZ} \equiv \left\{ \mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(d)} \geq 0 \right\} \tag{4.26}$$

represented in the top-left panel of Fig. 4.3. The reason for the name of the MBZ will be clear in Chap. 5. In the rest of the panels of Fig. 4.3 are reported examples of mixed symmetry bands, for positive components. The respective opposite sign plots are obtained by the following rules:

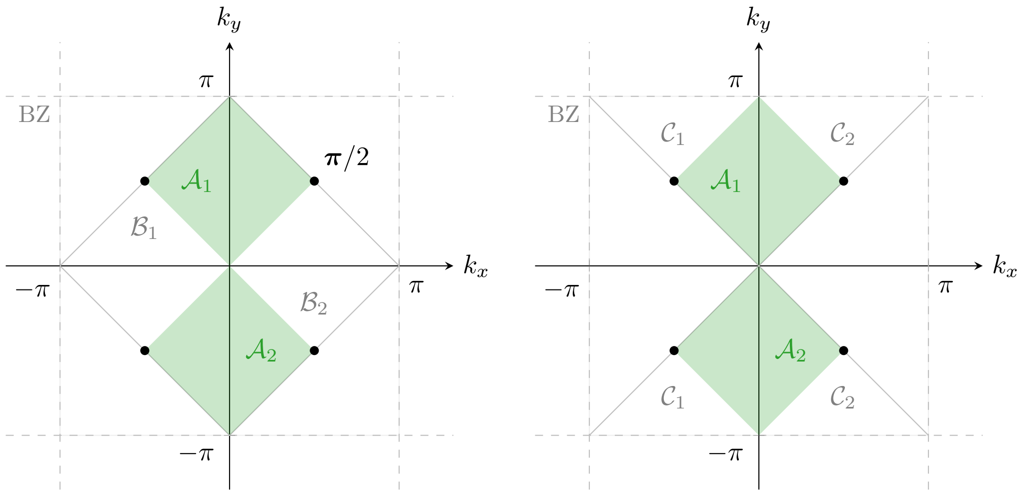
$\epsilon^{(s^*)} < 0$	and	$\epsilon^{(d)} < 0$	$\Longrightarrow$	Invert colors in Fig. 4.3
$\epsilon^{(s^*)} > 0$	and	$\epsilon^{(d)} < 0$	$\Longrightarrow$	Rotate plots by $\pi/2$ in Fig. 4.3
$\epsilon^{(s^*)} < 0$	and	$\epsilon^{(d)} > 0$	$\Longrightarrow$	Rotate plots by $\pi/2$ and invert colors in Fig. 4.3

Let us focus on the MBZ and the NBZ. Starting from the intersections of the two zones, both can be factored in two sectors of interest,

$$\mathrm{MBZ} = \mathcal{A} \cup \mathcal{B} \qquad \mathrm{NBZ} = \mathcal{A} \cup \mathcal{C}$$

being, accordingly to Fig. 4.4,

$$\mathcal{A} \equiv \mathcal{A}_1 \cup \mathcal{A}_2 \qquad \mathcal{B} \equiv \mathcal{B}_1 \cup \mathcal{B}_2 \qquad \mathcal{C} \equiv \mathcal{C}_1 \cup \mathcal{C}_2$$



a | MBZ sectors. b | NBZ sectors.

Figure 4.4 | Depiction of the sectors composing the MBZ (Fig. 4.4a) and the NBZ (Fig. 4.4b), based on the divisions shown in the top-left panel of Fig. 4.3. The full green zone composes $\mathcal{A}$, the region where both the s^*-wave and d-wave structure factors are positive defined; its MBZ complementary white counterpart in the left panel gives $\mathcal{B}$, for which the s^*-wave structure factors is positive-definite while the d-wave is negative. Similarly the NBZ complementary white counterpart in the right panel gives $\mathcal{C}$, for which signs are exchanged. The black dots, at the contours intersection, are band nodes.

These domains satisfy:

$$\begin{aligned}\mathcal{A} &\equiv \left\{\mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \geq 0\right\} \\ &= \mathrm{MBZ} \cap \mathrm{NBZ}\end{aligned} \tag{4.27}$$

$$\begin{aligned}\mathcal{B} &\equiv \left\{\mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \leq 0\right\} \\ &= \mathrm{MBZ} \setminus \mathrm{NBZ}\end{aligned} \tag{4.28}$$

$$\begin{aligned}\mathcal{C} &\equiv \left\{\mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \leq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \geq 0\right\} \\ &= \mathrm{NBZ} \setminus \mathrm{MBZ}\end{aligned} \tag{4.29}$$

and will be used later on. Whatever the shape of the mixed band, the following equation always holds

$$|k_x| = \frac{\pi}{2} \quad \text{and} \quad |k_y| = \frac{\pi}{2} \quad \implies \quad \tilde{\epsilon}_{(k_x,k_y)} = 0$$

since only cosines are involved. This identifies four node points, represented as black dots in Fig. 4.4.

4.5.1 Constraints in presence of a robust C_4 symmetry and bands shrinking

Consider the simpler situation where C_4 symmetry is imposed. This implies that the renormalised bands are

$$\tilde{\epsilon}_{\mathbf{k}} \equiv (u^{(s^*)}V - 2t)(\cos k_x + \cos k_y)$$

with a single self-consistency equation

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

The following chain of deductions leads us to a constraint on the possible values of $u^{(s^*)}$:

1. For $t > 0$, there is no self-consistent solution admitting $u^{(s^*)} \leq 0$. Let us prove this: suppose instead $u^{(s^*)} \leq 0$. For any function $g_{\mathbf{k}}$ defined on the BZ:

$$\sum_{\mathbf{k}\in\mathrm{BZ}} g_{\mathbf{k}} = \sum_{\mathbf{k}\in\mathrm{MBZ}} [g_{\mathbf{k}} + g_{\mathbf{k}+\boldsymbol{\pi}}]$$

where we defined the vector $\boldsymbol{\pi} \equiv (\pi, \pi)$. The self-consistency equation thus reads

$$\begin{aligned} u^{(s^*)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathbf{k}+\boldsymbol{\pi}}; \beta, \tilde{\mu}) \right] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] \end{aligned} \tag{4.30}$$

having we used the antisymmetry of s^*-wave functions by $\boldsymbol{\pi}$ shifts. Now, if $u^{(s^*)} \leq 0$ the renormalised band $\tilde{\epsilon}_{\mathbf{k}}$ is negative inside the MBZ and positive outside, implying

$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \geq f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

the equality only being satisfied on the MBZ contour. Then, since inside the MBZ it holds by definition $\cos k_x + \cos k_y \geq 0$, the above equations imply

$$u^{(s^*)} \leq 0 \quad \Longrightarrow \quad u^{(s^*)} \geq 0$$

To exclude the possibility of $u^{(s^*)} = 0$, it suffices to notice that from the argument above such a situation only arises when the MBZ contour degenerates to the entirety of the MBZ itself – which is, when the band is flat. But that cannot be the case for $u^{(s^*)} = 0$, as long as $t > 0$. Thus we have proven that

$$u^{(s^*)} > 0$$

2. For $t > 0$, there is no self-consistent solution admitting $u^{(s^*)} \geq 2t/V$. We proceed exactly as did for the previous point, up to the passage:

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right]$$

Now, if $u^{(s^*)} \geq 2t/V$ this means the prefactor in front of the renormalised bands is positive. Then $\tilde{\epsilon}_{\mathbf{k}}$ is positive inside the MBZ and negative outside, implying

$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \leq f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

thus we have

$$u^{(s^*)} \geq 2t/V \quad \Longrightarrow \quad u^{(s^*)} \leq 0$$

which is absurd. Thus

$$u^{(s^*)} < 2t/V$$

Collecting the two points above, we have the constraint on $u^{(s^*)}$

$$\mathrm{C}_4 \quad \Longrightarrow \quad 0 < u^{(s^*)} < 2t/V \tag{4.31}$$

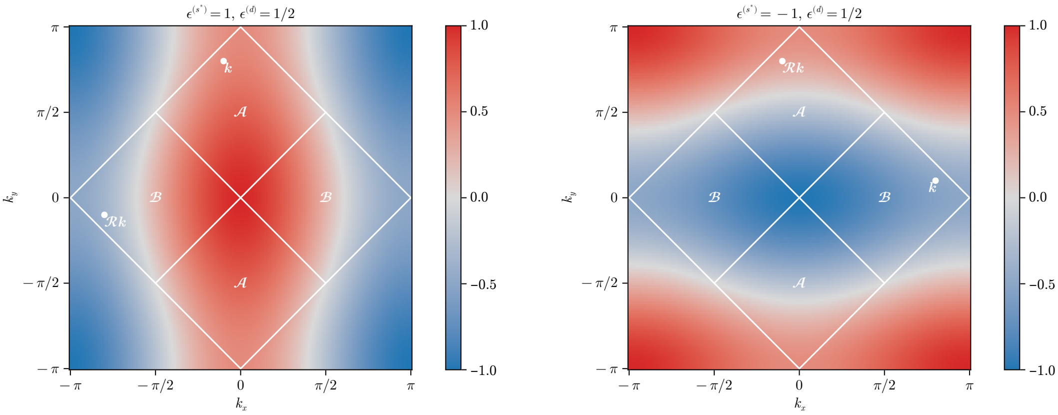
The effective hopping is smaller than the original hopping. We can denominate this net effect “bands shrinking”.

4.5.2 Natural suppression of nematicity within $\mathrm{C}_2/\mathbb{Z}_2$ symmetry

Let us extend the arguments above to a weaker symmetry, allowing for

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2/\mathbb{Z}_2$$

We work with the full, anisotropic band of Eq. (4.19). From the following chain of deductions we conclude that C_4 is not broken:



a | Same-sign mixed bands.

b | Opposite-sign mixed bands.

Figure 4.5 | Theoretical example of bands with $u^{(s^*)} > 2t/V$, $u^{(d)} > 0$ (top-left) and $u^{(s^*)} < 0$, $u^{(d)} > 0$ (top right). Two example points are reported, connected by a $\pi/2$ rotation as given in Eqns. (4.34), (4.32).

1. Let for instance

$$u^{(s^*)} \geq 2t/V > 0 \qquad u^{(d)} > 0 \qquad \Longrightarrow \qquad \epsilon^{(s^*)} \geq 0 \qquad \epsilon^{(d)} > 0$$

In Fig. 4.5a an example band is shown for $\epsilon^{(s^*)} = 1$, $\epsilon^{(d)} = 1/2$. Note that if $u^{(s^*)} = 2t/V$, the renormalised band is purely d-wave since $\epsilon^{(s^*)} = 0$.

Recall Eq. (4.30), and separate in two sums over $\mathcal{A}$ and $\mathcal{B}$:

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathcal{A}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] + \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathcal{B}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right]$$

Consider the sum over $\mathcal{B}$. Looking at the bottom-left panel of Fig. 4.3, such a region contains nodal curves across which bands change sign. Let $\mathcal{R}$ be the clockwise $\pi/2$ rotation: then, any point in $\mathcal{B}$ can be seen as the rotation of a point in $\mathcal{A}$

$$\forall \mathbf{k}' \in \mathcal{B} \quad \exists \quad \mathbf{k} \in \mathcal{A} \mid \mathbf{k}' = \mathcal{R}\mathbf{k} \tag{4.32}$$

A graphical rendition of this idea is given in Fig. 4.5a. Then we can rewrite $u^{(s^*)}$ as a single sum over $\mathcal{A}$,

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathcal{A}} (\cos k_x + \cos k_y) \times \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \right] \tag{4.33}$$

From Eq. (4.27), $\forall \mathbf{k} \in \mathcal{A}$ the form factors $\cos k_x \pm \cos k_y$ are positive by construction. By hypothesis $\epsilon^{(s^*)} \geq 0$ and $\epsilon^{(d)} > 0$. Hence:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \overbrace{\epsilon^{(s^*)} (\cos k_x + \cos k_y)}^{\geq 0} + \overbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}^{\geq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \underbrace{\epsilon^{(s^*)} (\cos k_x + \cos k_y)}_{\geq 0} - \underbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}_{\geq 0} \end{aligned}$$

Summing the two equations above we have:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} + \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} \geq 0$$

thus $\tilde{\epsilon}_{\mathbf{k}} \geq -\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}$. The Fermi-Dirac distribution is monotonically descendent. Then:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \begin{array}{r} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \leq f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \\ f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \geq f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \end{array}$$

This condition, inserted in Eq. (4.33), leads to an absurd:

$$u^{(s^*)} \geq 2t/V \qquad u^{(d)} > 0 \quad \Longrightarrow \quad u^{(s^*)} \leq 0$$

2. An identical argument, exchanging the roles of $\mathcal{A}$ and $\mathcal{B}$, can be carried out to show that

$$u^{(s^*)} \geq 2t/V > 0 \qquad u^{(d)} < 0 \quad \Longrightarrow \quad u^{(s^*)} \leq 0$$

3. The two points above imply that it is possible to have $u^{(s^*)} \geq 2t/V$ if and only if $u^{(d)} = 0$. But from Sec. 4.5.1, we conclude that the upper boundary $u^{(s^*)} < 2t/V$ always holds.

4. Let for instance

$$u^{(s^*)} \leq 0 \qquad u^{(d)} > 0 \quad \Longrightarrow \quad \epsilon^{(s^*)} < 0 \qquad \epsilon^{(d)} > 0$$

In Fig. 4.5b an example band is shown for $\epsilon^{(s^*)} = -1$, $\epsilon^{(d)} = 1/2$.

Proceed as for the first point. Similarly to Eq. (4.32), Any point in $\mathcal{A}$ can be obtained by rotating some point in $\mathcal{B}$,

$$\forall \mathbf{k}' \in \mathcal{A} \quad \exists \quad \mathbf{k} \in \mathcal{B} \mid \mathbf{k}' = \mathcal{R}\mathbf{k} \tag{4.34}$$

An equation similar to Eq. (4.33) holds, the sum being performed over $\mathcal{B}$

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{B}} (\cos k_x + \cos k_y) \times \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \right] \tag{4.35}$$

From Eq. (4.28), $\forall \mathbf{k} \in \mathcal{B}$ the form factor $\cos k_x + \cos k_y$ is positive by construction; the form factor $\cos k_x - \cos k_y$ is negative by construction. By hypothesis $\epsilon^{(s^*)} < 0$ and $\epsilon^{(d)} > 0$. Hence:

$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \begin{array}{r} \tilde{\epsilon}_{\mathbf{k}} = \overbrace{\epsilon^{(s^*)} (\cos k_x + \cos k_y)}^{\leq 0} + \overbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}^{\leq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} = \underbrace{\epsilon^{(s^*)} (\cos k_x + \cos k_y)}_{\leq 0} - \underbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}_{\leq 0} \end{array}$$

Summing the two equations above we have:

$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} + \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} \leq 0$$

thus $\tilde{\epsilon}_{\mathbf{k}} \leq -\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}$. The Fermi-Dirac distribution is monotonically descendent. Then:

$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \begin{array}{r} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \geq f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \\ f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \leq f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \end{array}$$

This condition, inserted in Eq. (4.33), leads to an absurd:

$$u^{(s^*)} \leq 0 \qquad u^{(d)} > 0 \quad \Longrightarrow \quad u^{(s^*)} > 0$$

5. can be carried out to show that

$$u^{(s^*)} \leq 0 \qquad u^{(d)} < 0 \quad \Longrightarrow \quad u^{(s^*)} > 0$$

6. The two points above imply that it is possible to have $u^{(s^*)} \leq 0$ if and only if $u^{(d)} = 0$. But from Sec. 4.5.1, we conclude that the lower boundary $u^{(s^*)} > 0$ always holds.

From the chain of derivations above, we can conclude self-consistently that also in the nematic structure the constraint of Eq. (4.31),

$$0 < u^{(s^*)} < 2t/V$$

holds. In the s^*-wave channel, partial breaking of rotational symmetry is not enough to modify the main effect identified under C_4 symmetry, "bands shrinking". What is missing is the behaviour of the d-wave channel when said effect takes place. Continuing the enumeration above,

7. Let $0 < u^{(s^*)} < 2t/V$ and $u^{(d)} > 0$. Then $\epsilon^{(d)} > 0$. The self-consistency equation for $u^{(d)}$ reads

$$\begin{aligned} u^{(d)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{NBZ}} (\cos k_x - \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \right] \end{aligned}$$

   where we recognized, as is shown in Fig. 4.4b, that $\mathrm{BZ} \setminus \mathrm{NBZ} = \mathcal{R}\mathrm{NBZ}$. It holds:

$$\forall \mathbf{k} \in \mathrm{NBZ} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \epsilon^{(s^*)} (\cos k_x + \cos k_y) + \overbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}^{\geq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \epsilon^{(s^*)} (\cos k_x + \cos k_y) - \underbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}_{\geq 0} \end{aligned}$$

   thus $\tilde{\epsilon}_{\mathcal{R}\mathbf{k}} < \tilde{\epsilon}_{\mathbf{k}}$, and inserting this fact in the self-consistency equation above we get

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} > 0 \quad \Longrightarrow \quad u^{(d)} < 0$$

   which is absurd.

8. An identical argument, with inverted inequalities, can be carried out to show that

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} < 0 \quad \Longrightarrow \quad u^{(d)} > 0$$

9. The two points above imply that it is possible to satisfy the constraint $0 < u^{(s^*)} < 2t/V$ if and only if $u^{(d)} = 0$.

All the deductions above indicate that $u^{(d)} = 0$ everywhere. Nematic effects are always absent,

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} = 0 \tag{4.36}$$

This is physically expected but non trivial. The normal phase does not break C_4 and we can ignore $u^{(d)}$ at all. In next section we identify the expected self-consistent value for $u^{(s^*)}$.

4.5.3 Expected dependence on δ at low temperature

We understood that $u^{(d)} = 0$. Moreover, the unique HFP $u^{(s^*)}$ is independent of U, but in general it will depend in some way on the doping level. We work at low-temperature, $\beta \to \infty$. Consider the self-consistency equation of Tab. 4.2 as a function of temperature and chemical potential,

$$u^{(s^*)}(\beta, \tilde{\mu}) = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

in the zero-temperature limit, as long as the band is not flat, it holds

$$\lim_{\beta \to \infty} u^{(s^*)}(\beta, \tilde{\mu}) = \frac{1}{L_x L_y} \sum_{\epsilon_{\mathbf{k}} \leq \tilde{\mu}} (\cos k_x + \cos k_y)$$

Let FS(δ) be the Fermi surface for a given doping level δ. We denote with Int(FS) be its internal part, the Fermi sea:

$$\text{FS} = \{\mathbf{k} \in \text{BZ} \mid \tilde{\epsilon}_{\mathbf{k}} = \tilde{\mu}\}$$
$$\text{Int(FS)} = \{\mathbf{k} \in \text{BZ} \mid \tilde{\epsilon}_{\mathbf{k}} \leq \tilde{\mu}\}$$

At half filling, the Fermi surface coincides with the boundary of the MBZ:

$$\delta = 0 \qquad \Longrightarrow \qquad \text{FS} = \partial\text{MBZ}$$

This is implied by the fact that renormalised bands are antisymmetric by $\boldsymbol{\pi}$ shifts,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = -\tilde{\epsilon}_{\mathbf{k}+\boldsymbol{\pi}\sigma}$$

The portion of an s^*-wave band outside the MBZ is mirrored with respect to the portion inside the MBZ. This implies that half the states lie in the MBZ, half out.

At $\delta > 0$ the FS expands in the region BZ $\setminus$ MBZ where the form factor $\cos k_x + \cos k_y$ becomes negative. This gives the self-consistency equation

$$\lim_{\beta\to\infty} u^{(s^*)}[\beta, \tilde{\mu}] \simeq \frac{1}{L_x L_y} \sum_{\epsilon_{\mathbf{k}} \leq 0} |\cos k_x + \cos k_y| - \frac{1}{L_x L_y} \sum_{0<\epsilon_{\mathbf{k}} \leq \tilde{\mu}} |\cos k_x + \cos k_y|$$

which, being the sum of a positive and a negative term, is maximised whenever $\tilde{\mu} = 0$, at half filling. This holds as long as the band is not flat. Indeed, if the effective hopping is not zero, to rise the doping level means to

$$\delta_1 > \delta_2 \qquad \Longrightarrow \qquad \text{Int}(\text{FS}_1) \subset \text{Int}(\text{FS}_2)$$

with obvious meaning of the notation. Using this property we conclude

$$\tilde{t} > 0 \quad \text{and} \quad \delta_1 > \delta_2 \qquad \Longrightarrow \qquad \lim_{\beta\to\infty} u^{(s^*)}[\beta, \tilde{\mu}]\Big|_{\delta_1} < \lim_{\beta\to\infty} u^{(s^*)}[\beta, \tilde{\mu}]\Big|_{\delta_2}$$

This is the first expected behaviour of the self-consistent functional $u^{(s^*)}$: if there exists a region of parametric space where the shrinking effect is not strong enough to flatten the band, a rise in the doping level must lower $u^{(s^*)}$. Such a region must exist: at least all the segment $(V = 0, \delta)$ is included, since on said region of parametric space the model is just the common free electronic model.

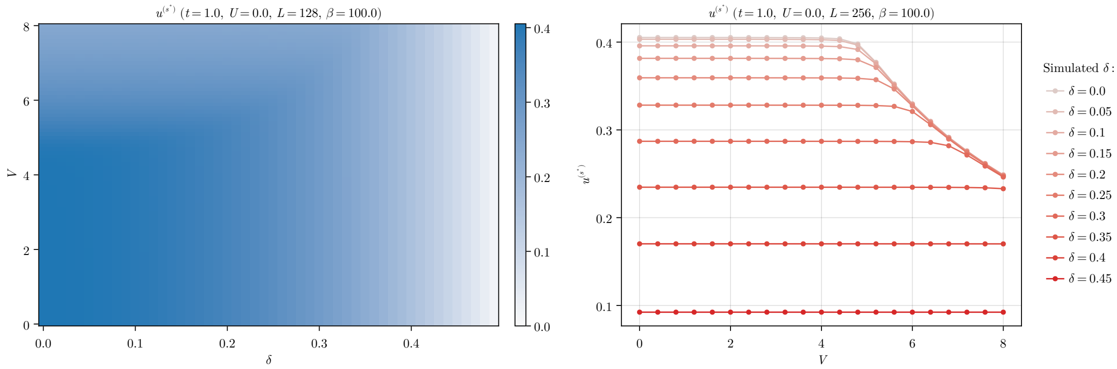
Following this line of reasoning, we can extract the two low-temperature limits of $u^{(s^*)}$ at half-filling and unitary filling.

1. At half-filling, the self-consistency equation reduces to

$$u^{(s^*)}[\beta, 0]\Big|_{\beta\gg 1} \simeq \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{MBZ}} (\cos k_x + \cos k_y)$$

In thermodynamic limit, performing integration on the MBZ shown in the top-left panel of Fig. 4.3:

$$\begin{aligned}
\lim_{\beta\to\infty} u^{(s^*)}[\beta, 0] &\simeq \frac{1}{(2\pi)^2} \int_{-\pi}^{\pi} \mathrm{d}k_y \int_{-\pi+|k_y|}^{\pi-|k_y|} \mathrm{d}k_x \cos k_x + \frac{1}{(2\pi)^2} \int_{-\pi}^{\pi} \mathrm{d}k_x \int_{-\pi+|k_x|}^{\pi-|k_x|} \mathrm{d}k_y \cos k_y \\
&= \frac{1}{2\pi^2} \int_{-\pi}^{0} \mathrm{d}k_y \int_{-\pi-k_y}^{\pi+k_y} \mathrm{d}k_x \cos k_x + \frac{1}{2\pi^2} \int_{0}^{\pi} \mathrm{d}k_y \int_{-\pi+k_y}^{\pi-k_y} \mathrm{d}k_x \cos k_x \\
&= \frac{1}{\pi^2} \int_{-\pi}^{0} \mathrm{d}k_y \sin(\pi + k_y) + \frac{1}{\pi^2} \int_{0}^{\pi} \mathrm{d}k_y \sin(\pi - k_y) \\
&= \left(\frac{2}{\pi}\right)^2 \approx 0.405
\end{aligned} \tag{4.37}$$



a | Heatmap of $u^{(s^*)}$ in δV plane.

b | V-dependency of $u^{(s^*)}$ varying δ.

Figure 4.6 | Plot of the self-consistent convergence values for the HFP $u^{(s^*)}$ in the normal phase. To extract the data on the right side coarse-graining precision of the left-hand side data has been exchanged with higher lattice dimensions. As is evident, based on doping level $u^{(s^*)}$ exhibits non-trivial regions of V-independence.

2. At unitary filling we have

$$\lim_{\beta\to\infty} u^{(s^*)}\left[\beta, 2\tilde{t}^{(s^*)}\right] = \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\cos k_x + \cos k_y) = 0$$

which vanishes due to the structure factor periodicity by $\boldsymbol{\pi}$.

To summarise: around $V = 0$, $u^{(s^*)}$ is expected to descend monotonically from $4/\pi^2$ to 0 at increasing doping.

4.6 Discussion of numeric results

In this section we present and discuss the numeric results, the physical relevance of the found phenomena and the boundaries of the algorithm. Let us start by sketching the algorithm used to determine self-consistently the mean-field structure of the normal phase in parameter space. For the normal phase, the HFPs “vector” defined in Sec. 2.3.2 has just one component:

$$\mathbf{v} = \left[u^{(s^*)}\right]$$

As before, we treat spin as a pure degeneracy. The only HFP is characterised by the first self-consistency equation of Tab. 4.2. The resulting model has two RMPs:

$$\tilde{t}_{ij} \equiv \left(t - \frac{u^{(s^*)}V}{2}\right)\varphi_{ij}^{(s^*)}$$
$$\tilde{\mu} \equiv \mu + n(2zV - U)$$

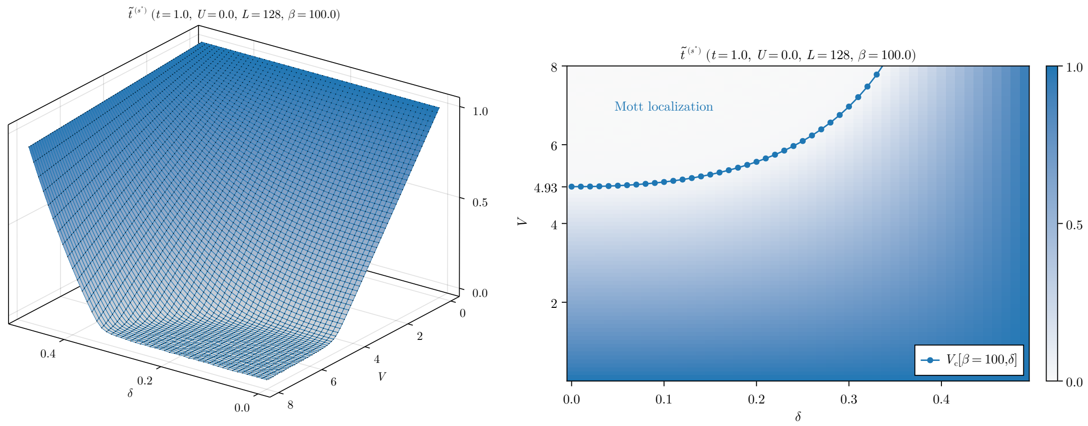
the second one being only dependent on the model parameters and not on HFPs. Finally, the phase free energy is given by Eq. (4.24). The normal phase is completely independent of U. Because of this we present the results varying the other parameters, keeping everywhere $U = 0$. We ran simulations in the (V, δ) plane, with the setup:

$$U = 0 \qquad \Delta n = 10^{-2} \qquad \Delta\mathbf{v} = \left[u^{(s^*)} : 5\times 10^{-4}\right]$$

with $\Delta\mathbf{v}$ the tolerance vector defined as in Sec. 2.3.2.

In Fig. 4.6a are reported the results obtained for the HFP $u^{(s^*)}$ on a 128×128 lattice and “low” temperature $\beta = 100$, varying the non-local attraction and the doping level

$$V = 0,\ 0.1,\ 0.2,\ \dots,\ 4.0 \qquad \delta = 0,\ 0.01,\ 0.02,\ \dots,\ 0.49$$



a | Simulated values of $\tilde{t}^{(s^*)}$

b | Critical Mott boundary.

Figure 4.7 | Plot of the self-consistent convergence values for the HFP $u^{(s^*)}$ in the normal phase and the Mott-boundary V_c, computed as is described in text. In the right panel, the tick $V = 4.93$ is specified explicitly as a graphic confirm of the computation of Eq. (4.38). For graphic reasons the points at $V_c > 8$ have been omitted.

The expectations of Sec. 4.5.3 are satisfied: at low V (indicatively $V \lesssim 5$) the trend is monotonically descendent at increasing doping. As V gets bigger, however, $u^{(s^*)}$ starts decreasing in the top-left corner of Fig. 4.6a. This effect is more visible in Fig. 4.6b, where we doubled L and ran simulations with a smaller δ resolution: the convergence value of $u^{(s^*)}$ is approximately independent of V up to some point, where this behaviour is broken. Note that, as was expected from the discussion of Sec. 4.5.3, the half-filled convergence value of $u^{(s^*)}$ in the V-independent region is approximately given by Eq. (4.37).

This effect can only be interpreted in one way, as we discuss in next section: $u^{(s^*)}$ maintains a constant value at fixed doping, as is described in Sec. 4.5.3, as long as $\tilde{t}^{(s^*)} > 0$. When V gets large enough, $u^{(s^*)}$ starts decreasing because of its constraints. The net effect is the nullification of $\tilde{t}^{(s^*)}$, which is, a total localisation effect.

4.6.1 Mott localisation effect

The most interesting result we obtained for the normal phase is the possibility of localising electrons by tuning the V/δ ratio. In Fig. 4.6b, at low V the convergence value of $u^{(s^*)}$ is approximately V-independent at sufficiently low temperatures. Let $V_c[\beta,\delta]$ be the critical value such that

$$0 \leq V < V_c[\beta,\delta] \quad : \quad u^{(s^*)} \text{ is } V \text{ independent.}$$

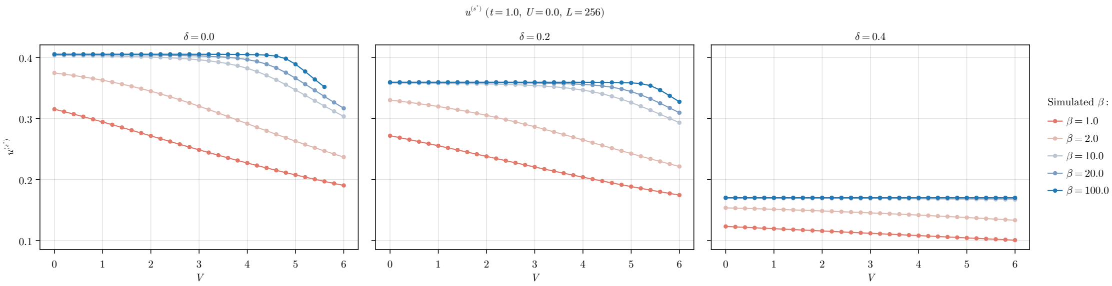
Then within the entire segment $[0, V_c]$ the convergence value is (approximately) the same obtained for $V = 0$. We can define in first instance the critical V as the value such that the hopping correction is maximized,

$$V_c[\beta,\delta] \equiv \frac{2t}{u^{(s^*)}[\beta,\tilde{\mu}(\delta)]\big|_{V=0}}$$

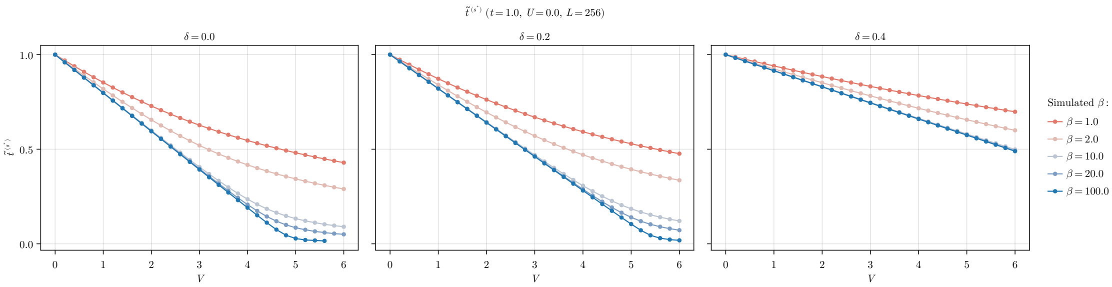
When V becomes larger than this critical value, the resulting RMP $\tilde{t}^{(s^*)}$ drops to 0 and due to the constraint of Eq. (4.31) the convergence value of $u^{(s^*)}$ is forced to decrease. Note that the constraint of Eq. (4.31) does not prevent $u^{(s^*)}$ from saturating to its upper boundary from below. In Fig. 4.7a the converged values of $\tilde{t}^{(s^*)}$, extracted from the same data of Fig. 4.6a, are plotted in parametric space.

While to extract the value of $V_c[\beta,\delta]$ is a non-trivial analytic problem due to the insurgence of elliptic integrals and the smearing effects of temperature, for the high-β half-filled case it is immediate from Eq. (4.37) to see that

$$V_c[\beta \to \infty, 0] = \frac{2t}{(2/\pi)^2} \approx 4.93t \tag{4.38}$$



a | Temperature dependence of $u^{(s^*)}$ plots at various dopings.



b | Temperature dependence of $\tilde{t}^{(s^*)}$ plots at various dopings.

Figure 4.8 | Comparative plots of the HFP $u^{(s^*)}$ (top panel) and RMP $\tilde{t}^{(s^*)}$ (bottom panel) in null-doped (left plots), "medium-doped" ($\delta = 0.2$, central plots) and "heavily-doped" ($\delta = 0.4$, right plots) contexts. Increasing temperature progressively destroys Mott localization, reducing the bands shrinking effect.

At half-filling, the effective bands amplitude reduction is so strong that small values of V (compared to t) are sufficient to nullify kinetics. In Fig. 4.7b we report, superimposed, the convergence values of the RMP $\tilde{t}^{(s^*)}$ and $V_c[\beta = 100, \delta]$ computed with the simple procedure hereby described. Our prediction of the critical value for V approximates reasonably the localisation region.

Increasing temperature Mott localisation is suppressed. Also, out precedent argument were valid only at low temperature. In Fig. 4.8a we report results for the HFP $u^{(s^*)}$ computed at decreasingly lower temperature and three example doping levels,

$$\beta = 1.0, 2.0, 10.0, 20.0, 100.0 \qquad \delta = 0.0, 0.2, 0.4$$

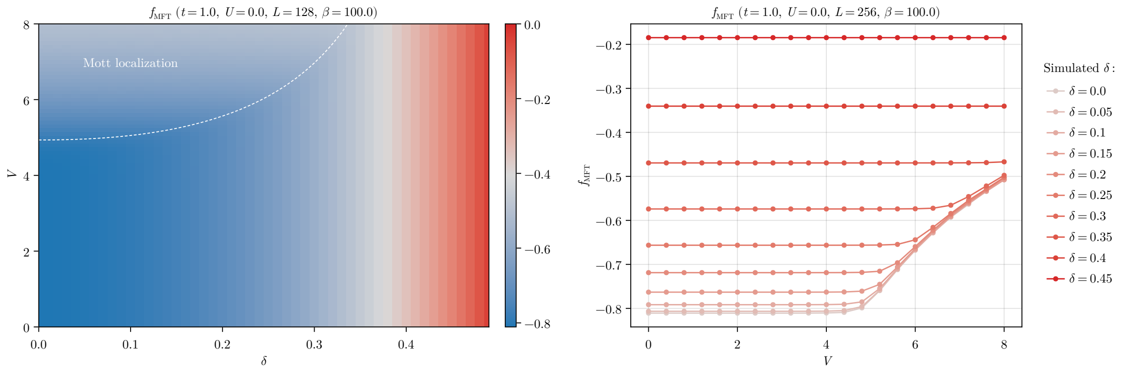
At high temperature the V-independence region that was visible in Fig. 4.6b completely disappears, making $u^{(s^*)}$ a monotonically descendent function. In Fig. 4.8a the same dataset is plotted with focus on the RMP $\tilde{t}^{(s^*)}$. Temperature increase disrupts localisation effects, at least at small values of V. However, the bands shrinking effect is still relevant at $\beta = 1.0$, the hopping parameter approximately halving at $V \approx 5 \div 6$.

4.6.2 Solution free energy and entropy densities

In order to compare phases, we need an estimation of free energy of each. The plots of Fig. 4.9 report our findings on the normal phase free energy, computed based on Eq. (4.24). For a non-flat band, doping increases energy because the band fills up and electrons populate higher energy levels.

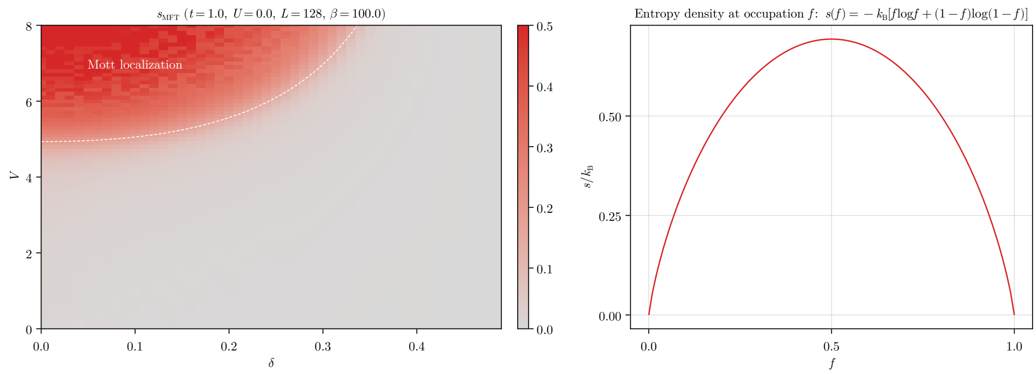
Looking the data of Fig. 4.9a, at fixed δ free energy density remains pretty much constant up to the Mott boundary. This is simply because as long as a band with non-zero amplitude is present, the HFP correction of Eq. (4.24) compensates the free energy increase due to bands shrinking,

$$\begin{aligned} -V\left|u^{(s^*)}\right|^2 &= -\frac{V u^{(s^*)*}}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} (\epsilon_{\mathbf{k}} - \tilde{\epsilon}_{\mathbf{k}})\, f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \end{aligned}$$



a | Heatmap of $f_{\rm MFT}$ in δV plane. b | V-dependency of $f_{\rm MFT}$ varying δ.

Figure 4.9 | Free energy density of the normal phase. On the left-side panel, the dashed white line has the same meaning as the dotted line of Fig. 4.7b, indicating the Mott transition.



a | Heatmap of $s_{\rm MFT}$ in δV plane. b | V-dependency of $f_{\rm MFT}$ varying δ.

Figure 4.10 | Entropy density of the normal phase (left panel) and theoretical contribution to entropy for a state with occupation f. On the left-side panel, the dashed white line has the same meaning as the dotted line of Fig. 4.7b, indicating the Mott transition.

having we used in the second passage

$$\tilde{\epsilon}_{\mathbf{k}} = \tilde{\epsilon}_{\mathbf{k}} + u^{(s^*)*} V(\cos k_x + \cos k_y)$$

Recalling the first point of the derivation of Sec. 4.4, to add the HFP correction to free energy is the same as removing the *tilde* sign from single-state energies, thus using the free model ($V = 0$) free energy density expression.

Mott localisation is energetically expensive. This is visible in both panels of Fig. 4.9. The V-dependent energy increase is an entropic effect: indeed, for a flat band:

$$f(0; \beta, \tilde{\mu}(\delta)) = \frac{1}{2}$$

at any temperature. The expression for entropy density of a free Fermi gas [29] is given by

$$s(f) = -k_{\rm B}\left[f \log f + (1-f)\log(1-f)\right]$$

represented in Fig. 4.10b. The maximum is located at $f = 1/2$, then a flat band realises the maximally entropic condition. The Mott insulating region can be seen as a degeneration of the FS interface to the entire BZ, a process that maximises entropy.

All of this holds, as all the rest, at low temperature: by lowering β, finite-temperature effects become more and more relevant and the statements above need to be corrected.

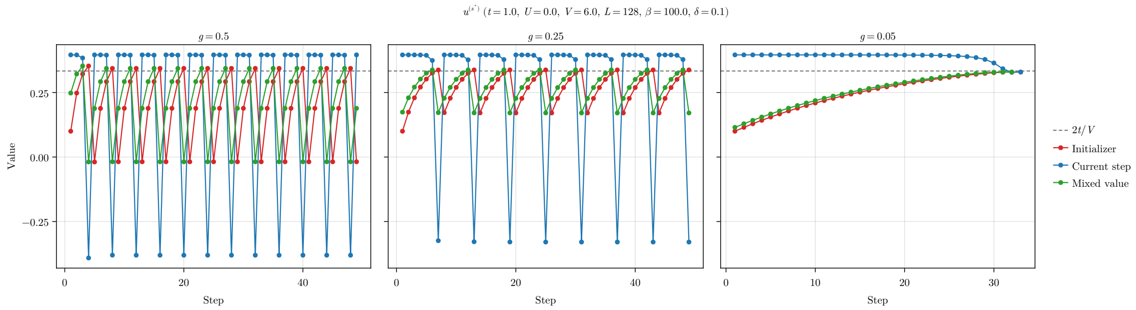


Figure 4.11 | Comparison of three example tracks for the single-valued HFPs vector of the normal phase, simulated at a "Mott-deep" point.

4.6.3 Convergence problems and mixing fine-tuning

The normal phase is simple enough to describe some of the typical convergence problems arising in HF iterative schemes anticipated in Sec. 2.3.2. Bands flattening is a source of algorithmic instability: numerics can lead to values of $u^{(s^*)}$, during iterations, slightly out of bounds with respect to Eq. (4.31). Even if the numeric effect is small, the consequence can be vast. To heal this effect and extract the actual convergence values, it is necessary to optimise the mixing parameter $g \in [0, 1]$.

We defined g in Sec. 2.3.2, and operatively in Eq. (2.21). This parameter controls deterministically how, from an initialiser (the HFPs vector we plug in the algorithm) and an output (the HFPs vector we get back from the algorithm) the new initialiser for the following step is generated. We show in Fig. 4.11 the evolutions at different mixing parameters of a simulation at

$$V = 6.0 \qquad \delta = 0.1 \qquad \beta = 100.0$$

This point is inside the Mott region, as can be checked in Fig. 4.6a. The HFP $u^{(s^*)}$ needs to converge at $2t/V = t/3$ in order to give complete localisation. The initialiser

$$\text{Step 1} \quad : \quad u^{(s^*)} = 0.1$$

was arbitrarily chosen. In each panel of Fig. 4.11, the red track represents the value of $u^{(s^*)}$ we plug in the self-consistency equations, the blue the result of the self-consistent equation, the green one the mixed value of the two based on different gs. The gray dashed line is the theoretical convergence target.

In the first two panels, there is an evident infinite loop occurring whenever the green line crosses the threshold $2t/V$. At those steps, the green point is out of bounds with respect to Eq. (4.31). When this happens, the following red point gives a band with inverted sign, a feature that "breaks" the self-consistency equation and generates a blue point with no physical significance.

To overcome this problem it is necessary to move "slowly". As is shown in the right panel, the choice of a small enough mixing parameter ensures convergence. In the track, each blue dot is above the target and would produce a band with inverted sign. Mixing blue and red dots with a 95% unbalance on the red ones, we ensure that the green track approaches the target from below. Note, however, that lowering g cannot be reiterated too many times. Indeed, for $g \to 0$, any point of our map becomes a fixed point.

This kind of problems can turn out to be pretty common for iHF schemes, and to understand how to knob the algorithmic parameters without sacrificing accurateness and precision, as well as avoiding computationally costly and frustrating infinite loops, was an essential part of this algorithm design.

Chapter 5

Antiferromagnetic instability

In this chapter we apply HF method to the AF phase, widely studied for the HM [31]. We focus on one particular realisation of antiferromagnetism, referred to as "commensurate". It is inherently metallic and only physically sensed at half-filling.

For the commensurate antiferromagnet, the non-local attraction $\hat{H}_V$ adds two main self-consistent effects. First, redefinition of the bands gap: an effect we denote "gap complexification" (Sec. 5.6) and later show is actually suppressed (Sec. 5.8). Second, bare bands shrinking (Sec. 5.5) similarly to what found for the normal phase.

As for the normal phase, this chapter is equally divided in the same three main parts. First: discussion of AF phase symmetries, based on Chap. 3, and implementation of HF methods. Second: theoretical constraints on the self-consistent solution. Third: numeric results.

5.1 The AF phase and its symmetries

The AF phase is not uniquely defined. Fig. 5.1 presents three (of many) types of antiferromagnets. We represent in the left panel the one we are interested in. The AF structure comes from an uneven and periodic distribution of spin densities with one site periodicity in both directions. The arrows in figure indicate what spin sector is most abundant on a given site. These SDW structures are 2-sites periodic, making the AF phase "commensurate". The central one presents another simple commensurate AF with higher periodicity. The right panel shows a possible "striped" antiferromagnet, where AF is effectively established along one particular direction; the example we report is 2 sites periodic in y direction, but any other periodicity could work. Striped antiferromagnetism is studied in HM physics [3] as a possible long-range ordering energetically competing

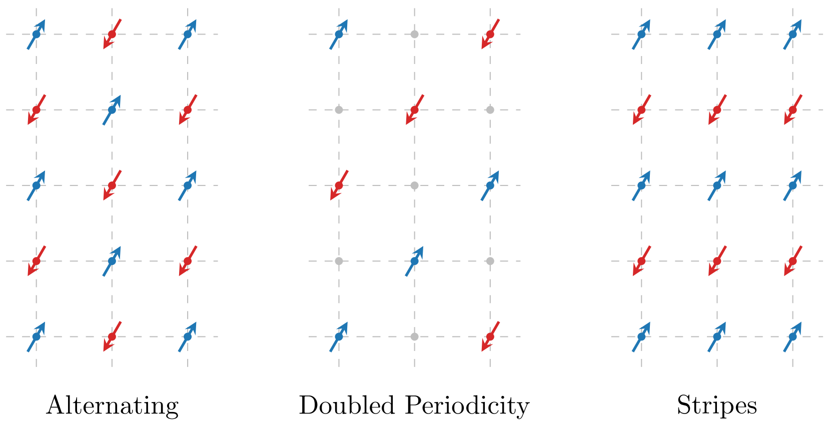


Figure 5.1 Three possible antiferromagnetic structures. Dots and arrows indicate spin abundance on a given site, while empty sites are intended as spin-balanced. Within this text we deal with the lowest-periodicity commensurate phase, the antiferromagnet on left panel.

with anisotropic superconductivity. Nematic antiferromagnets, establishing long-range magnetic ordering along one particular direction, are interesting due to competition and interplay against superconducting ordering at finite doping [24]. Geometrically non-trivial magnetic ordering are at the core of SDW formation in copper oxides at finite doping [25, 26].

In this thesis, the AF phase is described by simultaneous global spin rotation and translational symmetry SSBs:

$$\mathrm{SU}(2) \overset{\mathrm{SSB}}{\to} \mathrm{U}^z(1)$$

$$\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)} \overset{\mathrm{SSB}}{\to} \mathrm{T}_2^{(x)} \otimes \mathrm{T}_2^{(y)}$$

We describe a SDW whose period is of two sites, as in the left panel of Fig. 5.1. As done for the normal phase, we allow

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$$

in order to capture the same theoretical nematic effects seen in Sec. 4.3.1. The theoretical reason to do so for AF phase relies in our previous discussion. As said, planar Hubbard lattices exhibits tendency towards nematic magnetic ordering, thus (even if our Ansatz is the simplest and non evidently nematic) nematic effects are not to be excluded *a priori*.

Commensurate antiferromagnetism is specified by the Ansatz:

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n - (-1)^{x+y+\delta_{\sigma=\uparrow}} m \qquad n, m \in [0,1] \tag{5.1}$$

This Ansatz captures the idea of the staggered distribution of spin populations shown in the left panel of Fig. 5.1. At site $\mathbf{r} = (x, y)$ the two spin populations average to n and differ by $2m$. A detailed justification for this Ansatz in the HM is given in Sec. D.2. Note that $\langle \hat{n}_{\mathbf{r}\sigma} \rangle \leq 1$ due to Pauli exclusion principle. This argument is already sufficient to claim that doping disrupts commensurate antiferromagnetism.

5.1.1 Symmetries of the commensurate AF Ansatz

The Ansatz of Eq. (5.1) breaks translational invariance in each spin sector, but preserves $\mathrm{U}^z(1)$ and $\mathrm{U}^c(1)$ symmetries. Let us prove this. The Ansatz is defined by the following EV being non-zero,

$$\langle \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow} \rangle = 2m \tag{5.2}$$

ignoring the local sign factor. SSB happens establishes when $m \neq 0$. Then:

- Recall the definitions of $\hat{\Psi}_{\mathbf{r}}$ of Eq. (3.6)

  $$\hat{\Psi}_{\mathbf{r}} = \begin{bmatrix} \hat{c}_{\mathbf{r}\uparrow} \\ \hat{c}_{\mathbf{r}\downarrow} \end{bmatrix}$$

  and the local spin operators of Eq. (3.7),

  $$\hat{S}^+_{\mathbf{r}} = \frac{\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow}}{2} \qquad \hat{S}^-_{\mathbf{r}} = \frac{\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}}{2} \qquad \hat{S}^z_{\mathbf{r}} = \frac{\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow} + \hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}}{2}$$

  Consider Eq. (3.16) and its conjugate,

  $$\left[\hat{c}^\dagger_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}\right] = -\hat{c}^\dagger_{\mathbf{r}\sigma} \qquad [\hat{c}_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}] = \hat{c}_{\mathbf{r}\sigma}$$

  The operators $\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}$ and $\hat{S}^\pm_{\mathbf{r}}$ do not commutate. Indeed,

  $$\begin{aligned}\left[\hat{S}^+_{\mathbf{r}}, \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right] &= \frac{1}{2}\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}, \hat{n}_{\mathbf{r}\uparrow}\right]\hat{c}_{\mathbf{r}\downarrow} - \frac{1}{2}\hat{c}^\dagger_{\mathbf{r}\uparrow}[\hat{c}_{\mathbf{r}\downarrow}, \hat{n}_{\mathbf{r}\downarrow}] \\ &= -\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow} \\ \left[\hat{S}^-_{\mathbf{r}}, \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right] &= \frac{1}{2}\hat{c}^\dagger_{\mathbf{r}\downarrow}[\hat{c}_{\mathbf{r}\uparrow}, \hat{n}_{\mathbf{r}\uparrow}] - \frac{1}{2}\left[\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{n}_{\mathbf{r}\downarrow}\right]\hat{c}_{\mathbf{r}\uparrow} \\ &= \hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\end{aligned}$$

  Instead, $\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}$ and $\hat{S}^z_{\mathbf{r}}$ commutate, because all on-site density operators always do. Then, in order to have $m \neq 0$ in Eq. (5.2), SU(2) needs to be broken while $\mathrm{U}^z(1)$ needs not.

- From Eq. (3.5) we know that on-site density operators are gauge-invariant separately in both spin sectors. Then no selection rule applies to $\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}$: its EV can be non-zero without breaking $\mathrm{U}^z(1)$.

The chosen Ansatz is coherent with the symmetry requirements of Tab. 3.1. Let us rephrase this in the language of HF methods. We need an HFPs vector $\mathbf{v}$, whose components v_i are equipped with self-consistency equations. Commensurate antiferromagnetism is described by those HFPs emerging from Wick's theorem applied to the EHM respecting symmetry requirement. Among these we include explicitly the spin averaged unbalance has a two-sites periodicity,

$$\forall \mathbf{r} = (x, y) \in \mathcal{L} \quad : \quad m \equiv (-1)^{x+y} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \tag{5.3}$$

By allowing for this EV to be non-zero we respect phase symmetry and impose antiferromagnetism to be commensurate. Setting $m \neq 0$ is equivalent to imposing Eq. (5.1).

The on-site averaged density is given by:

$$\forall \mathbf{r} \in \mathcal{L} \quad : \quad n \equiv \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \tag{5.4}$$

Despite of similarity with m, this is not an HFP. First, on-site density respect all EHM symmetries. Second, we are imposing average density as a parametrisation of the model. There is nothing to be searched self-consistently here.

Note that the quantities n, m in Eqns. (5.4), (5.3) are $\mathbf{r}$-independent. Then it must hold

$$n = \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \tag{5.5}$$

$$m = \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} (-1)^{x+y} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \tag{5.6}$$

From the first equation we gain an operational method to compute density, from the second we get a self-consistency equation for magnetisation. Indeed, the ground-state depends on HFP m in some way to be determined explicitly. While this passage added nothing about the theoretical description of antiferromagnetism (we averaged constants over the lattice), it will be crucial in order to move the discussion to reciprocal space.

5.1.2 The AF Ansatz in reciprocal space

The core parts of the following derivations will be carried out in reciprocal space. It is then useful to comment the reciprocal structure of the AF Ansatz. First, let:

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \tag{5.7}$$

$$\hat{\psi}_{\mathbf{k}\sigma} \equiv \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{k}\sigma} \tag{5.8}$$

Note that $\hat{n}_{\mathbf{k}\sigma}$ is not the Fourier-transform of $\hat{n}_{\mathbf{r}\sigma}$. Recall also the γ-wave form factors of Tab. 3.5. Then:

$$\hat{\psi}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} = \hat{\psi}_{\mathbf{k}\sigma} \tag{5.9}$$

$$\varphi^{(\gamma)}_{\mathbf{k}+\boldsymbol{\pi}} = -\varphi^{(\gamma)}_{\mathbf{k}} \tag{5.10}$$

which, by the way, implies the property

$$\varphi^{(\gamma)}_{\mathbf{k}+\boldsymbol{\pi}} \psi^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} = -\varphi^{(\gamma)}_{\mathbf{k}} \psi_{\mathbf{k}\sigma} \tag{5.11}$$

that will result useful later.

In the AF phase, we have $\langle \hat{n}_{\mathbf{k}\sigma} \rangle \neq 0$ and $\langle \hat{\psi}_{\mathbf{k}\sigma} \rangle \neq 0$. This is proven by the following chains of equations:

$$\begin{aligned}
\sum_{\mathbf{r}\in\mathcal{L}} \hat{n}_{\mathbf{r}\sigma} &= \sum_{\mathbf{r}\in\mathcal{L}} \hat{c}^{\dagger}_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma} \\
&\stackrel{\mathcal{F}}{=} \sum_{\mathbf{r}\in\mathcal{L}} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in\mathrm{BZ}} e^{i\mathbf{k}\cdot\mathbf{r}}\hat{c}^{\dagger}_{\mathbf{k}\sigma} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}'\in\mathrm{BZ}} e^{-i\mathbf{k}'\cdot\mathbf{r}}\hat{c}_{\mathbf{k}'\sigma} \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\mathbf{k}'\in\mathrm{BZ}} \hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma} \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} e^{-i(\mathbf{k}'-\mathbf{k})\cdot\mathbf{r}} \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}} \hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}} \hat{n}_{\mathbf{k}\sigma}
\end{aligned} \tag{5.12}$$

where we used Eq. (3.26). Thus:

$$\begin{aligned}
n &= \frac{1}{2}\sum_{\sigma}\left[\frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}} n + \overbrace{\frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}}(-1)^{x+y} m}^{=0}\right] \\
&= \frac{1}{2L_x L_y}\sum_{\sigma}\sum_{\mathbf{r}\in\mathcal{L}}\langle \hat{n}_{\mathbf{r}\sigma}\rangle \\
&= \frac{1}{2L_x L_y}\sum_{\sigma}\sum_{\mathbf{k}\in\mathrm{BZ}}\langle \hat{n}_{\mathbf{k}\sigma}\rangle
\end{aligned} \tag{5.13}$$

We used the Ansatz of Eq. (5.1). Since $n \neq 0$, in the last sum $\langle \hat{n}_{\mathbf{k}\sigma} \rangle \neq 0$ is impossible for all $\mathbf{k}$s. Similarly, for the oscillatory part:

$$\begin{aligned}
\sum_{\mathbf{r}\in\mathcal{L}} (-1)^{x+y}\hat{n}_{\mathbf{r}\sigma} &= \sum_{\mathbf{r}\in\mathcal{L}} (-1)^{x+y}\hat{c}^{\dagger}_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma} \\
&\stackrel{\mathcal{F}}{=} \sum_{\mathbf{r}\in\mathcal{L}} e^{i\boldsymbol{\pi}\cdot\mathbf{r}} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in\mathrm{BZ}} e^{i\mathbf{k}\cdot\mathbf{r}}\hat{c}^{\dagger}_{\mathbf{k}\sigma} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}'\in\mathrm{BZ}} e^{-i\mathbf{k}'\cdot\mathbf{r}}\hat{c}_{\mathbf{k}'\sigma} \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\mathbf{k}'\in\mathrm{BZ}} \hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma} \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} e^{-i[\mathbf{k}'-(\mathbf{k}+\boldsymbol{\pi})]\cdot\mathbf{r}} \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}} \hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} && (5.14) \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}} \hat{\psi}_{\mathbf{k}\sigma} && (5.15)
\end{aligned}$$

where we defined the AF vector $\boldsymbol{\pi} \equiv (\pi, \pi)$. Thus, again using the Ansatz of Eq. (5.1),

$$\begin{aligned}
(-1)^{\delta_{\sigma=\downarrow}} m &= \overbrace{\frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}}(-1)^{x+y} n}^{=0} + \frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}}(-1)^{\delta_{\sigma=\downarrow}} m \\
&= \frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}}(-1)^{x+y}\langle \hat{n}_{\mathbf{r}\sigma}\rangle \\
&= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\langle \hat{\psi}_{\mathbf{k}\sigma}\rangle
\end{aligned} \tag{5.16}$$

where we used $1 = \left((-1)^{x+y}\right)^2$. If $m \neq 0$, in the last sum we cannot have $\langle \hat{\psi}_{\mathbf{k}\sigma} \rangle \neq 0$ for all $\mathbf{k}$s. We

Operator	Sector	RWCs	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
	s.s.	Hartree	μ shift by nzV
		Fock	Bands renormalisation and resymmetrization, gap complexification

Table 5.1 | Relevant Wick's contractions and net effect in the AF phase. Note the similarity with Tab. 4.1.

can restrict the sum to the MBZ and write it explicitly for both spins

$$m = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{MBZ}} \left[\hat{\psi}_{\mathbf{k}\uparrow} + \hat{\psi}^\dagger_{\mathbf{k}\uparrow}\right] \tag{5.17}$$

$$= -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{MBZ}} \left[\hat{\psi}_{\mathbf{k}\downarrow} + \hat{\psi}^\dagger_{\mathbf{k}\downarrow}\right] \tag{5.18}$$

where we used the property of Eq. (5.9).

Recall now Tabs. 3.2 and 3.3: the relevant Wick's contractions are, for each sector,

$\hat{H}_U$	Hartree contraction
$\hat{H}_V$ (o.s. sector)	Hartree contraction
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions

Their role is anticipated Tab. 5.1. In next sections we break down each one and derive the results by analytically applying HF methods.

5.2 Local Hartree effects: antiferromagnetism in the HM

For the AF phase, the derivation is more elaborated than for the normal phase. It is useful to separate the local and non-local interactions and apply to each one HF methods. Let us start from the local quartic interaction,

$$\hat{H}_U = U \sum_{\mathbf{r}\in\mathcal{L}} \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}$$

We only need to analyse the Hartree contraction channel, as is reported in Tab. 5.1. The HF decomposition and the Hartree sectors are given in Eq. (6.2).

Before extending to the non-local attraction, we dedicate the end of this section to complete the theoretical description of antiferromagnetism in the conventional, locally repulsive Hubbard model. This will allow us to identify easily what changes due to the presence of $\hat{H}_V$.

5.2.1 Local Hartree channel in the AF phase: analytic derivation

The HF method, applied to a quartic operator, gives the decomposition of Eq. (6.2). Decompose $\hat{H}_U$ in the Hartree channel:

$$\hat{H}_U \simeq U \sum_{\mathbf{r}\in\mathcal{L}} \left[\hat{n}_{\mathbf{r}\uparrow}\langle\hat{n}_{\mathbf{r}\downarrow}\rangle + \langle\hat{n}_{\mathbf{r}\uparrow}\rangle\hat{n}_{\mathbf{r}\downarrow} - \langle\hat{n}_{\mathbf{r}\uparrow}\rangle\langle\hat{n}_{\mathbf{r}\downarrow}\rangle\right]$$

Note that in the AF phase, as is reported in Tab. 3.2, the local repulsion decomposes just in the Hartree channel. Then no other term must be included in the above approximation of the quartic operator.

The one-body operators can be rewritten as:

$$\hat{n}_{\mathbf{r}\uparrow}\langle\hat{n}_{\mathbf{r}\downarrow}\rangle + \langle\hat{n}_{\mathbf{r}\uparrow}\rangle\hat{n}_{\mathbf{r}\downarrow} = \frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle + \langle\hat{n}_{\mathbf{r}\downarrow}\rangle}{2}(\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}) - \frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle - \langle\hat{n}_{\mathbf{r}\downarrow}\rangle}{2}(\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow})$$

From Eqns. (5.3) and (5.4), we recognise n and the HFP m. We have the decomposition:

$$\hat{H}_U \simeq nU \sum_{\mathbf{r}\in\mathcal{L}} [\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}] - mU \sum_{\mathbf{r}\in\mathcal{L}} (-1)^{x+y} [\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}] - E^{(\mathrm{AF})}_{\mathrm{H}/U} \tag{5.19}$$

with $E^{(\mathrm{AF})}_{\mathrm{H}/U}$ the HM Hartree contraction energy, discussed in Sec. 5.2.2. The first term of Eq. (5.19) gives a number operator,

$$nU \sum_{\mathbf{r}\in\mathcal{L}} [\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}] = nU \times \hat{N}$$

Then we get the first renormalisation of chemical potential:

$$\mu \to \mu - nU \tag{5.20}$$

We partially anticipated it in Tab. 5.1. Another shift to μ will come from the non-local attraction. In the second term of Eq. (5.19), apply Eq. (5.14). It follows:

$$mU \sum_{\mathbf{r}\in\mathcal{L}} (-1)^{x+y} [\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}] = mU \sum_{\mathbf{k}\in\mathrm{BZ}} \left[\hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\uparrow} - \hat{c}^\dagger_{\mathbf{k}\downarrow}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\downarrow}\right]$$

Recall the definition of the MBZ given in Eq. (4.25). The full BZ can be seen as the sum of two MBZ shifted by $\boldsymbol{\pi}$. Then, using $2\boldsymbol{\pi}$ periodicity, the following identity holds:

$$\begin{aligned}
\sum_{\mathbf{k}\in\mathrm{BZ}} \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} &= \sum_{\mathbf{k}\in\mathrm{MBZ}} \left[\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} + \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}+2\boldsymbol{\pi}\sigma}\right] \\
&= \sum_{\mathbf{k}\in\mathrm{MBZ}} \left[\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} + \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}\sigma}\right]
\end{aligned} \tag{5.21}$$

We write everything in terms of the operators $\hat{n}_{\mathbf{k}\sigma}$ and $\hat{\psi}_{\mathbf{k}\sigma}$, defined in Eqns. (5.7), (5.8). The local interaction is reduced to

$$\hat{H}_U \simeq -mU \sum_{\mathbf{k}\in\mathrm{MBZ}} \sum_{\sigma} (-1)^{\delta_{\sigma=\downarrow}} \left(\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^\dagger_{\mathbf{k}\sigma}\right) + nU\hat{N} - E^{(\mathrm{AF})}_{\mathrm{H}/U}$$

We define the gap function:

$$\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$$

This new object allows us to rewrite the local repulsion as:

$$\hat{H}_U \simeq -\sum_{\mathbf{k}\in\mathrm{MBZ}} \sum_{\sigma} \left(\Delta_{\mathbf{k}\sigma}\hat{\psi}_{\mathbf{k}\sigma} + \Delta^*_{\mathbf{k}\sigma}\hat{\psi}^\dagger_{\mathbf{k}\sigma}\right) + nU\hat{N} - E^{(\mathrm{AF})}_{\mathrm{H}/U} \tag{5.22}$$

The gap function is actually independent of $\mathbf{k}$. It is a constant in reciprocal space. However, we defined it as $\Delta_{\mathbf{k}\sigma}$: this notation will be useful later.

5.2.2 Local Hartree contraction energy

In last section we defined $E^{(\mathrm{AF})}_{\mathrm{H}/U}$ as the local Hartree contraction energy,

$$\begin{aligned}
E^{(\mathrm{AF})}_{\mathrm{H}/U} &\equiv U \sum_{\mathbf{r}\in\mathcal{L}} \langle\hat{n}_{\mathbf{r}\uparrow}\rangle \langle\hat{n}_{\mathbf{r}\downarrow}\rangle \\
&= U \sum_{\mathbf{r}\in\mathcal{L}} \left(n + (-1)^{x+y}m\right)\left(n - (-1)^{x+y}m\right)
\end{aligned}$$

where we used the Ansatz of Eq. (5.1). Expanding the product above we get:

$$\begin{aligned}
\left(n + (-1)^{x+y}m\right)\left(n - (-1)^{x+y}m\right) &= n^2 + \left((-1)^{x+y} - (-1)^{x+y}\right)nm - \left((-1)^{x+y}\right)^2 m^2 \\
&= n^2 - m^2
\end{aligned}$$

The sum over $\mathcal{L}$ gives $L_x L_y$ identical terms, thus we get:

$$E^{(\mathrm{AF})}_{\mathrm{H}/U} = L_x L_y \times U\left(n^2 - m^2\right) \tag{5.23}$$

which is the contraction energy in the local Hartree channel.

5.2.3 Antiferromagnetism in the HM

One of the aims of this projects is to identify, theoretically, what changes for each phase due to the presence of the non-local attraction $\hat{H}_V$ with respect to the conventional HM. In this section we give a description of antiferromagnetism in the conventional HM, and later we move to the EHM.

Matrix representation. The HM hamiltonian contains the kinetic operator and the local repulsion,

$$\hat{H}_t + \hat{H}_U$$

Let the reciprocal space spinors:

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix}$$

Despite sharing a similar notation with the spinors of Eq. (3.6), they are inherently different objects. Recall Eq. (3.28): the kinetic sector is represented in reciprocal space by

$$\begin{aligned}\hat{H}_t &= \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \\ &= \sum_{\mathbf{k}\in\mathrm{MBZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \left(\hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} - \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \right)\end{aligned}$$

where we used antiperiodicity by $\boldsymbol{\pi}$ of the bare bands $\epsilon_{\mathbf{k}}$ defined in Eq. (3.29),

$$\epsilon_{\mathbf{k}} = -\epsilon_{\mathbf{k}+\boldsymbol{\pi}}$$

The kinetic operator can be written in terms of the spinors $\hat{\Psi}_{\mathbf{k}\sigma}$,

$$\hat{H}_t = \sum_{\mathbf{k}\in\mathrm{MBZ}} \sum_{\sigma} \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \begin{bmatrix} \epsilon_{\mathbf{k}\sigma} & \\ & -\epsilon_{\mathbf{k}\sigma} \end{bmatrix} \hat{\Psi}_{\mathbf{k}\sigma}$$

The local repulsion $\hat{H}_U$ of Eq. (5.22) can also be written in terms of these spinors:

$$\hat{H}_U = \sum_{\mathbf{k}\in\mathrm{MBZ}} \sum_{\sigma} \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \begin{bmatrix} & -\Delta^*_{\mathbf{k}\sigma} \\ -\Delta_{\mathbf{k}\sigma} & \end{bmatrix} \hat{\Psi}_{\mathbf{k}\sigma} + nU\hat{N} - E^{(\mathrm{AF})}_{\mathrm{H}/U}$$

Collect everything. For the AF phase, the final form of the quadratic hamiltonian is

$$\hat{H}_t + \hat{H}_U \simeq \sum_{\mathbf{k}\in\mathrm{MBZ}} \sum_{\sigma} \hat{\Psi}^\dagger_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} + nU\hat{N} - E^{(\mathrm{AF})}_{\mathrm{H}/U} \tag{5.24}$$

where we defined the 2×2 matrix:

$$h_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \epsilon_{\mathbf{k}\sigma} & -\Delta^*_{\mathbf{k}\sigma} \\ -\Delta_{\mathbf{k}\sigma} & -\epsilon_{\mathbf{k}\sigma} \end{bmatrix}$$

This is a parametric hamiltonian. We need to identify its ground state for each choice of the parameters $\Delta_{\mathbf{k}\sigma}$. This is done by mapping the problem to the one of a set of spin subject to local magnetic fields: let us show this.

Pseudospin representation. The matrix $h_{\mathbf{k}\sigma}$ can be written in term of the Pauli matrices τ^α

$$h_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma}\tau^z - \Delta_{\mathbf{k}\sigma}\tau^x$$

Let us define the pseudospin operator $\hat{\mathbf{s}}_{\mathbf{k}\sigma}$

$$\hat{s}^\alpha_{\mathbf{k}\sigma} \equiv \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^\alpha \hat{\Psi}_{\mathbf{k}\sigma} \quad \text{with} \quad \hat{\mathbf{s}}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{s}^x_{\mathbf{k}\sigma} \\ \hat{s}^y_{\mathbf{k}\sigma} \\ \hat{s}^z_{\mathbf{k}\sigma} \end{bmatrix}$$

EV	Meaning
$\langle \hat{s}^x_{\mathbf{k}\sigma} \rangle$	$\left\langle \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^\dagger_{\mathbf{k}\sigma} \right\rangle$
$\langle \hat{s}^y_{\mathbf{k}\sigma} \rangle$	$i \left\langle \hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}^\dagger_{\mathbf{k}\sigma} \right\rangle$
$\langle \hat{s}^z_{\mathbf{k}\sigma} \rangle$	$\langle \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle$

Table 5.2 | List of the various expectation values (EVs) of each pseudo-spin component and their meaning in terms of fermionic operators.

and the pseudofield $\mathbf{b}_{\mathbf{k}\sigma}$:

$$\mathbf{b}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} -\Delta_{\mathbf{k}\sigma} \\ 0 \\ \epsilon_{\mathbf{k}\sigma} \end{bmatrix}$$

These operators satisfy a $\mathfrak{su}(2)$ algebra up to a constant C,

$$\left[\hat{s}^\alpha_{\mathbf{k}\sigma}, \hat{s}^\beta_{\mathbf{k}\sigma}\right] = C \times i \sum_\gamma \epsilon_{\alpha\beta\gamma} \hat{s}^\gamma_{\mathbf{k}\sigma}$$

The proof is conceptually identical to the one presented in Sec. 3.1.2. The actual value of C is irrelevant, as we will see in next paragraphs. **The two set of spin operators are unrelated.** Let us express the pseudospin operator in terms of the operators of Eqns. (5.7), (5.8):

$$\hat{s}^x_{\mathbf{k}\sigma} = \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^\dagger_{\mathbf{k}\sigma} \tag{5.25}$$

$$\hat{s}^y_{\mathbf{k}\sigma} = i \left(\hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}^\dagger_{\mathbf{k}\sigma}\right) \tag{5.26}$$

$$\hat{s}^z_{\mathbf{k}\sigma} = \hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \tag{5.27}$$

We report these relations also in Tab. 5.2. Let also:

$$\begin{aligned} \hat{\mathbb{I}}_{\mathbf{k}\sigma} &\equiv \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \mathbb{I}_{2\times 2} \hat{\Psi}_{\mathbf{k}\sigma} \\ &= \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{aligned} \tag{5.28}$$

In pseudospin representation we get:

$$\hat{H}_t + \hat{H}_U \simeq \sum_{\mathbf{k}\in \mathrm{MBZ}} \sum_\sigma \mathbf{b}_{\mathbf{k}\sigma} \cdot \hat{\mathbf{s}}_{\mathbf{k}\sigma} + nU\hat{N} - E^{(\mathrm{AF})}_{\mathrm{H}/U} \tag{5.29}$$

The hamiltonian represents a system of spins subject to local magnetic fields. The field $\mathbf{b}_{\mathbf{k}\sigma}$ is orthogonal to the y plane, and is tilted in the z plane by an angle $2\theta_{\mathbf{k}\sigma}$ with:

$$\cos(2\theta_{\mathbf{k}\sigma}) = \frac{\epsilon_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}} \qquad \sin(2\theta_{\mathbf{k}\sigma}) = \frac{\Delta_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}} \qquad E_{\mathbf{k}\sigma} \equiv \sqrt{\epsilon^2_{\mathbf{k}\sigma} + |\Delta_{\mathbf{k}\sigma}|^2}$$

A sketch of these angles and the pseudofield is given in Fig. 5.2.

Diagonalisation. To diagonalise the Hubbard hamiltonian $\hat{H}_t + \hat{H}_U$ of Eq. (5.29) means to perform some unitary operation on the $\hat{\Psi}_{\mathbf{k}\sigma}$ spinors which reduces the 2×2 matrix to a diagonal form. This means to diagonalise each $h_{\mathbf{k}\sigma}$. This is obtained through a rotation around the y axis:

$$d_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} W^\dagger_{\mathbf{k}\sigma} \quad \text{with} \quad d_{\mathbf{k}\sigma} \equiv \begin{bmatrix} E_{\mathbf{k}\sigma} & \\ & -E_{\mathbf{k}\sigma} \end{bmatrix} \tag{5.30}$$

The matrix $W_{\mathbf{k}\sigma}$ is unitary, and gives a counter-clockwise rotation by an angle $2\theta_{\mathbf{k}\sigma}$ around the y axis:

$$\begin{aligned} W_{\mathbf{k}\sigma} &\equiv e^{-i\theta_{\mathbf{k}\sigma}\tau^y} \\ &= \cos\theta_{\mathbf{k}\sigma} - i \sin\theta_{\mathbf{k}\sigma} \tau^y \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma} & -\sin\theta_{\mathbf{k}\sigma} \\ \sin\theta_{\mathbf{k}\sigma} & \cos\theta_{\mathbf{k}\sigma} \end{bmatrix} \end{aligned} \tag{5.31}$$

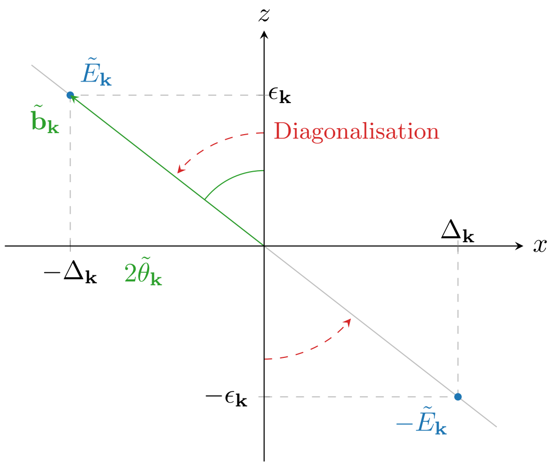


Figure 5.2 | Sketch of the diagonalisation procedure for the HM antiferromagnet. To diagonalise the HF hamiltonian, we align the z axis with the pseudo-field.

Its eigenvalues are $\pm E_{\mathbf{k}\sigma}$, with

$$E_{\mathbf{k}\sigma} = \sqrt{\epsilon_{\mathbf{k}\sigma}^2 + |\Delta_{\mathbf{k}\sigma}|^2} \tag{5.32}$$

In Fig. 5.2 we sketch the pseudo-magnetic field in the xz plane, and the diagonalisation procedure. $W_{\mathbf{k}\sigma}$ is a rotation by an angle $2\theta_{\mathbf{k}\sigma}$ around the y axis, and in the end aligns the original z axis with the pseudofield axis.

Applying the rotation matrix to the spinors we obtain new spinors of $\gamma^{(\pm)}$ fermions,

$$\hat{\Gamma}_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma}\hat{\Psi}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{\gamma}_{\mathbf{k}\sigma}^{(+)} \\ \hat{\gamma}_{\mathbf{k}\sigma}^{(-)} \end{bmatrix} \tag{5.33}$$

In the end we have the non-interacting hamiltonian:

$$\begin{aligned}
\hat{H}_t + \hat{H}_U &\simeq \sum_{\mathbf{k}\in\text{MBZ}} \sum_{\sigma} \hat{\Psi}_{\mathbf{k}\sigma}^{\dagger} h_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} + (\cdots) \\
&= \sum_{\mathbf{k}\in\text{MBZ}} \sum_{\sigma} \hat{\Psi}_{\mathbf{k}\sigma}^{\dagger} W_{\mathbf{k}\sigma}^{\dagger} W_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} W_{\mathbf{k}\sigma}^{\dagger} W_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} + (\cdots) \\
&= \sum_{\mathbf{k}\in\text{MBZ}} \sum_{\sigma} \hat{\Gamma}_{\mathbf{k}\sigma}^{\dagger} d_{\mathbf{k}\sigma} \hat{\Gamma}_{\mathbf{k}\sigma} + (\cdots) \\
&= \sum_{\mathbf{k}\in\text{MBZ}} \sum_{\sigma} E_{\mathbf{k}\sigma} \left(\hat{\gamma}_{\mathbf{k}\sigma}^{(+)\dagger} \hat{\gamma}_{\mathbf{k}\sigma}^{(+)} - \hat{\gamma}_{\mathbf{k}\sigma}^{(-)\dagger} \hat{\gamma}_{\mathbf{k}\sigma}^{(-)} \right) + (\cdots)
\end{aligned}$$

where we intend $(\cdots) = nU\hat{N} - E_{\text{H}/U}^{(\text{AF})}$. The final non-interacting hamiltonian describes a free Fermi gas of $\gamma^{(\pm)}$ fermions, respectively on the bands $\pm E_{\mathbf{k}\sigma}$. Note that these two bands are separated by a gap $\Delta_{\mathbf{k}\sigma}$.

We could redefine the pseudospin operators dividing them by an appropriate constant, in order to satisfy exactly the $\mathfrak{su}(2)$ algebra, and at the same time multiply the pseudo-magnetic field by the same constant. The product $\mathbf{b}_{\mathbf{k}\sigma} \cdot \hat{\mathbf{s}}_{\mathbf{k}\sigma}$ would be unchanged, and the diagonalisation matrix $W_{\mathbf{k}\sigma}$ would be the same. Because of this, the aforementioned constant C is irrelevant.

5.2.4 Features of the AF solution

This subsection is devoted to discussing briefly some features of the AF solution we found. It is organised as a series of unrelated paragraphs, each to derive a precise property or formula we use in the algorithmic computation. With these observations we conclude the part about the local repulsion and extend to full EHM.

Pseudospin EVs. The expectation value of the original pseudospin operators in the various directions can be expressed in terms of the tilted pseudospins $\hat{\Gamma}^\dagger_{\mathbf{k}\sigma}\tau^\alpha\hat{\Gamma}_{\mathbf{k}\sigma}$,

$$\begin{aligned}
\langle\hat{\Psi}^\dagger_{\mathbf{k}\sigma}\tau^x\hat{\Psi}_{\mathbf{k}\sigma}\rangle &= -\sin 2\theta_{\mathbf{k}\sigma}\langle\hat{\Gamma}^\dagger_{\mathbf{k}\sigma}\tau^z\hat{\Gamma}_{\mathbf{k}\sigma}\rangle \\
\langle\hat{\Psi}^\dagger_{\mathbf{k}\sigma}\tau^y\hat{\Psi}_{\mathbf{k}\sigma}\rangle &= 0 \\
\langle\hat{\Psi}^\dagger_{\mathbf{k}\sigma}\tau^z\hat{\Psi}_{\mathbf{k}\sigma}\rangle &= \cos 2\theta_{\mathbf{k}\sigma}\langle\hat{\Gamma}^\dagger_{\mathbf{k}\sigma}\tau^z\hat{\Gamma}_{\mathbf{k}\sigma}\rangle
\end{aligned}$$

where we used the angular relations of Fig. 5.2. Moving to the tilted frame, the z component of the pseudo-spin is just a thermal superposition of "up" and "down" states

$$\langle\hat{\Gamma}^\dagger_{\mathbf{k}\sigma}\tau^z\hat{\Gamma}_{\mathbf{k}\sigma}\rangle = f\left(E_{\mathbf{k}\sigma};\beta,\tilde{\mu}\right) - f\left(-E_{\mathbf{k}\sigma};\beta,\tilde{\mu}\right) \tag{5.34}$$

Let us define the thermal factors:

$$\mathrm{Th}^{(\pm)}_{\mathbf{k}\sigma}[\beta,\mu] \equiv f\left(-E_{\mathbf{k}\sigma};\beta,\mu\right) \pm f\left(E_{\mathbf{k}\sigma};\beta,\mu\right) \tag{5.35}$$

which will often appear in computations. We have:

$$\langle\hat{s}^x_{\mathbf{k}\sigma}\rangle = \frac{\Delta_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}}\mathrm{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu] \tag{5.36}$$

$$\langle\hat{s}^y_{\mathbf{k}\sigma}\rangle = 0 \tag{5.37}$$

$$\langle\hat{s}^z_{\mathbf{k}\sigma}\rangle = -\frac{\epsilon_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}}\mathrm{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu] \tag{5.38}$$

Gauge invariance guarantees that we can rotate the pseudo-field on the xy plane, thus obtaining $\langle\hat{s}^y_{\mathbf{k}\sigma}\rangle \neq 0$. From these EVs, we get the expression of the self-consistency equation for m, and the self-consistent formula to compute the density. Note that $E_{\mathbf{k}\sigma}$ is actually spin independent, and so is the thermal factor. We can neglect the spin index from both.

Magnetisation self-consistency equation. Using Eqns. (5.17), (5.18) we get:

$$m = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left\langle\left(\hat{\psi}_{\mathbf{k}\uparrow}+\hat{\psi}^\dagger_{\mathbf{k}\uparrow}\right)-\left(\hat{\psi}_{\mathbf{k}\downarrow}+\hat{\psi}^\dagger_{\mathbf{k}\downarrow}\right)\right\rangle$$

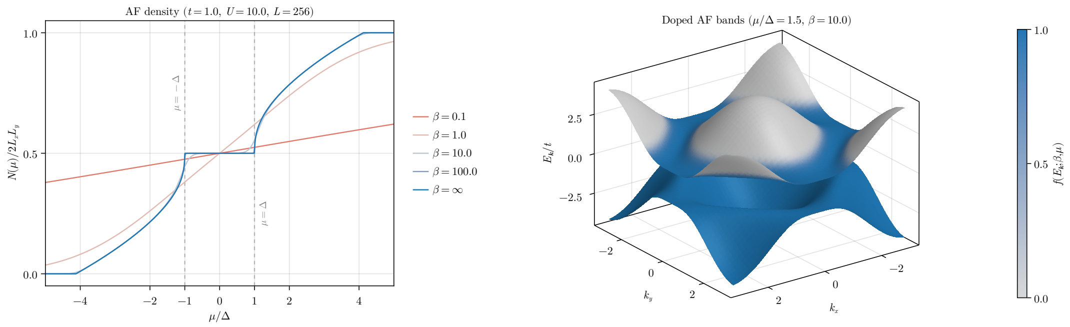
From Tab. 5.2, we recognise a pseudospin EV in direction x. From Eq. (5.36)

$$\begin{aligned}
m &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left[\langle\hat{s}^x_{\mathbf{k}\uparrow}\rangle-\langle\hat{s}^x_{\mathbf{k}\downarrow}\rangle\right] \\
&= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\mathrm{Re}\{\Delta_{\mathbf{k}\uparrow}\}-\mathrm{Re}\{\Delta_{\mathbf{k}\downarrow}\}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta,\mu] \\
&= \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\Delta_{\mathbf{k}\uparrow}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta,\mu]
\end{aligned} \tag{5.39}$$

In last passage we used $\Delta_{\mathbf{k}\uparrow} = -\Delta_{\mathbf{k}\downarrow}$, which follows from the definition of the gap function. The last passage gives the operational self-consistency equation for the HFP m.

Density. Restricting the sum of Eq. (5.13) to the MBZ and extending to both spins, we have:

$$\begin{aligned}
n &= \frac{1}{2L_xL_y}\sum_\sigma\sum_{\mathbf{k}\in\mathrm{MBZ}}\langle\hat{n}_{\mathbf{k}\sigma}+\hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle \\
&= \frac{1}{2L_xL_y}\sum_\sigma\sum_{\mathbf{k}\in\mathrm{MBZ}}\left\langle\Psi^\dagger_{\mathbf{k}\sigma}\hat{\Psi}_{\mathbf{k}\sigma}\right\rangle \\
&= \frac{1}{2L_xL_y}\sum_\sigma\sum_{\mathbf{k}\in\mathrm{MBZ}}\left\langle\Gamma^\dagger_{\mathbf{k}\sigma}\hat{\Gamma}_{\mathbf{k}\sigma}\right\rangle \\
&= \frac{1}{2L_xL_y}\sum_\sigma\sum_{\mathbf{k}\in\mathrm{MBZ}}\left(\left\langle\hat{\gamma}^{(+)\dagger}_{\mathbf{k}\sigma}\hat{\gamma}^{(+)}_{\mathbf{k}\sigma}\right\rangle+\left\langle\hat{\gamma}^{(-)\dagger}_{\mathbf{k}\sigma}\hat{\gamma}^{(-)}_{\mathbf{k}\sigma}\right\rangle\right) \\
&= \frac{1}{2L_xL_y}\sum_\sigma\sum_{\mathbf{k}\in\mathrm{MBZ}}\mathrm{Th}^{(+)}_{\mathbf{k}\sigma}[\beta,\mu]
\end{aligned}$$



a | Electronic density.

b | Example doped bands.

Figure 5.3 | Left panel: electronic density in the AF phase as a function of the chemical potential, measured in relation to the gap $\Delta = mU$, at various temperatures. Right panel: example doped bands, with $\mu/\Delta = 1.5$ set arbitrarily and temperature set to $\beta = 10$ to enhance smearing effects around FS. Single-state filling is proportional to color magnitude.

where we used unitarity of the $\hat{\Psi} \to \hat{\Gamma}$ transformation.

The thermal factor is spin-independent. Moreover, the Bogoliubov bands $E_{\mathbf{k}\sigma}$ are MBZ periodic. Thus we get the two equivalent formulas for density:

$$n = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{MBZ}} \text{Th}^{(+)}_{\mathbf{k}\uparrow}[\beta, \mu] \tag{5.40}$$

$$= \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \text{Th}^{(+)}_{\mathbf{k}\uparrow}[\beta, \mu] \tag{5.41}$$

The two are perfectly equivalent, but the second one is handier to implement in numeric computations. In Fig. 5.3a we sketch density as a function of the chemical potential. At low temperature, the gap plateau at zero energy is most visible. The lattice is half-filled for any chemical potential inside the plateau range.

Unconventional metallicity. Consider now the gran canonical hamiltonian $\hat{K} \equiv \hat{H} - \mu\hat{N}$ and use Eq. (5.28). The diagonalising matrix remains the same, being

$$-\mu\hat{N} = \sum_{\mathbf{k}\in\text{MBZ}} \begin{bmatrix} -\mu & \\ & -\mu \end{bmatrix} \implies W_{\mathbf{k}\sigma}\left[h_{\mathbf{k}\sigma} - \mu\mathbb{I}_{2\times 2}\right] W^{\dagger}_{\mathbf{k}\sigma} = d_{\mathbf{k}\sigma} - \mu\mathbb{I}_{2\times 2} \tag{5.42}$$

thus for non-null chemical potential (i.e. finite doping) the eigenvalues are just shifted, $\pm E_{\mathbf{k}} - \mu$. These two bands have the same number of single-particle states. In Fig. 5.3b we sketch the bands $\pm E_{\mathbf{k}\sigma}$. Intense colour indicates an almost filled single-particle state, lighter gray colour an almost empty single-particle state. The gap is at zero energy, but at any finite doping $\tilde{\mu}$ crosses the upper band.

Instability of the doped antiferromagnet. As we said at the beginning of this chapter, there exist several ways to realise an Hubbard antiferromagnet. For the conventional HM, it is known [1, 2] that the commensurate AF phase is unstable at any finite doping. This is discussed briefly in Sec. D.2. The key takeaway is that insulating antiferromagnets are stable with respect to spin fluctuations, while metallic antiferromagnets are not. The commensurate AF phase is insulating only at half-filling, because the chemical potential lies within the gap.

The mere existence of a self-consistent solution does not guarantee stability: indeed, when in Sec. 2.2.2 we derived the variational solution, we only required it to be a stationary point of the HF energy. In other words, there is no need for the solution to be a minimum. The commensurate AF phase is, in fact, a saddle point of HF energy [1] and any finite doping disrupts commensurate ordering towards uncommensurate phases.

These observations end the description of commensurate antiferromagnetism in the conventional HM. Some more details and observations are provided in App. D.

5.3 Extension to the EHM

We now expand to the EHM, including the non-local interaction $\hat{H}_V$. Our strategy is to generalise the discussion of the section above. A formally identical solution with renormalised quantities,

$$\epsilon_{\mathbf{k}\sigma} \to \tilde{\epsilon}_{\mathbf{k}\sigma} \qquad E_{\mathbf{k}\sigma} \to \tilde{E}_{\mathbf{k}\sigma} \qquad \Delta_{\mathbf{k}\sigma} \to \tilde{\Delta}_{\mathbf{k}\sigma} \qquad \mu \to \tilde{\mu}$$

is coherent by construction with the symmetry structure of the AF phase. Explicitly, the renormalised bands are $\pm\tilde{E}_{\mathbf{k}\sigma}$, with

$$\tilde{E}_{\mathbf{k}\sigma} \equiv \sqrt{\tilde{\epsilon}^2_{\mathbf{k}\sigma} + |\tilde{\Delta}_{\mathbf{k}\sigma}|^2}$$

Moreover, we allow $\Delta_{\mathbf{k}\sigma}$ to be a general complex function of $\mathbf{k}$ and σ,

$$\Delta_{\mathbf{k}\sigma} = \mathrm{Re}\{\Delta_{\mathbf{k}\sigma}\} + i\,\mathrm{Im}\{\Delta_{\mathbf{k}\sigma}\}$$

We define the renormalised angle $\tilde{\theta}_{\mathbf{k}\sigma}$ through the relations:

$$\sin 2\tilde{\theta}_{\mathbf{k}\sigma} = \frac{|\tilde{\Delta}_{\mathbf{k}\sigma}|}{\tilde{E}_{\mathbf{k}\sigma}} \qquad \cos 2\tilde{\theta}_{\mathbf{k}\sigma} = \frac{\tilde{\epsilon}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}} \tag{5.43}$$

and similarly the angle $\tilde{\zeta}_{\mathbf{k}\sigma}$,

$$\sin 2\tilde{\zeta}_{\mathbf{k}\sigma} = \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \qquad \cos 2\tilde{\zeta}_{\mathbf{k}\sigma} = \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \tag{5.44}$$

A sketch of these angles is given in Fig. 5.4. For a complex gap function, the new diagonalisation matrix is given by:

$$\begin{aligned}
\tilde{W}_{\mathbf{k}\sigma} &\equiv e^{-i\tilde{\theta}_{\mathbf{k}\sigma}\tau^y} e^{-i\tilde{\zeta}_{\mathbf{k}\sigma}\tau^z} \\
&= \left(\cos\tilde{\theta}_{\mathbf{k}\sigma} - i\sin\tilde{\theta}_{\mathbf{k}\sigma}\tau^y\right)\left(\cos\tilde{\zeta}_{\mathbf{k}\sigma} - i\sin\tilde{\zeta}_{\mathbf{k}\sigma}\tau^z\right) \\
&= \begin{bmatrix} \cos\tilde{\theta}_{\mathbf{k}\sigma} & -\sin\tilde{\theta}_{\mathbf{k}\sigma} \\ \sin\tilde{\theta}_{\mathbf{k}\sigma} & \cos\tilde{\theta}_{\mathbf{k}\sigma} \end{bmatrix} \begin{bmatrix} e^{-i\tilde{\zeta}_{\mathbf{k}\sigma}} & \\ & e^{i\tilde{\zeta}_{\mathbf{k}\sigma}} \end{bmatrix} \\
&= \begin{bmatrix} \cos\tilde{\theta}_{\mathbf{k}\sigma} e^{-i\tilde{\zeta}_{\mathbf{k}\sigma}} & -\sin\tilde{\theta}_{\mathbf{k}\sigma} e^{i\tilde{\zeta}_{\mathbf{k}\sigma}} \\ \sin\tilde{\theta}_{\mathbf{k}\sigma} e^{-i\tilde{\zeta}_{\mathbf{k}\sigma}} & \cos\tilde{\theta}_{\mathbf{k}\sigma} e^{i\tilde{\zeta}_{\mathbf{k}\sigma}} \end{bmatrix}
\end{aligned}$$

The following relations hold (*tilde* sign indicates the renormalised quantity):

$$\begin{aligned}
\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma}\tau^x\hat{\Psi}_{\mathbf{k}\sigma}\rangle &= -\sin 2\tilde{\theta}_{\mathbf{k}\sigma}\cos 2\tilde{\zeta}_{\mathbf{k}\sigma}\langle\hat{\Gamma}^\dagger_{\mathbf{k}\sigma}\tau^z\hat{\Gamma}_{\mathbf{k}\sigma}\rangle \\
\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma}\tau^y\hat{\Psi}_{\mathbf{k}\sigma}\rangle &= -\sin 2\tilde{\theta}_{\mathbf{k}\sigma}\sin 2\tilde{\zeta}_{\mathbf{k}\sigma}\langle\hat{\Gamma}^\dagger_{\mathbf{k}\sigma}\tau^z\hat{\Gamma}_{\mathbf{k}\sigma}\rangle \\
\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma}\tau^z\hat{\Psi}_{\mathbf{k}\sigma}\rangle &= \cos 2\tilde{\theta}_{\mathbf{k}\sigma}\langle\hat{\Gamma}^\dagger_{\mathbf{k}\sigma}\tau^z\hat{\Gamma}_{\mathbf{k}\sigma}\rangle
\end{aligned}$$

Let us extend Eq. (5.34) and define the renormalised thermal factors,

$$\langle\hat{\Gamma}^\dagger_{\mathbf{k}\sigma}\tau^z\hat{\Gamma}_{\mathbf{k}\sigma}\rangle = f\left(\tilde{E}_{\mathbf{k}\sigma};\beta,\tilde{\mu}\right) - f\left(-\tilde{E}_{\mathbf{k}\sigma};\beta,\tilde{\mu}\right) \tag{5.45}$$

$$\tilde{\mathrm{Th}}^{(\pm)}_{\mathbf{k}\sigma}[\beta,\tilde{\mu}] \equiv f\left(-\tilde{E}_{\mathbf{k}\sigma};\beta,\tilde{\mu}\right) \pm f\left(\tilde{E}_{\mathbf{k}\sigma};\beta,\tilde{\mu}\right) \tag{5.46}$$

which gives $\langle\hat{s}^z_{\mathbf{k}\sigma}\rangle = -\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu]$. Pseudospin expectation values are:

$$\langle\hat{s}^x_{\mathbf{k}\sigma}\rangle = \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{\tilde{E}_{\mathbf{k}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu] \tag{5.47}$$

$$\langle\hat{s}^y_{\mathbf{k}\sigma}\rangle = \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{\tilde{E}_{\mathbf{k}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu] \tag{5.48}$$

$$\langle\hat{s}^z_{\mathbf{k}\sigma}\rangle = -\frac{\tilde{\epsilon}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu] \tag{5.49}$$

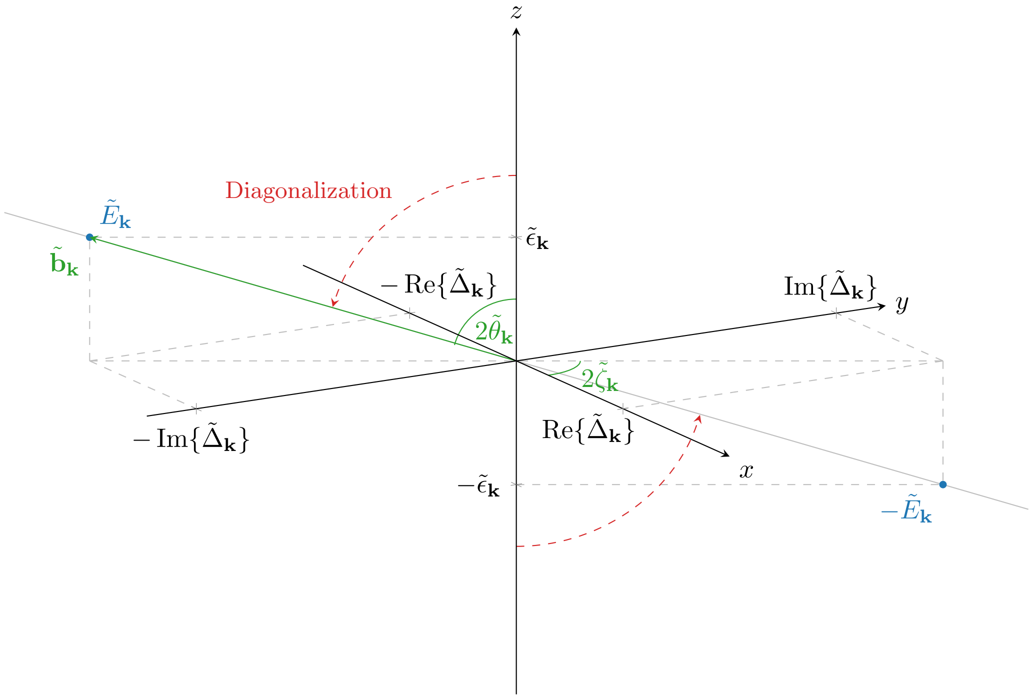


Figure 5.4 Sketch of the diagonalisation procedure for the EHM antiferromagnet. To diagonalise the HF hamiltonian, we align the z axis with the pseudo-field. This is an extesion of Fig. 5.2.

These equations are an extension of Eqns. (5.47), (5.48), (5.49) for the EHM. From now on, in the following sections, we deal with the problem of determining how the various quantities are renormalised.

By renormalising all quantities, the self-consistency equation for magnetisation must stay formally the same of Eq. (5.39)

$$m = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\tilde{\Delta}_{\mathbf{k}\uparrow}}{\tilde{E}_{\mathbf{k}\uparrow}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}\uparrow}[\beta,\mu] \tag{5.50}$$

as well as the formula for density:

$$n = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \mathrm{Th}^{(+)}_{\mathbf{k}\uparrow}[\beta,\mu] \tag{5.51}$$

With this strategy, we get back to the application of HF methods to the full EHM, including the non-local attraction.

5.4 Non-local Hartree effects: chemical potential renormalisation

In last sections we obtained the pseudospin representation of the HM hamiltonian for the AF phase. The main results were Eqns. (5.22), (5.29). Now we must approximate $\hat{H}_V$, reducing to a quadratic operator. We start from the Hartree channel.

The non-local attraction decomposes, in the non-local Hartree channel, in same-spin and opposite-spin sectors. Using Eq. (2.18) we get:

$$\hat{H}_V \simeq \overbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{n}_{i\sigma}\rangle\hat{n}_{j\sigma}+\hat{n}_{i\sigma}\langle\hat{n}_{j\sigma}\rangle-\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\sigma}\rangle\right]}^{\text{s.s.}}$$
$$\underbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{n}_{i\sigma}\rangle\hat{n}_{j\bar{\sigma}}+\hat{n}_{i\sigma}\langle\hat{n}_{j\bar{\sigma}}\rangle-\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\bar{\sigma}}\rangle\right]}_{\text{o.s.}}+(\text{all the rest})$$

Let $i \to \mathbf{r} = (x, y)$ and $j \to \mathbf{r}' = (x', y')$. We use Eq. (5.1), describing oscillatory density for each spin sector. This Ansatz is composed of two parts: the offset n, and the oscillatory part m. The algebraic derivation relative to n part is, of course, the same as for the normal phase. We now treat separately the "operatorial" part of the braced terms of Eq. (4.4),

$$-V\sum_{\langle\mathbf{rr}'\rangle}\sum_{\sigma}\left[\langle\hat{n}_{\mathbf{r}\sigma}\rangle\hat{n}_{\mathbf{r}'\sigma}+\hat{n}_{\mathbf{r}\sigma}\langle\hat{n}_{\mathbf{r}'\sigma}\rangle\right]-V\sum_{\langle\mathbf{rr}'\rangle}\sum_{\sigma}\left[\langle\hat{n}_{\mathbf{r}\sigma}\rangle\hat{n}_{\mathbf{r}'\bar{\sigma}}+\hat{n}_{\mathbf{r}\sigma}\langle\hat{n}_{\mathbf{r}'\bar{\sigma}}\rangle\right] \tag{5.52}$$

as well as the its scalar counterpart

$$V\sum_{\langle\mathbf{rr}'\rangle}\sum_{\sigma}\langle\hat{n}_{\mathbf{r}\sigma}\rangle\langle\hat{n}_{\mathbf{r}'\sigma}\rangle+V\sum_{\langle\mathbf{rr}'\rangle}\sum_{\sigma}\langle\hat{n}_{\mathbf{r}\sigma}\rangle\langle\hat{n}_{\mathbf{r}'\bar{\sigma}}\rangle \tag{5.53}$$

accounting for energy shift originated as "leftovers" of Wick's contractions.

5.4.1 Non-local Hartree channel in the AF phase: analytic derivation

Let us start from the operator of Expr. (5.52). By inserting the Ansatz of Eq. (5.1), we get

$$\overbrace{-nV\sum_{\langle\mathbf{rr}'\rangle}\sum_{\sigma}\left[\hat{n}_{\mathbf{r}'\sigma}+\hat{n}_{\mathbf{r}\sigma}\right]+mV\sum_{\langle\mathbf{rr}'\rangle}\sum_{\sigma}(-1)^{\delta_{\sigma=\uparrow}}\left[(-1)^{x+y}\hat{n}_{\mathbf{r}'\sigma}+(-1)^{x'+y'}\hat{n}_{\mathbf{r}\sigma}\right]}^{\text{s.s.}}$$
$$\underbrace{-nV\sum_{\langle\mathbf{rr}'\rangle}\sum_{\sigma}\left[\hat{n}_{\mathbf{r}'\bar{\sigma}}+\hat{n}_{\mathbf{r}\sigma}\right]+mV\sum_{\langle\mathbf{rr}'\rangle}\sum_{\sigma}\left[(-1)^{x+y+\delta_{\sigma=\uparrow}}\hat{n}_{\mathbf{r}'\bar{\sigma}}+(-1)^{x'+y'+\delta_{\bar{\sigma}=\uparrow}}\hat{n}_{\mathbf{r}\sigma}\right]}_{\text{o.s.}}$$

For a square lattice, if $\mathbf{r} = (x, y)$ and $\mathbf{r}' = (x', y')$ are NNs, both the following identities are true

$$(-1)^{x'+y'} = (-1)^{x+y+1}$$
$$(-1)^{\delta_{\bar{\sigma}=\uparrow}} = (-1)^{\delta_{\sigma=\uparrow}+1}$$

We obtain

$$\overbrace{-nV\sum_{\langle\mathbf{rr}'\rangle}\sum_{\sigma}\left[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\sigma}\right]}^{\text{s.s. (density)}}\overbrace{-mV\sum_{\langle\mathbf{rr}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}\left[\hat{n}_{\mathbf{r}\sigma}-\hat{n}_{\mathbf{r}'\sigma}\right]}^{\text{s.s. (magnetisation)}}$$
$$\underbrace{-nV\sum_{\langle\mathbf{rr}'\rangle}\sum_{\sigma}\left[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\bar{\sigma}}\right]}_{\text{o.s. (density)}}+\underbrace{mV\sum_{\langle\mathbf{rr}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}\left[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\bar{\sigma}}\right]}_{\text{o.s. (magnetisation)}} \tag{5.54}$$

In the above expressions we separated in "density" contributions and "magnetisation" ones. Let us deal with them separately.

Density terms. Consider s.s. and o.s. (density) terms of Expr. (5.54),

$$-nV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\sigma}\right]-nV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\overline{\sigma}}\right] \tag{5.55}$$

The discussion is identical to Sec. 4.2.1 for the normal phase, with renormalisation of μ being the same of Eq. (4.7).

$$\tilde{\mu}\equiv\mu+n(2zV-U) \tag{5.56}$$

We already included the shift to μ due to the local Hartree channel, anticipated in Eq. (5.20).

Magnetization terms. The magnetisation s.s. and o.s. terms of Expr. (5.54) perfectly cancel out:

$$mV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}\left[(\hat{n}_{\mathbf{r}\sigma}-\hat{n}_{\mathbf{r}'\sigma})-(\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\overline{\sigma}})\right]=0$$

The Hartree channel only affects μ.

5.4.2 Non-local Hartree contraction energy

We now collect double counting terms originated by Wick's contraction in Hartree sector. Consider the scalar Expr. (5.53): it's the sum of two parts, the first coming from s.s. sector, the second from o.s. sector. In computing the contraction energy from Hartree channel, being the operatorial result identical to the one obtained for the normal phase, we expect the same to happen for double counting terms.

s.s. sector double counting energy. For s.s. terms we get

$$\begin{aligned}V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\langle\hat{n}_{\mathbf{r}\sigma}\rangle\langle\hat{n}_{\mathbf{r}'\sigma}\rangle &= \overbrace{V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\left[(n+(-1)^{x+y}m)(n+(-1)^{x'+y'}m)\right]}^{\sigma=\uparrow}\\ &\quad+\underbrace{V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\left[(n-(-1)^{x+y}m)(n-(-1)^{x'+y'}m)\right]}_{\sigma=\downarrow}\end{aligned}$$

The mixed terms (those of order $\mathcal{O}(n)\times\mathcal{O}(m)$) cancel out, and since sign factors necessarily alternate on neighbouring sites, the remainder is just

$$V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\sigma}\rangle=V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}2(n^2-m^2)$$

o.s. sector double counting energy. Proceeding identically on the o.s. sector, we get

$$\begin{aligned}V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\langle\hat{n}_{\mathbf{r}\sigma}\rangle\langle\hat{n}_{\mathbf{r}'\overline{\sigma}}\rangle &= \overbrace{V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\left[(n+(-1)^{x+y}m)(n-(-1)^{x'+y'}m)\right]}^{\sigma=\uparrow}\\ &\quad+\underbrace{V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\left[(n-(-1)^{x+y}m)(n+(-1)^{x'+y'}m)\right]}_{\sigma=\downarrow}\end{aligned}$$

which gives

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\overline{\sigma}}\rangle=V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}2(n^2+m^2)$$

Hence, the total "contraction energy", given by sum of both contributions, is identical to the normal one of Eq. (4.9) indeed:

$$E^{(\mathrm{AF})}_{\mathrm{H}/V} = -2V \sum_{\langle ij \rangle} \sum_{\sigma} n^2 = 4n^2 V \times \frac{z}{2} L_x L_y \tag{5.57}$$

the minus sign coming from the fact that we are subtracting this energy shift.

5.5 Fock effects: gap and bands renormalisation

We explored the Hartree channel. Let us now move to the effects originated by Fock contractions. We can use the very beginning of Sec. 4.3 derivation, up to Eq. (4.11),

$$\begin{aligned} &V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} \rangle \right] \\ &\qquad \overset{\mathcal{F}}{=} \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left[\cos(\delta k_x) + \cos(\delta k_y) \right] \langle \hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \end{aligned} \tag{5.58}$$

where $E^{(\mathrm{AF})}_{\mathrm{F}/V}$ is defined as in Eq. (4.12), with the expectation value being computed on the AF (thermal) ground-state. As before, we discuss the operatorial part and the scalar part separately.

5.5.1 Fock channel in the AF phase: analytic derivation

The commensurate AF phase is described by two EVs being in principle non-zero,

$$\left| \langle \hat{n}_{\mathbf{k}\sigma} \rangle \right| \geq 0 \qquad \left| \langle \hat{\psi}_{\mathbf{k}\sigma} \rangle \right| \geq 0$$

as we specified in Sec. 5.1.2. This implies only two contributions in Eq. (5.58) are non-zero:

$$\mathbf{k} = -\mathbf{k}' \quad \text{or} \quad \mathbf{k} + \boldsymbol{\pi} = -\mathbf{k}'$$

Then we get:

$$\begin{aligned} (\cdots) \overset{\mathcal{F}}{=} &\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} \left[\cos(2k_x) + \cos(2k_y) \right] \\ &\left[\underbrace{\langle \hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}}_{\text{Diagonal terms}} - \underbrace{\langle \hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}}_{\text{Off-diagonal terms}} \right] \end{aligned} \tag{5.59}$$

Now, the above equation presents *diagonal* and *off-diagonal* terms.

Diagonal terms. The diagonal terms of Eq. (5.59) are products of $\hat{n}_{\mathbf{k}\sigma}$ operators. Consider Fig. 4.1a: $\hat{n}_{\mathbf{q}\sigma}$, with $\mathbf{q} \equiv \mathbf{K} + \mathbf{k}$, is coupled to $\langle \hat{n}_{\mathbf{q}'\sigma} \rangle$ with $\mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$. Apply to the diagonal part of Eq. (5.59) this change of variables represented in Fig. 4.1

$$\begin{aligned} &\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} \left[\cos(2k_x) + \cos(2k_y) \right] \langle \hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} \\ &\qquad = \frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} \left[\cos(\delta q_x) + \cos(\delta q_y) \right] \langle \hat{c}^{\dagger}_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \hat{c}^{\dagger}_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} \\ &\qquad = \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{n}_{\mathbf{q}\sigma} \rangle \right] \times V \sum_{\mathbf{q}'} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{n}_{\mathbf{q}'\sigma} \end{aligned} \tag{5.60}$$

where in the last passage we substituted the decomposition of Eq. (3.35). Define as in Eq. (4.17) the EV

$$u_{\mathbf{q}\sigma} \equiv \langle \hat{n}_{\mathbf{q}\sigma} \rangle \tag{5.61}$$

Symmetry	Self-consistency equation	Graph
s^*-wave	$u^{(s^*)} = -\dfrac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \dfrac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta, \tilde{\mu}]$	Fig. 3.4b
$d_{x^2-y^2}$-wave	$u^{(d)} = -\dfrac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x - \cos k_y) \dfrac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta, \tilde{\mu}]$	Fig. 3.4e

Table 5.3 | Symmetry and spin resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$.

The expression in square brackets of Eq. (5.60) is just its γ component, and is expanded in

$$\begin{aligned}
u^{(\gamma)}_{\sigma} &= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{n}_{\mathbf{q}\sigma} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \left[\varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{n}_{\mathbf{q}\sigma} \rangle + \varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}} \langle \hat{n}_{\mathbf{q}+\boldsymbol{\pi}\sigma} \rangle \right] \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}} \left[\langle \hat{n}_{\mathbf{q}\sigma} \rangle - \langle \hat{n}_{\mathbf{q}+\boldsymbol{\pi}\sigma} \rangle \right] \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{\Psi}^{\dagger}_{\mathbf{q}\sigma} \tau^z \hat{\Psi}_{\mathbf{q}\sigma} \rangle \\
&= -\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}} \frac{\tilde{\epsilon}_{\mathbf{q}\sigma}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta, \tilde{\mu}]
\end{aligned} \tag{5.62}$$

where, in the last passage, Eq. (5.49) has been used. We are not breaking inversion symmetry (and as a consequence, neither C_2): this implies that, whatever the shape of the renormalised bands, it must be

$$\tilde{\epsilon}_{\mathbf{q}\sigma} = \tilde{\epsilon}_{-\mathbf{q}\sigma} \quad \Longrightarrow \quad u^{(p_\ell)}_{\sigma} = 0 \quad \text{with} \quad \ell \in \{x, y\}$$

Renormalised bare bands need to be symmetric functions of the moment. Hence for $\gamma \in \{p_x, p_y\}$ the above sum vanishes. The remaining s^* and d symmetries give legitimate HFPs with relative self-consistency equations, reported in Tab. 5.3 using spin degeneracy.

Recognising harmonics, Eq. (5.60) becomes

$$\begin{aligned}
&\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} \left[\cos(2k_x) + \cos(2k_y)\right] \langle \hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} \\
&\qquad = V \sum_{\mathbf{q},\sigma} \left[u^{(s^*)}_{\sigma} (\cos q_x + \cos q_y) + u^{(d)}_{\sigma} (\cos q_x - \cos q_y) \right] \hat{c}^{\dagger}_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma}
\end{aligned}$$

These two factors, together, define the full bands renormalisation

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u^{(s^*)}_{\sigma} V (\cos k_x + \cos k_y) + u^{(d)}_{\sigma} V (\cos k_x - \cos k_y) \tag{5.63}$$

as was for the normal phase in Eq. (4.19). In Sec. 4.3.3 we discussed appearance of anisotropic hopping with d-wave extension of bare bands: here theory predicts something identical. These RMPs are, as before, controlled linearly by V and in principle can lead to relevant changes in the kinetic sector of the hamiltonian.

Off-diagonal terms. Consider off-diagonal terms of Eq. (5.59). These contribute instead to full gap, being out of diagonal in the 2×2 hamiltonian matrix of Eq. (5.24). In pseudo-spin picture, they account for a transverse field. Define $\mathbf{q}$, $\mathbf{q}'$ as in Fig. 4.1b,

$$\mathbf{q} \equiv \mathbf{K} + \mathbf{k} \qquad \mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$$

and rewrite as for Eq. (5.60)

$$
\begin{aligned}
&-\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}\\
&\qquad\qquad = -\sum_{\gamma,\sigma}\left[\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}\langle\hat{\psi}^\dagger_{\mathbf{q}\sigma}\rangle\right]\times V\sum_{\mathbf{q}'\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{q}'}\hat{\psi}^\dagger_{\mathbf{q}'\sigma}
\end{aligned}
$$

Now shift the first sum by an AF vector $\boldsymbol{\pi}$ defining $\mathbf{k}+\boldsymbol{\pi}\equiv\mathbf{q}$, and use Eq. (5.11) as well as reciprocal space periodicity by a BZ,

$$
\begin{aligned}
\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}\langle\hat{\psi}^\dagger_{\mathbf{q}\sigma}\rangle &= \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{k}+\boldsymbol{\pi}}\langle\hat{\psi}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle\\
&= -\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{k}}\langle\hat{\psi}_{\mathbf{k}\sigma}\rangle
\end{aligned}
$$

which finally gives

$$
\begin{aligned}
&-\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}\\
&\qquad\qquad = \sum_{\gamma,\sigma}\left[\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}\langle\hat{\psi}_{\mathbf{q}\sigma}\rangle\right]\times V\sum_{\mathbf{q}'\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{q}'}\hat{\psi}^\dagger_{\mathbf{q}'\sigma}
\end{aligned}
\tag{5.64}
$$

Proceed as done previously: define the expectation value

$$
v_{\mathbf{q}\sigma}\equiv i\langle\hat{\psi}_{\mathbf{q}\sigma}\rangle \tag{5.65}
$$

(which is just the AF order parameter with a complex factor) and its γ-wave component, with $\gamma\in\{s^*,d,p_x,p_y\}$:

$$
\begin{aligned}
v^{(\gamma)}_{\sigma} &= \frac{i}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}\langle\hat{\psi}_{\mathbf{q}\sigma}\rangle\\
&= \frac{i}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}\left[\varphi^{(\gamma)}_{\mathbf{q}}\langle\hat{\psi}_{\mathbf{q}\sigma}\rangle+\varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}}\langle\hat{\psi}_{\mathbf{q}+\boldsymbol{\pi}\sigma}\rangle\right]\\
&= \frac{i}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}\varphi^{(\gamma)}_{\mathbf{q}}\left[\langle\hat{\psi}_{\mathbf{q}\sigma}\rangle-\langle\hat{\psi}^\dagger_{\mathbf{q}\sigma}\rangle\right]\\
&= \frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}\varphi^{(\gamma)}_{\mathbf{q}}\langle\hat{\Psi}^\dagger_{\mathbf{q}\sigma}\tau^y\hat{\Psi}_{\mathbf{q}\sigma}\rangle\\
&= \frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}\varphi^{(\gamma)}_{\mathbf{q}}\frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta,\tilde{\mu}]
\end{aligned}
\tag{5.66}
$$

where we used Eq. (5.11), Tab. 5.2 and Eq. (5.48). There is no particular reason to exclude one component or another. In Tab. 5.4 the four derived self-consistency equations are reported, ignoring spin index.

From structure factors definitions of Tab. 3.5, we see that:

$$
v^{(s^*)}_{\sigma},v^{(d)}_{\sigma}\in\mathbb{R}\qquad v^{(p_x)}_{\sigma},v^{(p_y)}_{\sigma}\in i\mathbb{R}
$$

Getting back to Eq. (5.64) and inserting these components, we get

$$
\begin{aligned}
&-\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}\\
&\qquad\qquad = -iV\sum_{\gamma,\sigma}v^{(\gamma)}_{\sigma}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{q}}\hat{\psi}^\dagger_{\mathbf{q}\sigma}
\end{aligned}
\tag{5.67}
$$

Symmetry	Self-consistency equation	Graph
s^*-wave	$v^{(s^*)} = \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\text{MBZ}}(\cos k_x+\cos k_y)\frac{\text{Im}\{\tilde{\Delta}_\mathbf{k}\}}{\tilde{E}_\mathbf{k}}\tilde{\text{Th}}_\mathbf{k}^{(-)}[\beta,\tilde{\mu}]$	Fig. 3.4b
p_x-wave	$v^{(p_x)} = \frac{i\sqrt{2}}{L_xL_y}\sum_{\mathbf{k}\in\text{MBZ}}\sin k_x\frac{\text{Im}\{\tilde{\Delta}_\mathbf{k}\}}{\tilde{E}_\mathbf{k}}\tilde{\text{Th}}_\mathbf{k}^{(-)}[\beta,\tilde{\mu}]$	Fig. 3.4c
p_y-wave	$v^{(p_y)} = \frac{i\sqrt{2}}{L_xL_y}\sum_{\mathbf{k}\in\text{MBZ}}\sin k_y\frac{\text{Im}\{\tilde{\Delta}_\mathbf{k}\}}{\tilde{E}_\mathbf{k}}\tilde{\text{Th}}_\mathbf{k}^{(-)}[\beta,\tilde{\mu}]$	Fig. 3.4d
$d_{x^2-y^2}$-wave	$v^{(d)} = \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\text{MBZ}}(\cos k_x-\cos k_y)\frac{\text{Im}\{\tilde{\Delta}_\mathbf{k}\}}{\tilde{E}_\mathbf{k}}\tilde{\text{Th}}_\mathbf{k}^{(-)}[\beta,\tilde{\mu}]$	Fig. 3.4e

Table 5.4 | Symmetry and spin resolved self-consistency equations for the harmonics of $v_\mathbf{k}$.

We can restrict to MBZ as

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$= -iV\sum_{\gamma,\sigma}\sum_{\mathbf{q}\in\text{MBZ}}\left[v_\sigma^{(\gamma)}\varphi_\mathbf{q}^{(\gamma)*}\hat{\psi}^\dagger_{\mathbf{q}\sigma}+v_\sigma^{(\gamma)}\varphi_{\mathbf{q}+\boldsymbol{\pi}}^{(\gamma)*}\hat{\psi}^\dagger_{\mathbf{q}+\boldsymbol{\pi}\sigma}\right]$$

and use Eq. (5.11), exchanging terms positions

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$= iV\sum_{\gamma,\sigma}\sum_{\mathbf{q}\in\text{MBZ}}\left[v_\sigma^{(\gamma)}\varphi_\mathbf{q}^{(\gamma)*}\hat{\psi}_{\mathbf{q}\sigma}-v_\sigma^{(\gamma)}\varphi_\mathbf{q}^{(\gamma)*}\hat{\psi}^\dagger_{\mathbf{q}\sigma}\right] \tag{5.68}$$

Now: recalling Tab. 3.5, for the four symmetries we are considering, form factors are real-valued for symmetries s^* and d, and purely imaginary for p_x and p_y. As mentioned before, the same holds for the $v_\sigma^{(\gamma)}$ harmonics. Then

$$iv_\sigma^{(\gamma)}\varphi_\mathbf{q}^{(\gamma)*}\in i\mathbb{R} \tag{5.69}$$

and to conjugate a purely imaginary number it suffices to put a minus sign in front of it,

$$-iv_\sigma^{(\gamma)}\varphi_\mathbf{q}^{(\gamma)*} = \left(iv_\sigma^{(\gamma)}\varphi_\mathbf{q}^{(\gamma)*}\right)^*$$

Therefore, we conclude

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$= V\sum_{\gamma,\sigma}\sum_{\mathbf{q}\in\text{MBZ}}\left[iv_\sigma^{(\gamma)}\varphi_\mathbf{q}^{(\gamma)*}\hat{\psi}_{\mathbf{q}\sigma}+\text{h.c.}\right] \tag{5.70}$$

which is no surprise, since hamiltonian are hermitian operators.

5.5.2 Fock contraction energy

Let us now get back to $E^{(\text{AF})}_{\text{F}/V}$, the double counting energy of Eq. (5.58). As said before, its definition is the same as for the normal phase, Eq. (4.12), with averaging operation being performed on the AF ground state:

$$E^{(\text{AF})}_{\text{F}/V} \equiv \frac{V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\sum_{\sigma}[\cos(\delta k_x)+\cos(\delta k_y)]\,\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\rangle\langle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\rangle \tag{5.71}$$

Proceeding exactly as in Eq. (5.59) and making changing variables as in Fig. 4.1, we get

$$E_{\mathrm{F}/V}^{(\mathrm{AF})} = \frac{V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} \left[\cos\left(\delta q_x\right) + \cos\left(\delta q_y\right)\right] \Big[\underbrace{\langle \hat{n}_{\mathbf{q}\sigma}\rangle \langle \hat{n}_{\mathbf{q}'\sigma}\rangle}_{\text{Diagonal terms}} - \underbrace{\langle \hat{\psi}_{\mathbf{q}\sigma}\rangle \langle \hat{\psi}_{\mathbf{q}'\sigma}\rangle}_{\text{Off-diagonal terms}} \Big]$$

Apply the decomposition of Eq. (3.35) and use Eqns. (5.61), (5.65),

$$E_{\mathrm{F}/V}^{(\mathrm{AF})} = \frac{V}{2 L_x L_y} \sum_{\gamma,\sigma} \sum_{\mathbf{q},\mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} \left[u_{\mathbf{q}\sigma} u_{\mathbf{q}'\sigma} + v_{\mathbf{q}\sigma} v_{\mathbf{q}'\sigma} \right] \tag{5.72}$$

Now: it holds

$$u_{\mathbf{q}\sigma} = \langle \hat{n}_{\mathbf{q}\sigma} \rangle \qquad \Longrightarrow \qquad u_{\mathbf{q}\sigma} = u_{\mathbf{q}\sigma}^*$$

and then

$$\begin{aligned} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{q},\mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} u_{\mathbf{q}\sigma} u_{\mathbf{q}'\sigma} &= \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}\in \mathrm{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} u_{\mathbf{q}\sigma} \right] \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}'\in \mathrm{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} u_{\mathbf{q}'\sigma}^* \right] \\ &= \left| u_{\sigma}^{(\gamma)} \right|^2 \end{aligned}$$

At the same time, being $\hat{\psi}_{\mathbf{q}+\boldsymbol{\pi}\sigma} = \hat{\psi}_{\mathbf{q}\sigma}^\dagger$,

$$v_{\mathbf{q}\sigma} = i \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \qquad \Longrightarrow \qquad v_{\mathbf{q}+\boldsymbol{\pi}\sigma} = -v_{\mathbf{q}\sigma}^*$$

Since the form factors satisfy $\varphi_{\mathbf{q}+\boldsymbol{\pi}}^{(\gamma)} = -\varphi_{\mathbf{q}}^{(\gamma)}$, we get:

$$\begin{aligned} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{q},\mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} v_{\mathbf{q}\sigma} v_{\mathbf{q}'\sigma} &= \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}\in \mathrm{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} v_{\mathbf{q}\sigma} \right] \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}'\in \mathrm{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} v_{\mathbf{q}'\sigma} \right] \\ &= v_{\sigma}^{(\gamma)} \left[\frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} \varphi_{\mathbf{k}+\boldsymbol{\pi}}^{(\gamma)*} v_{\mathbf{k}+\boldsymbol{\pi}\sigma} \right] \\ &= v_{\sigma}^{(\gamma)} \left[\frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} \varphi_{\mathbf{k}}^{(\gamma)*} v_{\mathbf{k}\sigma}^* \right] \\ &= \left| v_{\sigma}^{(\gamma)} \right|^2 \end{aligned}$$

where in the second passage we employed the change of variables $\mathbf{q}' = \mathbf{k}+\boldsymbol{\pi}$ and used BZ periodicity to map the original sum onto a new sum for $\mathbf{k} \in \mathrm{BZ}$. Inserting above results in Eq. (5.72), the latter becomes:

$$E_{\mathrm{F}/V}^{(\mathrm{AF})} = L_x L_y \times \frac{V}{2} \sum_{\gamma,\sigma} \left[\left| u_{\sigma}^{(\gamma)} \right|^2 + \left| v_{\sigma}^{(\gamma)} \right|^2 \right] \tag{5.73}$$

The new HFPs amplitudes contribute positively to contraction energy. The consequences of this fact will be discussed in Sec. 5.7.1.

5.6 Full MFT hamiltonian and gap complexification

Finally, it is time to collect everything in a single hamiltonian operator. The approximated hamiltonian is:

$$\begin{aligned} \hat{H} - \mu \hat{N} \simeq \sum_{\sigma} \sum_{\mathbf{k}\in \mathrm{MBZ}} \tilde{\epsilon}_{\mathbf{k}\sigma} \left[\hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \right] - \sum_{\sigma} \sum_{\mathbf{k}\in \mathrm{MBZ}} \left[\tilde{\Delta}_{\mathbf{k}\sigma} \hat{\psi}_{\mathbf{k}\sigma} + \tilde{\Delta}_{\mathbf{k}\sigma}^* \hat{\psi}_{\mathbf{k}\sigma}^\dagger \right] \\ - \tilde{\mu} \hat{N} - \left[E_{\mathrm{H}/U}^{(\mathrm{AF})} + E_{\mathrm{H}/V}^{(\mathrm{AF})} + E_{\mathrm{F}/V}^{(\mathrm{AF})} \right] \end{aligned} \tag{5.74}$$

where renormalized bands are given by:

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma} + V \sum_{\gamma \in \{s^*, d\}} u_\sigma^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*}$$

and full gap function by

$$\tilde{\Delta}_{\mathbf{k}\sigma} = \Delta_{\mathbf{k}\sigma} - iV \sum_\gamma v_\sigma^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} \tag{5.75}$$

The gap correction is purely imaginary, compared to HM gap $\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$. Since there is no other contribution to the x pseudo-spin projections other than the pure HM ones, spin antisymmetry is preserved in this direction

$$\langle \hat{s}_{\mathbf{k}\sigma}^x \rangle = - \langle \hat{s}_{\mathbf{k}\bar{\sigma}}^x \rangle$$

This property confirms that Eq. (5.50) actually holds. Indeed, we anticipated that the self-consistency equation of the magnetisation should have been identical for the HM and the EHM. The fact that the x component EV of the pseudospin maintains this antisymmetry relation is a confirmation of that, since the proof that lead us to Eq. (5.39) for the HM was based on this property.

One of the net effects of $\hat{H}_V$ on the AF phase, with respect to the conventional HM, is "gap complexification". This is expressed by:

$$\left| \operatorname{Im}\{\Delta_{\mathbf{k}\sigma}\} \right| \geq 0 \quad \Longrightarrow \quad \left| \langle s_{\mathbf{k}\sigma}^y \rangle \right| \geq 0$$

while for HM the y pseudo-spin component was identically null.

5.6.1 The relative phase in the zero temperature half-filled ground state

In Sec. D.1, we derive analytically the AF ground state on conventional HM at zero temperature, as a simple Bloch's transform of a polarized (tilted) pseudo-spins collective pure state. Eq. (E.29) reports the final form of said ground-state. Said quantum state is in general pretty similar to conventional BCS state.

At half-filling and zero temperature we can extend that state to the new complexified gap. At zero temperature the AF ground-state for the EHM is given by

$$|\Psi\rangle = \bigotimes_\sigma \bigotimes_{\mathbf{k} \in \text{MBZ}} \left[\sin \tilde{\theta}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}^\dagger + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}} \cos \tilde{\theta}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}^\dagger \right] |\Omega_{\mathbf{k}\sigma}\rangle \tag{5.76}$$

The proof is identical to the one provided in Sec. D.1 for the HM, apart from renormalisations. Let $|\mathbf{k}\sigma\rangle \equiv \hat{c}_{\mathbf{k}\sigma}^\dagger |\Omega_{\mathbf{k}\sigma}\rangle$. Then we can write

$$|\Psi\rangle = \bigotimes_\sigma \bigotimes_{\mathbf{k} \in \text{MBZ}} \left[\sin \tilde{\theta}_{\mathbf{k}\sigma} + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}} \cos \tilde{\theta}_{\mathbf{k}\sigma} \hat{\psi}_{\mathbf{k}\sigma} \right] |\mathbf{k}\sigma\rangle$$

having we used $|\Omega_{\mathbf{k}\sigma}\rangle = \{\hat{c}_{\mathbf{k}\sigma}, \hat{c}_{\mathbf{k}\sigma}^\dagger\} |\Omega_{\mathbf{k}\sigma}\rangle$.

The angles appearing in above equations are the same of Eqns. (5.43), (5.44). The relative phase $2\tilde{\zeta}_{\mathbf{k}\sigma}$ is the gap function angle in complex plane, since

$$\begin{aligned}
\langle \Psi | \hat{s}_{\mathbf{k}\sigma}^x | \Psi \rangle &= \langle \Psi | \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^\dagger | \Psi \rangle \\
&= \sin \tilde{\theta}_{\mathbf{k}\sigma} \cos \tilde{\theta}_{\mathbf{k}\sigma} \left(e^{-2i\tilde{\zeta}_{\mathbf{k}\sigma}} + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}} \right) \\
&= \sin 2\tilde{\theta}_{\mathbf{k}\sigma} \cos 2\tilde{\zeta}_{\mathbf{k}\sigma}
\end{aligned}$$

$$\begin{aligned}
\langle \Psi | \hat{s}_{\mathbf{k}\sigma}^y | \Psi \rangle &= i \langle \Psi | \hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}_{\mathbf{k}\sigma}^\dagger | \Psi \rangle \\
&= i \sin \tilde{\theta}_{\mathbf{k}\sigma} \cos \tilde{\theta}_{\mathbf{k}\sigma} \left(e^{-2i\tilde{\zeta}_{\mathbf{k}\sigma}} - e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}} \right) \\
&= \sin 2\tilde{\theta}_{\mathbf{k}\sigma} \sin 2\tilde{\zeta}_{\mathbf{k}\sigma}
\end{aligned}$$

These equations are coherent with Eqns. (5.47), (5.48), thus the involved angles (apart from irrelevant 2π shifts) coincide.

Phase	HFPs
sAF	$\{m, u^{(s^*)}, u^{(d)}, v^{(s^*)}, v^{(d)}\}$
aAF	$\{m, u^{(s^*)}, u^{(d)}, -iv^{(p_x)}, -iv^{(p_y)}\}$

Table 5.5 Set of HFP for the two possible realization of the AF phase, respectively a symmetric and an antisymmetric imaginary part of the gap.

5.6.2 AF phase flavours and spin degeneracy

In this analysis, antiferromagnetism is realised without breaking inversion symmetry. This translates in the requirement that:

$$\tilde{E}_{\mathbf{k}\sigma} \stackrel{!}{=} \tilde{E}_{-\mathbf{k}\sigma}$$

We know that:

$$\tilde{E}_{\mathbf{k}\sigma} = \sqrt{\tilde{\epsilon}^2_{\mathbf{k}\sigma} + \mathrm{Re}^2\{\tilde{\Delta}_{\mathbf{k}\sigma}\} + \mathrm{Im}^2\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}$$

Renormalized bare bands as well as the real part of the gap satisfy said inversion symmetry requirement,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \tilde{\epsilon}_{-\mathbf{k}\sigma} \qquad \mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \mathrm{Re}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\}$$

hence inversion symmetry is respected only if

$$\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \pm\,\mathrm{Im}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\}$$

which is: either the imaginary part of the gap is inversion symmetric, or is antisymmetric. Based on this consideration, we are able two discard two of the four self-consistency equations for the v-related HFPs listed in Tab. 5.4: s^* and d-wave symmetries HFPs are nonzero if $\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is momentum-even; p_x and p_y-wave symmetries HFPs are nonzero if $\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is momentum-odd. The two options cannot coexist, and separate simulations can be implemented in the two symmetry sectors.

We name the two possible symmetries “sAF” (symmetric antiferromagnet) and “aAF” (antisymmetric antiferromagnet). The set of HFPs then restricts to these AF structures, and is reported for both cases in Tab. 5.5. The presence itself of two “AF flavours” is not-so-physical, given that up to now we have no clear physical interpretation for $\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}$. As we will discuss in Sec. 5.8, it is possible to prove that almost all these new HFPs actually vanish – a fact that is not directly inferrable right now, but can be seen only delving into the discussion of the non-linear map that defines the algorithm. For now, let us proceed integrally with all the HFPs we found and derive the results.

From now on we will omit σ index, working just in the $\uparrow$ sector. As said in Sec. 5.1, the AF phase does not break $\mathbb{Z}_2$ spin inversion symmetry, being the SDW ordering generated by a staggered but globally symmetric spin density distribution among sites. The approximated quadratic hamiltonian of Eq. 5.74 is invariant by a $\sigma \leftrightarrow \overline{\sigma}$ exchange, thus all measurable physical quantities must be.

5.7 AF free energy density

In order to derive the free energy density expression for AF phase, we proceed precisely as in Sec. 4.4. The only notable difference is in the expression of entropy density: Eq. (4.23) is written for two opposite bands on MBZ

$$s = s^{(+)} + s^{(-)}$$

with

$$s^{(\pm)} \equiv -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \Big[\Big(1 - f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) \ln \Big(1 - f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) + f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) \ln f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) \Big] \tag{5.77}$$

Consider also that we are computing free energy on the gran-canonical hamiltonian

$$\hat{H} - \mu\hat{N} \simeq \sum_{\sigma}\sum_{\mathbf{k}\in\mathrm{MBZ}} \left[(\tilde{E}_{\mathbf{k}} - \tilde{\mu})\hat{\gamma}^{(+)\dagger}_{\mathbf{k}\sigma}\hat{\gamma}^{(+)}_{\mathbf{k}\sigma} - (\tilde{E}_{\mathbf{k}} + \tilde{\mu})\hat{\gamma}^{(-)\dagger}_{\mathbf{k}\sigma}\hat{\gamma}^{(-)}_{\mathbf{k}\sigma}\right] + E_{\mathrm{c}}^{(\mathrm{AF})}$$

being $E_{\mathrm{c}}^{(\mathrm{AF})}$ the total AF contraction energy. Note that on the left-hand side we have μ and on the right-hand side its RMP counterpart $\tilde{\mu}$.

With identical meaning of the notation with respect to Sec. 4.4, the free energy contributions are:

1. Quasiparticles energy, taking care of spin degeneracy:

$$e_{\mathrm{g}}^{(\mathrm{AF})} = \frac{2}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) - \frac{2}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})$$

2. Contraction energy:

$$e_{\mathrm{c}}^{(\mathrm{AF})} = U(m^2 - n^2) + V\left[2zn^2 - \sum_{\gamma}\left(\left|u^{(\gamma)}\right|^2 + \left|v^{(\gamma)}\right|^2\right)\right]$$

3. Entropic energy

$$\begin{aligned} e_{s}^{(\mathrm{AF})} = \frac{2}{\beta L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \ln\left(1 - f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\right) - \frac{2}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) \\ + \frac{2}{\beta L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \ln\left(1 - f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\right) + \frac{2}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) \end{aligned}$$

4. Chemical potential correction

$$e_{n}^{(\mathrm{AF})} = \mu n$$

Compose everything

$$f_{\mathrm{MFT}}^{(\mathrm{AF})} \equiv e_{\mathrm{g}}^{(\mathrm{AF})} + e_{\mathrm{c}}^{(\mathrm{AF})} + e_{s}^{(\mathrm{AF})} + e_{n}^{(\mathrm{AF})}$$

and we get

$$\begin{aligned} f_{\mathrm{MFT}}^{(\mathrm{AF})} = \frac{2}{\beta L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \left[\ln\left(1 - f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\right) + \ln\left(1 - f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\right)\right] \\ + Um^2 - V\sum_{\gamma}\left[\left|u^{(\gamma)}\right|^2 + \left|v^{(\gamma)}\right|^2\right] + n\left[\mu + n(2zV - U)\right] \end{aligned}$$

As was for the normal phase, contraction energy and chemical potential energy corrections build up the RMP $\tilde{\mu}$ of Eq. (5.56),

$$\begin{aligned} f_{\mathrm{MFT}}^{(\mathrm{AF})} = \frac{2}{\beta L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \left[\ln\left(1 - f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\right) + \ln\left(1 - f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\right)\right] \\ + Um^2 - V\sum_{\gamma}\left[\left|u^{(\gamma)}\right|^2 + \left|v^{(\gamma)}\right|^2\right] + \tilde{\mu}n \end{aligned} \tag{5.78}$$

In this expression, we see that $u^{(\gamma)}$s, $v^{(\gamma)}$s contribution is negative, while magnetisation contributes positively. At a first glance, this suggests that a system with no magnetisation and non-zero $u^{(\gamma)}$s, $v^{(\gamma)}$s would be energetically favoured. In that situation we would have a purely imaginary gap for a non-magnetised phase, which would be not so intuitive to explain physically. In next section we explain how the above expression for the free energy density can be used to give a constraint that proves that said situation is impossible. HFPs are interrelated and the physical solution cannot admit arbitrary values.

5.7.1 Energetic constraints on the HFPs

Let us derive a constraint from the free energy density expression. Collecting the terms directly related to the gap function, the following identity holds:

$$Um^2 - V\sum_{\gamma}\left|v^{(\gamma)}\right|^2 = \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\left|\tilde{\Delta}_{\mathbf{k}}\right|^2}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] \tag{5.79}$$

The proof is simple and immediate: expanding right-hand side,

$$\begin{aligned}\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\left|\tilde{\Delta}_{\mathbf{k}}\right|^2}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] = \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\mathrm{Re}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}]\\ + \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\mathrm{Im}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}]\end{aligned}$$

Now:

$$\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\} = Um$$

which implies

$$\begin{aligned}\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\mathrm{Re}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] &= Um\times\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}]\\ &= Um^2\end{aligned}$$

having we recognised the expression for the magnetisation from Eq. (5.50). Equivalently,

$$\begin{aligned}\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\} &= -V\sum_{\gamma}v^{(\gamma)}\varphi_{\mathbf{k}}^{(\gamma)*}\\ &= -V\sum_{\gamma}v^{(\gamma)*}\varphi_{\mathbf{k}}^{(\gamma)}\end{aligned}$$

where we used Eq. (5.69) to exchange the conjugation operation in the last passage. This implies:

$$\begin{aligned}\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\mathrm{Im}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] &= -V\sum_{\gamma}v^{(\gamma)*}\sum_{\mathbf{k}\in\mathrm{MBZ}}\varphi_{\mathbf{k}}^{(\gamma)}\frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}]\\ &= -V\sum_{\gamma}v^{(\gamma)*}\times v^{(\gamma)}\\ &= -V\sum_{\gamma}\left|v^{(\gamma)*}\right|^2\end{aligned}$$

where in second passage we recognised $v^{(\gamma)}$ given in Eq. (5.66). Then Eq. (5.79) is verified.

A consequence of Eq. (5.79) is that, being the right-hand side positive, for $V > 0$ the total amplitude of the "complexifiers" vector is maximised,

$$\sum_{\gamma}\left|v^{(\gamma)}\right|^2 \leq M \quad \text{being} \quad M^{(\mathrm{AF})} \equiv m^2\frac{U}{V} \tag{5.80}$$

The above constraint predicts that:

- In the strong coupling limit $U \ll V$, gap complexification is suppressed;
- If magnetisation is null, so is the imaginary part of the gap

  $$\lim_{m\to 0} v^{(\gamma)} = 0$$

  which is coherent with the entire derivation and takes us back to the normal phase.

Eq. (5.79) is also solved by a null magnetisation and thus null complexifiers. This can be interpreted by the fact that when the constraint is impossible to solve self-consistently, the gap must close.

Eq. (5.80) provides a neat graphic interpretation for the constraint on HFPs in both sAF and aAF phases. At any competition ratio U/V, the product $m\sqrt{U/V}$ is the radius of a sphere. Inside this sphere must lie the vector defined by remaining $v^{(\gamma)}$ HFPs. For our AF flavours, the sphere is actually a circle, being in both cases the total number of HFPs equal to 3, counting also m. Note that the solution:

$$\forall\gamma \quad : \quad v^{(\gamma)} = 0$$

solves the constraint of Eq. (5.80). These considerations do not imply gap complexification is an actual, observed effect.

5.7.2 Free energy at half-filling

Let us discuss the half-filling condition. We start with the identity:

$$\begin{aligned}\ln\left(1-\frac{1}{e^x+1}\right)+\ln\left(1-\frac{1}{e^{-x}+1}\right) &= \ln\left[\frac{e^x}{(e^x+1)^2}\right]\\ &= x-2\ln(1+e^x)\end{aligned}$$

Let now $x=\beta\tilde{E}_{\mathbf{k}}$. At low temperature for a band with no nodes, we can approximate $x\to+\infty$ on the entire BZ. This gives

$$x-2\ln(1+e^x)\simeq -x$$

Hence we get the half-filling free energy density substituting ,

$$f^{(\mathrm{AF})}_{\mathrm{MFT}} \simeq -\frac{2}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\tilde{E}_{\mathbf{k}} + Um^2 - V\sum_{\gamma}\left[\left|u^{(\gamma)}\right|^2+\left|v^{(\gamma)}\right|^2\right]$$

The first term in this sum gives the energy of the fully occupied lower band.

This approximation makes sense when $\tilde{E}_{\mathbf{k}}\neq 0$ everywhere. This is always realised in AF phase: from the constraint of Eq. (5.80) we know magnetisation is necessary for all gap HFPs; for a magnetised system, whatever the symmetries,

$$\tilde{E}_{\mathbf{k}} = \sqrt{\tilde{\epsilon}^2_{\mathbf{k}}+(Um)^2+\left|\,\mathrm{Im}\{\Delta_{\mathbf{k}}\}\right|^2} \geq Um$$

At half filling and low temperature, the system is well approximated by a completely filled electronic band $-\tilde{E}_{\mathbf{k}}$, and this correctly gives a simple $f^{(\mathrm{AF})}_{\mathrm{MFT}}$ approximation.

5.8 Theoretical constraints on the AF phase HFPs

Up to now, all we derived was the most general HF mathematical description of a commensurate AF established on EHM, allowing for nematic spatial symmetry breaking as well as non-trivial renormalisation effects. This section, basically a clone of Sec. 4.5, is dedicated to understanding, given the algorithmic problem, how its solutions should behave. As we will see, based on this theoretical analysis we are able to reduce a lot the algorithmic complexity and limit the solution search to a tailored subspace of HFPs space.

We start from considering a robust C_4 antiferromagnet, thus setting $u^{(d)}=0$ by hand, and show how the constraints on $u^{(s^*)}$ mimic those posed for the normal phase in Eq. (4.31). Then we move to a rather convoluted but necessary analysis of the five-components HFPs vector, described for both AF flavours in Tab. 5.5. For the sake of generality, even if – as discussed in Sec. D.2 – at finite doping the system is naturally unstable, we take on all calculations at generic $\tilde{\mu}$.

5.8.1 Robust C_4 bands shrinking extension to AF phases

For the normal phase, we saw that when C_4 symmetry is preserved (Sec. 4.5.1) we can conclude immediately that bands renormalisation is actually a shrinking. An identical result holds here: if C_4 symmetry is imposed onto the renormalized bands,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv (u^{(s^*)}V-2t)(\cos k_x+\cos k_y)$$

of $u^{(\gamma)}$, just the $s*$-wave channel contributes with a single self-consistency equation

$$u^{(s^*)} = -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$$

In this situation, we are able to derive constraints for $u^{(s^*)}$ by analysing the self-consistency equation. As we did for the normal phase:

1. If $t > 0$ and $u^{(s^*)} \leq 0$

$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} \leq 0$$

   the equality only holding on the MBZ contour. Looking at the self-consistency equation, all other objects inside the sum are positive-defined. Which gives

$$u^{(s^*)} \leq 0 \quad \Longrightarrow \quad u^{(s^*)} \geq 0$$

   But $u^{(s^*)} \neq 0$ as long as $t > 0$, as can be seen by the self-consistency equation. We conclude necessarily

$$u^{(s^*)} > 0$$

2. If $t > 0$ and $u^{(s^*)} \geq 2t/V$

$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} \geq 0$$

   the equality only holding on the MBZ contour. Inverting inequalities from point above, we conclude necessarily

$$u^{(s^*)} < 2t/V$$

The final result,

$$0 < u^{(s^*)} < 2t/V \tag{5.81}$$

is identical to the constraint obtained for the normal phase. Also in AF phase the bare bands are effectively shrunk by $\hat{H}_V$.

5.8.2 HFPs vanishing in sAF phase

We applied HF methods to the EHM and obtained two flavours of antiferromagnetism, each of which involves 5 HFPs. The scenario appears to be rich. In this section we present an argument to prove that the actual number of HFPs is much smaller, as many of those are mathematically suppressed.

Let us start from the sAF phase, assuming

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$$

The system is described by the five HFPs on the first row of Tab. 5.5 and their self-consistency equations of Tabs. 5.3 and 5.4.

Let us define the functional

$$F[f(\mathbf{k})] \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} f(\mathbf{k}) \tag{5.82}$$

The functional F is linear:

$$F[af(\mathbf{k}) + bg(\mathbf{k})] = aF[f(\mathbf{k})] + bF[g(\mathbf{k})] \tag{5.83}$$

being $a, b \in \mathbb{C}$ and f, g two integrable functions in reciprocal space. Another handy property is the sign-preserving relation for sign-defined functions. Let for instance

$$\forall \mathbf{k} \quad : \quad f(\mathbf{k}) \geq 0 \quad \Longrightarrow \quad F[f(\mathbf{k})] \geq 0 \tag{5.84}$$

The relation above is implied by the fact that both $\tilde{\text{Th}}_{\mathbf{k}}^{(-)}$ and $\tilde{E}_{\mathbf{k}}$ are positive-defined,

$$\tilde{\text{Th}}_{\mathbf{k}}^{(-)} = \frac{1}{e^{\beta(\tilde{E}_{\mathbf{k}}-\mu)}+1} - \frac{1}{e^{\beta(-\tilde{E}_{\mathbf{k}}-\mu)}+1} \geq 0 \qquad \tilde{E}_{\mathbf{k}} = \sqrt{\tilde{\epsilon}_{\mathbf{k}}^2 + |\Delta_{\mathbf{k}\sigma}|^2} \geq 0$$

so, being the functional F essentially a weighted integral over MBZ, any positive-defined function has a positive image. An identical statement with inverted inequalities holds for negative-defined function.

Consider now the self-consistency equation for $u^{(s^*)}$: explicitly,

$$\begin{aligned}
u^{(s^*)} &= -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{MBZ}} (c_x + c_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}] \\
&= \frac{\left[2t - V u^{(s^*)}\right]}{L_x L_y} \sum_{\mathbf{k}\in\text{MBZ}} \frac{\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} (c_x + c_y)^2 - \frac{V u^{(d)}}{L_x L_y} \sum_{\mathbf{k}\in\text{MBZ}} \frac{\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} (c_x^2 - c_y^2)
\end{aligned}$$

having we used the notation $c_\ell \equiv \cos k_\ell$ and substituted the full bare bands

$$\tilde{\epsilon}_{\mathbf{k}} \equiv -\left[2t - u_\sigma^{(s^*)} V\right](c_x + c_y) + u_\sigma^{(d)} V (c_x - c_y)$$

In terms of the functional F, we can rewrite:

$$\begin{aligned}
u^{(s^*)} &= \left[2t - V u^{(s^*)}\right] \times F\left[(c_x + c_y)^2\right] - V u^{(d)} \times F\left[c_x^2 - c_y^2\right] \\
&= \left[2t - V u^{(s^*)}\right] \times \left\{F\left[c_x^2\right] + 2F\left[c_x c_y\right] + F\left[c_y^2\right]\right\} - V u^{(d)} \times \left\{F\left[c_x^2\right] - F\left[c_y^2\right]\right\}
\end{aligned}$$

having we used the linearity property of Eq. (5.83) in last passage.

Let now:

$$A \equiv F\left[c_x^2\right] \qquad B \equiv 2F\left[c_x c_y\right] \qquad C \equiv F\left[c_y^2\right] \tag{5.85}$$

which allows us to rewrite the self-consistency equation as:

$$u^{(s^*)} = \left[2t - V u^{(s^*)}\right](A + B + C) - V u^{(d)} (A - C)$$

Repeating an identical procedure on the three self-consistency equations for $u^{(d)}$, $v^{(s^*)}$ and $v^{(d)}$, we get the full system:

$$u^{(s^*)} = \left[2t - V u^{(s^*)}\right](A + B + C) - V u^{(d)} (A - C) \tag{5.86}$$

$$u^{(d)} = \left[2t - V u^{(s^*)}\right](A - C) - V u^{(d)} (A - B + C) \tag{5.87}$$

$$v^{(s^*)} = -V v^{(s^*)} (A + B + C) - V v^{(d)} (A - C) \tag{5.88}$$

$$v^{(d)} = -V v^{(s^*)} (A - C) - V v^{(d)} (A - B + C) \tag{5.89}$$

These equations define a four-components nonlinear map, whose parameters A, B, C depend on the HFPs themselves, being the functional F dependent on the HFPs. Because of this last point, the four equations are coupled to each other and the map does not decouple in two independent sub-maps. Note that the full map would include another equation for the magnetisation.

Aim of this section is to prove that no self-consistent solution exists with non-zero $u^{(d)}$, $v^{(s^*)}$ and $v^{(d)}$. Being the derivation convoluted and abstract enough, let us proceed by enumerating the relevant facts.

1. Let us start by applying a linear transform on the map: define the new "sum" and "difference" variables

$$S_u \equiv u^{(s^*)} + u^{(d)} \qquad D_u \equiv u^{(s^*)} - u^{(d)} \qquad S_v \equiv v^{(s^*)} + v^{(d)} \qquad D_v \equiv v^{(s^*)} - v^{(d)}$$

whose matrix representation is

$$
\begin{bmatrix} S_u \\ D_u \\ S_v \\ D_v \end{bmatrix} \equiv \begin{bmatrix} 1 & 1 & & \\ 1 & -1 & & \\ & & 1 & 1 \\ & & 1 & -1 \end{bmatrix} \begin{bmatrix} u^{(s^*)} \\ u^{(d)} \\ v^{(s^*)} \\ v^{(d)} \end{bmatrix}
$$

The map transform is obtained by summing and subtracting two couples of equations: Eq. (5.86) and (5.87) (the first two) as well as Eq. (5.88) and (5.89) (the second two). By doing this we get the simplified system:

$$(1+2AV)S_u + BVD_u = 2t(2A+B) \tag{5.90}$$
$$(1+2CV)D_u + BVS_u = 2t(B+2C) \tag{5.91}$$
$$(1+2AV)S_v + BVD_v = 0 \tag{5.92}$$
$$(1+2CV)D_v + BVS_v = 0 \tag{5.93}$$

2. Let $S_v = 0$. Then $D_v = 0$. From Eq. (5.93), using $C = F\left[c_y^2\right] \geq 0$ implied by Eq. (5.84):

$$\underbrace{(1+2CV)}_{>0} D_v = 0 \qquad \Longrightarrow \qquad D_v = 0$$

Identically, let $D_v = 0$. Then applying the same consideration on Eq. (5.92) with $C \leftrightarrow A$, we conclude $S_v = 0$. Either both are non-zero or both are zero.

3. Let $S_v \neq 0$ and $D_v \neq 0$. Then $B \neq 0$. To prove this, consider for example Eq. (5.92): we are working at $V > 0$, thus

$$B = -\frac{1+2AV}{V}\frac{S_v}{D_v}$$

which is non-zero because $A = F\left[c_x^2\right] \geq 0$, the latter following from Eq. (5.84). We could have used identically Eq. (5.93).

4. Let $S_v \neq 0$ and $D_v \neq 0$. Hence the following relation holds:

$$(BV)^2 = (1+2CV)(1+2AV) \tag{5.94}$$

To prove the latter, it is sufficient to multiply Eq. (5.92) by $BV \neq 0$ (see point above),

$$BV(1+2AV)S_v + (BV)^2 D_v = 0$$

and substitute from Eq. (5.93) $BVS_v = -(1+2CV)D_v$,

$$\left[-(1+2CV)(1+2AV) + (BV)^2\right] D_v = 0$$

Since $D_v \neq 0$ by hypothesis, Eq. (5.94) is implied.

5. Let $S_v \neq 0$ and $D_v \neq 0$. Hence the following relation holds:

$$B + 2C = V(B^2 - 4AC) \tag{5.95}$$

To prove this: multiply Eq. (5.90) by BV on both sides,

$$BV(1+2AV)S_u + (BV)^2 D_u = 2tBV(2A+B)$$

and substitute from Eq. (5.91) $BVS_v = -(1+2CV)D_v + 2t(B+2C)$,

$$
\begin{aligned}
-(1+2CV)(1+2AV)D_v + 2t(B+2C)(1+2AV) + (BV)^2 D_u &= 2tBV(2A+B) \\
2t(B+2C)(1+2AV) &= 2tBV(2A+B) \\
B + 2ABV + 2C + 4ACV &= 2ABV + B^2 V
\end{aligned}
$$

Passing from first to second line we used Eq. (5.94), and in the passage below $t > 0$. Last line implies Eq. (5.95).

6. Let $S_v \neq 0$ and $D_v \neq 0$. Then $B^2 \neq 4AC$. Indeed, let $B^2 = 4AC$. Then from Eq. (5.94),

$$(BV)^2 = 1 + 2V(A+C) + 4ACV^2 \qquad \Longrightarrow \qquad 1 + 2V(A+C) = 0$$

which is obviously absurd, being $A \geq 0$, $C \geq 0$ from Eq. (5.84). A consequence of this implication on Eq. (5.95) is that, at a given V, the map parameters should satisfy

$$V = \frac{B + 2C}{B^2 - 4AC} \tag{5.96}$$

7. Let $S_v \neq 0$ and $D_v \neq 0$. Then $A = C$. To prove this, substitute Eq. (5.96) in Eq. (5.94),

$$\begin{aligned}(B^2 - 4AC)\left(\frac{B+2C}{B^2-4AC}\right)^2 - 2\frac{B+2C}{B^2-4AC}(A+C) - 1 = 0 \\ (B+2C)^2 - 2(B+2C)(A+C) - (B^2 - 4AC) = 0 \\ B^2 + 4C^2 + 4BC - 2AB - 2BC - 4AC - 4C^2 - B^2 + 4AC = 0\end{aligned}$$

Simplifying all recurring terms, we get

$$2B(C - A) = 0$$

From previous points we know $B \neq 0$. Then $A = C$.

8. It's impossible to have $S_v \neq 0$ and $D_v \neq 0$. To prove this, assume the contrary and use the point above to reduce Eq. (5.96) to

$$A = C \qquad \Longrightarrow \qquad V = \frac{B+2C}{B^2 - 4C^2} = \frac{1}{B - 2C}$$

Inserting the latter in Eq. (5.93),

$$\begin{aligned}\left(1 + \frac{2C}{B-2C}\right) D_v + \frac{B}{B-2C} S_v = 0 \\ \frac{B}{B-2C} D_v + \frac{B}{B-2C} S_v = 0\end{aligned}$$

which implies

$$D_v + S_v = 0 \qquad \Longrightarrow \qquad v^{(s^*)} = 0$$

Recalling from Tab. 5.4 the explicit self-consistency equation for $v^{(d)}$ and setting $v^{(s^*)} = 0$,

$$v^{(d)} = -V v^{(d)} F\left[(c_x - c_y)^2\right]$$

and using the positivity rule of Eq. (5.84) we get

$$V > 0 \quad \text{and} \quad F\left[(c_x - c_y)^2\right] \geq 0 \qquad \Longrightarrow \qquad v^{(d)} = 0$$

because no finite positive number can equal any finite negative number, and *viceversa*. Thus

$$S_v \neq 0 \quad \text{and} \quad D_v \neq 0 \qquad \Longrightarrow \qquad v^{(s^*)} = v^{(d)} = 0 \qquad \Longrightarrow \qquad S_v = D_v = 0$$

Absurd. We explored any possibility for S_v, D_v: no stable solution with finite values of the HFPs $v^{(s^*)}$, $v^{(d)}$ exists, thus both must be zero and can be eliminated from the algorithm.

To gain full knowledge on the map fixed point's shape, we should analyze similarly the behaviour of the self-consistency equations for $u^{(s^*)}$, $u^{(d)}$. This is done later in Sec. 5.8.4. For now, it suffices to note that – limitedly to the sAF phase – gap complexification is not present. This effect is not a consequence of HF derivation, which remains self-coherent and general; it is instead of the peculiar form of the non-linear map defined by the set of self-consistency equations.

5.8.3 HFPs vanishing in aAF phase

Let us discuss the sAF phase. It is described by the five HFPs on the second row of Tab. 5.5, as well as the self-consistency equations of Tabs. 5.3 and 5.4. We continue using the functional F as defined in Eq. (5.82). We also define A, B, C as in Eq. (5.85). Let $s_\ell \equiv \sin k_\ell$, and

$$P \equiv F\left[s_x^2\right] \qquad Q \equiv 2F\left[s_x s_y\right] \qquad R \equiv F\left[s_y^2\right] \tag{5.97}$$

Using these definitions, four of five self-consistency equations are given by

$$u^{(s^*)} = \left[2t - Vu^{(s^*)}\right](A+B+C) - Vu^{(d)}(A-C) \tag{5.98}$$

$$u^{(d)} = \left[2t - Vu^{(s^*)}\right](A-C) - Vu^{(d)}(A-B+C) \tag{5.99}$$

$$v^{(p_x)} = -Vv^{(p_x)}P - Vv^{(p_y)}Q \tag{5.100}$$

$$v^{(p_y)} = -Vv^{(p_x)}Q - Vv^{(p_y)}R \tag{5.101}$$

Let us prove that both $v^{(p_\ell)} = 0$.

1. For the aAF phase, $Q = 0$. We write Q explicitly

$$\begin{aligned} Q &= F\left[s_x s_y\right] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta,\tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} \sin k_x \sin k_y \\ &= \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta,\tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} \sin k_x \sin k_y \\ &= \frac{1}{2L_x L_y} \sum_{k_x=-\pi}^{+\pi} \sin k_x \sum_{k_y=-\pi}^{+\pi} \frac{\tilde{\mathrm{Th}}^{(-)}_{(k_x,k_y)}[\beta,\tilde{\mu}]}{\tilde{E}_{(k_x,k_y)}} \sin k_y \end{aligned}$$

From second to third line, we used MBZ periodicity. The k_y sum at last line is null: even if using antiperiodic harmonics for $\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}$, we did not break inversion symmetry neither in x nor y direction. Bogoliubov bands $\tilde{E}_{\mathbf{k}}$ and thermal factors must show said symmetry. The product of a symmetric function $(\tilde{\mathrm{Th}}/\tilde{E})$ with $\sin k_y$ integrates to zero. This ends the point.

2. $v^{(p_x)} = 0$. To see this, just plug $Q = 0$ in Eq. (5.100),

$$v^{(p_x)} = -Vv^{(p_x)}F\left[s_x^2\right]$$

Using the positivity rule of Eq. (5.84) we get

$$V > 0 \quad \text{and} \quad F\left[s_x^2\right] \geq 0 \qquad \Longrightarrow \qquad v^{(p_x)} = 0$$

3. $v^{(p_y)} = 0$. Same proof as point above.

This ends this short section and shows that also for the aAF phase all gap complexifiers actually vanish. This is coherent with the constraint of Eq. (5.80). As a consequence, there is no need to distinguish two AF phases,

$$\mathrm{sAF} = \mathrm{aAF}$$

and gap complexification is a completely "ghost" effect, even if theoretically allowed. All is left is to check for the behaviour of $u^{(d)}$.

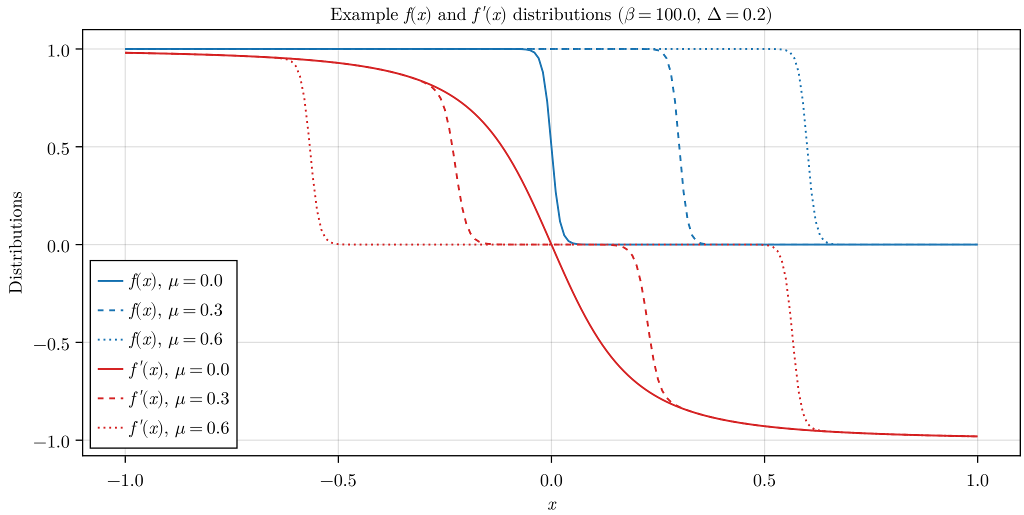


Figure 5.5 Sketch of the Fermi-Dirac distribution $f(x)$ and the pseudo-distribution $f'(x)$ described in text for the normal phase. These example data have been plotted arbitrarily at $mU = \Delta = 0.2$ for $\mu = 0.0, 0.3, 0.6$.

5.8.4 Natural suppression of nematicity within $\mathrm{C}_2/\mathbb{Z}_2$ symmetry

A consequence of the two sections above is that the gap function is purely real, non-renormalized and constant across all reciprocal space,

$$\tilde{\Delta}_{\mathbf{k}} = mU \quad \Longrightarrow \quad \tilde{E}_{\mathbf{k}} = \sqrt{\tilde{\epsilon}_{\mathbf{k}}^2 + (mU)^2}$$

Bogoliubov bands maintain renormalisation through the $u^{(\gamma)}$ HFPs channel. Aim of this section is to prove that, as was for the normal phase, no nematicity occurs in the AF phase, giving $u^{(d)} = 0$.

While the proof given in Sec. 4.5.2 was long and quite elaborated, this one is actually very short. Let us start from recalling the normal phase context:

$$\text{Normal phase:} \quad \begin{aligned} u^{(s^*)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\ u^{(d)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \end{aligned}$$

while, using $\boldsymbol{\pi}$ periodicity, in the present context we can re-write all equations doubling the integration domain,

$$\text{AF phase:} \quad \begin{aligned} u^{(s^*)} &= -\frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}] \\ u^{(d)} &= -\frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}] \end{aligned}$$

Let us define the pseudo distribution

$$f'(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \equiv -\frac{\tilde{\epsilon}_{\mathbf{k}}}{2\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$$

which, substituted in the self-consistency equations, makes them very similar to the normal ones, up to a $f \to f'$ exchange.

The fact that $u^{(d)} = 0$ follows from the simple consideration that in Sec. 4.5.2 (when showing the same for the normal phase) we never used the specific form of f; instead, we only made use of it being monotonically descendent as a function of $\epsilon_{\mathbf{k}}$. If f' also is a monotonically descendent function of $\epsilon_\mathbf{k}$, we can re-use the entire derivation of Sec. 4.5.2. Let us show that using a somewhat lighter notation,

$$\epsilon_\mathbf{k} \to x \qquad mU \to \Delta$$

and omitting parametric arguments. The function f' is given by the product

$$f'(x) = g(x)h(x)$$

with

$$\begin{aligned} g(x) &= -\frac{x}{\sqrt{x^2+\Delta^2}} \\ h(x) &= \left(\frac{1}{e^{\beta(\sqrt{x^2+\Delta^2}-\mu)}+1} - \frac{1}{e^{\beta(-\sqrt{x^2+\Delta^2}-\mu)}+1} \right) \end{aligned}$$

The following points are true:

1. The function is antisymmetric about $x = 0$, because $g(x)$ is antisymmetric, while $h(x)$ is symmetric.
2. $g(x)$ is monotonically descendent. Its derivative is given by

$$\begin{aligned} \frac{\mathrm{d}g}{\mathrm{d}x} &= -\frac{\sqrt{x^2+\Delta^2} - x^2/\sqrt{x^2+\Delta^2}}{x^2+\Delta^2} \\ &= -\frac{\Delta^2}{(x^2+\Delta^2)^{3/2}} \end{aligned}$$

which is always negative.

3. The product $g(x)h(x)$ is monotonically descendent at positive finite doping. To prove this, we can compute the derivative of f',

$$\frac{\mathrm{d}f'}{\mathrm{d}x} = \frac{\mathrm{d}g}{\mathrm{d}x}h(x) + g(x)\frac{\mathrm{d}h}{\mathrm{d}x}$$

We have $h(x) > 0$, and as we said $\mathrm{d}g/\mathrm{d}x < 0$, which makes the first term negative. The second is negative too: let

$$E(x) \equiv \sqrt{x^2+\Delta^2} \qquad \Longrightarrow \qquad \frac{\mathrm{d}E}{\mathrm{d}x} = \frac{x}{E(x)} = -g(x)$$

Then:

$$\begin{aligned} \frac{\mathrm{d}h}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{1}{e^{\beta(-E(x)-\mu)}+1} - \frac{1}{e^{\beta(E(x)-\mu)}+1} \right] \\ &= \beta\frac{\mathrm{d}E}{\mathrm{d}x} \underbrace{\left[\frac{e^{\beta(-E(x)-\mu)}}{\left[e^{\beta(-E(x)-\mu)}+1\right]^2} + \frac{e^{\beta(E(x)-\mu)}}{\left[e^{\beta(E(x)-\mu)}+1\right]^2} \right]}_{k(x)} \end{aligned}$$

The term in square brackets is obviously positive, $k(x) > 0$. Taking all together, we have

$$g(x)\frac{\mathrm{d}h}{\mathrm{d}x} = -\beta g^2(x)k(x) < 0 \qquad \Longrightarrow \qquad \frac{\mathrm{d}f'}{\mathrm{d}x} < 0$$

making f' descendent monotonicity proven.

These points end the proof: as anticipated, f' is a monotonically descendent function of $\epsilon_\mathbf{k}$, a feature that allows us to re-use the entire derivation of Sec. 4.5.2, and conclude that also for the AF phase

$$u^{(d)} = 0$$

while $u^{(s^*)}$ remains constrained in the limits of Eq. (5.81). Starting from a rather complicated scenario, we were able to reduce all the AF HFPs down to the pair

$$m, u^{(s^*)}$$

Perhaps not-so-excitingly, the real net effect of the non-local extension of the HM is, once again, the shrinking effect on bands. Now it is a matter of numeric analysis to assess if Mott localization takes place.

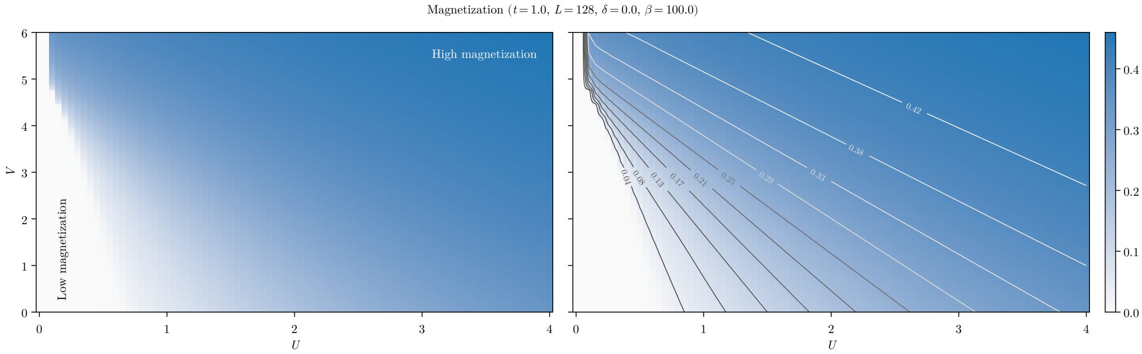


Figure 5.6 Plot of the self-consistent convergence values for m (left panel) and isomagnetic curves computed on the same data (right panel), along which magnetisation is constant. For the sake of graphic clarity, only the points at $0 \leq U \leq 4$ and $0 \leq V \leq 6$ are plotted. [Something broke at $U = 0.05$, $V \geq 5$, change with new data.]

5.9 Discussion of numeric results

In this section we present numeric results. For the AF phase, the HFPs vector, as we thoroughly showed, has two components

$$\mathbf{v} = \begin{bmatrix} m \\ u^{(s^*)} \end{bmatrix} \qquad \mathbf{v} = \mathbb{R}^2$$

The resulting model RMPs are, as before:

$$\tilde{t}_{ij} \equiv \left(t - \frac{u^{(s^*)}V}{2} \right) \varphi_{ij}^{(s^*)}$$
$$\tilde{\mu} \equiv \mu + n(2zV - U)$$

In the end, the computational scenario is not so different from the normal phase. The only real difference is our choice of parametric boundaries for simulations:

$$U = 0,\ 0.05,\ 0.1,,\ \ldots,\ 8.0 \qquad V = 0,\ 0.05,\ 0.1,,\ \ldots,\ 8.0 \qquad \delta = 0.0$$

We set $\delta = 0.0$. As we anticipated in Sec. 5.2.4 (and discuss in Sec. D.2), any finite doping implies immediate disruption of commensurate antiferromagnetism, a consideration that makes any simulation at $\delta > 0$ of low physical significance. Following this consideration, we omit data at finite $\delta \neq 0$.

5.9.1 Magnetization boost

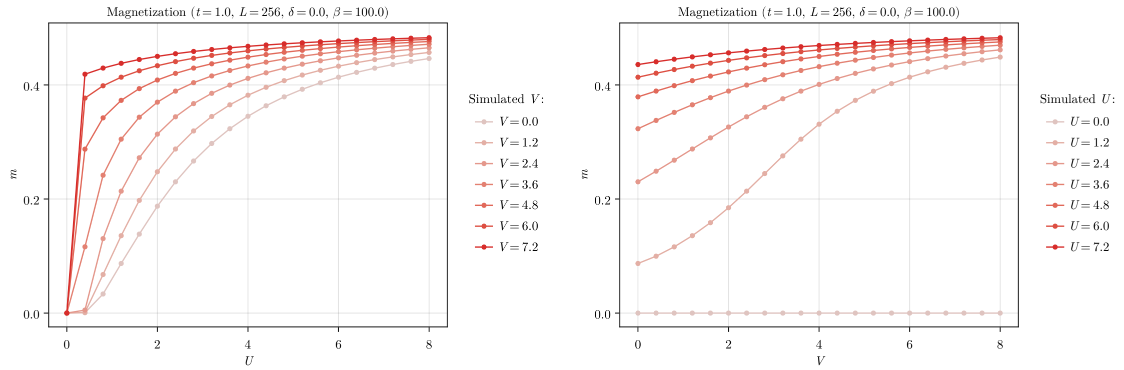
The convergence values for m computed in (U, V) plane are shown in the panels of Fig. 5.6, along with a level curves plot to identify isomagnetic curves (right panel). Note that, to increase graphic clarity in heatmaps bottom left sector, from the plots are omitted the points at $U \geq 4$ and $V \geq 6$. Tolerances on density and HFPs have been set to

$$\Delta n = 10^{-2} \qquad \Delta \mathbf{v} = \begin{bmatrix} m\colon 5 \times 10^{-4} \\ u^{(s^*)}\colon 5 \times 10^{-4} \end{bmatrix}$$

From heatmaps we see that $\hat{H}_V$ actually boosts magnetisation. Since the only observable net effect of the non-local interaction is to lower t by shrinking bands, at any fixed point (t, U, V) the model can be effectively mapped to

$$\text{EHM: } (t, U, V) \quad \rightarrow \quad \text{HM: } (\tilde{t}, U, 0)$$

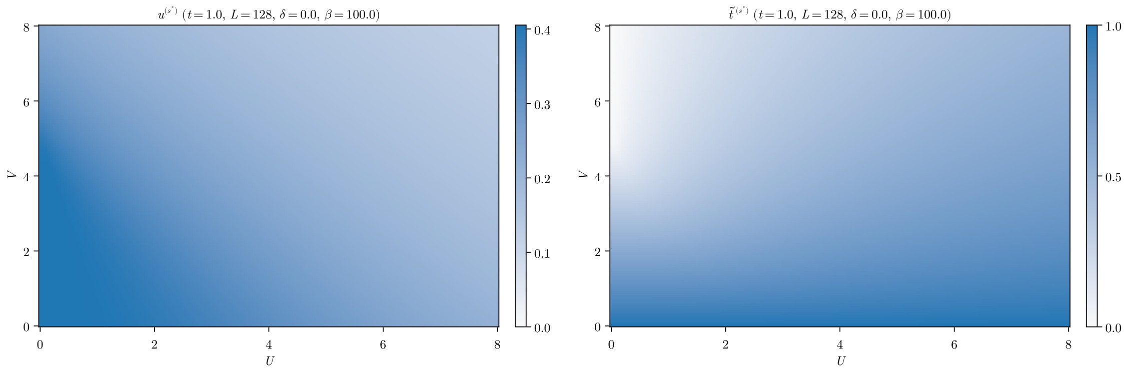
being $\tilde{t}$ dependent on V. For the pure HM, magnetisation is controlled by the ratio U/t. To lower t is physically the same as to increase U, and from this comes the magnetisation precession visible in



a | U-dependency of m varying V.

b | V-dependency of m varying U.

Figure 5.7 | Plot of the self-consistent convergence values for the HFPs m and $u^{(s^*)}$ in the AF phase.



a | Heatmap of $u^{(s^*)}$ in UV plane.

b | Heatmap of $t^{(s^*)}$ in UV plane.

Figure 5.8 | Data for the HFP $u^{(s^*)}$ and $\tilde{t}^{(s^*)}$ varying both U and V.

Fig. 5.6. Isomagnetic lines on the right panel show, for U significantly greater than 0, linear trends whose angular coefficient depend on m itself. The low-U limit of isomagnetic lines presents some oscillations that are imputable to numeric discretisation of parametric space. [Why isomagnetic curves are lines? Something about $U/(V-V_c)$ ratio being constant?]

Data shown in Fig. 5.7 confirm clearly this V-controlled magnetisation “boost”. The effect is more enhanced at low values of U. As is visible from Fig. 5.7b, magnetisation does not always saturate the same way. For example, considering the curve at $U = 1.2$, magnetisation varies following a somewhat tilted sigmoid shape, making boost most effective around $V \approx 2 \div 4$. Because of m saturation, this behaviour is lost at higher Us.

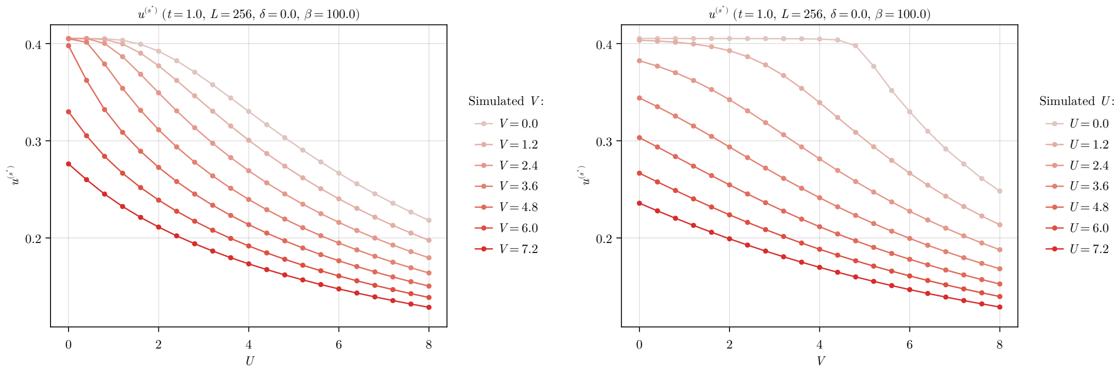
5.9.2 Bands shrinking

The bands shrinking effect was discussed in Sec. 5.8.1 within C_4 symmetry, and then confirmed in Sec. 5.8.4 when trying to extend to nematic structures. As was for the normal phase, we have proven that $\hat{H}_V$ net effect on the hamiltonian kinetic part is to shrink bands,

$$t \quad \rightarrow \quad \tilde{t} < t$$

For the normal phase, in Sec. 4.6.1 we presented numeric results showing a Mott-localized region appears in (δ, V) plane, whose $\delta \to 0$ limit was approximately $V = 4.93t$, as shown in Fig. 4.7b.

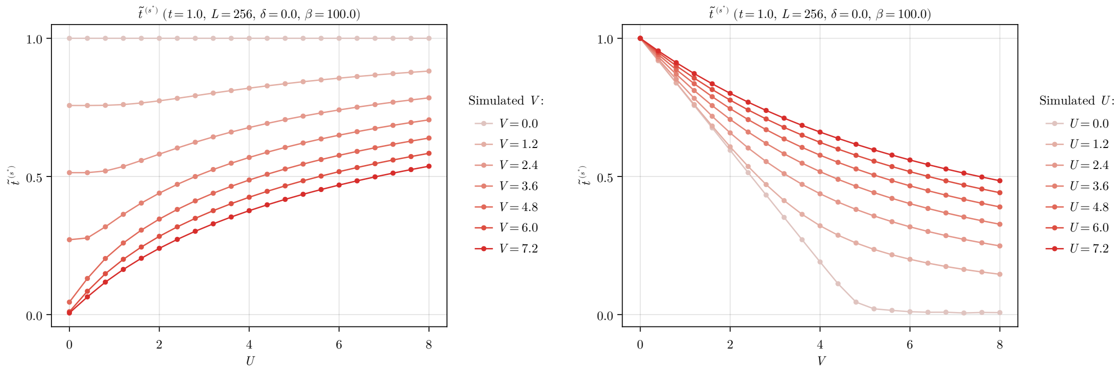
The convergence values for the HFP $u^{(s^*)}$ are shown in Fig. 5.8a, and more in detail in the two panels of Fig. 5.9. In general, $u^{(s^*)}$ decreases monotonically both in U and in V. As we already discussed in normal phase, the $U = 0$ data of Fig. 5.9b present a plateau up to a Mott-critical value. This effect is absent at positive Us. Both interactions disrupt the HFP; however, the two



a | U-dependency of $u^{(s^*)}$ varying V.

b | V-dependency of $u^{(s^*)}$ varying U.

Figure 5.9 | Plot of the self-consistent convergence values for the HFPs m and $u^{(s^*)}$ in the AF phase.



a | U-dependency of $t^{(s^*)}$ varying V.

b | V-dependency of $t^{(s^*)}$ varying U.

Figure 5.10 | Plot of the self-consistent convergence values for the HFPs m and $u^{(s^*)}$ in the AF phase.

limits

$$U \gg 1 \quad \text{and} \quad V \gg 1$$

are much different. By computing the value of $\tilde{t}^{(s^*)}$,

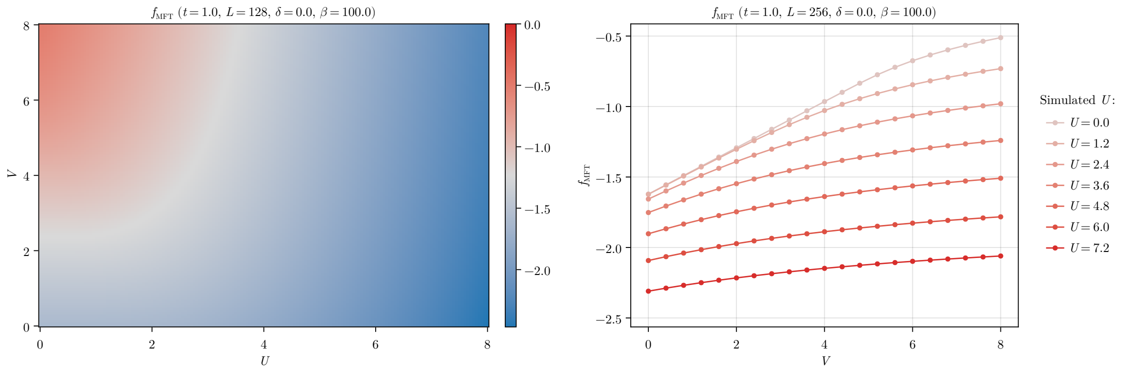
$$\tilde{t}^{(s^*)} = 2t - u^{(s^*)}V$$

we obtain the data of Fig. 5.8b. The limit $U \gg 1$ taken at fixed V, also shown in Fig. 5.10a, provides "U-induced antiferromagnetism", with bands shrinking effect being almost irrelevant as V lowers; on the contrary, the opposite limit $V \gg 1$ taken at fixed U, also shown in Fig. 5.10b, shrinks bands strongly providing a "Mott-induced antiferromagnetism".

In (U, V) space there is not Mott-localised region of the kind we had in (δ, V) space for the normal phase (see Fig. 4.7b). At half filling, complete localisation only occurs at $U = 0$, as can be seen in the bottom left sector of Fig. 5.10a. Any finite value of U disrupts bands flattening, thus – being the local repulsion an essential ingredient in Hubbard antiferromagnetism – we can claim, based on numeric evidence, that bands flattening is incompatible with a commensurate AF phase on the half-filled EHM.

5.9.3 Solution free energy and entropy densities

In Fig. 5.11 the solution free energy is plotted. In the right panel, Fig. 5.11b, we see that antiferromagnetism becomes more stable at low V, high U. The opposite limit of weak local repulsion, strong non-local attraction (top-left corner of Fig. 5.11a) instead increases f_{MFT}, reaching zero



a | Heatmap of f_{MFT} in δV plane.

b | V-dependency of f_{MFT} varying δ.

Figure 5.11 | Free energy density of the AF phase.

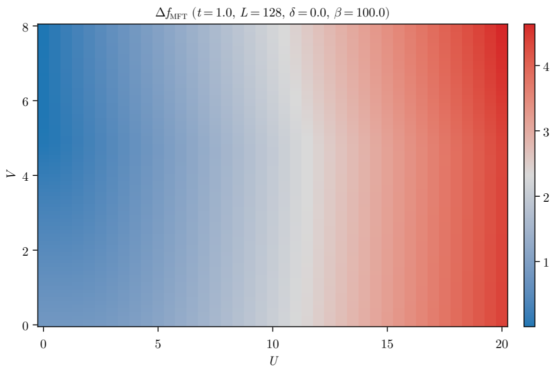


Figure 5.12 | Difference of free energies for the normal phase minus the AF phase.

energy at $U = 0$, $V > 4.93t$. The latter isz the critical value discussed in Sec. 4.6.1 and computed in Eq. (4.38).

It is then evident how, even if the macroscopic magnetic properties are the same along one isomagnetic line of those plotted in Fig. 5.6, right panel, and thus the realized state exhibits an identical value of m, much different is the energetic content of the state itself. While in the $U \gg 1$, fixed V limit free energy gain comes from gap amplification – which moves down the Bogoliubov bands $\tilde{E}_{\mathbf{k}}$ –, in the $V \gg 1$, fixed U limit energies of the lower Bogoliubov band (full at low temperature) are shifted upwards via the Mott localization effect.

The AF phase is always energetically favoured at half-filling, with respect to the normal phase. Considering the energy difference,

$$\Delta f_{\text{MFT}} \equiv f_{\text{MFT}}^{(\text{N})} - f_{\text{MFT}}^{(\text{AF})}$$

shown for a slightly more coarse-grained dataset in Fig. 5.12, we see immediately how $\Delta f_{\text{MFT}} \geq 0$ in all (U, V) plane. The normal phase data of Sec. 4.6 having been simulated at $U = 0$ due to their intrinsic U independence. The AF phase is simulated in the same range $0 \leq V \leq 8$, but for $0 \leq U \leq 10$. The comparable normal phase dataset was just a repetition of the $U = 0$ data at all Us. As we expect, in the $V \gg 1$ and low U limit the energy difference is null: in this limit the two phases tend to coincide, since at $U = 0.0$ no magnetisation is physically possible. Moreover, the $U \gg 1$ limit makes the AF phase undoubtedly energetically favoured. Overall, at half filling the interesting free fermions Mott-localization effect is, at least when comparing these phases, existent but surpassed by the high energetic gain of establishing an AF structure at infinitesimal U. The situation could change at finite doping, a regime we do not investigate limitedly to AF Physics.

At low temperature and half filling, it is not worth plotting entropy density. Half-filled commensurate antiferromagnets are insulators, with two gapped Bogoliubov bands. At low temperature, electronic distribution fills completely the lower band and empties the upper one. Because of this, insulating gapped systems inherently are low entropic.

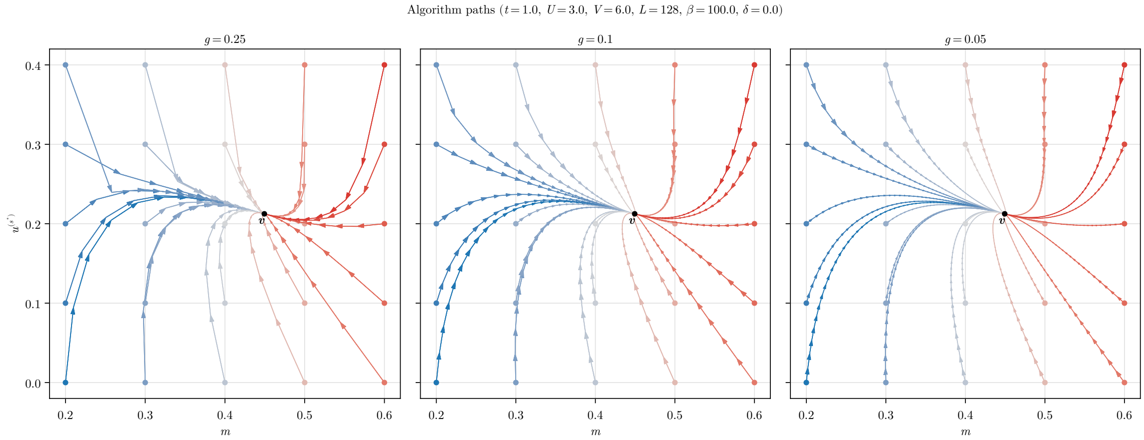


Figure 5.13 Quiver plot of algorithmic evolution in HFPs space at fixed model parametrization. The black dot $\mathbf{v}$ represents the nonlinear map fixed point, while the colored paths the sequences of HFPs coordinates simulated via iterative HF procedure, starting from different HFPs initializers (colored dots).

5.9.4 Fixed-point interpretation of HF solution

As a final remark discussing the mathematical structure of the algorithm, we anticipated in Sec. 2.3.2 how the reject-mix-iterate scheme upon which our HF algorithm is built can be seen as an effective search for the singular fixed point of a nonlinear map. A physical solution is represented as a HFP vector $\mathbf{v}$ such that the algorithm is a local identity,

$$\mathrm{A}(\mathbf{v}) = \mathbf{v}$$

To predict for a given initializer $\mathbf{v}_0$ the path it will follow through HFPs space is, in general, highly non-trivial due to the intrinsic nonlinearity of the algorithmic map defined by self-consistency equations for the various HFPs. An example of such nonlinearity is given in the derivation of Secs. 5.8.2, 5.8.3.

In Fig. 5.13 we sketch this concept in a practical situation. In figure are plotted various algorithmic paths through AF HFPs space, with operational setup at

$$U = 3 \qquad V = 6 \qquad L = 128 \qquad \beta = 100 \qquad \delta = 0$$

At this point, the self-consistent HF "fixed-point" solution is given by

$$m \approx 0.449 \qquad u^{(s^*)} \approx 0.212$$

indicated in panels as the black dot. The three panels show different map evolutions varying the mixing parameter within three choices, $g \in \{0.25, 0.1, 0.05\}$. We chose various initializers in the set:

$$\mathbf{v}_0 = \begin{bmatrix} m_0 \\ u_0^{(s^*)} \end{bmatrix} \quad : \quad m_0 \in \{0.2,\ 0.3,\ ...,\ 0.6\} \quad \text{and} \quad u_0^{(s^*)} \in \{0.0,\ 0.1,\ ...,\ 0.4\}$$

In figures, different colours indicate different evolutions, while the tip size indicates the magnitude of the algorithmic step in passing from one iteration to the following.

All paths are able to converge to $\mathbf{v}$, and the quantity of steps necessary to reach the fixed point, a sort of attractor, increases as we lower g. To select the correct g turns out to be the real algorithm fine-tuning knob: as we saw for the normal phase, some model parametrisation could lead to unavoidable algorithmic instabilities. The problem is that we do not know beforehand where the instabilities will occur, and how. The correct g is the compromise between keeping low the number of steps (larger mixing) and moving slow enough to not diverge when encountering unstable points (smaller mixing). By recursively fine-tuning g repeating non-converged simulation with more care, we are essentially making the nonlinear map "softer" and curvier, as is in the right panel of Fig. 5.13. It's not always resource convenient to do so, so for most simulations we stick to a rougher map (like the left panel one) by setting $g = 0.5$.

Chapter 6

Superconducting instability

In this chapter the HF methods introduced in Sec. 2.2 are applied to the SC phase. We study a translationally invariant superconductor, which breaks the conservation of total charge. Analytic calculations will lead us to various effects: bare bands renormalisation, anisotropic Cooper pairing and nematicity. Our description of the SC phase is closely related to the BCS solution for pure s-wave superconductivity.

As for previous phases, this chapter is divided in three parts. First, analysis of the SC phase symmetries based on the discussion of Chap. 3, and implementation of HF methods. Second, search for theoretical constraints on the self-consistent solution. Third, numeric results.

6.1 The SC phase and its symmetries

As for antiferromagnetism, superconductivity can establish in a variety of ways. We deal with the simplest realisation. In Sec. 3.2 we anticipated that the SC phase is described by the two SSBs

$$\mathrm{U}^z(1) \overset{\mathrm{SSB}}{\to} 0$$

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$$

Translational invariance is unbroken. Then electronic density needs to be uniform across all sites, similarly to the normal phase discussed in Chap. 4. Non-uniform density distributions are inconsistent with translational invariance in both directions. We have, then

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n \tag{6.1}$$

which is identical to Eq. (4.1) for the normal phase.

Recall Tabs. 3.2 and 3.3, collecting the selection rules for the local repulsion $\hat{H}_U$ and the non-local attraction $\hat{H}_V$. From Eqns. (6.2), (2.18) for the SC phase we need to account for the following contractions:

$\hat{H}_U$	Hartree and Cooper contractions
$\hat{H}_V$ (o.s. sector)	Hartree and Cooper contractions
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions

We anticipate their roles in Tab. 6.1.

The SSB of $\mathrm{U}^z(1)$ allows for EVs like $\langle \hat{c}^\dagger \hat{c}^\dagger \rangle$ to be non-zero. This leads to the formation of Cooper pairs. These are composite quantum objects with a specific spin-spatial structure. Consider the generic Cooper fluctuation

$$\langle \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{j\bar{\sigma}} \rangle \quad \text{with } i, j \text{ nearest neighbours}$$

We work in o.s. sector because we know from Tab. 3.3 that the s.s. sector is prohibited. From basic Quantum Mechanics we know the summation rules of the spin algebra $\mathfrak{su}(2)$,

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$$

Operator	Sector	RWCs	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
		Cooper	Isotropic pairing in singlet channel
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
		Cooper	Anisotropic pairing in singlet and triplet channels
	s.s.	Hartree	μ shift by nzV
		Fock	Band renormalisation and resymmetrisation

Table 6.1 Relevant Wick's contractions and net effect in the SC phase. This RWCs table is more structured than Tabs. 4.1 and 5.1.

Spatial structure	Pairing channel
Symmetric wave function	Singlet pairing
Anti-symmetric wave function	Triplet pairing

Table 6.2 Relation of the Cooper pairing channel with the wavefunction spatial symmetry (intended as the inversion $(x, y) \to (-x, -y)$).

The two pairing channels are, at this level, the singlet channel associated to total spin 0 and the triplet channel associated to total spin 1. Now, when applying HF method to the quartic operators, we get the decomposition of Eq. (**??**). We can impose to the approximated HF ground-state a particular spatial structure – say, inversion symmetry or antisymmetry. This reflects in the requirement that Cooper pairs show the same spatial structure.

This gives a selection rule over Cooper pairings: if we work with inversion-symmetric structures – say, s^*-wave or d-wave – the pairing will take place in singlet channel, allowing us to eliminate the triplet pairing. Working with antisymmetric structures, inversely, we can eliminate the singlet channel. This concept is summarised in Tab. 6.2.

We proceed as we did for the AF phase: in order to highlight the physical effects of $\hat{H}_V$, we first analyse the SC phase limitedly to the conventional HM. In next section we prove that BCS-like superconductivity is impossible at HF level on the pure HM, with just the local repulsion $\hat{H}_U$. It is the presence of the non-local attraction that enables Cooper pairing mechanism. Before starting, let

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \tag{6.2}$$

$$\hat{\phi}_{\mathbf{k}} \equiv \hat{c}_{-\mathbf{k}\downarrow}\hat{c}_{\mathbf{k}\uparrow} \tag{6.3}$$

where $\hat{n}_{\mathbf{k}\sigma}$ was defined as for the AF phase, in Eq. (5.7). The HF approximation in reciprocal space to the EHM will be written in terms of these operators.

6.2 Local effects: (impossibility of) superconductivity in the HM

Let us start from the local effects. Following Tab. 3.2 and Eqns. (6.2), (2.18), we know the local repulsion decomposes in Hartree and Cooper channels,

$$\begin{aligned}
\hat{H}_U &= U \sum_{\mathbf{r}\in\mathcal{L}} \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\downarrow} \\
&\simeq U \sum_{\mathbf{r}\in\mathcal{L}} \Big(\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\langle\hat{c}^{\dagger}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\rangle + \langle\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\hat{c}^{\dagger}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow} - \langle\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle\hat{c}^{\dagger}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\rangle && \text{(Hartree)} \\
&\qquad + \hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}^{\dagger}_{\mathbf{r}\downarrow}\langle\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle + \langle\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}^{\dagger}_{\mathbf{r}\downarrow}\rangle\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow} - \langle\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}^{\dagger}_{\mathbf{r}\downarrow}\rangle\langle\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle \Big) && \text{(Cooper)}
\end{aligned}$$

In next sections we deal with each channel separately.

6.2.1 Local Hartree channel in the SC phase: analytic derivation

For the Hartree channel, being the density uniform across sites, we can re-use the entirety of Sec. 4.2.1 for the normal phase. Substituting Eq. (6.1), we get the approximation of Eq. (4.2),

$$\hat{H}_U \simeq nU \times \hat{N} - E^{(\mathrm{SC})}_{\mathrm{H}/U} + (\text{Cooper channel}) \tag{6.4}$$

thus chemical potential renormalization $\tilde{\mu} \to \mu - nU$ is present also in the SC phase.

6.2.2 Local Hartree contraction energy

As for the section above, we use the result of Sec. 4.2.2 for the normal phase. From Eq. (4.8)

$$E^{(\mathrm{SC})}_{\mathrm{H}/U} = L_x L_y \times n^2 U \tag{6.5}$$

For a locally repulsive model, the Hartree channel contributes negatively to ground state energy. Indeed, contraction energy is subtracted from the hamiltonian. We now move to the Cooper channel.

6.2.3 Local Cooper channel in the SC phase: analytic derivation

We move to reciprocal space, through the transformation of Eq. (3.30). Consider the Cooper channel of Eq. (6.2). In reciprocal space we get:

$$\begin{aligned}\hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \Big[&\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \\ &+ \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \rangle - \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \rangle \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \rangle \Big]\end{aligned} \tag{6.6}$$

We are not breaking translational invariance, thus only Cooper fluctuations with net zero total momentum are allowed. This means only $\mathbf{K} = \mathbf{0}$ contributes. Let us prove this: consider the EV of the Cooper pair

$$\langle \hat{c}^\dagger_{\mathbf{k}_1\uparrow} \hat{c}^\dagger_{\mathbf{k}_2\downarrow} \rangle$$

for two momenta $\mathbf{k}_1$, $\mathbf{k}_2$. Moving to real space via an $\mathcal{F}$-transform, we get

$$\langle \hat{c}^\dagger_{\mathbf{k}_1\uparrow} \hat{c}^\dagger_{\mathbf{k}_2\downarrow} \rangle \overset{\mathcal{F}}{=} \frac{1}{L_x L_y} \sum_{\mathbf{r}_1 \in \mathcal{L}} \sum_{\mathbf{r}_2 \in \mathcal{L}} e^{-i(\mathbf{k}_1 \cdot \mathbf{r}_1 + \mathbf{k}_2 \cdot \mathbf{r}_2)} \langle \hat{c}^\dagger_{\mathbf{r}_1\uparrow} \hat{c}^\dagger_{\mathbf{r}_2\downarrow} \rangle$$

where we used the convention of Eq. (3.25). Let now $\mathbf{r}_1 \to \mathbf{r}$ and $\mathbf{r}_2 \to \Delta\mathbf{r}$. Translational invariance implies that

$$g(\Delta\mathbf{r}) = \langle \hat{c}^\dagger_{\mathbf{r}\uparrow} \hat{c}^\dagger_{\mathbf{r}+\Delta\mathbf{r}\downarrow} \rangle$$

is independent of $\mathbf{r}$. Thus we have:

$$\begin{aligned}\langle \hat{c}^\dagger_{\mathbf{k}_1\uparrow} \hat{c}^\dagger_{\mathbf{k}_2\downarrow} \rangle &\overset{\mathcal{F}}{=} \sum_{\Delta\mathbf{r} \in \mathcal{L}} e^{-i\mathbf{k}_2 \cdot \Delta\mathbf{r}} g(\Delta\mathbf{r}) \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} e^{-i(\mathbf{k}_1 + \mathbf{k}_2) \cdot \mathbf{r}} \\ &= \sum_{\Delta\mathbf{r} \in \mathcal{L}} e^{-i\mathbf{k}_2 \cdot \Delta\mathbf{r}} g(\Delta\mathbf{r}) \delta_{\mathbf{k}_1 = \mathbf{k}_2}\end{aligned}$$

where we used the identity of Eq. (3.26). Then the above EV vanishes for $\mathbf{k}_1 \neq \mathbf{k}_2$. Getting back to Eq. (6.6), this proves that only pairs at null total momentum can have nonzero EVs.

Let now $w_{\mathbf{k}}$ be the EV of $\phi^\dagger_{\mathbf{k}}$, and $w^{(s)}$ its s-wave projection:

$$w_{\mathbf{k}} \equiv \langle \phi^\dagger_{\mathbf{k}} \rangle \tag{6.7}$$

$$w^{(s)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \mathrm{BZ}} w_{\mathbf{k}} \tag{6.8}$$

Eq. (6.6) reduces to

$$\hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} \left[\langle \hat{\phi}^\dagger_{\mathbf{k}} \rangle \hat{\phi}_{\mathbf{k}'} + \hat{\phi}^\dagger_{\mathbf{k}} \langle \hat{\phi}_{\mathbf{k}'} \rangle \right] - E^{(\mathrm{SC})}_{\mathrm{C}/U} \tag{6.9}$$

being $E^{(\mathrm{SC})}_{\mathrm{C}/U}$ the HM Cooper contraction energy, written explicitly in Sec. 6.2.4. Let now

$$\Delta^{(s)} \equiv -U w^{(s)} \tag{6.10}$$

This is the s-wave component of the gap function. Note that, in general, $\Delta^{(s)} \in \mathbb{C}$. However, we can always choose $\Delta^{(s)}$ as purely real. This is due to the fact that the hamiltonian is gauge-invariant, then it suffices to apply the gauge map of Eq. (3.4) to the fermionic operators with

$$\eta \equiv \frac{\arg \Delta^{(s)}}{2}$$

to obtain $\Delta^{(s)} \in \mathbb{R}$. For the sake of generality, we proceed with $\Delta^{(s)} \in \mathbb{C}$. Although this choice gives no advantage now, we implement it anyway in order to derive the most general form of the equations that follow. This is useful for the generalisation to the EHM.

In the end, we reduce $\hat{H}_U$ from Eq. (6.9) to

$$\hat{H}_U \simeq (\text{Hartree channel}) - \sum_{\mathbf{k}\in\mathrm{BZ}} \left[\Delta^{(s)} \hat{\phi}_{\mathbf{k}} + \Delta^{(s)*} \hat{\phi}^\dagger_{\mathbf{k}} \right] - E^{(\mathrm{SC})}_{\mathrm{C}/U} \tag{6.11}$$

This last decomposition resembles Eq. (5.22), obtained for the AF phase in the Hartree channel.

6.2.4 Local Cooper contraction energy

The Cooper contraction energy in the Cooper channel is given by:

$$\begin{aligned} E^{(\mathrm{SC})}_{\mathrm{C}/U} &\equiv \frac{U}{L_x L_y} \sum_{\mathbf{k}\mathbf{k}'} \langle \hat{\phi}^\dagger_{\mathbf{k}} \rangle \langle \hat{\phi}_{\mathbf{k}'} \rangle \\ &= L_x L_y \times U \left| w^{(s)} \right|^2 \end{aligned} \tag{6.12}$$

As for the Hartree channel, also the Cooper channel contributes negatively to ground-state energy.

6.2.5 Superconductivity in the HM

We proceed here as we did for the AF phase in Sec. 5.2.3. We start by giving a matrix representation of the approximated hamiltonian, then map it to a pseudospin problem and finally specify how to diagonalise it to retrieve its ground state.

Matrix representation. Collecting the results of the above paragraphs, we obtain the quadratic approximation to the Hubbard hamiltonian in the SC phase,

$$\begin{aligned} \hat{H}_t + \hat{H}_U \simeq & \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \\ & - \sum_{\mathbf{k}\in\mathrm{BZ}} \left[\Delta^{(s)} \hat{\phi}_{\mathbf{k}} + \Delta^{(s)*} \hat{\phi}^\dagger_{\mathbf{k}} \right] + nU \times \hat{N} - \left[E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U} \right] \end{aligned} \tag{6.13}$$

We define the shifted bare bands:

$$\xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}$$

Notice we are using the renormalised chemical potential. Due to inversion symmetry, it holds $\xi_{\mathbf{k}} = \xi_{-\mathbf{k}}$. Using Fermi anticommutation rules, the kinetic operator can be written as:

$$
\begin{aligned}
\hat{H}_t - \tilde{\mu}\hat{N} &= \sum_{\sigma}\sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}\sigma}\hat{n}_{\mathbf{k}\sigma} \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}} \left(\xi_{\mathbf{k}}\hat{n}_{\mathbf{k}\uparrow} + \xi_{-\mathbf{k}}\hat{n}_{-\mathbf{k}\downarrow}\right) \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}} \left(\hat{c}^{\dagger}_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^{\dagger}_{-\mathbf{k}\downarrow} + 1\right) \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}} \left(\hat{c}^{\dagger}_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^{\dagger}_{-\mathbf{k}\downarrow}\right) + E^{(\mathrm{SC})}_{\mathrm{a}}
\end{aligned}
$$

being $E^{(\mathrm{SC})}_{\mathrm{a}}$ the "anticommutation energy"

$$
E^{(\mathrm{SC})}_{\mathrm{a}} \equiv \sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}} \tag{6.14}
$$

In the end, the hamiltonian is approximated by:

$$
\begin{aligned}
\hat{H}_t + \hat{H}_U \simeq \sum_{\mathbf{k}\in\mathrm{BZ}} \epsilon_{\mathbf{k}} \left[\hat{c}^{\dagger}_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{\mathbf{k}\downarrow}\hat{c}^{\dagger}_{\mathbf{k}\downarrow}\right] + E^{(\mathrm{SC})}_{\mathrm{a}} \\
- \sum_{\mathbf{q}\in\mathrm{BZ}} \left[\Delta^{(s)}\hat{\phi}_{\mathbf{q}} + \Delta^{(s)*}\hat{\phi}^{\dagger}_{\mathbf{q}}\right] + nU \times \hat{N} - \left[E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U}\right]
\end{aligned}
$$

with $E^{(\mathrm{SC})}_{\mathrm{H}/U}$, $E^{(\mathrm{SC})}_{\mathrm{C}/U}$ the contraction energies arising from Hartree and Cooper channels, and $E^{(\mathrm{SC})}_{\mathrm{a}}$ the energy shift resulting from anticommutation rules, given respectively in Eqns. (6.5), (6.12), and (6.14). We recognised the s-wave gap $\Delta^{(s)}$ from Eq. (6.10).

Let the Nambu spinor

$$
\hat{\Phi}_{\mathbf{k}} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}^{\dagger}_{-\mathbf{k}\downarrow} \end{bmatrix}
$$

The approximated hamiltonian can be written in matrix form as:

$$
\hat{H}_t + \hat{H}_U - \mu\hat{N} \simeq \sum_{\mathbf{k}\in\mathrm{BZ}}\sum_{\sigma} \hat{\Phi}^{\dagger}_{\mathbf{k}} h_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}} + E^{(\mathrm{SC})}_{\mathrm{a}} - \left[E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U}\right]
$$

being

$$
h_{\mathbf{k}} \equiv \begin{bmatrix} \xi_{\mathbf{k}} & -\Delta^*_{\mathbf{k}} \\ -\Delta_{\mathbf{k}} & -\xi_{\mathbf{k}} \end{bmatrix} \quad \text{with} \quad \xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}
$$

and $\Delta_{\mathbf{k}} = \Delta^{(s)}$.

Pseudospin representation. The hamiltonian can be written equivalently as

$$
h_{\mathbf{k}} = \xi_{\mathbf{k}}\tau^z - \mathrm{Re}\{\Delta_{\mathbf{k}}\}\tau^x - \mathrm{Im}\{\Delta_{\mathbf{k}}\}\tau^y
$$

Let now the pseudospin operator $\hat{\mathbf{s}}_{\mathbf{k}}$

$$
\hat{s}^{\alpha}_{\mathbf{k}} \equiv \hat{\Phi}^{\dagger}_{\mathbf{k}}\tau^{\alpha}\hat{\Phi}_{\mathbf{k}} \quad \text{with} \quad \hat{\mathbf{s}}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{s}^x_{\mathbf{k}\sigma} \\ \hat{s}^y_{\mathbf{k}\sigma} \\ \hat{s}^z_{\mathbf{k}\sigma} \end{bmatrix}
$$

and the pseudofield $\mathbf{b}_{\mathbf{k}}$:

$$
\mathbf{b}_{\mathbf{k}} \equiv \begin{bmatrix} -\mathrm{Re}\{\Delta_{\mathbf{k}}\} \\ -\mathrm{Im}\{\Delta_{\mathbf{k}}\} \\ \xi_{\mathbf{k}} \end{bmatrix}
$$

EV	Meaning
$\langle \hat{s}^x_{\mathbf{k}} \rangle$	$\left\langle \hat{\phi}_{\mathbf{k}} + \hat{\phi}^\dagger_{\mathbf{k}} \right\rangle$
$\langle \hat{s}^y_{\mathbf{k}} \rangle$	$i \left\langle \hat{\phi}_{\mathbf{k}} - \hat{\phi}^\dagger_{\mathbf{k}} \right\rangle$
$\langle \hat{s}^z_{\mathbf{k}} \rangle$	$\langle \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow} \rangle - 1$

Table 6.3 | List of the various expectation values (EVs) of each pseudospin component and their meaning in terms of fermionic operators.

For a purely real gap, the pseudofield has no y component. In general, different gauges rotate the pseudofield around the z axis.

As was for the AF phase, these operators satisfy a $\mathfrak{su}(2)$ algebra up to an constant, irrelevant in the context of this discussion. They satisfy the identities:

$$\hat{s}^x_{\mathbf{k}} = \hat{\phi}_{\mathbf{k}} + \hat{\phi}^\dagger_{\mathbf{k}} \tag{6.15}$$

$$\hat{s}^y_{\mathbf{k}} = i \left(\hat{\phi}_{\mathbf{k}} - \hat{\phi}^\dagger_{\mathbf{k}} \right) \tag{6.16}$$

$$\hat{s}^z_{\mathbf{k}} = \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{\mathbf{k}\downarrow} - 1 \tag{6.17}$$

We report the EVs related to these identities Tab. 6.3.

We express the approximated hamiltonian in pseudospin representation as:

$$\hat{H}_t + \hat{H}_U - \mu \hat{N} \simeq \sum_{\mathbf{k}\in\mathrm{BZ}} \mathbf{b}_{\mathbf{k}} \cdot \hat{\mathbf{s}}_{\mathbf{k}} + nU\hat{N} + E^{(\mathrm{SC})}_{\mathrm{a}} - \left[E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U} \right] \tag{6.18}$$

This operator is similar enough to the one found for the AF phase, in Eq. (5.29). It describes a set of pseudospin subject to a uniform field. As for the AF phase, we define the Bogoliubov bands $E_{\mathbf{k}}$

$$E_{\mathbf{k}} \equiv \sqrt{\xi^2_{\mathbf{k}} + |\Delta_{\mathbf{k}}|^2}$$

The key difference with the AF Bogoliubov bands of Eq. (5.32) is that $E_{\mathbf{k}}$ now are functions of $\tilde{\mu}$, since $\xi_{\mathbf{k}}$ depends on the chemical potential. In other words, now the Bogoliubov bands depend on the shifted bands $\xi_{\mathbf{k}}$ instead of the bare bands $\epsilon_{\mathbf{k}}$.

Moreover, let us define the angles $\theta_{\mathbf{k}}$, $\zeta_{\mathbf{k}}$ as

$$\cos 2\theta_{\mathbf{k}} = \frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}} \qquad \sin 2\theta_{\mathbf{k}} \cos 2\zeta_{\mathbf{k}} = \frac{\mathrm{Re}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}} \qquad \sin 2\theta_{\mathbf{k}} \sin 2\zeta_{\mathbf{k}} = \frac{\mathrm{Im}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}}$$

Note that choosing the gauge $\eta = \arg \Delta^{(s)}/2$ the imaginary part of the gap is zero, and this reflects in $2\zeta_{\mathbf{k}} = k \times 2\pi$ with $k \in \mathbb{Z}$.

Diagonalisation. For the SC phase, diagonalisation of the HF hamiltonian follows the same procedure as was for the AF phase. Indeed, the diagonalising matrix is given by

$$\begin{aligned} W_{\mathbf{k}} &\equiv e^{-i\theta_{\mathbf{k}}\tau^y} e^{-i\zeta_{\mathbf{k}}\tau^z} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}} & -\sin\theta_{\mathbf{k}} \\ \sin\theta_{\mathbf{k}} & \cos\theta_{\mathbf{k}} \end{bmatrix} \begin{bmatrix} e^{-i\zeta_{\mathbf{k}}} & \\ & e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \end{aligned} \tag{6.19}$$

and the graphic procedure is the same of Fig. 5.2, with $\epsilon \to \xi$. We get:

$$d_{\mathbf{k}} = W_{\mathbf{k}} h_{\mathbf{k}} W^\dagger_{\mathbf{k}} \quad \text{with} \quad d_{\mathbf{k}} \equiv \begin{bmatrix} E_{\mathbf{k}} & \\ & -E_{\mathbf{k}} \end{bmatrix} \tag{6.20}$$

In the tilted frame we have the expression for the Nambu spinors:

$$\hat{\Gamma}_{\mathbf{k}} = W_{\mathbf{k}} \hat{\Psi}_{\mathbf{k}} \tag{6.21}$$

Explicitly:

$$
\begin{aligned}
\hat{\Gamma}_{\mathbf{k}} &= \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}^{\dagger}_{-\mathbf{k}\downarrow} \end{bmatrix} \\
&= \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} \hat{c}_{\mathbf{k}\uparrow} - \sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \hat{c}^{\dagger}_{-\mathbf{k}\downarrow} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} \hat{c}_{\mathbf{k}\uparrow} + \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \hat{c}^{\dagger}_{-\mathbf{k}\downarrow} \end{bmatrix} \equiv \begin{bmatrix} \hat{\gamma}^{(+)}_{\mathbf{k}} \\ \hat{\gamma}^{(-)}_{\mathbf{k}} \end{bmatrix}
\end{aligned} \tag{6.22}
$$

The new Fermionic operators describe a Fermi gas on distributed on two mirrored bands,

$$
\hat{H}_t + \hat{H}_U - \tilde{\mu}\hat{N} \simeq \sum_{\mathbf{k}\in \mathrm{BZ}} \sum_{\sigma} E_{\mathbf{k}} \left(\hat{\gamma}^{(+)\dagger}_{\mathbf{k}} \hat{\gamma}^{(+)}_{\mathbf{k}} - \hat{\gamma}^{(-)\dagger}_{\mathbf{k}} \hat{\gamma}^{(-)}_{\mathbf{k}} \right) + (\cdots)
$$

with $(\cdots)$ the various energy shifts.

6.2.6 Features of the SC solution

In this section we give some details about the HF solution for the translationally-invariant SC phase. We prove that in the conventional HM, with local repulsion $\hat{H}_U$, isotropic BCS-like superconductivity is forbidden. This serves as ground to motivate the passage to the EHM.

Pseudospin EVs. The EVs of pseudospin operators in the various directions are given by:

$$
\begin{aligned}
\langle \hat{\Psi}^{\dagger}_{\mathbf{k}} \tau^x \hat{\Psi}_{\mathbf{k}} \rangle &= -\sin 2\theta_{\mathbf{k}} \cos 2\zeta_{\mathbf{k}} \langle \hat{\Gamma}^{\dagger}_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \\
\langle \hat{\Psi}^{\dagger}_{\mathbf{k}} \tau^y \hat{\Psi}_{\mathbf{k}} \rangle &= -\sin 2\theta_{\mathbf{k}} \sin 2\zeta_{\mathbf{k}} \langle \hat{\Gamma}^{\dagger}_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \\
\langle \hat{\Psi}^{\dagger}_{\mathbf{k}} \tau^z \hat{\Psi}_{\mathbf{k}} \rangle &= \cos 2\theta_{\mathbf{k}} \langle \hat{\Gamma}^{\dagger}_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle
\end{aligned}
$$

In the tilted frame, the averaged z projection of pseudospin is:

$$
\langle \hat{\Gamma}^{\dagger}_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle = f\left(E_{\mathbf{k}}; \beta, 0\right) - f\left(-E_{\mathbf{k}}; \beta, 0\right) \tag{6.23}
$$

Compared with the respective relation for the AF phase, Eq. (5.34), the main difference here is that the chemical potential inside the Fermi-Dirac distribution is set to zero. This is due to the fact that $\tilde{\mu}$ is already included in the renormalised bands $E_{\mathbf{k}}$. The approximated quadratic hamiltonian describes a gas of γ fermions at null chemical potential.

Let us define the thermal factors:

$$
\mathrm{Th}^{(\pm)}_{\mathbf{k}}[\beta] \equiv f\left(-E_{\mathbf{k}}; \beta, 0\right) \pm f\left(E_{\mathbf{k}}; \beta, 0\right) \tag{6.24}
$$

In the end we have the EVs:

$$
\langle \hat{s}^x_{\mathbf{k}} \rangle = \frac{\mathrm{Re}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}} \mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{6.25}
$$

$$
\langle \hat{s}^y_{\mathbf{k}} \rangle = \frac{\mathrm{Im}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}} \mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{6.26}
$$

$$
\langle \hat{s}^z_{\mathbf{k}} \rangle = -\frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}} \mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{6.27}
$$

Note that, as is expressed in Tab. 6.3, the z component of the pseudospin is related to density, being $1 + \langle \hat{s}^z_{\mathbf{k}} \rangle = \langle \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow} \rangle$.

s-wave gap self-consistency equation. Applying the HF method to the conventional HM in SC phase, we obtained just one HFP: $w^{(s)}$. Let us derive its self-consistency equation. First, let

$$
\tau^{\pm} \equiv \frac{\tau^x \pm i\tau^y}{2}
$$

Recall Eq. (6.8) and expand it, to obtain

$$
\begin{aligned}
w^{(s)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} \langle \hat{\phi}^{\dagger}_{\mathbf{k}} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} \langle \hat{\Phi}^{\dagger}_{\mathbf{k}} \tau^{+} \hat{\Phi}_{\mathbf{k}} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} \frac{\langle \hat{s}^{x}_{\mathbf{k}} \rangle + i \langle \hat{s}^{y}_{\mathbf{k}} \rangle}{2} \\
&= \frac{1}{2 L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} \frac{\Delta_{\mathbf{k}}}{E_{\mathbf{k}}} \mathrm{Th}^{(-)}_{\mathbf{k}}[\beta]
\end{aligned} \tag{6.28}
$$

Last line is the explicit form of the self-consistency equation for $w^{(s)}$.

Impossibility of s-wave superconductivity. Let us define the critical U,

$$
U_{\mathrm{c}}[\beta] \equiv \left[\frac{1}{2 L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} \frac{\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta]}{E_{\mathbf{k}}} \right]^{-1} \tag{6.29}
$$

We call it "critical" because its value gives a sort of boundary for convergence. Indeed, as we show in App. F, for the SC phase in conventional HM, we are able to optimise parametrically the mixing parameter g in order to ensure convergence, based on the ratio $U/U_{\mathrm{c}}[\beta]$.

Let us show that superconductivity is not possible in the conventional HM. Eq. (6.28) can be rewritten as:

$$
w^{(s)} = -\frac{U}{U_{\mathrm{c}}[\beta]} w^{(s)}
$$

By construction $U_{\mathrm{c}}[\beta] > 0$ (the thermal factor is negative-defined). Hence the unique possible self-consistent solution for the equation above is $w^{(s)} = 0$. This implication terminates the argument: superconductivity cannot establish self-consistently at HF level in the HM.

Density. In order to estimate density, we can use the z projection of the pseudospin operators. From Eq. (6.17) we know

$$
\hat{s}^{z}_{\mathbf{k}} + 1 = \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{\mathbf{k}\downarrow}
$$

Then, density can be estimated as:

$$
\begin{aligned}
n &= \frac{1}{2 L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} \langle \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow} \rangle \\
&= \frac{1}{2 L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} \left(\langle \hat{s}^{z}_{\mathbf{k}} \rangle + 1 \right) \\
&= \frac{1}{2 L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} \left(1 - \frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}} \mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \right)
\end{aligned} \tag{6.30}
$$

We can now move to the EHM.

6.3 Extension to the EHM

Add now the non-local interaction $\hat{H}_V$. As we did for the AF phase, we look for a solution with an identical mathematical structure as the one here described for the conventional HM. We define the renormalised quantities

$$
\xi_{\mathbf{k}} \to \tilde{\xi}_{\mathbf{k}} \qquad E_{\mathbf{k}} \to \tilde{E}_{\mathbf{k}} \qquad \Delta_{\mathbf{k}} \to \tilde{\Delta}_{\mathbf{k}} \qquad \mu \to \tilde{\mu}
$$

To be completely correct, the chemical potential was already renormalized in the HM derivation in Sec. E.2.1. It here acquires yet another renormalization. The band energies renormalization is simply

$$\tilde{E}_{\mathbf{k}} \equiv \sqrt{\tilde{\xi}_{\mathbf{k}}^2 + |\tilde{\Delta}_{\mathbf{k}}|^2}$$

Let the renormalised angle:

$$\sin 2\tilde{\theta}_{\mathbf{k}} = \frac{|\tilde{\Delta}_{\mathbf{k}}|}{\tilde{E}_{\mathbf{k}}} \qquad \cos 2\tilde{\theta}_{\mathbf{k}} = \frac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tag{6.31}$$

and similarly the angle $\tilde{\zeta}_{\mathbf{k}}$,

$$\sin 2\tilde{\zeta}_{\mathbf{k}} = \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{|\tilde{\Delta}_{\mathbf{k}}|} \qquad \cos 2\tilde{\zeta}_{\mathbf{k}} = \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{|\tilde{\Delta}_{\mathbf{k}}|} \tag{6.32}$$

The pseudospin EVs are renormalised too,

$$\langle \hat{\Phi}_{\mathbf{k}}^{\dagger} \tau^x \hat{\Phi}_{\mathbf{k}} \rangle = -\sin 2\tilde{\theta}_{\mathbf{k}} \sin 2\tilde{\zeta}_{\mathbf{k}} \langle \hat{\Gamma}_{\mathbf{k}}^{\dagger} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \tag{6.33}$$

$$\langle \hat{\Phi}_{\mathbf{k}}^{\dagger} \tau^y \hat{\Phi}_{\mathbf{k}} \rangle = -\sin 2\tilde{\theta}_{\mathbf{k}} \cos 2\tilde{\zeta}_{\mathbf{k}} \langle \hat{\Gamma}_{\mathbf{k}}^{\dagger} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \tag{6.34}$$

$$\langle \hat{\Phi}_{\mathbf{k}}^{\dagger} \tau^z \hat{\Phi}_{\mathbf{k}} \rangle = \cos 2\tilde{\theta}_{\mathbf{k}} \langle \hat{\Gamma}_{\mathbf{k}}^{\dagger} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \tag{6.35}$$

Let also:

$$\langle \hat{\Gamma}_{\mathbf{k}}^{\dagger} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle = f\left(\tilde{E}_{\mathbf{k}}; \beta, 0\right) - f\left(-\tilde{E}_{\mathbf{k}}; \beta, 0\right) \tag{6.36}$$

$$\tilde{\mathrm{Th}}_{\mathbf{k}}^{(\pm)}[\beta, \tilde{\mu}] \equiv f\left(-\tilde{E}_{\mathbf{k}}; \beta, 0\right) \pm f\left(\tilde{E}_{\mathbf{k}}; \beta, 0\right) \tag{6.37}$$

For the SC phase, the thermal factors reduce to:

$$\tilde{\mathrm{Th}}_{\mathbf{k}}^{(+)} = 1$$

$$\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)} = -\tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)$$

Hence, pseudospin expectation values are:

$$\langle \hat{s}_{\mathbf{k}}^x \rangle = -\frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{6.38}$$

$$\langle \hat{s}_{\mathbf{k}}^y \rangle = -\frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{6.39}$$

$$\langle \hat{s}_{\mathbf{k}}^z \rangle = \frac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{6.40}$$

These equations are an extension of Eqns. (6.38), (6.39), (6.40) for the EHM.

We can rewrite the equations for $w^{(s)}$ and n accordingly,

$$w^{(s)} = -\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{6.41}$$

$$n = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} \left[1 + \cos 2\tilde{\theta}_{\mathbf{k}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)\right] \tag{6.42}$$

where we used Eqns. (6.28), (6.30).

From Tabs. 3.2 and 3.3 we know none of the three channels (Hartree, Fock, Cooper) is subject to selection rules. Thus all terms of the decomposition of $\hat{H}_V$ given in Eq. (2.18) contribute. As always, we break down each.

6.4 Non-local Hartree effects: chemical potential renormalisation

We start by analysing the non-local Hartree channel,

$$\hat{H}_V \simeq -V \sum_{\langle ij \rangle} \sum_{\sigma,\sigma'} \Big(\hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}\big\langle \hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'}\big\rangle + \big\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}\big\rangle \hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'} - \big\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}\big\rangle\big\langle \hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'}\big\rangle \Big)$$
$$+ \,(\text{Fock channel}) + (\text{Cooper channel})$$

where we used Eq. (2.18). This discussion is based on that for the normal phase in Sec. 4.2.

6.4.1 Non-local Hartree channel in the SC phase: analytic derivation

Without repeating the calculations, the non-local attraction $\hat{H}_V$ is reduced in the Hartree channel to

$$\hat{H}_V \simeq -2nzV \times \hat{N} - E^{(\text{SC})}_{\text{H}/V} + (\text{all the rest})$$

This, together with the $\hat{H}_U$ Hartree channel decomposition

$$\hat{H}_U \simeq nU \times \hat{N} - E^{(\text{SC})}_{\text{H}/U}$$

gives the chemical potential full renormalization, identical to Eq. (4.7),

$$\tilde{\mu} \equiv \mu + n(2zV - U) \tag{6.43}$$

6.4.2 Non-local Hartree contraction energy

The contraction energy for the non-local sector was derived in Eq. (4.9),

$$E^{(\text{SC})}_{\text{H}/V} = -L_x L_y \times 2zn^2 V$$

and together with the local counterpart of Eq. (4.8)

$$E^{(\text{SC})}_{\text{H}/U} = L_x L_y \times n^2 U$$

Now we move to the more interesting Fock sector.

6.5 Fock effects: bare bands renormalization

As done for the other phases, we here discuss the bands renormalization effect derived from the Fock sector of the non-local attraction. From Eq. (2.18), we have

$$\hat{H}_V \simeq (\text{Hartree channel})$$
$$V \sum_{\langle ij \rangle} \sum_{\sigma,\sigma'} \Big(\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'}\big\langle \hat{c}^\dagger_{j\sigma'}\hat{c}_{i\sigma}\big\rangle + \big\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'}\big\rangle \hat{c}^\dagger_{j\sigma'}\hat{c}_{i\sigma} - \big\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'}\big\rangle\big\langle \hat{c}^\dagger_{j\sigma'}\hat{c}_{i\sigma}\big\rangle \Big) + (\text{Cooper channel})$$

This discussion is essentially analogous the one done for the normal phase in Sec. 4.3. As for previous phases, following the selection rules of Tab. 3.3, only the s.s. channel contributes. Therefore we use from now on, in this section, $\sigma = \sigma'$.

6.5.1 Fock channel in the SC phase: analytic derivation

Let us define, as done for the normal phase in Eq. (4.17) and for the AF phase in Eq. (5.61),

$$u_{\mathbf{k}\sigma} \equiv \langle \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \rangle \tag{6.44}$$

Eq. (4.18) stands perfectly valid, translated to the SC phase

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}+\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}\langle\hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}\rangle-\langle\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}\rangle\langle\hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}\rangle\right]$$

$$\stackrel{\mathcal{F}}{=}\sum_{\gamma,\sigma}\left[\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{k}}u_{\mathbf{k}\sigma}\right]\times V\sum_{\mathbf{k}'\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{k}'}\hat{c}^{\dagger}_{\mathbf{k}'\sigma}\hat{c}_{\mathbf{k}'\sigma}-E^{(\mathrm{SC})}_{\mathrm{F}/V}\tag{6.45}$$

being

$$E^{(\mathrm{SC})}_{\mathrm{F}/V}\equiv\frac{V}{L_xL_y}\sum_{\mathbf{k},\mathbf{k}'}\sum_{\sigma}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\langle\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}\rangle\langle\hat{c}^{\dagger}_{\mathbf{k}'\sigma}\hat{c}_{\mathbf{k}'\sigma}\rangle\tag{6.46}$$

To get to this point, we employed the assumption

$$\langle\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma'}\rangle=\delta_{\mathbf{k}=\mathbf{k}'}\delta_{\sigma=\sigma'}\langle\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}\rangle$$

which essentially states translational invariance and spin degeneracy. This assumption is verified by the BCS ground state of Eq. E.51. Let us define each γ-wave component as

$$u^{(\gamma)}_{\sigma}=\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{k}}\langle\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}\rangle$$

These are our HFPs in the Fock channel. The SC phase does not break inversion symmetry: then, as we concluded for the other phases, p_ℓ components are null. The renormalised bands are, as in Eqns. (4.19), (5.63),

$$\tilde{\epsilon}_{\mathbf{k}}\equiv\epsilon_{\mathbf{k}}+u^{(s^*)}V(\cos k_x+\cos k_y)+u^{(d)}V(\cos k_x-\cos k_y)$$

Let us compute the EVs of $u_{\mathbf{k}\sigma}$ in the two spin sectors,

$$\begin{aligned}u_{\mathbf{k}\uparrow}&=\langle\hat{c}^{\dagger}_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow}\rangle\\&=\left\langle\hat{\Phi}^{\dagger}_{\mathbf{k}}\frac{\mathbb{I}+\tau^z}{2}\hat{\Phi}_{\mathbf{k}}\right\rangle\\&=\frac{1}{2}\left[\left\langle\hat{\Gamma}^{\dagger}_{\mathbf{k}}\hat{\Gamma}_{\mathbf{k}}\right\rangle+\langle\hat{s}^z_{\mathbf{k}}\rangle\right]\\&=\frac{1}{2}\left[\tilde{\mathrm{T}}\mathrm{h}^{(+)}_{\mathbf{k}}[\beta]-\cos(2\tilde{\theta}_{\mathbf{k}})\tilde{\mathrm{T}}\mathrm{h}^{(-)}_{\mathbf{k}}[\beta]\right]\\&=\frac{1}{2}\left[1-\cos(2\tilde{\theta}_{\mathbf{k}})\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\right]\end{aligned}\tag{6.47}$$

having we used Eq. (**??**) and following. For the spin $\downarrow$ EV,

$$\begin{aligned}u_{-\mathbf{k}\downarrow}&=\langle\hat{c}^{\dagger}_{-\mathbf{k}\downarrow}\hat{c}_{-\mathbf{k}\downarrow}\rangle\\&=1-\langle\hat{c}_{-\mathbf{k}\downarrow}\hat{c}^{\dagger}_{-\mathbf{k}\downarrow}\rangle\\&=1-\left\langle\hat{\Phi}^{\dagger}_{\mathbf{k}}\frac{\mathbb{I}-\tau^z}{2}\hat{\Phi}_{\mathbf{k}}\right\rangle\\&=1-\frac{1}{2}\left[\left\langle\hat{\Gamma}^{\dagger}_{\mathbf{k}}\hat{\Gamma}_{\mathbf{k}}\right\rangle-\langle\hat{s}^z_{\mathbf{k}}\rangle\right]\\&=1-\frac{1}{2}\left[\tilde{\mathrm{T}}\mathrm{h}^{(+)}_{\mathbf{k}}[\beta]+\cos2\tilde{\theta}_{\mathbf{k}}\tilde{\mathrm{T}}\mathrm{h}^{(-)}_{\mathbf{k}}[\beta]\right]\\&=1-\frac{1}{2}\left[1+\cos(2\tilde{\theta}_{\mathbf{k}})\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\right]\end{aligned}\tag{6.48}$$

From the above results, note first that

$$u_{\mathbf{k}\uparrow}=u_{-\mathbf{k}\downarrow}$$

Symmetry	Self-consistency equation	Graph
s^*-wave	$u^{(s^*)} = \dfrac{1}{2L_xL_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) \left[1 - \dfrac{\tilde{\xi}_\mathbf{k}}{\tilde{E}_\mathbf{k}} \tanh\left(\dfrac{\beta\tilde{E}_\mathbf{k}}{2}\right)\right]$	Fig. 3.4b
$d_{x^2-y^2}$-wave	$u^{(d)} = \dfrac{1}{2L_xL_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y) \left[1 - \dfrac{\tilde{\xi}_\mathbf{k}}{\tilde{E}_\mathbf{k}} \tanh\left(\dfrac{\beta\tilde{E}_\mathbf{k}}{2}\right)\right]$	Fig. 3.4e

Table 6.4 | Symmetry resolved self-consistency equations for the harmonics of $u_\mathbf{k}$.

which is coherent with our requirement of a uniform-density, spin uniform and inversion symmetric phase. Note also that Eqns. (6.47), (6.48) combine to give the expression for density of Eq. (6.42).

If we take the $\tilde{\theta}_\mathbf{k} \to 0$ limit, we must retrieve the Fermi-Dirac distribution. Indeed, this limit coincides with $U \to 0$. Explicitly:

$$
\begin{aligned}
\lim_{\tilde{\theta}_\mathbf{k}\to 0} u_{\mathbf{k}\sigma} &= \frac{1}{2}\left[1 + \tanh\left(\frac{\beta\tilde{\xi}_\mathbf{k}}{2}\right)\right] \\
&= \frac{1}{2}\left[\frac{e^{\beta\tilde{\xi}_\mathbf{k}/2} + e^{-\beta\tilde{\xi}_\mathbf{k}/2}}{e^{\beta\tilde{\xi}_\mathbf{k}/2} + e^{-\beta\tilde{\xi}_\mathbf{k}/2}} - \frac{e^{\beta\tilde{\xi}_\mathbf{k}/2} - e^{-\beta\tilde{\xi}_\mathbf{k}/2}}{e^{\beta\tilde{\xi}_\mathbf{k}/2} + e^{-\beta\tilde{\xi}_\mathbf{k}/2}}\right] \\
&= f(\tilde{\epsilon}_\mathbf{k}; \beta, \tilde{\mu})
\end{aligned}
$$

Correctly, the null-gap limit (for which the angle reduces to zero) gives back a perfectly degenerate Fermi gas.

We can then derive the $u_\mathbf{k}$-related HFPs self-consistency equations. In next passages, we use $\sigma = \uparrow$ but could have used the opposite due to spin degeneracy discussed above. Then:

$$
\begin{aligned}
u^{(\gamma)} &\equiv \frac{1}{L_xL_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_\mathbf{k}^{(\gamma)} \langle \hat{n}_{\mathbf{k}\uparrow} \rangle \\
&= \frac{1}{2L_xL_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_\mathbf{k}^{(\gamma)} \left[1 - \cos(2\tilde{\theta}_\mathbf{k}) \tanh\left(\frac{\beta\tilde{E}_\mathbf{k}}{2}\right)\right]
\end{aligned}
\tag{6.49}
$$

reported explicitly for the only two inversion symmetric structures in Tab. 6.4.

6.5.2 Fock contraction energy

In terms of the energy shift due to Wick's contraction, Eq. (4.20) holds identically

$$
E_{\mathrm{F}/V}^{(\mathrm{SC})} = L_xL_y \times V \left[\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right]
$$

with the only obvious difference that the $u^{(\gamma)}$ are computed on the SC ground state by the means of Eqns. (6.47), (6.48).

6.6 Non-local Cooper effects: anisotropic pairing

In this section we deal with the Cooper part of Eq. (2.18),

$$
\begin{aligned}
\hat{H}_V =\simeq & \text{ (Hartree channel)} + \text{(Fock channel)} \\
& - V \sum_{\langle ij \rangle} \sum_{\sigma,\sigma'} \left(\hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma'} \langle \hat{c}_{j\sigma'}\hat{c}_{i\sigma} \rangle + \langle \hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma'} \rangle \hat{c}_{j\sigma'}\hat{c}_{i\sigma} - \langle \hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma'} \rangle \langle \hat{c}_{j\sigma'}\hat{c}_{i\sigma} \rangle \right)
\end{aligned}
$$

We know from Eq. (3.17) that the non-local attraction can be separated in o.s. and s.s. channels. As we reported in Tab. 3.3 and stated in Sec. 6.1, the only non-zero contribution comes from the o.s. sector.

6.6.1 Non-local Cooper o.s. channel in the SC phase: analytic derivation

The o.s. Cooper sector contributes both to singlet and triplet pairing. Note that in singlet pairing channel the spatial wavefunction must be inversion-symmetric. On the contrary, in triplet pairing channel it needs to be inversion-antisymmetric.

We move to reciprocal space. From Eq. (3.32), we get

$$\begin{aligned}\hat{H}_V^{(\text{o.s.})} \simeq & \,(\text{Hartree channel}) \\ & - \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos\left(\delta k_x\right) + \cos\left(\delta k_y\right)\right] \times \Big[\big\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \\ & + \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \big\rangle - \big\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\rangle \big\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \big\rangle \Big]\end{aligned}$$

Identical considerations as for the local repulsion hold, and uniquely the $\mathbf{K} = \mathbf{0}$ term gives a non-zero contribution, in general. Therefore:

$$\begin{aligned}\hat{H}_V^{(\text{o.s.})} \simeq & \,(\text{Hartree channel}) \\ & - \frac{2V}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} \left[\cos\left(\delta k_x\right) + \cos\left(\delta k_y\right)\right] \times \Big[\big\langle \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \big\rangle \hat{c}_{-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{k}'\uparrow} \\ & + \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \big\langle \hat{c}_{-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{k}'\uparrow} \big\rangle - \big\langle \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \big\rangle \big\langle \hat{c}_{-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{k}'\uparrow} \big\rangle \Big]\end{aligned}$$

Then, using the harmonics decomposition of Eq. (3.35), we end up with:

$$\hat{H}_V^{(\text{o.s.})} \simeq (\text{Hartree channel}) - \frac{V}{L_x L_y} \sum_{\gamma} \sum_{\mathbf{k},\mathbf{k}'} \varphi^{(\gamma)}_{\mathbf{k}} \varphi^{(\gamma)*}_{\mathbf{k}'} \left[w_{\mathbf{k}} \hat{\phi}_{\mathbf{k}'} + w^*_{\mathbf{k}} \hat{\phi}^\dagger_{\mathbf{k}'} \right] - E^{(\text{SC})}_{\text{C}/V/\text{o.s.}} \tag{6.50}$$

In the equation above the contraction energy is given by

$$E^{(\text{SC})}_{\text{C}/V/\text{o.s.}} \equiv - \frac{V}{L_x L_y} \sum_{\gamma} \sum_{\mathbf{k},\mathbf{k}'} \varphi^{(\gamma)}_{\mathbf{k}} \varphi^{(\gamma)*}_{\mathbf{k}'} w_{\mathbf{k}} w^*_{\mathbf{k}'} \tag{6.51}$$

Then, let the harmonics $w^{(\gamma)}$ be

$$w^{(\gamma)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} w_{\mathbf{k}}$$

Eq. (6.50) becomes

$$\hat{H}_V^{(\text{o.s.})} \simeq (\text{Hartree channel}) - V \sum_{\gamma} \sum_{\mathbf{k}\in\text{BZ}} \left[w^{(\gamma)} \varphi^{(\gamma)*}_{\mathbf{k}} \hat{\phi}_{\mathbf{k}} + \text{h.c.} \right] - E^{(\text{SC})}_{\text{C}/V/\text{o.s.}}$$

extending the symmetry sum to $\gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\}$.

For the γ-wave component of the superconducting order parameter, the following chain, similar to Eq. (6.28), holds:

$$\begin{aligned} w^{(\gamma)} & \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \langle \hat{\phi}^\dagger_{\mathbf{k}} \rangle \\ & = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \langle \hat{\Phi}^\dagger_{\mathbf{k}} \tau^+ \hat{\Phi}_{\mathbf{k}} \rangle \\ & = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \frac{\langle \hat{s}^x_{\mathbf{k}} \rangle + i \, \langle \hat{s}^y_{\mathbf{k}} \rangle}{2} \\ & = \frac{1}{2 L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \text{Th}^{(-)}_{\mathbf{k}} [\beta] \end{aligned} \tag{6.52}$$

Symmetry	Self-consistency equation	Graph
s-wave	$w^{(s)} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. 3.4a
s^*-wave	$w^{(s^*)} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y)\frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. 3.4b
p_x-wave	$w^{(p_x)} = \frac{i\sqrt{2}}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \sin k_x\frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. 3.4c
p_y-wave	$w^{(p_y)} = \frac{i\sqrt{2}}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \sin k_y\frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. 3.4d
$d_{x^2-y^2}$-wave	$w^{(d)} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y)\frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. 3.4e

Table 6.5 | Symmetry resolved self-consistency equations for the harmonics of $w_{\mathbf{k}}$.

where $\tau^+ = (\tau^x + i\tau^y)/2$. We used Eqns. (6.33), (6.34) and (6.36) as well as the thermal factor defined in Eq. (6.24). The derivation above holds for all symmetries, (including s-wave). Explicitly, five self-consistency equations are then derived and listed in Tab. 6.5.

Of these, the singlet channel admits for non-zero fluctuations of the superconducting order parameter uniquely in the s^* and d channels. This is because the gap function is inversion-symmetric and p_ℓ channels equations integrate it with an antisymmetric function.

6.7 Full MFT hamiltonian and anisotropic pairing

Let now the total contraction energy

$$E_{\mathrm{c}}^{(\mathrm{SC})} \equiv -\left[E_{\mathrm{H}/U}^{(\mathrm{SC})} + E_{\mathrm{C}/U}^{(\mathrm{SC})} + E_{\mathrm{H}/V}^{(\mathrm{SC})} + E_{\mathrm{F}/V}^{(\mathrm{SC})} + E_{\mathrm{C}/V/\mathrm{o.s.}}^{(\mathrm{SC})}\right]$$

By composing all three channels we are able to derive the final form of the MFT gran-canonical hamiltonian,

$$\hat{H} - \mu\hat{N} \simeq \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{\xi}_{\mathbf{k}}\left[\hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} + \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^\dagger_{-\mathbf{k}\downarrow}\right] - \sum_{\mathbf{k}\in\mathrm{BZ}}\left[\tilde{\Delta}_{\mathbf{k}}\hat{\phi}_{\mathbf{k}} + \tilde{\Delta}^*_{\mathbf{k}}\hat{\phi}^\dagger_{\mathbf{k}}\right] + E_{\mathrm{a}}^{(\mathrm{SC})} + E_{\mathrm{c}}^{(\mathrm{SC})}$$

having we included the "anticommutation energy" also defined for the conventional HM in Eq. (E.44),

$$E_{\mathrm{a}}^{(\mathrm{SC})} \equiv \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{\xi}_{\mathbf{k}}$$

coming from fermionic anticommutation relations that are employed in order to get the kinetic term. The renormalized bands are

$$\tilde{\xi}_{\mathbf{k}} = \epsilon_{\mathbf{k}} + V\sum_{\gamma\in\{s^*,d\}} u^{(\gamma)}\varphi^{(\gamma)*}_{\mathbf{k}} - \tilde{\mu} \tag{6.53}$$

and the full gap function by

$$\tilde{\Delta}_{\mathbf{k}} = -Uw^{(s)} + V\sum_{\gamma} w^{(\gamma)}\varphi^{(\gamma)*}_{\mathbf{k}} \tag{6.54}$$

We can identify the gap harmonics by

$$\tilde{\Delta}^{(s)} = -Uw^{(s)} \qquad \tilde{\Delta}^{(\gamma)} = Vw^{(\gamma)} \quad \text{being} \quad \gamma\in\{s^*, p_x, p_y, d\}$$

As anticipated, the singlet channel admits three components only,

$$\text{Singlet channel} \quad : \quad \tilde{\Delta}_{\mathbf{k}} = \tilde{\Delta}^{(s)} + \tilde{\Delta}^{(s^*)}(\cos k_x + \cos k_y) + \tilde{\Delta}^{(d)}(\cos k_x - \cos k_y)$$

making the gap function purely real. An inverse statement holds for the triplet pairing, for which

$$\text{Triplet channel} \quad : \quad \tilde{\Delta}_{\mathbf{k}} = i\sqrt{2}\left[\tilde{\Delta}^{(p_x)} \sin k_x + \tilde{\Delta}^{(p_y)} \sin k_y\right]$$

In this case, the gap is purely imaginary. When working in a specific channel, the gap function is either purely real or purely imaginary. This lets us discuss the shape of the renormalised ground-state.

6.7.1 The relative phase in the zero temperature half-filled ground state

The conventional BCS ground state is widely known and derived for HM superconductivity in Sec. E.2.4. Jumping to Eq. (E.51) and extending it to the present context of the EHM, we get

$$|\Psi\rangle = \bigotimes_{\mathbf{k}\in\mathrm{BZ}} \left[\cos\tilde{\theta}_{\mathbf{k}} + e^{2i\tilde{\zeta}_{\mathbf{k}}} \sin\tilde{\theta}_{\mathbf{k}} \hat{\phi}^{\dagger}_{\mathbf{k}}\right] |\Omega_{\mathbf{k}}\rangle$$

a BCS state described by the RMPs. Within this short section, we just illustrate some features of this state. First, the phase factor angle is the gap "tilt" in the complex plane. This is proven by:

$$\begin{aligned}
\langle\Psi|\hat{s}^x_{\mathbf{k}}|\Psi\rangle &= \langle\Psi|\hat{\phi}_{\mathbf{k}} + \hat{\phi}^{\dagger}_{\mathbf{k}}|\Psi\rangle \\
&= \sin\tilde{\theta}_{\mathbf{k}} \cos\tilde{\theta}_{\mathbf{k}} \left(e^{2i\tilde{\zeta}_{\mathbf{k}}} + e^{-2i\tilde{\zeta}_{\mathbf{k}}}\right) \\
&= \sin(2\tilde{\theta}_{\mathbf{k}}) \cos(2\tilde{\zeta}_{\mathbf{k}})
\end{aligned}$$

$$\begin{aligned}
\langle\Psi|\hat{s}^y_{\mathbf{k}}|\Psi\rangle &= i\,\langle\Psi|\hat{\phi}_{\mathbf{k}} - \hat{\phi}^{\dagger}_{\mathbf{k}}|\Psi\rangle \\
&= i\sin\tilde{\theta}_{\mathbf{k}} \cos\tilde{\theta}_{\mathbf{k}} \left(e^{-2i\tilde{\zeta}_{\mathbf{k}}} - e^{2i\tilde{\zeta}_{\mathbf{k}}}\right) \\
&= \sin(2\tilde{\theta}_{\mathbf{k}}) \sin(2\tilde{\zeta}_{\mathbf{k}})
\end{aligned}$$

These results are coherent with Eqns. (6.33) as well as (6.34) and allow us to identify the phase with angle of the pseudofield in the xy plane.

6.8 SC free energy density

Free energy density of the SC phase is derived following the procedure used for the other two phases. As for the AF phase of Sec. 5.7, two fermionic bands are present; however, recalling the antiferromagnet entropy density of Eq. (5.77), the key differences are here given by the fact that the chemical potential inside the Fermi-Dirac factors is null and the sum is extended on the full BZ (but not later on spin index),

$$s^{(\pm)} \equiv -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \Big[\left(1 - f(\pm\tilde{E}_{\mathbf{k}};\beta,0)\right) \ln\left(1 - f(\pm\tilde{E}_{\mathbf{k}};\beta,0)\right) + f(\pm\tilde{E}_{\mathbf{k}};\beta,0) \ln f(\pm\tilde{E}_{\mathbf{k}};\beta,0)\Big] \quad (6.55)$$

with identical meaning of the notation. Using the relation

$$\frac{1}{e^x + 1} + \frac{1}{e^{-x} + 1} = 1$$

we conclude that

$$s^{(-)} = s^{(+)} \quad \Longrightarrow \quad s \equiv s^{(-)} + s^{(+)} = 2s^{(+)}$$

being s the total entropy density. The five contributions to free energy are:

1. Quasiparticles energy:
$$e_{\mathrm{g}}^{(\mathrm{SC})} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{E}_{\mathbf{k}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta]$$
$$= -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{E}_{\mathbf{k}} \tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$$

2. Anticommutation energy:
$$e_{\mathrm{a}}^{(\mathrm{SC})} \equiv \frac{E_{\mathrm{a}}^{(\mathrm{SC})}}{L_x L_y}$$
$$= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{\xi}_{\mathbf{k}}$$

3. Contraction energy:
$$e_{\mathrm{c}}^{(\mathrm{SC})} = -U\left[n^2 + \left|w^{(s)}\right|^2\right] + V\left[2zn^2 - \sum_{\gamma}\left|u^{(\gamma)}\right|^2 + \sum_{\gamma}\left|w^{(\gamma)}\right|^2\right]$$

4. Entropic energy
$$e_{s}^{(\mathrm{SC})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln\left(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\right) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{E}_{\mathbf{k}} f(\tilde{E}_{\mathbf{k}};\beta,0)$$

5. Chemical potential correction
$$e_{n}^{(\mathrm{SC})} = \mu n$$

Compose everything
$$f_{\mathrm{MFT}}^{(\mathrm{SC})} \equiv e_{\mathrm{g}}^{(\mathrm{SC})} + e_{\mathrm{a}}^{(\mathrm{SC})} + e_{\mathrm{c}}^{(\mathrm{SC})} + e_{s}^{(\mathrm{SC})} + e_{n}^{(\mathrm{SC})}$$
and use that:
$$\tanh\left(\frac{x}{2}\right) + \frac{2}{e^x+1} = 1$$
In the end we get:
$$f_{\mathrm{MFT}}^{(\mathrm{SC})} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \left[\tilde{\xi}_{\mathbf{k}} - \tilde{E}_{\mathbf{k}}\right] + \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln\left(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\right)$$
$$- U\left[n^2 + \left|w^{(s)}\right|^2\right] + V\left[2zn^2 - \sum_{\gamma}\left|u^{(\gamma)}\right|^2 + \sum_{\gamma}\left|w^{(\gamma)}\right|^2\right] + \mu$$

As was for the normal and AF phases, we recognize the chemical potential shift of Eq. (6.43),
$$f_{\mathrm{MFT}}^{(\mathrm{SC})} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \left[\tilde{\xi}_{\mathbf{k}} - \tilde{E}_{\mathbf{k}}\right] + \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln\left(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\right)$$
$$- U\left|w^{(s)}\right|^2 - V\left[\sum_{\gamma}\left|u^{(\gamma)}\right|^2 - \sum_{\gamma}\left|w^{(\gamma)}\right|^2\right] + \tilde{\mu}n \quad (6.56)$$

6.8.1 The gap energy

Similarly to Sec. ??, let us prove the following equation about the gap energy:
$$-U\left|w^{(s)}\right|^2 + V\sum_{\gamma}\left|w^{(\gamma)}\right|^2 = \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\left|\tilde{\Delta}_{\mathbf{k}}\right|^2}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right) \quad (6.57)$$

Expand the right-hand side through Eq. (6.54),

$$\begin{aligned}
\frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\frac{\left|\tilde{\Delta}_{\mathbf{k}}\right|^2}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right) &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\left[-Uw^{(s)*}+V\sum_{\gamma}w^{(\gamma)*}\varphi^{(\gamma)}_{\mathbf{k}}\right]\frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\\
&= -Uw^{(s)*}\times\overbrace{\frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)}^{w^{(s)}}\\
&\qquad + V\sum_{\gamma}w^{(\gamma)*}\times\underbrace{\frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{k}}\frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)}_{w^{(\gamma)}}
\end{aligned}$$

having we used the self-consistency equations of Tab. 6.5. This proves Eq.(6.57), which is similar to Eq. (5.79) with one exception. Since the right-hand side of Eq.(6.57) is positive, for $V>0$ we have the constraint

$$\sum_{\gamma}\left|w^{(\gamma)}\right|^2\geq M\quad\text{being}\quad M^{(\mathrm{SC})}\equiv\left|w^{(s)}\right|^2\frac{U}{V}$$

Now, as we state in Sec. 6.2.6, for the pure HM we have identically $w^{(s)}=0$. The pure HM cannot exhibit BCS-like superconductivity. For the EHM instead we see that:

- In the strong coupling limit $U\ll V$, even small s-wave gaps impose larger $\gamma\neq s$ components;
- The s-wave gap $w^{(s)}$ cannot converge to a finite value if for any $\gamma\neq s$ we have $w^{(\gamma)}=0$.

Of course, Eq. (6.57) is also solved by a null gap on both sides: this indicates that when the constraint is impossible to fulfil self-consistently, the gap closes.

Appendix A

Superexchange and virtual hopping in Hubbard lattices

A key mechanism in AF phase formation in Hubbard lattice is superexchange. The AF phase is stabilized by spin fluctuations and second-order virtual hopping. The mechanism becomes clear enough by considering a 2-sites Hubbard toy model.

A.1 Virtual hopping in the 2-sites Hubbard lattice

Consider the toy model:

$$\hat{H} = -t\left\{\hat{c}^{\dagger}_{1\uparrow}\hat{c}_{2\uparrow} + \hat{c}^{\dagger}_{1\downarrow}\hat{c}_{2\downarrow} + \text{h.c.}\right\} + U\left\{\hat{n}_{1\uparrow}\hat{n}_{1\downarrow} + \hat{n}_{2\uparrow}\hat{n}_{2\downarrow}\right\}$$

with $i = 1, 2$ the site index. The two sites are represented in Fig. A.1. The two competing processes are:

1. electrons inter-sites hopping with amplitude $-t$;

2. local repulsion $+U$, acting when two anti-aligned electrons reside on the same site;

For an half-filled system, the Hilbert space is six-dimensional. I use the notation $|n_{1\uparrow}n_{1\downarrow}n_{2\uparrow}n_{2\downarrow}\rangle$ to indicate the six computational basis states:

$$\begin{array}{lll} |\psi_1\rangle \equiv |1010\rangle & |\psi_3\rangle \equiv |1001\rangle & |\psi_5\rangle \equiv |0011\rangle \\ |\psi_2\rangle \equiv |1100\rangle & |\psi_4\rangle \equiv |0110\rangle & |\psi_6\rangle \equiv |0101\rangle \end{array}$$

For example, the top panel of Fig. A.1 shows state $|\psi_2\rangle$.

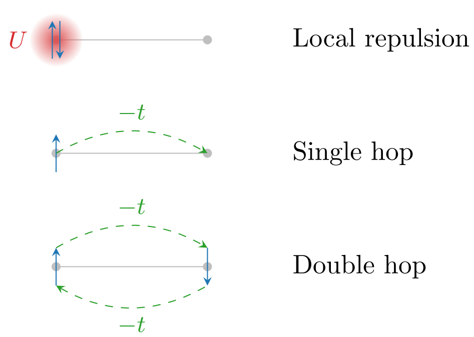


Figure A.1 | Two sites Hubbard model.

Structure	Eigenstate	Energy
Spin-1/2 singlet	$\dfrac{\|\phi_3\rangle - \|\phi_4\rangle}{\sqrt{2}}$	$E^- = \dfrac{U}{2} - \sqrt{\dfrac{U^2}{4} + 4t^2}$
Spin-1/2 triplet	$\|\phi_1\rangle, \dfrac{\|\phi_3\rangle + \|\phi_4\rangle}{\sqrt{2}}, \|\phi_6\rangle$	0
	$\dfrac{\|\phi_2\rangle - \|\phi_5\rangle}{\sqrt{2}}$	U
	$\dfrac{\|\phi_2\rangle + \|\phi_5\rangle}{\sqrt{2}}$	$E^+ = \dfrac{U}{2} + \sqrt{\dfrac{U^2}{4} + 4t^2}$

Table A.1 | List of exact eigenstates and relative energies for the 2-sites half-filled Hubbard model.

A.1.1 Exact solution of the half-filled model

The hamiltonian matrix is directly evaluated in this basis

$$H_{ij} = \langle\psi_i|\,\hat{H}\,|\psi_j\rangle \qquad \Longrightarrow \qquad H = \begin{bmatrix} 0 & & & & & \\ & U & -t & -t & & \\ & -t & & & -t & \\ & -t & & & -t & \\ & & -t & -t & U & \\ & & & & & 0 \end{bmatrix}$$

Empty slots in the matrix stand for zeros. Evidently the states $|\psi_1\rangle$ (both up) and $|\psi_6\rangle$ (both down) are zero-energy eigenstates. These states cannot realize electrons hopping because of Pauli principle. The internal 4×4 matrix is readily diagonalized by the means of a change of basis V,

$$V \begin{bmatrix} U & -t & -t & \\ -t & & & -t \\ -t & & & -t \\ & -t & -t & U \end{bmatrix} V^\dagger = \begin{bmatrix} E^- & & & \\ & 0 & & \\ & & U & \\ & & & E^+ \end{bmatrix}$$

Tab. A.1 shows the eigenvectors and relative eigenvalues obtained from diagonalization. The ground-state is the singlet state,

$$\frac{|\phi_3\rangle - |\phi_4\rangle}{\sqrt{2}} = \frac{|1010\rangle - |0101\rangle}{\sqrt{2}} = \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}}$$

of energy

$$E^- = \frac{U}{2} - \sqrt{\frac{U^2}{4} + 4t^2} \simeq -\frac{4t^2}{U}$$

the latter equality being true if $U \gg t$ (strong repulsion limit). The singlet state pairs with a spatially-symmetric (nodeless) wavefunction. The entire triplet (second row of Tab. A.1) remains at zero energy. Excited states are anti-symmetrized and symmetrized version of the polarized states $|\phi_1\rangle$ and $|\phi_6\rangle$.

A.1.2 Virtual hopping

The key feature of the singlet state is the one represented in the bottom panel of Fig. A.1: if the two electrons occupy separate sites and are anti-aligned, both "see" the other site as empty, thus free to hop to. [To be continued...]

Appendix B

Gaussian states

In this appendix we discuss the properties of the many-body states we use to approximate the real ground-state of the system within HF method. We introduce Slater determinants to grasp the general idea behind HF variational method, and then extend to a larger set of states.

B.1 Slater determinants

A many-body wavefunction describing a collective electronic state needs to be antisymmetric under interchange of coordinates and spins $(\mathbf{r}_i, \sigma_i)$, $(\mathbf{r}_j, \sigma_j)$ of any couple of electrons. The simplest way to build a many-body wavefunction satisfying this rule is "to anti-symmetrise" a set of orthonormal N one-body spin-orbitals, being N the total number of electrons and D the total number of spin-orbitals. Let $\{\psi_i\}$ be said set,

$$\frac{1}{2\mathcal{V}} \sum_{\sigma \in \{\uparrow,\downarrow\}} \int \mathrm{d}\mathbf{r}\ \psi_i(\mathbf{r}, \sigma)\psi_j(\mathbf{r}, \sigma) = \delta_{ij}$$

being $\mathcal{V}$ the total volume.

Let S_N be the $N!$ elements symmetric group of permutations of N objects and $\mathrm{P} \in \mathrm{S}_N$ a specific permutation of a given N-uple,

$$\mathrm{P} \quad : \quad (x_1, x_2, \cdots, x_N) \to \left(x_{\mathrm{P}(1)}, x_{\mathrm{P}(2)}, \cdots, x_{\mathrm{P}(N)}\right)$$

Let F_P be the operator that implements the permutation P on the N variables of a function f,

$$F_\mathrm{P}\left[f(x_1, x_2, \cdots, x_N)\right] = f\left(x_{\mathrm{P}(1)}, x_{\mathrm{P}(2)}, \cdots, x_{\mathrm{P}(N)}\right)$$

We define the anti-symmetrisation operator:

$$\mathcal{A}[f] \equiv \frac{1}{\sqrt{N!}} \sum_{\mathrm{P} \in \mathrm{S}_N} (-1)^{\sigma(\mathrm{P})} F_\mathrm{P}[f]$$

being $\sigma(\mathrm{P})$ the "parity factor" of the permutation $\mathrm{P} \in \mathcal{S}_N$. The factor $\sigma(\mathrm{P})$ counts the number of exchanges needed to obtain P: this way, an even number of exchanges gives a total sign $+1$, an odd number gives -1. By construction, the function $\mathcal{A}[f]$ is antisymmetric under exchange of a pair of coordinates.

Consider now the many-body wavefunction

$$\Psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2; \cdots; \mathbf{r}_N, \sigma_N) = \mathcal{A}\left[\prod_{i=1}^{N} \psi_i(\mathbf{r}_i, \sigma_i)\right]$$

This class of wavefunctions, obtained as antisymmetric combinations of product states, are representable as so-called Slater determinants,

$$\Psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2; \cdots; \mathbf{r}_N, \sigma_N) = \frac{1}{\sqrt{N!}} \det \begin{bmatrix} \psi_1(\mathbf{r}_1, \sigma_1) & \psi_1(\mathbf{r}_2, \sigma_2) & \cdots & \psi_1(\mathbf{r}_N, \sigma_N) \\ \psi_2(\mathbf{r}_1, \sigma_1) & \psi_2(\mathbf{r}_2, \sigma_2) & \cdots & \psi_2(\mathbf{r}_N, \sigma_N) \\ \vdots & \vdots & & \vdots \\ \psi_N(\mathbf{r}_1, \sigma_1) & \psi_N(\mathbf{r}_2, \sigma_2) & \cdots & \psi_N(\mathbf{r}_N, \sigma_N) \end{bmatrix}$$

This representation includes automatically Pauli exclusion principle: determinants of matrices with two equal rows or two equal columns are null. Thus, all spin-orbitals must differ, as well as no single spin-orbital ψ_i can be doubly occupied (which is, all spin-coordinate couples must be different).

Any normalised combinations of Slater determinants gives an antisymmetric wavefunction, although not necessarily representable as a single Slater determinant. In HF theory, we search for the best possible choice of orthonormal one-body spin orbitals $\{\psi_1, \psi_2, \cdots, \psi_N\}$ to approximate the true ground-state of the model through a single Slater determinant. The space of possible "Slater-like" many-body wavefunctions is the parametric space where we perform the variational constrained search.

The EHM is formulated in second-quantisation. It is then useful to represent Slater-like states in this language. Let us map all the spin-orbitals from the discussion above to pairs of annihilation-creation fermionic operators,

$$\psi_i \quad \rightarrow \quad \hat{c}_i, \hat{c}_i^\dagger$$

Any Slater state can be expressed in second-quantisation as:

$$|\Psi\rangle = \prod_{i=1}^{N} \hat{c}_i^\dagger |\Omega\rangle \tag{B.1}$$

being $|\Omega\rangle$ the vacuum. Antisymmetry is guaranteed by Fermi anticommutation rules.

B.2 Wick's theorem applied to Slater determinants

The concept of vacuum state needs to be interpreted in relation to an appropriate choice of second-quantisation annihilation and creation operators. For a given set of fermion operators, the vacuum is that state mapped to zero by any annihilation operator. As we will see, to correctly define the vacuum state is crucial in order to apply Wick's theorem and develop an HF approximation of EHM. Before moving to generalisation, let us discuss a playground setup.

Consider some unspecified model whose ground state at null temperature is given in Eq. (B.1) for a given set of electronic operators $\hat{c}_i$, $\hat{c}_i^\dagger$. Let $\hat{c}_i$ with $N < i < D$ identify all spin-orbitals not already included in the definition of $|\Psi\rangle$ of Eq. (B.1) (thus, non populated). Then it holds:

$$\begin{aligned} 0 \le i \le N \qquad & \hat{c}_i^\dagger |\Psi\rangle = 0 \\ i > N \qquad & \hat{c}_i |\Psi\rangle = 0 \end{aligned}$$

Let:

$$\hat{a}_i \equiv \begin{cases} \hat{c}_i^\dagger & \text{for } 0 \le i \le N \\ \hat{c}_i & \text{for } i > N \end{cases}$$

These new operators satisfy a Fermi algebra $\{\hat{a}_i, \hat{a}_j^\dagger\} = \delta_{ij}$ and annihilate the "vacuum" state $|\Psi\rangle$. In other words, any Slater state like Eq. (B.1) can be seen as a vacuum state for a suitably chosen set of fermionic operators.

This property is handy for evaluation of expectation values (EV). Let $\hat{O}^{(\ell)}$ be a linear combination of $\hat{c}_i$ or $\hat{c}_i^\dagger$,

$$\hat{O}^{(\ell)} \equiv \sum_{i=1}^{D} \left(u_i^{(\ell)} \hat{c}_i + v_i^{(\ell)} \hat{c}_i^\dagger \right) \qquad u_i^{(\ell)}, v_i^{(\ell)} \in \mathbb{C} \tag{B.2}$$

Consider the product operator:

$$\hat{\mathcal{O}} \equiv \hat{O}^{(1)} \hat{O}^{(2)} \cdots \hat{O}^{(2k)} \qquad k \in \mathbb{N}$$

Inserting the above decomposition, we get a large sum of many terms, each containing some sequence of annihilations and creations. Using linearity we can concentrate on one specific term of these, derive results and then recompose them for the original operator. Consider one of these terms,

$$\hat{\mathcal{O}} = \cdots + (\text{coefficient}) \times \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} + \cdots$$

with $\hat{C}_i$ some fermionic operator, either $\hat{c}_j$ or $\hat{c}_j^\dagger$ for some j. Now, Wick's theorem states that

$$\begin{aligned}\hat{C}_1\hat{C}_2\cdots\hat{C}_{2k} = \mathrm{N}\left\{\hat{C}_1\hat{C}_2\cdots\hat{C}_{2k}\right\}\\ &+ \sum_{(\alpha\beta)}(-1)^{\zeta_{\alpha\beta}}D_{\alpha\beta}\mathrm{N}\left\{\prod_{i\neq\alpha,\beta}\hat{C}_i\right\}\\ &+ \sum_{(\alpha\beta)(\gamma\delta)}(-1)^{\zeta_{\alpha\beta}}D_{\alpha\beta}(-1)^{\zeta_{\gamma\delta}}D_{\gamma\delta}\mathrm{N}\left\{\prod_{i\neq\alpha,\beta,\gamma,\delta}\hat{C}_i\right\}\\ &+\cdots\end{aligned}$$

where $\mathrm{N}\{\cdots\}$ indicates normal ordering, $D_{\alpha\beta}$ is the contraction of the pair $\hat{C}_\alpha\hat{C}_\beta$ and $\zeta_{\alpha\beta}$ is a symmetry factor included to take care of the number of fermionic algebra. In this context contractions are just expectation values over $|\Psi\rangle$. Sums are performed over pairs of indices, implicitly taken to be different from each other. To take averages of this last operator over $|\Psi\rangle$ is a not-so-hard task, given the simple expression of the state in terms of electronic annihilations and creations. However it can be made simpler: if we change basis and express all operators in terms of $\hat{a}$, $\hat{a}^\dagger$,

$$\hat{C}_i \quad\rightarrow\quad \hat{A}_i$$

the above decomposition holds identically, with slight change of notation. Now, taking a zero temperature average on $|\Psi\rangle$ (the vacuum with respect to the new operators) all normal-ordered terms disappear,

$$\langle\Psi|\mathrm{N}\{\hat{A}_i\hat{A}_j\cdots\}|\Psi\rangle = 0$$

thus in the decomposition above only fully contracted terms survive. Going back to $\hat{c}$ operators, we can re-express fully contracted terms in the original basis. Thus:

$$\left\langle\hat{C}_1\hat{C}_2\cdots\hat{C}_{2k}\right\rangle = \sum(\text{total sign factor})\times(\text{product of complete contractions})$$

where sum is intended, implicitly, over all possible ways of choosing k couples of indices from a starting set of $2k$ elements.

Taking another step back, due to linearity we can apply this result to all terms of $\mathcal{O}$ and conclude that the EV of $\mathcal{O}$ decomposes as a sum of 2-points EVs. This can be seen as the defining property of gaussian states. In this procedure, the only key property of Slater states we used was the fact that they are vacuum to some fermionic basis. Finite-temperature extension of Wick's theorem as well as more complicated change of basis are possible, but the concept remains the same: on a Slater state, which is essentially a non-interacting state for some set of fermions, Wick's theorem is particularly well-behaved.

From the point above we can conclude that:

$$\langle\hat{c}_\alpha^\dagger\hat{c}_\beta^\dagger\hat{c}_\gamma\hat{c}_\delta\rangle = \langle\hat{c}_\alpha^\dagger\hat{c}_\delta\rangle\langle\hat{c}_\beta^\dagger\hat{c}_\gamma\rangle - \langle\hat{c}_\alpha^\dagger\hat{c}_\beta\rangle\langle\hat{c}_\gamma^\dagger\hat{c}_\delta\rangle$$

being the average intended on a Slater state. In this context, only contractions with an equal number of creation and annihilation operators contribute.

B.3 Generalisation to Bogoliubov-Valatin states

In HF theory we look for a good approximation of the true many-body wavefunction constraining search to a class of states. Introducing Slater determinants was useful to grasp the concrete idea behind the method. However, in order to treat some physical phenomena we need to extend the variational procedure to a other sets of states. For example, Bardeen-Cooper-Shrieffer (BCS) theory of superconductivity gives a many-body ground-state which is not representable as a Slater determinant of "standard" electronic operators. In this section we extend the notions introduced before and set grounds for the generalised Hartree-Fock-Bogoliubov (HFB) method we use in this thesis.

Consider the Nambu spinor

$$\hat{\Phi} \equiv \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^\dagger \end{bmatrix}$$

being $\hat{\mathbf{c}}$ the vector of all annihilation operators, ordered, and $\hat{\mathbf{c}}^\dagger$ of all creation operators. Let us assume the vectors have D entries. Consider an unitary map M, such that

$$\hat{\Gamma} \equiv M\hat{\Phi} \quad \text{being} \quad \hat{\Gamma} \equiv \begin{bmatrix} \hat{\boldsymbol{\gamma}} \\ \hat{\boldsymbol{\gamma}}^\dagger \end{bmatrix}$$

The new Nambu vectors $\hat{\boldsymbol{\gamma}}$ and $\hat{\boldsymbol{\gamma}}^\dagger$ contain our new, generalised fermionic operators in second quantisation. Let:

$$M \equiv \begin{bmatrix} U & V \\ U^\dagger & V^\dagger \end{bmatrix}$$

If M is unitary, it follows:

$$\begin{aligned} \mathbb{I}_{2D\times 2D} &= MM^\dagger \\ &= \begin{bmatrix} U & V \\ U^\dagger & V^\dagger \end{bmatrix} \begin{bmatrix} U & U^\dagger \\ V^\dagger & V \end{bmatrix} \\ &= \begin{bmatrix} UU^\dagger + VV^\dagger & U^2 + V^2 \\ \left(U^\dagger\right)^2 + \left(V^\dagger\right)^2 & U^\dagger U + V^\dagger V \end{bmatrix} \end{aligned}$$

thus, among other conditions, $UU^\dagger + VV^\dagger = \mathbb{I}_{D\times D}$.

The new operators in $\hat{\Gamma}$ satisfy Fermi anticommutation rules. In fact, writing the mapping explicitly for any of the new operators and some index $i \in [1, D]$,

$$\hat{\gamma}_i \equiv \sum_{j=1}^{D} U_{ij}\hat{c}_j + \sum_{j=1}^{D} V_{ij}\hat{c}_j^\dagger \qquad \hat{\gamma}_i^\dagger \equiv \sum_{j=1}^{D} U_{ji}^*\hat{c}_j^\dagger + \sum_{j=1}^{D} V_{ji}^*\hat{c}_j$$

and substituting in the anti-commutator

$$\begin{aligned} \{\hat{\gamma}_i, \hat{\gamma}_j^\dagger\} &= \sum_{i'=1}^{D} U_{ii'} \sum_{j'=1}^{D} U_{j'j}^* \{\hat{c}_{i'}, \hat{c}_{j'}^\dagger\} + \sum_{i'=1}^{D} V_{ii'} \sum_{j'=1}^{D} V_{j'j}^* \{\hat{c}_{i'}^\dagger, \hat{c}_{j'}\} \\ &= \sum_{i'=1}^{D} U_{ii'} \sum_{j'=1}^{D} U_{j'j}^* \delta_{i'j'} + \sum_{i'=1}^{D} V_{ii'} \sum_{j'=1}^{D} V_{j'j}^* \delta_{i'j'} \\ &= \left[UU^\dagger + VV^\dagger\right]_{ij} \\ &= \delta_{ij} \end{aligned}$$

being M unitary.

Assume now $|\Psi\rangle$ to be the vacuum state for the new operators,

$$\forall i \quad : \quad \hat{\gamma}_i |\Psi\rangle = 0$$

Suppose we want to evaluate, with identical meaning of the notation as in previous section, something like

$$\left\langle \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} \right\rangle$$

We can apply the result obtained above. Indeed, inverse maps linking old and new operators read:

$$\hat{c}_i \equiv \sum_{j=1}^{D} U_{ij}\hat{\gamma}_j + \sum_{j=1}^{D} U_{ji}^*\hat{\gamma}_j^\dagger \qquad \hat{c}_i^\dagger \equiv \sum_{j=1}^{D} V_{ji}^*\hat{\gamma}_j^\dagger + \sum_{j=1}^{D} V_{ij}\hat{\gamma}_j$$

In Sec. **??** we argued that, when averaging a product of operators like the one of Eq. (B.2), only 2-points EVs contribute. This time the idea is the same, substituting $\mathcal{O}^{(\ell)} \to \hat{C}_\ell$ and $\hat{c}_i, \hat{c}_i^\dagger \to \hat{\gamma}_i, \hat{\gamma}_i^\dagger$.

Thus, also here we can decompose the product operator, apply Wick's theorem, discard all non-fully-contracted terms and recompose the results. All we used was just the unitary mapping between $\hat{\Phi}$ and $\hat{\Gamma}$, the fact that $\gamma_i, \hat{\gamma}_i^\dagger$ obey Fermi anticommutation rules and "see" $|\Psi\rangle$ as a vacuum state.

As in Sec. **??**, we conclude that:

$$\langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\gamma \hat{c}_\delta \rangle = \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\delta \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\gamma \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle \langle \hat{c}_\gamma^\dagger \hat{c}_\delta \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \rangle \langle \hat{c}_\gamma \hat{c}_\delta \rangle}_{\text{Cooper}} \tag{B.3}$$

The subscripts indicate the name we will give to these terms throughout the text. Note that, differently from "pure Slater states" EVs with two annihilations or two creations do not vanish, in general.

All of this holds when averaging is performed on a state, $|\Psi\rangle$, which is vacuum for a set of operators linked to fermionic Nambu spinor $\hat{\Phi}$ by some unitary operation. This set of states is the domain of our variational constrained search. This kind of unitary mapping is often referred to as Bogoliubov-Valatin transformation, and the method hereby presented as Hartree-Fock-Bogoliubov. For simplicity, in this thesis we stick to the name "HF method" referring to this discussion. This argument has been carried out at zero temperature, but can be extended to finite temperature.

Appendix C

Connection with mean-field theories

In this appendix we bridge the gap between HF methods and mean-field theory (MFT). **In general, in MFT approaches we substitute the two-body (or more) interactions of the hamiltonian with a single-body interaction with a mean field, determined locally via self-consistency equations. This is, for example, the Weiß mean-field solution to the quantum Ising model. The HF method we sketched here does not work like that. In the context of Hubbard models, the analogous of Weiß MFT would be dynamical mean-field theory (DMFT) [10].**

However, there are some points of connection. Let us sketch first the general scheme behind MFT. Suppose we have some operator $\hat{A} = \hat{B}\hat{C}$. Let

$$\delta\hat{O} \equiv \hat{O} - \langle\hat{O}\rangle \qquad \hat{O} \in \{\hat{A}, \hat{B}, \hat{C}\}$$

Then we have:

$$\begin{aligned}\hat{A} &= \left(\langle\hat{B}\rangle + \delta\hat{B}\right)\left(\langle\hat{C}\rangle + \delta\hat{C}\right)\\ &= \langle\hat{B}\rangle\langle\hat{C}\rangle + \langle\hat{B}\rangle\delta\hat{C} + \delta\hat{B}\langle\hat{C}\rangle + \delta\hat{B}\delta\hat{C}\\ &= \langle\hat{B}\rangle\langle\hat{C}\rangle + \langle\hat{B}\rangle\left(\hat{C} - \langle\hat{C}\rangle\right) + \left(\hat{B} - \langle\hat{B}\rangle\right)\langle\hat{C}\rangle + \delta\hat{B}\delta\hat{C}\\ &= -\langle\hat{B}\rangle\langle\hat{C}\rangle + \langle\hat{B}\rangle\hat{C} + \hat{B}\langle\hat{C}\rangle + \delta\hat{B}\delta\hat{C}\end{aligned}$$

Up to now, this relation is exact. MFT relies on the idea that the last term, of order $\mathcal{O}(\delta^2)$, can be neglected. This amounts to say that the operator fluctuations about its average are "small". This gives the approximation:

$$\hat{A} \simeq \langle\hat{B}\rangle\hat{C} + \hat{B}\langle\hat{C}\rangle - \langle\hat{B}\rangle\langle\hat{C}\rangle \tag{C.1}$$

Within the assumption that independently $\hat{B}$ and $\hat{C}$ fluctuations are "small", we are able to simplify $\hat{A}$ with a sum of three terms: the product of each operator with the averaged value of the other, minus the product of the averages.

In the context of HF methods, we are in an slightly different situation. We are not making assumptions on the fluctuations of the quartic sectors of the hamiltonian. We are restricting the space of wavefunctions and minimising energy there.

Let us restart from the interacting hamiltonian of Eq. (2.2). In this thesis we deal with the EHM, thus (limitedly to this section) let us restrict to the two-body generic quartic interaction:

$$\hat{V} = \sum_{\alpha,\beta} V_{\alpha\beta}\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\hat{c}_\beta\hat{c}_\alpha$$

with $\forall\alpha,\beta\colon V_{\alpha\beta} = V^*_{\beta\alpha}$. In terms of the general hamiltonian of Eq. (2.2), we have:

$$B_{\alpha\beta\gamma\delta} = 2V_{\alpha\beta}\delta_{\alpha\delta}\delta_{\beta\gamma} \tag{C.2}$$

We work with HF states. Thus Eq. (2.1) holds exactly (adapted for two indices instead of four)

$$\alpha \neq \beta \quad : \quad \langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\hat{c}_\beta\hat{c}_\alpha\rangle = \underbrace{\langle\hat{c}^\dagger_\alpha\hat{c}_\alpha\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\beta\rangle}_{\text{Hartree}} - \underbrace{\langle\hat{c}^\dagger_\alpha\hat{c}_\beta\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\alpha\rangle}_{\text{Fock}} + \underbrace{\langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\rangle\langle\hat{c}_\beta\hat{c}_\alpha\rangle}_{\text{Cooper}}$$

This gives:

$$\begin{aligned}\langle\hat{V}\rangle &= \sum_{\alpha,\beta} V_{\alpha\beta}\langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\hat{c}_\beta\hat{c}_\alpha\rangle \\ &= \sum_{\alpha,\beta} V_{\alpha\beta}\left(\langle\hat{c}^\dagger_\alpha\hat{c}_\alpha\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\beta\rangle - \langle\hat{c}^\dagger_\alpha\hat{c}_\beta\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\alpha\rangle + \langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\rangle\langle\hat{c}_\beta\hat{c}_\alpha\rangle\right) \end{aligned} \tag{C.3}$$

from which we have the equations:

$$\frac{\partial\langle\hat{V}\rangle}{\partial\langle\hat{c}^\dagger_\alpha\hat{c}_\beta\rangle} = \delta_{\alpha\beta}\left[\sum_\gamma V_{\alpha\gamma}\langle\hat{c}^\dagger_\gamma\hat{c}_\gamma\rangle\right] - V_{\alpha\beta}\langle\hat{c}^\dagger_\beta\hat{c}_\alpha\rangle \tag{C.4}$$

$$\frac{\partial\langle\hat{V}\rangle}{\partial\langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\rangle} = V_{\alpha\beta}\langle\hat{c}_\beta\hat{c}_\alpha\rangle \tag{C.5}$$

Now, our aim is to approximate $\hat{V}$ as:

$$\hat{V} \simeq \sum_{\alpha\beta}\left[x_{\alpha\beta}\hat{c}^\dagger_\alpha\hat{c}_\beta + y_{\alpha\beta}\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta + y^*_{\beta\alpha}\hat{c}_\alpha\hat{c}_\beta\right] + (\text{some constant}) \tag{C.6}$$

where $x_{\alpha\beta} = x^*_{\beta\alpha}$ to ensure hermiticity. Note that we are actively ignoring terms like $\hat{c}\hat{c}^\dagger$. Anti-commuting these, we get sums of operators like $\hat{c}^\dagger\hat{c}$ and constants. The above expression is the most general. Take its EV:

$$\langle\hat{V}\rangle \simeq \sum_{\alpha\beta}\left[x_{\alpha\beta}\langle\hat{c}^\dagger_\alpha\hat{c}_\beta\rangle + y_{\alpha\beta}\langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\rangle + y^*_{\beta\alpha}\langle\hat{c}_\alpha\hat{c}_\beta\rangle\right] + (\text{some constant})$$

Now, taking the derivatives:

$$x_{\alpha\beta} = \frac{\partial\langle\hat{V}\rangle}{\partial\langle\hat{c}^\dagger_\alpha\hat{c}_\beta\rangle} \qquad y_{\alpha\beta} = \frac{\partial\langle\hat{V}\rangle}{\partial\langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\rangle}$$

Applying Eqns. (C.4) and (C.5), we get the identities:

$$x_{\alpha\beta} = \delta_{\alpha\beta}\left[\sum_\gamma V_{\alpha\gamma}\langle\hat{c}^\dagger_\gamma\hat{c}_\gamma\rangle\right] - V_{\alpha\beta}\langle\hat{c}^\dagger_\beta\hat{c}_\alpha\rangle \tag{C.7}$$

$$y_{\alpha\beta} = V_{\alpha\beta}\langle\hat{c}_\beta\hat{c}_\alpha\rangle \tag{C.8}$$

These equations are a rephrasing of Eqns. (2.14), (2.15) for the fields $h^{pq}_{\alpha\beta}$, setting $A_{\alpha\beta} = 0$. This can be checked by using Eq. (C.2).

Now: consider the special case where $\forall\alpha\colon V_{\alpha\alpha} = 0$. This is the case for the EHM. Then the above equations imply:

$$\begin{aligned}\hat{V} &= \sum_{\alpha\neq\beta} V_{\alpha\beta}\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\hat{c}_\beta\hat{c}_\alpha \\ &\simeq \sum_{\alpha\neq\beta} V_{\alpha\beta}\Big(\overbrace{\hat{c}^\dagger_\alpha\hat{c}_\alpha\langle\hat{c}^\dagger_\beta\hat{c}_\beta\rangle + \langle\hat{c}^\dagger_\alpha\hat{c}_\alpha\rangle\hat{c}^\dagger_\beta\hat{c}_\beta}^{\text{Hartree}}\overbrace{-\hat{c}^\dagger_\alpha\hat{c}_\beta\langle\hat{c}^\dagger_\beta\hat{c}_\alpha\rangle - \langle\hat{c}^\dagger_\alpha\hat{c}_\beta\rangle\hat{c}^\dagger_\beta\hat{c}_\alpha}^{\text{Fock}} \\ &\qquad + \underbrace{\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\langle\hat{c}_\beta\hat{c}_\alpha\rangle + \langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\rangle\hat{c}_\beta\hat{c}_\alpha}_{\text{Cooper}}\Big) + (\text{some constant})\end{aligned}$$

To determine the constant, we take the EV of the above equation. The result should equal Eq. (C.3). Take e.g. the Hartree sector:

$$\langle\hat{c}^\dagger_\alpha\hat{c}_\alpha\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\beta\rangle + \langle\hat{c}^\dagger_\alpha\hat{c}_\alpha\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\beta\rangle + C = \langle\hat{c}^\dagger_\alpha\hat{c}_\alpha\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\beta\rangle \quad\Longrightarrow\quad C = -\langle\hat{c}^\dagger_\alpha\hat{c}_\alpha\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\beta\rangle$$

with C a constant. A similar implication holds for the Fock and the Cooper sectors. Then:

$$
\begin{aligned}
\hat{V} &= \sum_{\alpha\neq\beta} V_{\alpha\beta}\hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\beta \hat{c}_\alpha && \text{(C.9)} \\
&\overset{\text{HF}}{\simeq} \sum_{\alpha\neq\beta} V_{\alpha\beta}\Big(\hat{c}^\dagger_\alpha \hat{c}_\alpha \langle \hat{c}^\dagger_\beta \hat{c}_\beta\rangle + \langle \hat{c}^\dagger_\alpha \hat{c}_\alpha\rangle \hat{c}^\dagger_\beta \hat{c}_\beta - \langle \hat{c}^\dagger_\alpha \hat{c}_\alpha\rangle\langle \hat{c}^\dagger_\beta \hat{c}_\beta\rangle && \text{(Hartree)} \\
&\qquad - \hat{c}^\dagger_\alpha \hat{c}_\beta \langle \hat{c}^\dagger_\beta \hat{c}_\alpha\rangle - \langle \hat{c}^\dagger_\alpha \hat{c}_\beta\rangle \hat{c}^\dagger_\beta \hat{c}_\alpha + \langle \hat{c}^\dagger_\alpha \hat{c}_\beta\rangle\langle \hat{c}^\dagger_\beta \hat{c}_\alpha\rangle && \text{(Fock)} \\
&\qquad + \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \langle \hat{c}_\beta \hat{c}_\alpha\rangle + \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta\rangle \hat{c}_\beta \hat{c}_\alpha - \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta\rangle\langle \hat{c}_\beta \hat{c}_\alpha\rangle\Big) && \text{(Cooper)}
\end{aligned}
$$

Passing from the first row to the following ones, we decomposed the original quartic operator to a sum of three decomposition much similar to Eq. (C.1). This is the main connection with MFT. In HF theory, we do not write the quartic operator as a product of two quadratic operators and decompose it as we would do in MFT. Instead, a very similar decomposition (summed over the three sectors) is given by the fact that EVs are computed over HF states.

Appendix D

Antiferromagnetism in the HM

In this appendix we give some details about the antiferromagnetic phase in the conventional Hubbard model,

$$\hat{H} = -t\sum_{\langle ij\rangle}\sum_{\sigma}\left[\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma} + \text{h.c.}\right] + U\sum_{i}\hat{n}_{i\uparrow}\hat{n}_{i\downarrow} \qquad t, U > 0 \tag{D.1}$$

approximated by HF methods. We use the results discussed in Sec. 5.2.3.

Let the Nambu spinors:

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix}\hat{c}_{\mathbf{k}\sigma}\\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\end{bmatrix}$$

Let also the gap function:

$$\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$$

We know from Eq. (5.29) that the approximated hamiltonian is:

$$\hat{H}_t + \hat{H}_U \simeq \sum_{\mathbf{k}\in\text{MBZ}}\sum_{\sigma}\mathbf{b}_{\mathbf{k}\sigma}\cdot\hat{\mathbf{s}}_{\mathbf{k}\sigma} + nU\hat{N} - E^{(\text{AF})}_{\text{H}/U}$$

where:

$$\mathbf{b}_{\mathbf{k}\sigma} \equiv \begin{bmatrix}-\Delta_{\mathbf{k}\sigma}\\ 0\\ \epsilon_{\mathbf{k}\sigma}\end{bmatrix} \qquad \hat{s}^{\alpha}_{\mathbf{k}\sigma} \equiv \hat{\Psi}^{\dagger}_{\mathbf{k}\sigma}\tau^{\alpha}\hat{\Psi}_{\mathbf{k}\sigma} \qquad \hat{\mathbf{s}}_{\mathbf{k}\sigma} \equiv \begin{bmatrix}\hat{s}^{x}_{\mathbf{k}\sigma}\\ \hat{s}^{y}_{\mathbf{k}\sigma}\\ \hat{s}^{z}_{\mathbf{k}\sigma}\end{bmatrix}$$

Through the unitary matrix $W_{\mathbf{k}\sigma}$ given in Eq. (5.31), the hamiltonian is mapped to a gas of non-interacting fermions collected in the $\hat{\Gamma}$ spinors of Eq. (5.33),

$$\hat{\Gamma}_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma}\hat{\Psi}_{\mathbf{k}\sigma} = \begin{bmatrix}\hat{\gamma}^{(+)}_{\mathbf{k}\sigma}\\ \hat{\gamma}^{(-)}_{\mathbf{k}\sigma}\end{bmatrix}$$

The final non-interacting hamiltonian is:

$$\hat{H}_t + \hat{H}_U \simeq \sum_{\mathbf{k}\in\text{MBZ}}\sum_{\sigma}E_{\mathbf{k}\sigma}\left(\hat{\gamma}^{(+)\dagger}_{\mathbf{k}\sigma}\hat{\gamma}^{(+)}_{\mathbf{k}\sigma} - \hat{\gamma}^{(-)\dagger}_{\mathbf{k}\sigma}\hat{\gamma}^{(-)}_{\mathbf{k}\sigma}\right) + nU\hat{N} - E^{(\text{AF})}_{\text{H}/U}$$

with $E_{\mathbf{k}\sigma}$ the Bogoliubov bands of Eq. (5.32). In next section we give the explicit form of the many-body ground-state at zero temperature and half-filling. From now on, limitedly to this appendix, we use the simplified notation:

$$\Delta_{\mathbf{k}\uparrow} \quad\to\quad \Delta$$

with $\Delta_{\mathbf{k}\uparrow} = -\Delta_{\mathbf{k}\downarrow}$, and $\Delta = mU$.

D.1 Explicit AF ground state at half-filling and $\beta\to\infty$

Let $|\Uparrow_{\mathbf{k}\sigma}\rangle$ and $|\Downarrow_{\mathbf{k}\sigma}\rangle$ be respectively the "up" and "down" $\mathbf{k}$-th pseudo-spin states. Let $|\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle$ be the "down" state in the tilted frame. The ground state is given by

$$|\Psi\rangle \equiv \bigotimes_{\sigma}\bigotimes_{\mathbf{k}\in\text{MBZ}}|\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle \tag{D.2}$$

We can link the "down" states via an inverse rotation,

$$\begin{aligned}|\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle &= e^{-i\theta_{\mathbf{k}\sigma}\hat{s}^y_{\mathbf{k}\sigma}}\,|\Downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos\theta_{\mathbf{k}\sigma}\,|\Downarrow_{\mathbf{k}\sigma}\rangle - i\sin\theta_{\mathbf{k}\sigma}\hat{s}^y_{\mathbf{k}\sigma}\,|\Downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos\theta_{\mathbf{k}\sigma}\,|\Downarrow_{\mathbf{k}\sigma}\rangle - \sin\theta_{\mathbf{k}\sigma}(\hat{s}^+_{\mathbf{k}\sigma} - \hat{s}^-_{\mathbf{k}\sigma})\,|\Downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos\theta_{\mathbf{k}\sigma}\,|\Downarrow_{\mathbf{k}\sigma}\rangle - \sin\theta_{\mathbf{k}\sigma}\,|\Uparrow_{\mathbf{k}\sigma}\rangle \end{aligned} \tag{D.3}$$

having we used

$$\hat{s}^{\pm}_{\mathbf{k}\sigma} \equiv \frac{\hat{s}^x_{\mathbf{k}\sigma} \pm i\hat{s}^y_{\mathbf{k}\sigma}}{2}$$

Use now Eqns. (5.28) and (5.27),

$$\frac{\hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}^z_{\mathbf{k}\sigma}}{2} = \hat{n}_{\mathbf{k}\sigma} \qquad \frac{\hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}^z_{\mathbf{k}\sigma}}{2} = \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}$$

Evaluating the above operators on the "up" and "down" states, we get

$$\begin{aligned}\frac{1}{2}\langle\Uparrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}^z_{\mathbf{k}\sigma}|\Uparrow_{\mathbf{k}\sigma}\rangle = 1 &\qquad \frac{1}{2}\langle\Uparrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}^z_{\mathbf{k}\sigma}|\Uparrow_{\mathbf{k}\sigma}\rangle = 0 \\ \frac{1}{2}\langle\Downarrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}^z_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle = 0 &\qquad \frac{1}{2}\langle\Downarrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}^z_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle = 1\end{aligned}$$

Thus, up to a phase

$$|\Uparrow_{\mathbf{k}\sigma}\rangle \sim \hat{c}^\dagger_{\mathbf{k}\sigma}\,|\Omega_{\mathbf{k}\sigma}\rangle \qquad |\Downarrow_{\mathbf{k}\sigma}\rangle \sim \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\,|\Omega_{\mathbf{k}\sigma}\rangle$$

being $|\Omega_{\mathbf{k}\sigma}\rangle$ the $(\mathbf{k},\sigma)$ Hilbert subspace vacuum.

It follows that composing Eqns. (E.26) and (E.27) the ground state $|\Psi\rangle$ can be written as:

$$|\Psi\rangle = \bigotimes_{\sigma}\bigotimes_{\mathbf{k}\in\mathrm{MBZ}}\left[\sin\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}\sigma} + e^{2i\zeta_{\mathbf{k}\sigma}}\cos\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]|\Omega_{\mathbf{k}\sigma}\rangle \tag{D.4}$$

having we absorbed in the relative phase $e^{2i\zeta_{\mathbf{k}\sigma}}$ the two states phases differences as well as the original sign difference of Eq. (E.27). Now, inserting the explicit form of the ground state from equation above:

$$\begin{aligned}\langle\overline{\Downarrow}_{\mathbf{k}\sigma}|\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^\dagger_{\mathbf{k}\sigma}|\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle &= \left(e^{2i\zeta_{\mathbf{k}\sigma}} + e^{-2i\zeta_{\mathbf{k}\sigma}}\right)\sin\theta_{\mathbf{k}\sigma}\cos\theta_{\mathbf{k}\sigma} \\ &= \cos 2\zeta_{\mathbf{k}\sigma}\sin 2\theta_{\mathbf{k}\sigma}\end{aligned}$$

but from Eq. (5.25)

$$\begin{aligned}\langle\overline{\Downarrow}_{\mathbf{k}\sigma}|\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^\dagger_{\mathbf{k}\sigma}|\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle &= \langle\overline{\Downarrow}_{\mathbf{k}\sigma}|\hat{s}^x_{\mathbf{k}\sigma}|\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle \\ &= \sin 2\theta_{\mathbf{k}\sigma}\end{aligned}$$

since the pseudofield has no y component. This implies necessarily

$$\cos 2\zeta_{\mathbf{k}\sigma} \quad\Longrightarrow\quad \zeta_{\mathbf{k}\sigma} = k\pi \quad \text{with} \quad k\in\mathbb{Z}$$

and lets us conclude from Eq. (E.28)

$$|\Psi\rangle = \bigotimes_{\sigma}\bigotimes_{\mathbf{k}\in\mathrm{MBZ}}\left[\sin\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}\sigma} + \cos\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]|\Omega_{\mathbf{k}\sigma}\rangle \tag{D.5}$$

For low-temperature physics, we can expect the ground-state density matrix to be highly dominated by this BCS-like zero-temperature ground-state, which highlights the essential features of the system.

Finally, using Eq. (5.33), we check that correctly the ground-state describes a perfect degenerate $\gamma^{(-)}$ quasiparticles Fermi gas,

$$|\Psi\rangle = \bigotimes_{\sigma}\bigotimes_{\mathbf{k}\in\mathrm{MBZ}}\gamma^{(-)\dagger}_{\mathbf{k}\sigma}\,|\Omega_{\mathbf{k}\sigma}\rangle \tag{D.6}$$

The AF phase can be thought of as a long-range coordination between fermionic states related by a $\boldsymbol{\pi}$ shift in reciprocal space, producing a new free gas where all interactions have been "condensed" in these AF pairs.

D.2 Instability of the commensurate AF phase

The commensurate AF phase is unstable at any finite doping. This result is discussed in [1, 2], and here we recall just the main passages. Consider the local spin operator defined in Eq. (3.7),

$$\hat{S}_j^\alpha = \frac{1}{2}\hat{\Psi}_j^\dagger \tau^\alpha \hat{\Psi}_j \qquad \hat{S}_j^+ \equiv \frac{\hat{S}_j^x + i\hat{S}_j^y}{2} \qquad \hat{S}_j^- \equiv \frac{\hat{S}_j^x - i\hat{S}_j^y}{2}$$

Let us take a continuum limit, $L_x \to \infty$ and $L_y \to \infty$, while the lattice spacing goes to zero.

Continuous SSBs are connected to the presence of a stable gapless Golstone mode. Such a Goldstone mode is essentially a sharp peak at zero frequency for some dressed response function. For the commensurate AF phase, the low-energy collective excitations are transverse spin fluctuations about the antiferromagnetic ground-state, describing SDWs. In order to see if the Golstone mode is stable for a given doping δ, we need to investigate whether the respective response function diverges at zero frequency. If it does, we have a gapless Goldstone mode and the phase is stable; if not, the phase is unstable. We work at $\omega = 0$ in order to capture gapless modes.

Consider the interaction representation of spin operators. Our new local spin operators are given by

$$\hat{\mathbf{S}}(\mathbf{r}, t)$$

The retarded transverse spin-spin response is given by:

$$\chi^{-+}(\mathbf{r}, t; \mathbf{r}', t') =\equiv i\mathrm{H}(t - t') \langle \Psi | \left[\hat{S}^-(\mathbf{r}, t), \hat{S}^+(\mathbf{r}', t') \right] | \Psi \rangle$$

being $\mathrm{H}(x)$ the Heaviside function. The argument carried by Singh and Tešanović relies on random phase approximation (RPA). Let χ_0^{-+} be the non-interacting transverse response function. Then

$$\chi^{-+} \simeq \frac{\chi_0^{-+}}{1 - U\chi_0^{-+}} \quad \text{(RPA)}$$

We move to momentum-frequency space. Within RPA approximation, low-lying spin waves collective excitations at wavevector $\mathbf{q}$ give a stable gapless Goldstone mode if:

$$1 - U\chi_0^{-+}(\mathbf{q}, 0) = 0 \tag{D.7}$$

Commensurate antiferromagnetism is stable if the above equation is satisfied at $\mathbf{q} \to \boldsymbol{\pi}$. Indeed, the wavevector $\boldsymbol{\pi}$ expresses two-sites periodicity in both directions.

Now, as we specified in Sec. 5.1, commensurate antiferromagnetism is described by the simultaneous SSB:

$$\mathrm{SU}(2) \overset{\mathrm{SSB}}{\to} \mathrm{U}^z(1)$$
$$\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)} \overset{\mathrm{SSB}}{\to} \mathrm{T}_2^{(x)} \otimes \mathrm{T}_2^{(y)}$$

Thus translational invariance is halved in both directions. The lattice $\mathcal{L}$ is separated in the two sublattices $\mathcal{S}_1$, $\mathcal{S}_2$ of Fig. 3.1. As a consequence, the non-interacting response function χ_0^{+-} does not depend strictly on $\mathbf{r}$, $\mathbf{r}'$ themselves. Instead, it depends only on which sublattice $\mathbf{r}$ and $\mathbf{r}'$ belong to. Because of this χ_0^{+-} is representable as a 2×2 matrix $\boldsymbol{\chi}_0^{+-}$. Then Eq. (D.7) is actually an eigenvalue equation,

$$\mathbb{I} - U\boldsymbol{\chi}_0^{+-}(\mathbf{q}, 0) = \mathbf{0}$$

with $\mathbf{0}$ the 2×2 zero matrix. Diagonalising $\boldsymbol{\chi}_0^{+-}(\mathbf{q}, 0)$ (which is symmetric due to lattice exchange symmetry) we can express the above equation in terms of the diagonal matrix containing its eigenvalues.

Consider the largest eigenvalue $\lambda_\delta(\mathbf{k}, \omega)$ of the matrix $[\chi_0^{+-}]$ at doping δ. We work in momentum-frequency space. The phase is unstable if

$$1 - U\lambda_\delta \neq 0 \tag{D.8}$$

because, as we said, in that case we do not have a pole for the transverse spin-spin response function. Singh and Tešanović [1] found

$$\lambda_0(\mathbf{k} \to \boldsymbol{\pi}, 0) = \frac{1}{U} \qquad \text{(half filling)} \tag{D.9}$$

$$\lambda_\delta(\mathbf{k} \to \boldsymbol{\pi}, 0) = \frac{1}{U} + \frac{2\epsilon_\mathrm{F}}{\Delta} N(\epsilon_\mathrm{F}) \qquad \text{(finite doping)} \tag{D.10}$$

The reason for the distinction above is the intrinsic metallic nature of the system for infinitesimal doping. The half-filled AF phase is insulating, because the chemical potential lies within the gap. For an insulator Fermi energy is not defined. Instead, at finite filling the phase is metallic because the chemical potential crosses the upper band.

From Eq. (D.8) and (D.9), we see that the half filled antiferromagnet is stable towards transverse spin fluctuations, because χ^{+-} develops a divergence at zero frequency. At any finite doping, since we are describing a metal, a FS exists and thus $N(\epsilon_\mathrm{F})$ is finite. This makes the phase unstable, and this is the reason we do not discuss the AF phase outside of half-filling.

As we discussed at the beginning of Chap. 5, antiferromagnetism can establish in many ways. For instance, striped antiferromagnets are of particular interest [3]. In order to investigate the most stable physical realisation of an antiferromagnet on the conventional HM, on a pure HF level, we should:

1. Choose an initialiser wavevector, say $\mathbf{k} = \boldsymbol{\pi}$;
2. Solve the self-consistency equations for the HFPs vector for an antiferromagnet oscillating at wavevector $\mathbf{k}$;
3. Compute the free energy F as a functional of the wavevector around $\mathbf{k}$;
4. Look for gradient descent of the free energy, $\boldsymbol{\nabla} F < 0$;
5. If the gradient descent is found, update $\mathbf{k} \to \mathbf{k} + \delta\mathbf{k}$ and repeat from step 2, otherwise halt.

The sketched procedure can lead to a lot of complications. For instance, the fact that at incommensurate wavevector $\mathbf{k} \neq \boldsymbol{\pi}$ the unit cell is not as simple as a pair of sites, but can become much larger requiring more refined computational solutions than our pseudospin representation of the HM hamiltonian in the commensurate AF channel.

[Extend to EHM: is it true that:

$$\chi^{-+} \simeq \frac{\chi_0^{-+}}{1 - [U - 2V(\cos q_x + \cos q_y)]\chi_0^{-+}} \qquad \text{(RPA)}$$

?]

D.3 iHF results

For the conventional HM, antiferromagnetism is described by a single HFP, the magnetisation m. Its self-consistency Eq. (5.39) reads:

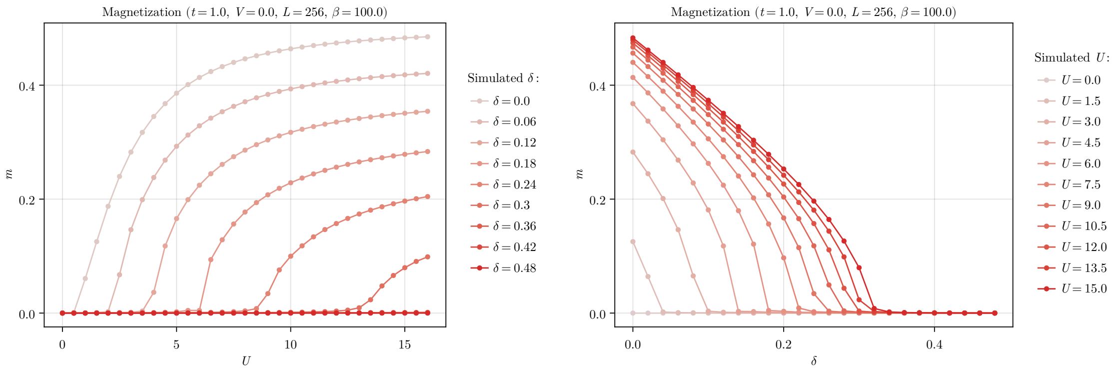
$$m_{i+1} = \frac{U}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{MBZ}} \frac{m_i}{2E_\mathbf{k}[\Delta_i]} \left[f(-E_\mathbf{k}[m_i]; \beta, \mu) - f(E_\mathbf{k}[m_i]; \beta, \mu) \right] \tag{D.11}$$

where we indicated as m_i the value of magnetisation at step i. We ran iHF simulations in the ranges:

$$0 \leq U \leq 16 \qquad 0 \leq \delta \leq 0.48$$

We considered also $\delta > 0$, although – as we specified in previous section – a doped commensurate antiferromagnet is actually unstable.

In Fig. D.1 magnetisation m at $\beta = 100$ is plotted. As is visible in Fig. D.1a, ignoring instability issues, doping disrupts AF ordering; the horizontal asymptote lowers as δ gets bigger. Also, for



a | Local repulsion U/t depencence.

b | Doping $\delta = n - 0.5$ dependence.

Figure D.1 | Plots of the zero-temperature AF instability order parameter m as a function of both the local repulsion U/t (Fig. D.1a) and the doping $\delta = n - 1/2$ (Fig. D.1b).

$\delta > 0$, a finite magnetisation m exists only for $U \geq U_c[\delta]$ (a critical value parametrically dependent on doping).

Fig. D.1b shows the dependence of the magnetisation on doping varying the local repulsion U. Identical analysis have been performed for different temperatures, leading to analogous results as long as $\beta \gtrsim 10$ and to $m \simeq 0$ for higher temperatures.

D.4 Analytic expression of self-consistent magnetisation

In this subsection we provide an analytic expression to compute m self-consistently at half-filling. Multiply Eq. (5.39) on both sides by U. Since due to symmetry $\mu = 0$, we have the finite-temperature self-consistent equation

$$\Delta = \frac{U}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{MBZ}} \frac{\Delta}{\sqrt{\epsilon_{\mathbf{k}}^2 + \Delta^2}} \tanh\left(\frac{\beta}{2}\sqrt{\epsilon_{\mathbf{k}}^2 + \Delta^2}\right)$$

where we used the relation

$$\mathrm{Th}_{\mathbf{k}}^{(-)}[\beta, 0] = \tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right)$$

In thermodynamic limit $L \to \infty$

$$\frac{1}{U} = \frac{1}{2} \int_{\mathrm{BZ}} \frac{\mathrm{d}\mathbf{k}}{(2\pi)^2} \frac{1}{\sqrt{\epsilon_{\mathbf{k}}^2 + \Delta^2}} \tanh\left(\frac{\beta}{2}\sqrt{\epsilon_{\mathbf{k}}^2 + \Delta^2}\right)$$

Substitute the momentum integral with an integral over energies:

$$\frac{2}{U} = \int_{-4t}^{4t} \frac{\mathrm{d}\epsilon \rho(\epsilon)}{\sqrt{\epsilon^2 + \Delta^2}} \tanh\left(\frac{\beta}{2}\sqrt{\epsilon^2 + \Delta^2}\right)$$

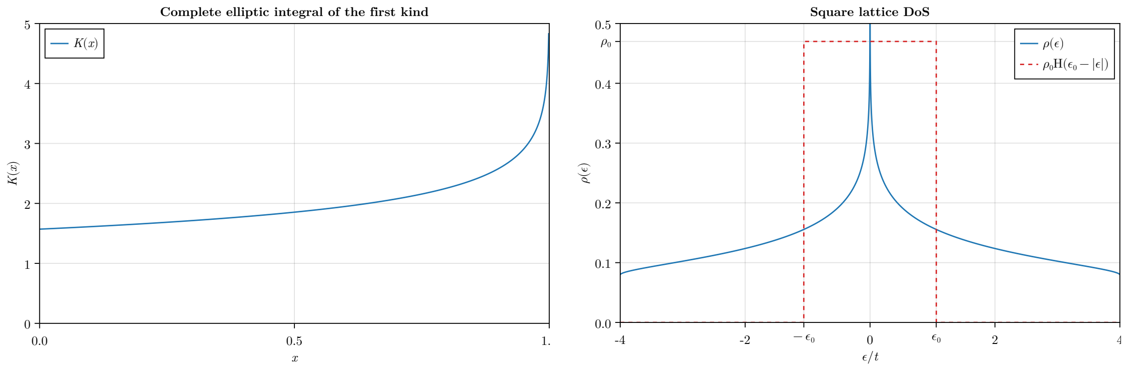
being $\rho(\epsilon)$ the 2D square lattice DoS. Its explicit form is given by [4]

$$\rho(\epsilon) = \frac{1}{2\pi^2} K\left(\sqrt{1 - \left(\frac{\epsilon}{4t}\right)^2}\right) \qquad K(x) \equiv \int_0^{\pi/2} \frac{\mathrm{d}\theta}{\sqrt{1 - x^2 \sin^2\theta}}$$

$K(x)$ is the complete elliptic integral of the first kind, shown in Fig. D.2a.

As is plotted in Fig. D.2b, the resulting DoS is logarithmically peaked at $\epsilon = 0$, on the MBZ. An extremely rough but common approximation is to substitute the actual DoS with a step function,

$$\rho(\epsilon) \simeq \rho_0 \mathrm{H}\left(\epsilon_0 - |\epsilon|\right) \quad \text{with} \quad 2\rho_0\epsilon_0 = 1$$



a | Plot of $K(x)$.   b | DoS and its approximation.

Figure D.2 | Plots of the complete elliptic integral of the first kind $K(x)$ and the density of states for the planar square lattice.

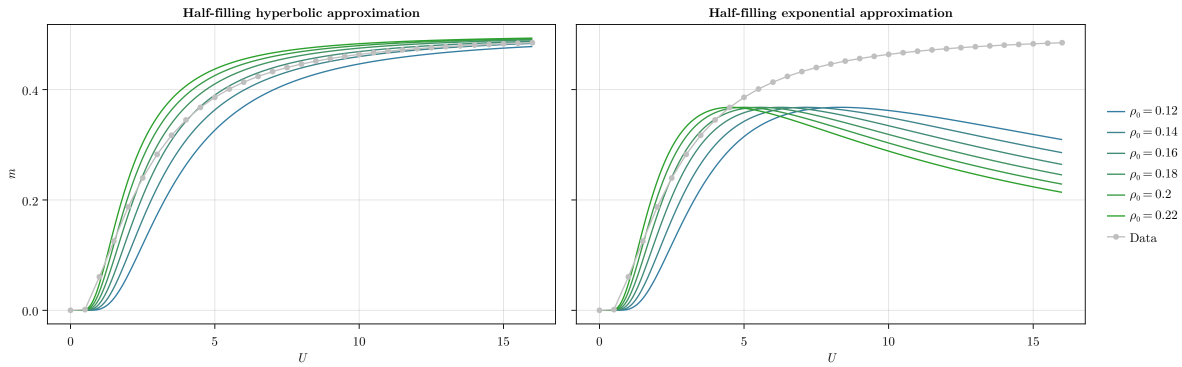


Figure D.3 | Sketch of the analytic expression for m. The left panel shows a family of coloured curves described by Eq. (D.12). The right panel shows a family of coloured curves described by Eq. (D.13), only reasonable at $U \to 0$.

Here, the peak parameters ϵ_0 and ρ_0 are chosen in order to maintain a correct DoS integral, considering also the 2 pre-factor of spin degeneracy. Within this extremely rough approximation, we have

$$\frac{2}{U} \simeq \rho_0 \int_{-\epsilon_0}^{\epsilon_0} \frac{\mathrm{d}\epsilon}{\sqrt{\epsilon^2 + \Delta^2}} \tanh\left(\frac{\beta}{2}\sqrt{\epsilon^2 + \Delta^2}\right)$$

Suppose the system is magnetized, $\Delta > 0$. At low temperature, $\beta \gg 1$, as a first approximation we can substitute the hyperbolic tangent with unity, obtaining

$$\begin{aligned}
\frac{2}{\rho_0 U} &\simeq \int_{-\epsilon_0}^{\epsilon_0} \frac{\mathrm{d}\epsilon}{\sqrt{\epsilon^2 + \Delta^2}} \\
&= \int_{-\epsilon_0/\Delta}^{\epsilon_0/\Delta} \frac{\mathrm{d}x}{\sqrt{1 + x^2}} \\
&= \sinh^{-1}\left(\frac{\epsilon_0}{\Delta}\right) - \sinh^{-1}\left(-\frac{\epsilon_0}{\Delta}\right) \\
&= 2\sinh^{-1}\left(\frac{\epsilon_0}{\Delta}\right)
\end{aligned}$$

thus

$$m \simeq \frac{\epsilon_0}{U \sinh(1/\rho_0 U)} \tag{D.12}$$

having we used $\Delta = mU$. In the left panel of Fig. D.3 we sketch this function of U varying ρ_0, ϵ_0. The same data of Fig. D.1a, $\delta = 0$ are plotted as gray dots. The shown curves are not fits.

Now: at large U,

$$\lim_{U \to +\infty} m \simeq \epsilon_0 \rho_0 = \frac{1}{2}$$

which gives the correct saturation limit at high local repulsion. In the opposite limit,

$$
\begin{aligned}
\lim_{U\to 0} m &\simeq \lim_{U\to 0} \frac{2\epsilon_0/U}{\exp(1/\rho_0 U) - \exp(-1/\rho_0 U)} \\
&\simeq \lim_{U\to 0} \frac{2\epsilon_0}{U} \exp\left(-\frac{1}{\rho_0 U}\right)
\end{aligned} \tag{D.13}
$$

giving a near-0 exponential suppression of magnetisation, typical of MFT self-consistent solutions. In the right panel of Fig. D.3 we sketch this approximation.

Appendix E

Mean-Field Theory in Hubbard lattices

In this Appendix the MFT solutions to the conventional Hubbard hamiltonian of Eq. (1.1),

$$\hat{H} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \left[\hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] + U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \qquad t, U > 0 \tag{E.1}$$

are described. The discussion is limited to the two-dimensional square lattice. The results of ths appendix are relevant and used throughout the text as the groundwork to be extended by the means of MFT applied to the non-local part of the EHM.

E.1 Antiferromagnetic phase

We work in a commensurate AF phase ordering, which is, we explicitly break symmetry by realizing a SDW phase with spin density unevenly distributed on sites with periodicity of two sites, each direction. Recalling the discussion of Sec. **??**, AF long-range ordering we deal with here is described by the composite SSB

$$\begin{gathered} \mathrm{SU}(2) \overset{\mathrm{SSB}}{\to} \mathrm{U}^{z}(1) \\ \mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)} \overset{\mathrm{SSB}}{\to} \mathrm{T}_2^{(x)} \otimes \mathrm{T}_2^{(y)} \end{gathered}$$

Recalling Tab. **??**, for such a physical phase the only Wick contraction channel available is the Hartree one. Let us proceed in such sense. As in Eqns. (5.7), (5.8), it is useful to define the density and AF fluctuation operators:

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \tag{E.2}$$

$$\hat{\psi}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{k}\sigma} \tag{E.3}$$

whose EVs are relevant order parameters for the normal and AF phases.

E.1.1 MFT treatment of the Hartree channel

In terms of the physical magnetization m, the MFT Ansatz is given by

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n - (-1)^{x+y+\delta_{\sigma=\uparrow}} m \tag{E.4}$$

Decoupling the HM hamiltonian and substituting,

$$\begin{aligned} \hat{H} &\simeq -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \left[\hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] + U \sum_{\mathbf{r}} \left[\hat{n}_{\mathbf{r}\uparrow} \langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \hat{n}_{\mathbf{r}\downarrow} - \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle \right] \\ &= -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \left[\hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] + nU \sum_{\mathbf{r}} \left[\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow} \right] - mU \sum_{\mathbf{r}} (-1)^{x+y} \left[\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow} \right] - E^{(\mathrm{AF})}_{\mathrm{H}/U} \end{aligned} \tag{E.5}$$

with $E^{(\mathrm{AF})}_{\mathrm{H}/U}$ the HM Hartree contraction energy,

$$\begin{aligned}E^{(\mathrm{AF})}_{\mathrm{H}/U} &\equiv U\sum_{\mathbf{r}}\langle\hat{n}_{\mathbf{r}\uparrow}\rangle\langle\hat{n}_{\mathbf{r}\downarrow}\rangle\\ &= U\sum_{\mathbf{r}}\left(n+(-1)^{x+y}m\right)\left(n-(-1)^{x+y}m\right)\\ &= L_xL_y\times U\left(n^2-m^2\right)\end{aligned}\tag{E.6}$$

By a Fourier transform, the phase factor of Eq. (E.5) can be absorbed in the destruction operator inside of $\hat{n}_{\mathbf{r}\sigma}$:

$$\begin{aligned}\sum_{\mathbf{r}}(-1)^{x+y}\hat{n}_{\mathbf{r}\sigma} &= \sum_{\mathbf{r}}(-1)^{x+y}\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma}\\ &\overset{\mathcal{F}}{=}\sum_{\mathbf{r}}e^{i\boldsymbol{\pi}\cdot\mathbf{r}}\frac{1}{\sqrt{L_xL_y}}\sum_{\mathbf{k}\in\mathrm{BZ}}e^{i\mathbf{k}\cdot\mathbf{r}}\hat{c}^\dagger_{\mathbf{k}\sigma}\frac{1}{\sqrt{L_xL_y}}\sum_{\mathbf{k}'\in\mathrm{BZ}}e^{-i\mathbf{k}'\cdot\mathbf{r}}\hat{c}_{\mathbf{k}'\sigma}\\ &= \sum_{\mathbf{k}\in\mathrm{BZ}}\sum_{\mathbf{k}'\in\mathrm{BZ}}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma}\frac{1}{L_xL_y}\sum_{\mathbf{r}}e^{-i[\mathbf{k}'-(\mathbf{k}+\boldsymbol{\pi})]\cdot\mathbf{r}}\\ &= \sum_{\mathbf{k}\in\mathrm{BZ}}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\end{aligned}\tag{E.7}$$

where $\boldsymbol{\pi}=(\pi,\pi)$. It follows:

$$mU\sum_{\mathbf{r}}(-1)^{x+y}\left[\hat{n}_{\mathbf{r}\uparrow}-\hat{n}_{\mathbf{r}\downarrow}\right] = mU\sum_{\mathbf{k}\in\mathrm{BZ}}\left[\hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\uparrow}-\hat{c}^\dagger_{\mathbf{k}\downarrow}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\downarrow}\right]$$

Note that the Hartree channel renormalizes chemical potential,

$$nU\sum_{\mathbf{r}}\left[\hat{n}_{\mathbf{r}\uparrow}+\hat{n}_{\mathbf{r}\downarrow}\right] = nU\times\hat{N}\qquad\Longrightarrow\qquad\mu\to\tilde{\mu}\equiv\mu-nU$$

an effect that is present also in the EHM. The kinetic part of the hamiltonian transforms as in Eq. (3.28),

$$-t\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}'\sigma}+\mathrm{h.c.}\right]\overset{\mathcal{F}}{=}\sum_{\mathbf{k}\in\mathrm{BZ}}\sum_{\sigma}\epsilon_{\mathbf{k}}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}$$

being $\epsilon_{\mathbf{k}}$ the bare bands of Eq. (3.29).

Take everything together. Using the property:

$$\begin{aligned}\sum_{\mathbf{k}\in\mathrm{BZ}}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} &= \sum_{\mathbf{k}\in\mathrm{MBZ}}\left[\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}+\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}+2\boldsymbol{\pi}\sigma}\right]\\ &= \sum_{\mathbf{k}\in\mathrm{MBZ}}\left[\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}+\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}\sigma}\right]\end{aligned}\tag{E.8}$$

and reduce Eq. (E.5) to its reciprocal form

$$\hat{H}\simeq\sum_{\mathbf{k}\in\mathrm{MBZ}}\sum_{\sigma}\epsilon_{\mathbf{k}}\left[\hat{n}_{\mathbf{k}\sigma}+\hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]-mU\sum_{\mathbf{k}\in\mathrm{MBZ}}\sum_{\sigma}(-1)^{\delta_{\sigma=\downarrow}}\left[\hat{\psi}_{\mathbf{k}\sigma}+\hat{\psi}^\dagger_{\mathbf{k}\sigma}\right]+nU\hat{N}-E^{(\mathrm{AF})}_{\mathrm{H}/U}$$

E.1.2 More on AF Ansatz

Before moving on, let us give a little detail on the AF Ansatz. The decoupling in Hartree sector of the density-density interaction can be treated with more care: denoting by d.c. the double counting terms we ignore,

$$\begin{aligned}\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\downarrow}-\mathrm{d.c.} &= \hat{n}_{\mathbf{r}\uparrow}\langle\hat{n}_{\mathbf{r}\downarrow}\rangle+\langle\hat{n}_{\mathbf{r}\uparrow}\rangle\hat{n}_{\mathbf{r}\downarrow}\\ &= \hat{n}_{\mathbf{r}\uparrow}\frac{\langle\hat{n}_{\mathbf{r}\downarrow}\rangle+\langle\hat{n}_{\mathbf{r}\uparrow}\rangle}{2}+\hat{n}_{\mathbf{r}\uparrow}\frac{\langle\hat{n}_{\mathbf{r}\downarrow}\rangle-\langle\hat{n}_{\mathbf{r}\uparrow}\rangle}{2}+\frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle+\langle\hat{n}_{\mathbf{r}\uparrow}\rangle}{2}\hat{n}_{\mathbf{r}\downarrow}+\frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle-\langle\hat{n}_{\mathbf{r}\uparrow}\rangle}{2}\hat{n}_{\mathbf{r}\downarrow}\\ &= \frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle+\langle\hat{n}_{\mathbf{r}\downarrow}\rangle}{2}\left(\hat{n}_{\mathbf{r}\uparrow}+\hat{n}_{\mathbf{r}\downarrow}\right)-\frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle-\langle\hat{n}_{\mathbf{r}\downarrow}\rangle}{2}\left(\hat{n}_{\mathbf{r}\uparrow}-\hat{n}_{\mathbf{r}\downarrow}\right)\end{aligned}$$

Our commensurate AF Ansatz essentially states that each site has the same spin-averaged density,

$$n \equiv \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \qquad \text{is } \mathbf{r}\text{-independent}$$

while the spin averaged unbalance has a two-sites periodicity

$$m \equiv (-1)^{x+y} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \qquad \text{is } \mathbf{r}\text{-independent}$$

This way we reduce translational invariance. Of course, if the two quantities above are **r**-independent indeed, it must hold

$$n = \frac{1}{L_x L_y} \sum_{\mathbf{r}} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \tag{E.9}$$

$$m = \frac{1}{L_x L_y} \sum_{\mathbf{r}} (-1)^{x+y} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \tag{E.10}$$

While from the first equation we gain an operational method to compute density, from the second we get a magnetization self-consistency equation.

E.1.3 Nambu representation of the AF phase and diagonalization

We now map the reciprocal version of the HM onto a simpler problem. Throughout this section we use the notation $\epsilon_{\mathbf{k}} \to \epsilon_{\mathbf{k}\sigma}$ with identical meaning. Let the Nambu spinors:

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix}$$

Let also the gap function:

$$\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$$

which gives the matrix decomposition

$$\hat{H} \simeq \sum_{\mathbf{k}\in \mathrm{MBZ}} \sum_{\sigma} \hat{\Psi}^{\dagger}_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} - E^{(\mathrm{AF})}_{\mathrm{H}/U} \quad \text{being} \quad h_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \epsilon_{\mathbf{k}\sigma} & -\Delta^*_{\mathbf{k}\sigma} \\ -\Delta_{\mathbf{k}\sigma} & -\epsilon_{\mathbf{k}\sigma} \end{bmatrix} \tag{E.11}$$

Notice: the Nambu hamiltonian is a 2×2 matrix over the MBZ – which is half the full BZ, coherently with a solution which essentially bipartites the lattice giving back a double sized unit cell.

The system ground-state is obtained by the means of a Bogoliubov rotation. The hamiltonian maps onto the simple one of a spin in a magnetic field (the gap is purely real here),

$$h_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma} \tau^z - \Delta_{\mathbf{k}\sigma} \tau^x$$

being τ^α the Pauli matrices. Then, defining

$$\hat{s}^{\alpha}_{\mathbf{k}\sigma} \equiv \hat{\Psi}^{\dagger}_{\mathbf{k}\sigma} \tau^\alpha \hat{\Psi}_{\mathbf{k}\sigma} \quad \text{and} \quad \mathbf{b}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} -\Delta_{\mathbf{k}\sigma} \\ 0 \\ \epsilon_{\mathbf{k}\sigma} \end{bmatrix}$$

one gets:

$$\hat{H} \simeq \sum_{\mathbf{k}\in \mathrm{MBZ}} \sum_{\sigma} \mathbf{b}_{\mathbf{k}\sigma} \cdot \hat{\mathbf{s}}_{\mathbf{k}\sigma} - E^{(\mathrm{AF})}_{\mathrm{H}/U} \quad \text{where} \quad \hat{\mathbf{s}}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{s}^x_{\mathbf{k}\sigma} \\ \hat{s}^y_{\mathbf{k}\sigma} \\ \hat{s}^z_{\mathbf{k}\sigma} \end{bmatrix} \tag{E.12}$$

The hamiltonian represents a system of spins subject to local magnetic fields, each tilted by an angle $\tan 2\theta_{\mathbf{k}\sigma} = \Delta_{\mathbf{k}\sigma}/\epsilon_{\mathbf{k}\sigma}$, as sketched in Fig. E.1. Evidently the following relations, also reported in Tab. 5.2, hold:

$$\hat{s}^x_{\mathbf{k}\sigma} = \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^{\dagger}_{\mathbf{k}\sigma} \tag{E.13}$$

$$\hat{s}^y_{\mathbf{k}\sigma} = i\left(\hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}^{\dagger}_{\mathbf{k}\sigma}\right) \tag{E.14}$$

$$\hat{s}^z_{\mathbf{k}\sigma} = \hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \tag{E.15}$$

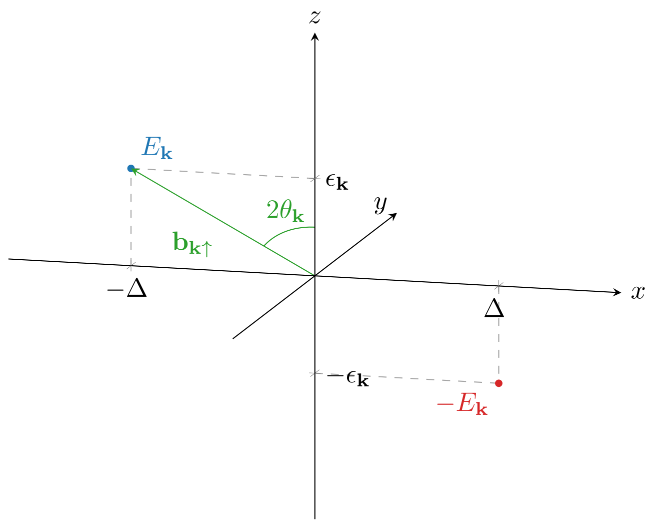


Figure E.1 | Pseudo-magnetic field originating from mean-field treatment of the square Hubbard hamiltonian. Here, only the $\sigma = \uparrow$ situation is plotted.

Let also:

$$\begin{aligned} \hat{\mathbb{I}}_{\mathbf{k}\sigma} &\equiv \hat{\Psi}^{\dagger}_{\mathbf{k}\sigma} \mathbb{I}_{2\times 2} \hat{\Psi}_{\mathbf{k}\sigma} \\ &= \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{aligned} \tag{E.16}$$

Diagonalization of each $h_{\mathbf{k}\sigma}$ is obtained trivially by a simple rotation around the y axis:

$$d_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} W^{\dagger}_{\mathbf{k}\sigma} \quad \text{with} \quad d_{\mathbf{k}\sigma} \equiv \begin{bmatrix} E_{\mathbf{k}\sigma} & \\ & -E_{\mathbf{k}\sigma} \end{bmatrix}$$

and where the diagonalization unitary matrix is simply a rotation by an angle $2\theta_{\mathbf{k}\sigma}$ around the y axis, counter-clockwise

$$\begin{aligned} W_{\mathbf{k}\sigma} &\equiv e^{-i\theta_{\mathbf{k}\sigma}\tau^y} \\ &= \cos\theta_{\mathbf{k}\sigma} - i\sin\theta_{\mathbf{k}\sigma}\tau^y \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma} & -\sin\theta_{\mathbf{k}\sigma} \\ \sin\theta_{\mathbf{k}\sigma} & \cos\theta_{\mathbf{k}\sigma} \end{bmatrix} \end{aligned} \tag{E.17}$$

Eigenvalues are:

$$E_{\mathbf{k}\sigma} \equiv \sqrt{\epsilon^2_{\mathbf{k}\sigma} + |\Delta_{\mathbf{k}\sigma}|^2} \tag{E.18}$$

At any finite temperature, the ground-state of such a system is well-known: a thermal superposition of "down" spins with "up" spins. In the pseudo-spin picture, at each "site" $(\mathbf{k}\sigma)$, the lowest-energy state is the one anti-aligned with the pseudo-magnetic field. It is useful now to define the thermal factors,

$$\mathrm{Th}^{(\pm)}_{\mathbf{k}\sigma}[\beta,\mu] \equiv f\left(-E_{\mathbf{k}\sigma};\beta,\mu\right) \pm f\left(E_{\mathbf{k}\sigma};\beta,\mu\right) \tag{E.19}$$

that will appear quite often. Then it's immediate to derive the various expectation values.

$$\langle \hat{s}^x_{\mathbf{k}\sigma} \rangle = \frac{\mathrm{Re}\{\Delta_{\mathbf{k}\sigma}\}}{E_{\mathbf{k}\sigma}} \mathrm{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu] \tag{E.20}$$

$$\langle \hat{s}^y_{\mathbf{k}\sigma} \rangle = \frac{\mathrm{Im}\{\Delta_{\mathbf{k}\sigma}\}}{E_{\mathbf{k}\sigma}} \mathrm{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu] \tag{E.21}$$

$$\langle \hat{s}^z_{\mathbf{k}\sigma} \rangle = -\frac{\epsilon_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}} \mathrm{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu] \tag{E.22}$$

Note that considering the gran canonical hamiltonian $\hat{K} \equiv \hat{H} - \mu\hat{N}$ and using Eq. (E.16) the diagonalizing matrix remains the same, being

$$-\mu\hat{N} = \sum_{\mathbf{k}\in\mathrm{MBZ}} \begin{bmatrix} -\mu & \\ & -\mu \end{bmatrix} \quad \Longrightarrow \quad W_{\mathbf{k}\sigma}\left[h_{\mathbf{k}\sigma} - \mu\mathbb{I}_{2\times 2}\right] W^{\dagger}_{\mathbf{k}\sigma} = d_{\mathbf{k}\sigma} - \mu\mathbb{I}_{2\times 2} \tag{E.23}$$

thus for non-null chemical potential (i.e. finite doping) the eigenvalues are just shifted, $\pm E_{\mathbf{k}} - \mu$. In other words, the new bands retain the same chemical potential of the bare bands. Notice also that the presence of an absolute value makes the eigenvalues independent of the σ index. Eigenvector fermion operators are obtained simply as:

$$\hat{\Gamma}_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma}\hat{\Psi}_{\mathbf{k}\sigma} \tag{E.24}$$

The $\hat{\Gamma}$ spinor operators are effectively free fermionic spinor fields and as such must be treated. Explicitly,

$$\begin{aligned}\hat{\Gamma}_{\mathbf{k}\sigma} &= \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma} & -\sin\theta_{\mathbf{k}\sigma} \\ \sin\theta_{\mathbf{k}\sigma} & \cos\theta_{\mathbf{k}\sigma} \end{bmatrix} \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} - \sin\theta_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \\ \sin\theta_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} + \cos\theta_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix} \equiv \begin{bmatrix} \hat{\gamma}^{(+)}_{\mathbf{k}\sigma} \\ \hat{\gamma}^{(-)}_{\mathbf{k}\sigma} \end{bmatrix}\end{aligned} \tag{E.25}$$

The consequence of Eq. (E.23) is interesting by itself: within this HF solution, the gap opens always at $E = 0$. In other words, the bands sketched in Fig. E.2b retain the same geometric features regardless of the doping, what changes is the bands width and the gap between them. The very key point here is that, being the two bands symmetric (thus containing each half of the electronic states) and defined over the MBZ, any finite doping $\delta > 0$ will produce a metallic behaviour, making chemical potential crossing the upper band. It is widely observed and confirmed by more refined numerical procedures, however, that AF establishes an insulating phase – here only reproduced at half-filling, where chemical potential lies within the gap. As will be detailed in D.2, the commensurate nature of AF SSB at wavevector $\boldsymbol{\pi}$ of this self-consistent solution is highly unstable and atypical at finite doping [1].

E.1.4 Explicit zero-temperature AF ground state at half-filling

The AF ground state can be expressed easily at $n = 1/2$ and $\beta \to +\infty$. Let $|\Uparrow_{\mathbf{k}\sigma}\rangle$ and $|\Downarrow_{\mathbf{k}\sigma}\rangle$ be respectively the "up" and "down" $\mathbf{k}$-th pseudo-spin states. Let $|\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle$ be the "down" state in the tilted frame. The ground state is given by

$$|\Psi\rangle \equiv \bigotimes_{\sigma} \bigotimes_{\mathbf{k}\in\mathrm{MBZ}} |\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle \tag{E.26}$$

We can link the "down" states via an inverse rotation,

$$\begin{aligned}|\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle &= e^{-i\theta_{\mathbf{k}\sigma}\hat{s}^{y}_{\mathbf{k}\sigma}}|\Downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos\theta_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle - i\sin\theta_{\mathbf{k}\sigma}\hat{s}^{y}_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos\theta_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle - \sin\theta_{\mathbf{k}\sigma}(\hat{s}^{+}_{\mathbf{k}\sigma} - \hat{s}^{-}_{\mathbf{k}\sigma})|\Downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos\theta_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle - \sin\theta_{\mathbf{k}\sigma}|\Uparrow_{\mathbf{k}\sigma}\rangle\end{aligned} \tag{E.27}$$

having we used

$$\hat{s}^{\pm}_{\mathbf{k}\sigma} \equiv \frac{\hat{s}^{x}_{\mathbf{k}\sigma} \pm i\hat{s}^{y}_{\mathbf{k}\sigma}}{2}$$

Use now Eqns. (E.16) and (E.15),

$$\frac{\hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}^{z}_{\mathbf{k}\sigma}}{2} = \hat{n}_{\mathbf{k}\sigma} \qquad \frac{\hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}^{z}_{\mathbf{k}\sigma}}{2} = \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}$$

Evaluating the above operators on the "up" and "down" states, we get immediately

$$\frac{1}{2}\langle\Uparrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma}+\hat{s}^z_{\mathbf{k}\sigma}|\Uparrow_{\mathbf{k}\sigma}\rangle = 1 \qquad \frac{1}{2}\langle\Uparrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma}-\hat{s}^z_{\mathbf{k}\sigma}|\Uparrow_{\mathbf{k}\sigma}\rangle = 0$$
$$\frac{1}{2}\langle\Downarrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma}+\hat{s}^z_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle = 0 \qquad \frac{1}{2}\langle\Downarrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma}-\hat{s}^z_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle = 1$$

Thus, up to a phase

$$|\Uparrow_{\mathbf{k}\sigma}\rangle \sim \hat{c}^\dagger_{\mathbf{k}\sigma}|\Omega_{\mathbf{k}\sigma}\rangle \qquad |\Downarrow_{\mathbf{k}\sigma}\rangle \sim \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}|\Omega_{\mathbf{k}\sigma}\rangle$$

being $|\Omega_{\mathbf{k}\sigma}\rangle$ the $(\mathbf{k},\sigma)$ Hilbert subspace vacuum.

It follows that composing Eqns. (E.26) and (E.27) the ground state $|\Psi\rangle$ can be written as:

$$|\Psi\rangle = \bigotimes_{\sigma}\bigotimes_{\mathbf{k}\in\mathrm{MBZ}}\left[\sin\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}\sigma} + e^{2i\zeta_{\mathbf{k}\sigma}}\cos\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]|\Omega_{\mathbf{k}\sigma}\rangle \tag{E.28}$$

having we absorbed in the relative phase $e^{2i\zeta_{\mathbf{k}\sigma}}$ the two states phases differences as well as the original sign difference of Eq. (E.27). Now, inserting the explicit form of the ground state from equation above:

$$\begin{aligned}\langle\Downarrow_{\mathbf{k}\sigma}|\hat{\psi}_{\mathbf{k}\sigma}+\hat{\psi}^\dagger_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle &= \left(e^{2i\zeta_{\mathbf{k}\sigma}}+e^{-2i\zeta_{\mathbf{k}\sigma}}\right)\sin\theta_{\mathbf{k}\sigma}\cos\theta_{\mathbf{k}\sigma}\\ &= \cos 2\zeta_{\mathbf{k}\sigma}\sin 2\theta_{\mathbf{k}\sigma}\end{aligned}$$

but from Eq. (E.13)

$$\begin{aligned}\langle\Downarrow_{\mathbf{k}\sigma}|\hat{\psi}_{\mathbf{k}\sigma}+\hat{\psi}^\dagger_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle &= \langle\Downarrow_{\mathbf{k}\sigma}|\hat{s}^x_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle\\ &= \sin 2\theta_{\mathbf{k}\sigma}\end{aligned}$$

since the pseudofield has no y component. This implies necessarily

$$\cos 2\zeta_{\mathbf{k}\sigma} \quad\Longrightarrow\quad \zeta_{\mathbf{k}\sigma} = k\pi \quad \text{with} \quad k\in\mathbb{Z}$$

and lets us conclude from Eq. (E.28)

$$|\Psi\rangle = \bigotimes_{\sigma}\bigotimes_{\mathbf{k}\in\mathrm{MBZ}}\left[\sin\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}\sigma} + \cos\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]|\Omega_{\mathbf{k}\sigma}\rangle \tag{E.29}$$

For low-temperature physics, we can expect the ground-state density matrix to be highly dominated by this BCS-like zero-temperature ground-state, which highlights the essential features of the system.

Finally, using Eq. (E.25), we check that correctly the ground-state describes a perfect degenerate $\gamma^{(-)}$ quasiparticles Fermi gas,

$$|\Psi\rangle = \bigotimes_{\sigma}\bigotimes_{\mathbf{k}\in\mathrm{MBZ}}\gamma^{(-)\dagger}_{\mathbf{k}\sigma}|\Omega_{\mathbf{k}\sigma}\rangle \tag{E.30}$$

The AF phase can be thought of as a long-range coordination between fermionic states related by a $\boldsymbol{\pi}$ shift in reciprocal space, producing a new free gas where all interactions have been "condensed" in these AF pairs.

E.1.5 Spin degeneracy removal

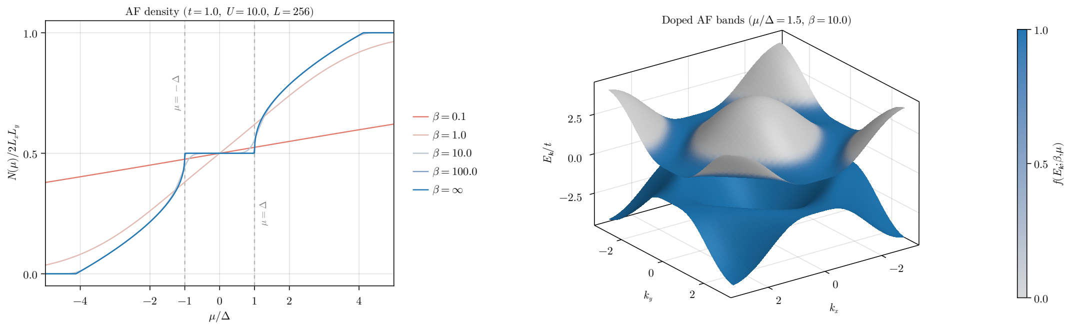
Up to now, we wrote explicitly the σ index. However,

$$\epsilon_{\mathbf{k}\uparrow} = \epsilon_{\mathbf{k}\downarrow} \qquad \Delta_{\mathbf{k}\uparrow} = -\Delta_{\mathbf{k}\downarrow}$$

thus $E_{\mathbf{k}\uparrow} = E_{\mathbf{k}\downarrow}$. Since the gap function sign does has not a measurable impact, we might as well take the spin index as a pure degeneracy and from now on work in the $\sigma=\uparrow$ sector, dropping the σ index. We can further simplify by dropping all indices from the gap function,

$$\Delta_{\mathbf{k}\uparrow} = mU \equiv \Delta$$

being the latter purely s-wave. This is a property of the pure HM: for the EHM, the gap acquires a non-trivial topological structure.



a | Electronic density. b | Example doped bands.

Figure E.2 | Left panel: electronic density in the AF phase as a function of the chemical potential, measured in relation to the gap $\Delta = mU$, at various temperatures. Right panel: example doped bands, with $\mu/\Delta = 1.5$ set arbitrarily and temperature set to $\beta = 10$ to enhance smearing effects around FS. Single-state filling is proportional to color magnitude.

E.1.6 Chemical potential and system density

Using the fact that

$$\left\langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \hat{\Gamma}_{\mathbf{k}\sigma} \right\rangle = \left\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} \right\rangle$$

due to the $W_{\mathbf{k}\sigma}$ unitarity properties, the total number of electrons is given by (also recalling Eq. (E.9))

$$\begin{aligned} N &= \sum_{\mathbf{k}\sigma} \langle \hat{n}_{\mathbf{k}\sigma} \rangle \\ &= \sum_{\sigma} \sum_{\mathbf{k}\in\text{MBZ}} \langle \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \\ &= \sum_{\sigma} \sum_{\mathbf{k}\in\text{MBZ}} \left\langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \hat{\Gamma}_{\mathbf{k}\sigma} \right\rangle \\ &= 2 \sum_{\mathbf{k}\in\text{MBZ}} \text{Th}^{(+)}_{\mathbf{k}}[\beta,\mu] \end{aligned} \tag{E.31}$$

being f be the Fermi-Dirac distribution at inverse temperature β and chemical potential μ,

$$f(\epsilon;\beta,\mu) = \frac{1}{e^{\beta(\epsilon-\mu)} + 1}$$

The 2 prefactor appeared because of spin degeneracy. Since the gapped bands refer to the smaller MBZ, thus exhibit the periodicity

$$E_{\mathbf{k}} = E_{\mathbf{k}+\boldsymbol{\pi}}$$

it holds equivalently

$$N = \sum_{\mathbf{k}\in\text{BZ}} \text{Th}^{(+)}_{\mathbf{k}}[\beta,\mu] \tag{E.32}$$

as is obvious. The 2 prefactor was absorbed in doubling the integration region, MBZ $\to$ BZ. Next sections are devoted to simplify the above theoretical results in order to obtain an algorithmic estimation for the magnetization m, which is just the AF instability order parameter.

In Fig. E.2a the density $n(\mu)$ is plotted as a function of μ. As is evident, at nearly zero temperature ($\beta \to \infty$) the system is half filled for $-\Delta < \mu < \Delta$. In order to obtain a finite doping, it must be $|\mu| > \Delta$; which means, the chemical potential needs to cross the electronic bands. This HF solution leads invariably to a rather unusual metallic AF at any finite doping.

E.1.7 Self-consistency gap equation and magnetization

A convergence algorithm can be designed to find the Hartree-Fock solution to the model. Ultimately, we aim to extract m self-consistently. Combining Eqns. (E.7), (E.8) and the definition of Eq. (E.3), we get

$$\frac{1}{2L_xL_y}\sum_{\mathbf{r}}(-1)^{x+y}\hat{n}_{\mathbf{r}\sigma} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left[\hat{\psi}_{\mathbf{k}\sigma}+\hat{\psi}^{\dagger}_{\mathbf{k}\sigma}\right]$$

thus employing Eq. (E.10)

$$\begin{aligned} m &= \frac{1}{2L_xL_y}\sum_{\mathbf{r}}(-1)^{x+y}\langle\hat{n}_{\mathbf{r}\uparrow}-\hat{n}_{\mathbf{r}\downarrow}\rangle \\ &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left\langle\left(\hat{\psi}_{\mathbf{k}\uparrow}+\hat{\psi}^{\dagger}_{\mathbf{k}\uparrow}\right)-\left(\hat{\psi}_{\mathbf{k}\downarrow}+\hat{\psi}^{\dagger}_{\mathbf{k}\downarrow}\right)\right\rangle \end{aligned}$$

Using now Eq. (E.13) we get the self-consistency equation for the HFP m,

$$\begin{aligned} m &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left[\langle\hat{s}^x_{\mathbf{k}\uparrow}\rangle-\langle\hat{s}^x_{\mathbf{k}\downarrow}\rangle\right] \\ &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\mathrm{Re}\{\Delta_{\mathbf{k}\uparrow}\}-\mathrm{Re}\{\Delta_{\mathbf{k}\downarrow}\}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta,\mu] \\ &= \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\Delta}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta,\mu] \end{aligned} \tag{E.33}$$

The magnetization m is the physical order parameter of the phase transition from the normal phase to the antiferromagnetic one. Within such phase transition, in fact, it is the expectation value

$$\langle\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle$$

that goes from zero (normal phase, free Fermi gas) to a non-zero value (AF). As is evident from equations above, m is just its (only) s-wave component.

E.2 (Impossibility of a stable) superconducting phase

Let us now move to the discussion of superconductivity in the HM. As for the AF phase, we use the results and analyses of this section as groundwork for the more sophisticated case of SC phase in the EHM. We anticipate here that BCS-like superconductivity cannot develop in the pure HM, due to the inherently repulsive nature of $\hat{H}_U$. BCS theory admits, for an half filled model, finite Cooper fluctuations for an infinitesimal attractive interaction. However, as will be discussed, by the means of MFT it is perfectly possible and coherent to derive a self-consistent superconducting solution to the HM. The point here is that such self-consistent solution sets to zero Cooper fluctuations.

As done for the AF phase in Sec. E.1 and following the introductory discussion of Sec. [Missing], let us start from the discussion of the phase symmetries. We aim to describe a uniform-density superconductor,

$$\langle\hat{n}_{i\sigma}\rangle = n$$

taking into account Cooper RWCs,

$$\langle\hat{c}^{\dagger}_{i\sigma}\hat{c}^{\dagger}_{i\bar{\sigma}}\rangle$$

which, having we broken $U^c(1)$ charge symmetry, we allow to fluctuate to non-zero values. For the pure HM, we know from Tab. **??** that both Hartree and Cooper pairing channels contribute to the RWCs set. We handle them separately.

E.2.1 MFT treatment of the Hartree channel

For the Hartree channel, knowing the deal with a uniform-density phase, we can re-use the derivation of Sec. E.1.1 simply by setting $m = 0$. We immediately get from Eq. (E.5):

$$\begin{aligned}\hat{H}_U &\simeq U \sum_{\mathbf{r}} \left[\hat{n}_{\mathbf{r}\uparrow} \langle \hat{n}_{\mathbf{r}\downarrow}\rangle + \langle \hat{n}_{\mathbf{r}\uparrow}\rangle \hat{n}_{\mathbf{r}\downarrow} - \langle \hat{n}_{\mathbf{r}\uparrow}\rangle \langle \hat{n}_{\mathbf{r}\downarrow}\rangle\right] + \text{(Cooper channel)} \\ &= nU \times \hat{N} - E^{(\mathrm{SC})}_{\mathrm{H}/U} + \text{(Cooper channel)} \end{aligned} \tag{E.34}$$

thus chemical potential renormalization $\tilde{\mu} \equiv \mu - nU$ is present also for the SC phase. The contraction energy comes immediately from Eq. (E.6),

$$E^{(\mathrm{SC})}_{\mathrm{H}/U} = L_x L_y \times n^2 U \tag{E.35}$$

We now move to Cooper channel.

E.2.2 MFT treatment of the Cooper channel

Recall the result of Eq. (**??**) which gave the Fourier-transformed version of the local repulsion, and perform a Cooper class Wick's contraction on it,

$$\begin{aligned}\hat{H}_U \simeq \text{(Hartree channel)} + \frac{U}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \Big[&\big\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \\ &+ \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \big\rangle - \big\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\rangle \big\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \big\rangle \Big] \end{aligned} \tag{E.36}$$

We are not breaking translational invariance, thus only Cooper fluctuations with net zero total momentum are allowed. This means only $\mathbf{K} = \mathbf{0}$ contributes. Define as in Eq. (6.3) the pairing operator

$$\hat{\phi}_{\mathbf{k}} \equiv \hat{c}_{-\mathbf{k}\downarrow} \hat{c}_{\mathbf{k}\uparrow} \tag{E.37}$$

and define $w_{\mathbf{q}}$ as in Eq. (6.7),

$$w_{\mathbf{q}} \equiv \langle \phi^\dagger_{\mathbf{q}} \rangle$$

Its s-wave part is given by:

$$w^{(s)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} w_{\mathbf{q}} \tag{E.38}$$

Let $\hat{n}_{\mathbf{k}\sigma}$ as in Eq. (E.2). Eq. (E.34) reduces to

$$\hat{H}_U \simeq \text{(Hartree channel)} + \frac{U}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \Big[\langle \hat{\phi}^\dagger_{\mathbf{q}} \rangle \hat{\phi}_{\mathbf{q}'} + \hat{\phi}^\dagger_{\mathbf{q}} \langle \hat{\phi}_{\mathbf{q}'} \rangle \Big] - E^{(\mathrm{SC})}_{\mathrm{C}/U} \tag{E.39}$$

being $E^{(\mathrm{SC})}_{\mathrm{C}/U}$ the HM Cooper contraction energy,

$$\begin{aligned} E^{(\mathrm{SC})}_{\mathrm{C}/U} &\equiv \frac{U}{L_x L_y} \sum_{\mathbf{q}\mathbf{q}'} \langle \hat{\phi}^\dagger_{\mathbf{q}} \rangle \langle \hat{\phi}_{\mathbf{q}'} \rangle \\ &= L_x L_y \times U \big| w^{(s)} \big|^2 \end{aligned} \tag{E.40}$$

Let now

$$\Delta^{(s)} \equiv -U w^{(s)}$$

We finally reduce $\hat{H}_U$ to

$$\hat{H}_U \simeq \text{(Hartree channel)} - \sum_{\mathbf{q}\in\mathrm{BZ}} \Big[\Delta^{(s)} \hat{\phi}_{\mathbf{q}} + \Delta^{(s)*} \hat{\phi}^\dagger_{\mathbf{q}} \Big] - E^{(\mathrm{SC})}_{\mathrm{C}/U} \tag{E.41}$$

which can be easily mapped on a pseudo-spin model.

Composing Eqns. (E.34) and (E.41), we get the full MFT hamiltonian

$$\hat{H} \simeq \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}\sigma} \left[\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} - \hat{c}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right] - \sum_{\mathbf{q}\in\mathrm{BZ}} \left[\Delta^{(s)}\hat{\phi}_{\mathbf{q}} + \Delta^{(s)*}\hat{\phi}^{\dagger}_{\mathbf{q}}\right] + nU \times \hat{N} - \left[E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U}\right] \tag{E.42}$$

E.2.3 Nambu representation of the SC phase and diagonalization

We now proceed precisely as in Sec. E.1.3. Define the SC Nambu spinor

$$\hat{\Phi}_{\mathbf{k}} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}^{\dagger}_{-\mathbf{k}\downarrow} \end{bmatrix}$$

Define the pseudo-spin operator:

$$\hat{s}^{\alpha}_{\mathbf{k}} \equiv \hat{\Phi}^{\dagger}_{\mathbf{k}}\tau^{\alpha}\hat{\Phi}_{\mathbf{k}}$$

and the shifted bare bands,

$$\xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}$$

Notice we are using the chemical potential RMP. We employ from the start the spin-independency of the bare bands, as well as their pure s^*-wave structure, since no Fock interaction in the present context chan change that. The following chain holds:

$$\begin{aligned} \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow} &= \hat{c}^{\dagger}_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} + \hat{c}^{\dagger}_{-\mathbf{k}\downarrow}\hat{c}_{-\mathbf{k}\downarrow} \\ &= \hat{c}^{\dagger}_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^{\dagger}_{-\mathbf{k}\downarrow} + \hat{\mathbb{I}}_{\mathbf{k}} \\ &= \hat{\Phi}^{\dagger}_{\mathbf{k}}\tau^{z}\hat{\Phi}_{\mathbf{k}} + \hat{\mathbb{I}}_{\mathbf{k}} \\ &= \hat{s}^{z}_{\mathbf{k}} + \hat{\mathbb{I}}_{\mathbf{k}} \end{aligned} \tag{E.43}$$

Then, considering the kinetic part of the gran-canonical hamiltonian,

$$\begin{aligned} \hat{H}_t - \tilde{\mu}\hat{N} &= \sum_{\sigma}\sum_{\mathbf{k}\in\mathrm{BZ}} \left(\epsilon_{\mathbf{k}\sigma} - \tilde{\mu}\right)\hat{n}_{\mathbf{k}\sigma} \\ &= \sum_{\mathbf{k}\in\mathrm{BZ}} \left[\xi_{\mathbf{k}}\hat{n}_{\mathbf{k}\uparrow} + \xi_{-\mathbf{k}}\hat{n}_{-\mathbf{k}\downarrow}\right] \\ &= \sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}}\hat{s}^{z}_{\mathbf{k}} + E^{(\mathrm{SC})}_{\mathrm{a}} \end{aligned}$$

being $E^{(\mathrm{SC})}_{\mathrm{a}}$ the "anticommutation energy" coming from the kinetic term,

$$E^{(\mathrm{SC})}_{\mathrm{a}} \equiv \sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}} \tag{E.44}$$

The scalar term in the above equation is essential for estimating correctly the phase ground state free energy density. The full MFT gran-canonical hamiltonian in Nambu representation is obtained including Eq. (E.39),

$$\hat{H} - \mu\hat{N} \simeq \sum_{\mathbf{k}\in\mathrm{BZ}}\sum_{\sigma} \hat{\Phi}^{\dagger}_{\mathbf{k}}h_{\mathbf{k}}\hat{\Phi}_{\mathbf{k}} + E^{(\mathrm{SC})}_{\mathrm{a}} - \left[E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U}\right] \tag{E.45}$$

(note the non-renormalized chemical potential on the left) being

$$h_{\mathbf{k}} \equiv \begin{bmatrix} \xi_{\mathbf{k}} & -\Delta^{*}_{\mathbf{k}} \\ -\Delta_{\mathbf{k}} & -\xi_{\mathbf{k}} \end{bmatrix}$$

while the off-diagonal gap is a constant,

$$\Delta_{\mathbf{k}} \equiv \Delta^{(s)}$$

This hamiltonian behaves as an ensemble of spins in local magnetic pseudofields

$$\mathbf{b_k} \equiv \begin{bmatrix} -\operatorname{Re}\{\Delta_\mathbf{k}\} \\ -\operatorname{Im}\{\Delta_\mathbf{k}\} \\ \xi_\mathbf{k} \end{bmatrix} \qquad \tan 2\theta_\mathbf{k} \equiv \frac{\xi_\mathbf{k}}{\sqrt{\xi_\mathbf{k}^2 + |\Delta_\mathbf{k}|^2}} \qquad \tan 2\zeta_\mathbf{k} \equiv \frac{\operatorname{Im}\{\Delta_\mathbf{k}\}}{\operatorname{Re}\{\Delta_\mathbf{k}\}} \tag{E.46}$$

precisely as was for the AF phase, with the essential difference that now the chemical potential RMP lies inside the z component of the pseudofield itself. This is a consequence of the Nambu spinors definition. We have the spin hamiltonian

$$\hat{H} - \mu\hat{N} \simeq \sum_{\mathbf{k}\in \mathrm{BZ}} \mathbf{b_k}\cdot\hat{\mathbf{s}}_\mathbf{k} + E_\mathrm{a}^{(\mathrm{SC})} - \left[E_{\mathrm{H}/U}^{(\mathrm{SC})} + E_{\mathrm{C}/U}^{(\mathrm{SC})}\right] \quad \text{where} \quad \hat{\mathbf{s}}_\mathbf{k} = \begin{bmatrix}\hat{s}_\mathbf{k}^x \\ \hat{s}_\mathbf{k}^y \\ \hat{s}_\mathbf{k}^z\end{bmatrix} \tag{E.47}$$

As was for the AF phase, to diagonalize the hamiltonian means essentially to align the spin z axis with the direction of the pseudofield. Then, we diagonalize the hamiltonian via a set of rotations similarly to Eq. (E.17). The main difference is that in general the pseudofield can have a y component, thus the diagonalizing matrix is made of a sequence of two rotations, both counter-clockwise, the first around the z axis by an angle $2\zeta_\mathbf{k}$ to place the pseudofield in the xz plane, the second around the new y axis by an angle $2\theta_\mathbf{k}$ to align the z axis with the pseudofield,

$$\begin{aligned} W_\mathbf{k} &\equiv e^{-i\theta_\mathbf{k}\tau^y} e^{-i\zeta_\mathbf{k}\tau^z} \\ &= \begin{bmatrix}\cos\theta_\mathbf{k} & -\sin\theta_\mathbf{k} \\ \sin\theta_\mathbf{k} & \cos\theta_\mathbf{k}\end{bmatrix}\begin{bmatrix} e^{-i\zeta_\mathbf{k}} & \\ & e^{i\zeta_\mathbf{k}}\end{bmatrix} \\ &= \begin{bmatrix}\cos\theta_\mathbf{k} e^{-i\zeta_\mathbf{k}} & -\sin\theta_\mathbf{k} e^{i\zeta_\mathbf{k}} \\ \sin\theta_\mathbf{k} e^{-i\zeta_\mathbf{k}} & \cos\theta_\mathbf{k} e^{i\zeta_\mathbf{k}}\end{bmatrix} \end{aligned} \tag{E.48}$$

obtaining the diagonal matrix

$$d_\mathbf{k} \equiv W_\mathbf{k} h_\mathbf{k} W_\mathbf{k}^\dagger \quad \text{where} \quad d_\mathbf{k} \equiv \begin{bmatrix} E_\mathbf{k} & \\ & -E_\mathbf{k}\end{bmatrix}$$

and the bands

$$E_\mathbf{k} \equiv \sqrt{\xi_\mathbf{k}^2 + |\Delta_\mathbf{k}|^2}$$

Finally, the Nambu spinor for Bogoliubov quasiparticles is given similarly to Eq. (E.24)

$$\hat{\Gamma}_\mathbf{k} = W_\mathbf{k}\hat{\Phi}_\mathbf{k} \tag{E.49}$$

Explicitly,

$$\begin{aligned} \hat{\Gamma}_\mathbf{k} &= \begin{bmatrix}\cos\theta_\mathbf{k} e^{-i\zeta_\mathbf{k}} & -\sin\theta_\mathbf{k} e^{i\zeta_\mathbf{k}} \\ \sin\theta_\mathbf{k} e^{-i\zeta_\mathbf{k}} & \cos\theta_\mathbf{k} e^{i\zeta_\mathbf{k}}\end{bmatrix}\begin{bmatrix}\hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}_{-\mathbf{k}\downarrow}^\dagger\end{bmatrix} \\ &= \begin{bmatrix}\cos\theta_\mathbf{k} e^{-i\zeta_\mathbf{k}}\hat{c}_{\mathbf{k}\uparrow} - \sin\theta_\mathbf{k} e^{i\zeta_\mathbf{k}}\hat{c}_{-\mathbf{k}\downarrow}^\dagger \\ \sin\theta_\mathbf{k} e^{-i\zeta_\mathbf{k}}\hat{c}_{\mathbf{k}\uparrow} + \cos\theta_\mathbf{k} e^{i\zeta_\mathbf{k}}\hat{c}_{-\mathbf{k}\downarrow}^\dagger\end{bmatrix} \equiv \begin{bmatrix}\hat{\gamma}_\mathbf{k}^{(+)} \\ \hat{\gamma}_\mathbf{k}^{(-)}\end{bmatrix} \end{aligned} \tag{E.50}$$

These operators act as free fermion fields on the Bogoliubov bands $\pm E_\mathbf{k}$.

E.2.4 Explicit zero-temperature SC ground state at half-filling

Reasoning precisely as in Sec. E.1.4, we can extract the half-filling zero temperature ground state, commonly referred to as the "BCS state". Eqns. (E.26) and (E.27) hold equivalently (ignoring the σ index and extending momentum tensor product to the entire BZ),

$$|\Psi\rangle = \bigotimes_{\mathbf{k}\in\mathrm{BZ}} \left[\cos\theta_\mathbf{k}\,|\Downarrow_\mathbf{k}\rangle - \sin\theta_\mathbf{k}\,|\Uparrow_\mathbf{k}\rangle\right]$$

Using now Eq. (E.43), we have

$$\langle\Downarrow_\mathbf{k}|\hat{s}_\mathbf{k}^z + \hat{\mathbb{I}}_\mathbf{k}|\Downarrow_\mathbf{k}\rangle = 0$$
$$\langle\Uparrow_\mathbf{k}|\hat{s}_\mathbf{k}^z + \hat{\mathbb{I}}_\mathbf{k}|\Uparrow_\mathbf{k}\rangle = 2$$

so up to a phase

$$|\Downarrow_{\mathbf{k}}\rangle \sim |\Omega_{\mathbf{k}}\rangle \qquad |\Uparrow_{\mathbf{k}}\rangle \sim \hat{\phi}^{\dagger}_{\mathbf{k}}\,|\Omega_{\mathbf{k}}\rangle$$

Hence, we have

$$|\Psi\rangle = \bigotimes_{\mathbf{k}\in\mathrm{BZ}} \left[\cos\theta_{\mathbf{k}} + e^{2i\zeta_{\mathbf{k}}}\sin\theta_{\mathbf{k}}\hat{\phi}^{\dagger}_{\mathbf{k}}\right]|\Omega_{\mathbf{k}}\rangle \tag{E.51}$$

The argument $2\zeta_{\mathbf{k}}$ of the phase factor is precisely the phase of the gap function, as can be shown easily.

E.2.5 Superconducting fluctuations suppression in the conventional HM

Let us report the SC analogous of the AF Eqns. (E.13) and (E.14):

$$\hat{s}^{x}_{\mathbf{k}} = \hat{\phi}_{\mathbf{k}} + \hat{\phi}^{\dagger}_{\mathbf{k}} \tag{E.52}$$

$$\hat{s}^{y}_{\mathbf{k}} = i\left(\hat{\phi}_{\mathbf{k}} - \hat{\phi}^{\dagger}_{\mathbf{k}}\right) \tag{E.53}$$

while the analogous of Eq. (E.15) was given in Eq. (E.43). Let also the thermal factor

$$\mathrm{Th}^{(\pm)}_{\mathbf{k}}[\beta] \equiv f\left(-E_{\mathbf{k}};\beta,0\right) \pm f\left(E_{\mathbf{k}};\beta,0\right) \tag{E.54}$$

and the EVs:

$$\langle\hat{s}^{x}_{\mathbf{k}}\rangle = \frac{\mathrm{Re}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{E.55}$$

$$\langle\hat{s}^{y}_{\mathbf{k}}\rangle = \frac{\mathrm{Im}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{E.56}$$

$$\langle\hat{s}^{z}_{\mathbf{k}}\rangle = -\frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{E.57}$$

Up to now, the derivation was general. However, it is immediate to show that for the pure HM the SC phase can never establish. Recall Eq. (E.38): explicitly

$$\begin{aligned}
w^{(s)} &= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \langle\hat{\phi}^{\dagger}_{\mathbf{k}}\rangle \\
&= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \langle\hat{\Phi}^{\dagger}_{\mathbf{k}}\tau^{+}\hat{\Phi}_{\mathbf{k}}\rangle \\
&= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\langle\hat{s}^{x}_{\mathbf{k}}\rangle + i\,\langle\hat{s}^{y}_{\mathbf{k}}\rangle}{2} \\
&= \frac{1}{2L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\Delta_{\mathbf{k}}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta]
\end{aligned} \tag{E.58}$$

Now: $\Delta_{\mathbf{k}} = \Delta^{(s)} = -Uw^{(s)}$, thus if we define

$$U_{\mathrm{c}}[\beta] \equiv \left[\frac{1}{2L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta]}{E_{\mathbf{k}}}\right]^{-1} \tag{E.59}$$

(the reason for this notation will be clear in Sec. [Missing]) we end with the equation:

$$w^{(s)} = -\frac{U}{U_{\mathrm{c}}[\beta]}w^{(s)}$$

Since evidently $U_{\mathrm{c}}[\beta] > 0$ by definition (the thermal factor is negative by construction), the above equation can be solved in $\mathbb{C}$ only by $w^{(s)} = 0$. This implication terminates the argument: superconductivity cannot establish self-consistently at this level in the HM.

Appendix F

Mixing optimization in HM model

This short appendix is meant to give a practical example on a topic we often referenced to in text: how optimization of g, the mixing parameter defined in Sec. 2.3.2 and given operatively in Eq. (2.21) as

$$\mathbf{v}_{i+1} \equiv g\mathrm{A}\left(\mathbf{v}_i\right) + (1-g)\mathbf{v}_i$$

can be relevant in order to ensure convergence of an iterative HF scheme as presented in Sec. **??**.

F.1 Map analysis for s-wave superconductivity in HM

As we thoroughly discussed in Sec. E.2, conventional repulsive HM cannot exhibit BCS s-wave superconductivity within a MFT general scheme. This is proven trivially at the end of Sec. E.2.5, where we end up withe the self-consistency equation

$$w^{(s)} = -\frac{U}{U_\mathrm{c}} w^{(s)} \tag{F.1}$$

only solved at $U > 0$ by $w^{(s)} = 0$. Now, our aim is the following: given any initializer $w_0^{(s)}$, our iterative HF algorithm should give

$$w_0^{(s)} \quad \overset{\mathrm{Alg}}{\to} \quad \cdots \quad \overset{\mathrm{Alg}}{\to} \quad \sim 0$$

The point is the following: is this behaviour possible at any model parametrization?

Consider the nonlinear map of Eq. (F.1) as "seen by the algorithm": at step i, we compute the left side by using the value of $w^{(s)}$ at step $i-1$ (also for computing $U_\mathrm{c}[\beta]$). Let us drop the superscript (s) for the sake of readability. Seen this way, we are effectively moving along the numeric succession

$$w_{i+1} = -g\left(\frac{U}{U_\mathrm{c}[\beta]}\right)_i w_i + (1-g)w_i$$

the subscript indicating iteration. In previous equation we made use of Eq. (2.21) to generate the $i+1$-th step from step i. Let now:

$$U_\mathrm{s} \equiv \left(\frac{1}{g} - 1\right) U_\mathrm{c}$$
$$U_\mathrm{h} \equiv \left(\frac{2}{g} - 1\right) U_\mathrm{c}$$

the subscript indicating, for a reason clear soon, a "soft boundary" (first) and an "hard boundary" (second). Let us derive some trivial facts true whenever $w_i > 0$:

1. It always holds $w_{i+1} \leq w_i$. This is simple to see: if $w_{i+1} > w_i$,

$$-g\left(\frac{U}{U_\mathrm{c}[\beta]}\right)_i w_i + (1-g)w_i > w_i$$
$$-g\left(\frac{U}{U_\mathrm{c}[\beta]}\right)_i w_i - gw_i > 0$$

which is impossible because all involved terms are positive.

2. For $0 < U < U_\mathrm{s}$, we have $0 < w_{i+1} < w_i$. Indeed w_{i+1} is greater than zero when:

$$
\begin{aligned}
-g\left(\frac{U}{U_\mathrm{c}}\right)_i w_i + (1-g)w_i &> 0 \\
\frac{1-g}{g} &> \frac{U}{U_\mathrm{c}} \\
U_\mathrm{s} &> U
\end{aligned}
$$

3. For $U_\mathrm{s} < U < U_\mathrm{h}$, we have $-w_i < w_{i+1} < 0$. From point above, we know $U > U_\mathrm{s}$ implies $w_{i+1} < 0$ by simply inverting inequalities. Moreover,

$$
\begin{aligned}
-g\left(\frac{U}{U_\mathrm{c}}\right)_i w_i + (1-g)w_i &> -w_i \\
\frac{2-g}{g} &> \frac{U}{U_\mathrm{c}} \\
U_\mathrm{h} &> U
\end{aligned}
$$

4. For $U > U_\mathrm{h}$, we have $w_{i+1} < -w_i$. This is trivially implied by inverting inequalities in last point.

If we know U_c, for any given g we can predict the algorithmic image of a given step.

Now, consider U_c as a function of the gap Δ,

$$
U_\mathrm{c}(\Delta) = \left[\frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\frac{1}{\sqrt{\xi_\mathbf{k}^2 + |\Delta|^2}}\tanh\left(\frac{\beta}{2}\sqrt{\xi_\mathbf{k}^2 + |\Delta|^2}\right)\right]^{-1}
$$

[To be continued...]

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